



UOP Phenol Process With Huels MSHP™ Process

General Operating Manual

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I. Introduction

A. Scope

UOP issues this General Operating Manual (“GOM”) to provide background process information to the engineering group of the operating company or detail design engineering contractor (“Contractor”) as an aid to the detailed design activities and development of operating procedures and practices. The GOM is not considered a part of the Project Specifications but they do provide information related to the specifications included in the Schedule A. This material is informational purposes only and is intended to augment the information provided in Project Specifications. This information is expected to be reasonably consistent with the Project Specifications; it may not represent exact consistency with the Project Specifications. Any major inconsistencies should be brought to the attention of UOP for clarification. In all situations, the information provided in the Project Specifications take precedence over information in the GOM. Accordingly, UOP assumes no responsibility for the mis-application of this information to a specific process, particularly the use of such information to prepare detailed operating instructions for a specific process.

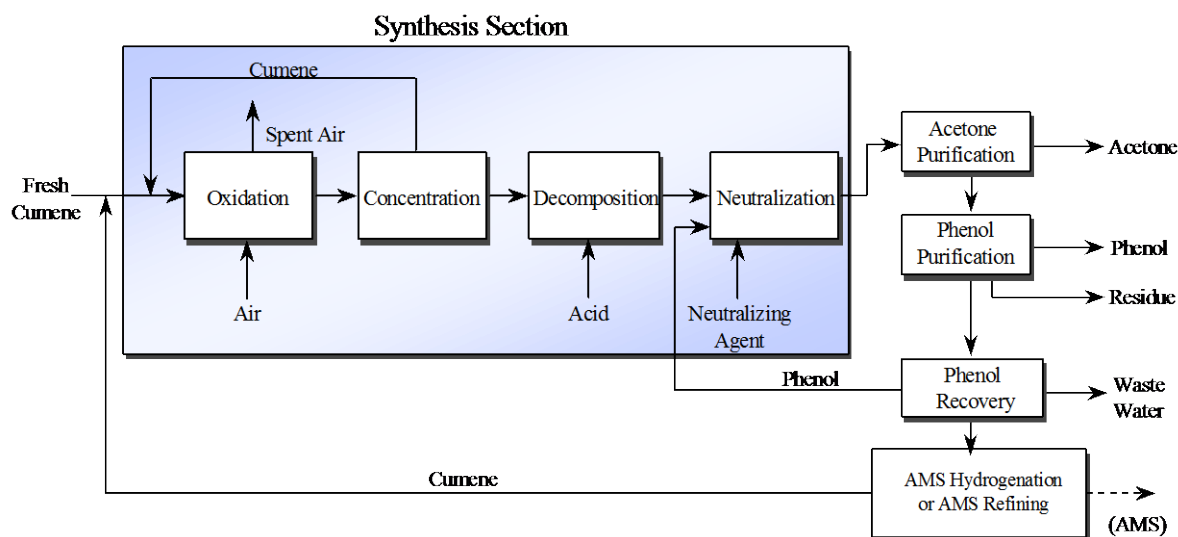
B. Purpose

The UOP Phenol Process with Huels™ MSHP Process produces phenol and acetone from cumene safely and efficiently. The UOP Phenol unit comprises of seven sections: oxidation, concentration, decomposition, neutralization, fractionation, phenol recovery and AMS hydrogenation. The first section, oxidation, auto-catalytically reacts cumene with oxygen to generate CHP. The oxygen source is ambient air. The spent air from the unit is processed to recover hydrocarbons consistent with local regulations prior to release to the environment. The second section, concentration, concentrates CHP as requires for safe and efficient decomposition. Next, in decomposition, concentrated CHP is reacted in the presence of sulfuric acid catalyst and decomposes to phenol, acetone and AMS at high yield. Following decomposition the sulfuric acid is neutralized with a special amine to avoid fouling and corrosion in downstream equipment. In fractionation the phenol and acetone products are purified to sales specifications. The phenol recovery section separates phenol from cumene and AMS and aqueous streams and recycles the phenol back to neutralization for optimum yield, and to meet local waste water regulations. The Huels MSHP™ process hydrogenates AMS to cumene for recycle back to oxidation.

C. Block Flow Diagram

Figure I-1

UOP Phenol Process



D. History

Until the advent of synthetic phenol just before World War I, natural phenol recovered from coal, tar and petroleum streams was the only source of phenol available. Since that time, synthetic phenol has increased in importance to the extent that it now accounts for 98% of the total phenol production.

The first synthetic phenol was produced by sulfonation of benzene using concentrated sulfuric acid, H_2SO_4 . The process uses large amounts of sulfuric acid (1.4:1 H_2SO_4 :Phenol) and required that the sulfuric acid concentration be held high. Large quantities of sodium hydroxide (NaOH), sodium sulfate (Na_2SO_3) or sodium carbonate (Na_2CO_3) were needed to neutralize and hydrolyze the benzene-sulfonate. Then additional sodium hydroxide was required to spring the sodium or calcium phenate.

Two competing processes with benzene sulfonation before World War II were based on chloro-benzene reaction pathway. There was both a liquid phase process and a vapor phase (regenerative) process. The liquid phase process required high pressure, 350 kg/cm²g (5000 psig) and temperature 360°C (680°F) to form the chloro-benzene. Poly-chlorinated benzene byproducts were produced as well. A silica catalyst was used to hydrolyze the chloro-benzene to phenol. The vapor phase process was also called the Raching process after its inventor. The vapor phase process was a capital intensive process since tantalum metallurgy was

required for the hot hydrochloric acid (HCl) in the presence of steam and air. The steam hydrolyzes the chloro-benzene at high temperature. Hooker Chemical made some important innovations with this process to reduce the capital cost and increase yields.

Dow invented a process for the oxidation and decomposition of toluene to manufacture phenol after World War II. The oxidation process was performed at a positive pressure in the presence of sodium carbonate (Na_2CO_3). Due to the high temperatures required in the oxidizers, the oxide yield is low compared to the cumene peroxidation route. Dow continues to use this process.

During World War II, the production of cumene expanded rapidly to fill the needs of the aviation gasoline consumption. As with most discoveries, the peroxidation route of cumene to phenol was observed by chance in the laboratories of Professor Hock in Germany during the war¹. With the end of the war, the consumption and price of cumene fell rapidly. The Hock laboratory discovery was used to develop the process in the USA and England. Other technologies based on cumene synthesis were developed independently in France and Germany. The process was first commercialized in 1952.

Allied (now Honeywell) was one of the pioneers in producing phenol. Its first plant, which came on-stream in 1954, was the second large-scale commercial plant in the world to use the cumene peroxidation technology. Eight phenol units having a combined capacity of over 910,000 metric tons per annum (MTA) were licensed by Allied prior to the merger of the technologies with UOP.

The first UOP designed unit producing phenol using the UOP Cumox Process (formerly UNISIR Cumene Oxidation Process) began operation in 1965. A total of three units with a combined capacity of 205,000 MTA were licensed by UOP prior to the merger.

The latest advance in phenol technology has come as a direct result of the merger of the Signal Companies, UOP's parent company, with the Allied Corporation. In mid-1986, Allied and UOP agreed to combine the best of their respective technologies into a single package licensed exclusively through UOP. Allied-Signal was merged with Honeywell to form the Honeywell Corporation. The phenol production facilities and technology have since been bought by Sunoco, and then resold back to Honeywell. The result is the UOP Phenol Process, which offers the highest yields in the industry. Since the merger of the two technologies, 22 projects representing over 3.4 million MTA of new phenol capacity have been licensed based on the combined UOP technology.

Of the various commercial routes to phenol, only with cumene peroxidation route is acetone produced as a co-product. The trend to phenol production via cumene

¹ Hock & Lange, Ber.dt. chem. Ges., 1944,77 257

peroxidation should continue since comparative economics among known processing routes, including some newer development processes, still show favorable advantages for the cumene route from the standpoint of cost of production, transfer price, raw material and by-product acetone market strength.

Several processes have been used for commercial production of acetone. The earliest were based on fermentation of carbon hydrates contained in materials such as kelp or molasses; these processes operated in the US until about 1963. Use of synthetic processes based on hydrocarbons or derivative began about 1930, and they now account for all US output and almost all of the output of the rest of the world.

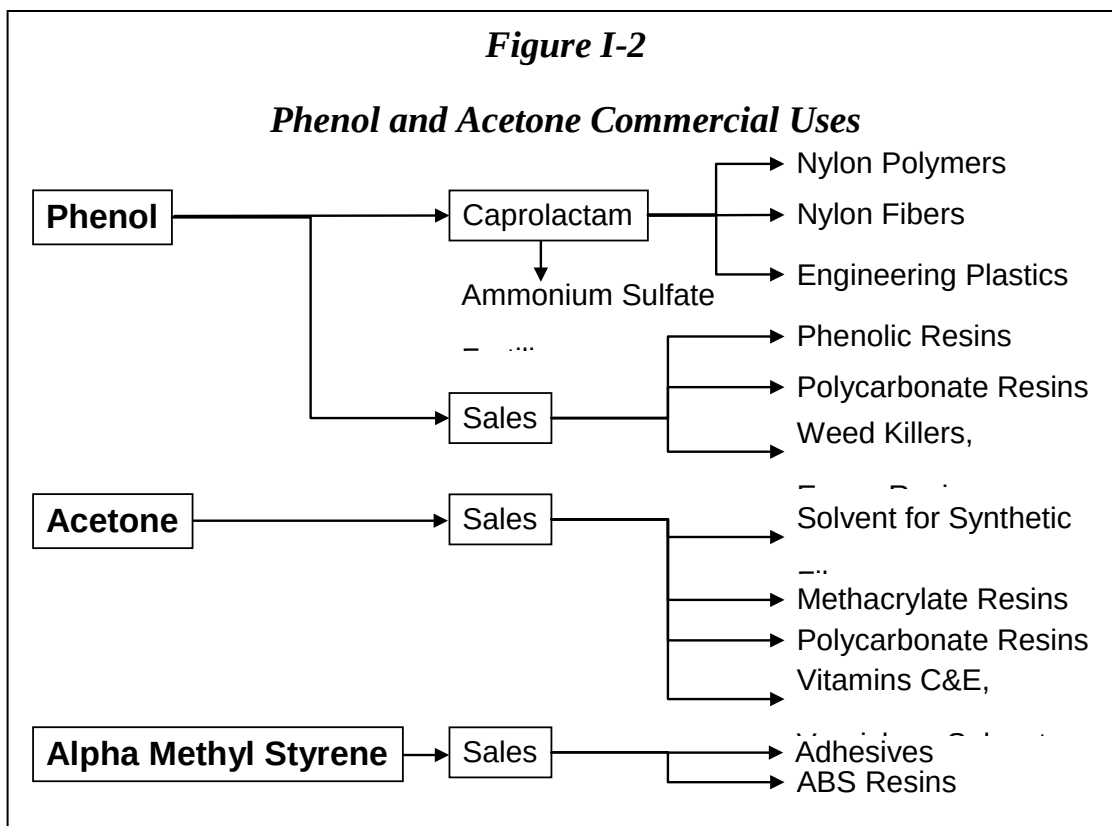
The two principal processes used in the US and elsewhere for acetone production are processes based on isopropyl alcohol (IPA) or cumene. As technology has changed, the relative importance of these two raw materials as feed for acetone has shifted greatly. The fraction of the world capacity of phenol unit derived acetone continues to increase and is a dominant factor in the price of acetone.

E. Process Overview

Phenol is an intermediate chemical that used in a wide variety of applications, such as:

- a. Phenolic and epoxy resins such as used in plywood manufacture
- b. Fibers such as nylon 6
- c. Pharmaceuticals such as salicylic acid for aspirin, antiseptics, throat sprays and cosmetic surgery
- d. Plastics such as bullet proof glass, copier/computer parts, compact discs
- e. Pesticides
- f. Lube oils

The UOP Phenol Process is a cost-effective technology for the entire range of markets, from low-purity, low-cost resin-grade phenol to high-purity phenol for caprolactam, chlorination, and epoxy-grade BPA to ultra-high purity premium-grade phenol required for poly-carbonate Bis-Phenol-A (BPA). See Figure I-2 for examples of Honeywell's end uses for its Frankford plant's phenol.



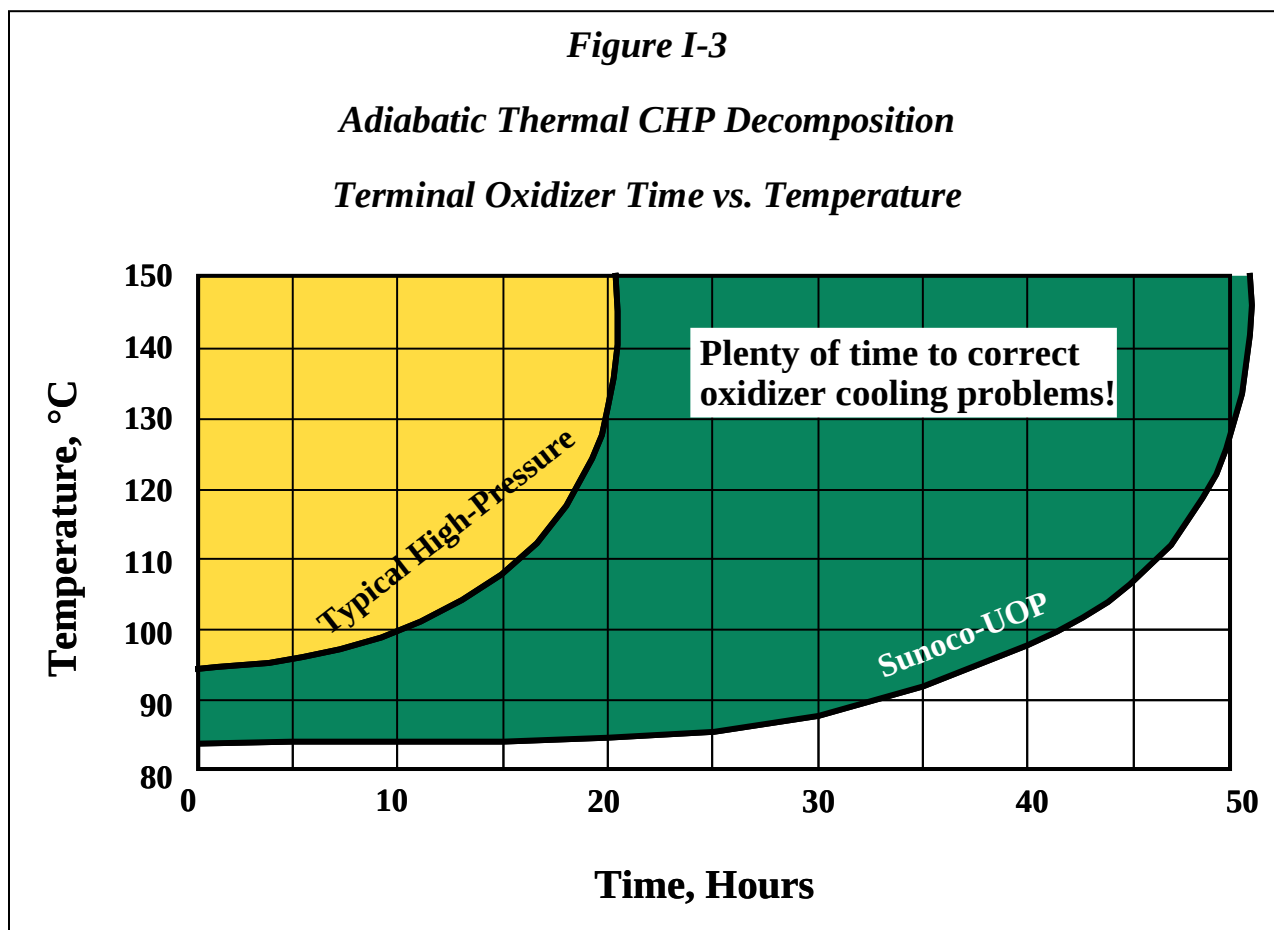
With demand projected to increase at a high rate, polycarbonate-grade BPA is the fastest growing phenol application. Polycarbonates are largely used as substitutes for glass in such applications as automobile light lenses, compact disc coatings, and electronic circuit boards. In all these applications, colorless transparency is one of the most critical properties. Thus, the phenol specification used for the production of polycarbonate-grade BPA is restricted to extremely low concentrations of those impurities that cause color in the BPA. The exact requirements depend on the BPA process used and are in most cases considered a proprietary aspect of the BPA technology.

Several innovative technologies are utilized in the UOP Phenol Process to maximize cumene efficiency. These are:

1. Low Pressure/Low Temperature Oxidation

The low pressure/low temperature oxidation results in vastly reduced by-product formation. A side benefit, but no less important is the increased safety and operability of the mild oxidizer conditions compared to other technology. As shown in Figure I-3, the low temperature oxidizer gives the user a much more time to react to an upset condition that negates the need for an automatic dump system that is prevalent with the other technologies. Also, due to the larger size of the UOP

oxidizers they are easier to operate and maintain constant CHP production and CHP concentration.



2. Low Residue Decomposition Technology

The low residue decomposition technology results in lower residue (heavies) make per ton of phenol produced, by performing the decomposition in two steps that minimize by-product formation from dimethylphenylcarbinol (DMPC). As a bonus, for those phenol producers who sell alpha-methylstyrene (AMS) as a separate product, the low residue decomposition technology produces a higher percentage of AMS per ton of phenol than the competing processes.

Figure I-1 shows the block flow diagram of the UOP Phenol Process. The numerous sections can be divided into three basic areas:

- a. Synthesis
- b. Products Purification and Recovery
- c. Effluent Treatment (Phenol Recovery)

3. Synthesis

The synthesis of phenol and acetone from cumene and air occurs in the oxidation, concentration, and decomposition sections. Synthesis follows two basic reactions:

Oxidation: Cumene + Oxygen (from Air) $\longrightarrow$ Cumene Hydroperoxide (CHP)

Decomposition: Cumene Hydroperoxide $\longrightarrow$ Phenol + Acetone

The most significant of the side reactions forms dimethylphenylcarbinol (DMPC) in the oxidizers, which dehydrates to alpha-methyl styrene (AMS) in the decomposition section.

The oxidation section synthesizes the CHP as well as treating the cumene and air feed and the spent air. The pretreatment of the cumene and air involves dilute caustic and water scrubbing. Spent air is treated to remove the volatile organic compounds before being vented to atmosphere. CHP synthesis is performed in two or more continuously stirred tank reactors (CSTR) to which cumene and air are added. CHP and other peroxides from oxidation and concentration can be found in small amounts in the unit waste water which are removed in the peroxide destruction reactor, introduced in 2013, prior to aerobic biological treatment.

Due to limitations caused by the reaction chemistry, the oxidation section cannot react the CHP to the concentration required for the decomposition reactions to take place. The concentration section safely and efficiently removes the bulk of the unreacted cumene and concentrates the CHP to the concentration needed for the decomposition section. The main pieces of equipment in the concentration section are the two deep vacuum flash columns. The rest of the equipment serves to heat, cool and safeguard the operation of the columns.

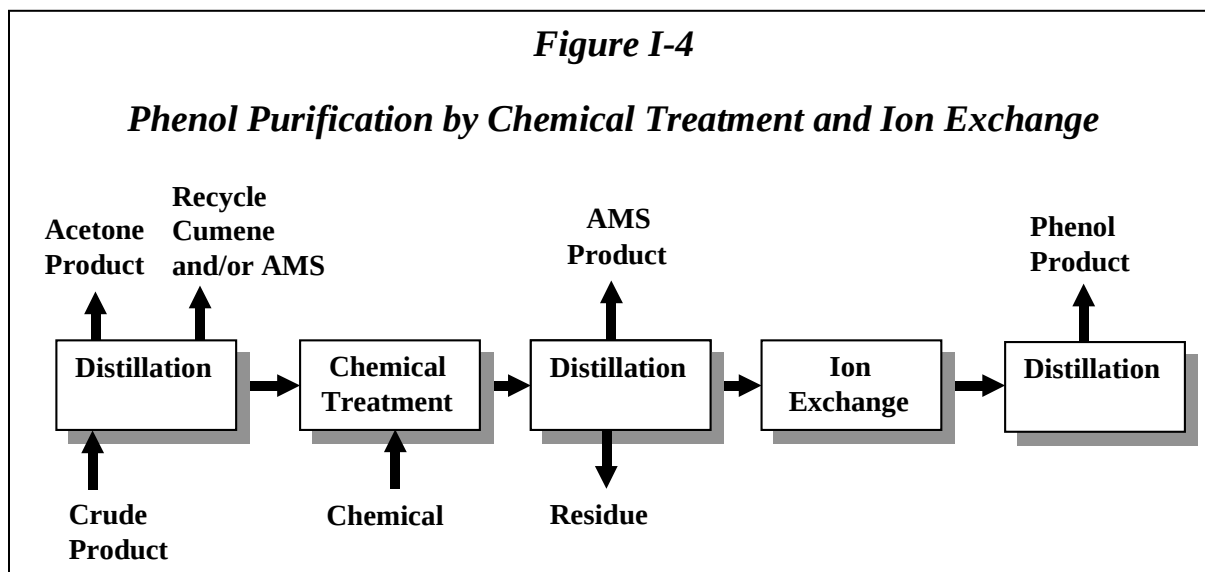
The decomposition section uses sulfuric acid, water and temperature to control the exothermic reaction. The main parts to the system are the decomposer, which acts as a surge drum, the circulation pumps, the cooler, and dehydrator. The dehydrator thermally removes water from the unreacted intermediate to produce AMS at a higher yield than possible by the acidic driven reaction.

The crude product is then treated to remove the trace sulfuric acid prior to purification and fractionation. An intermediate treatment step is required between synthesis and fractionation to minimize the corrosive environment in the fractionation section. This section is called the neutralization section. A chemical is injected to remove the sulfuric acid that catalyzes the decomposition reaction.

4. Product Purification and Recovery

Crude product purification and fractionation are integrated into one section. The acetone product is purified in a reactive distillation column. The current technology

purifies phenol in a two-step process. First, a reactive additive is added to the phenol to remove the bulk of the phenol co-boiling contaminants by forming a heavy compound that can be fractionated. A finishing step of phenol treatment with acidic ion exchange resin reacts the remaining carbonyls and methylbenzofuran to heavy compounds that can be fractionated as well as stabilizes phenol color. A block flow diagram is shown below in Figure I-4. The UOP design and operating conditions can be adjusted to achieve less than 30 wt-ppm of total GC impurities and less than 15 wt-ppm of total carbonyls as mesityl oxide.



5. Phenol Recovery

This section provides an efficient management of the plant recycle and waste streams which reduces load on the downstream aerobic wastewater treatment plant while at the same time maximizing product recovery. Phenol recovery utilizes phenol's slightly polar and acidic properties to remove phenol from cumene streams by caustic extraction followed by recovery of that phenol from the aqueous phase by contact with acid. The key to optimal performance is successful separation of the resulting salt solution from the organic phase. Additionally UOP's technology utilizes supersaturation of phenol to achieve low caustic consumption.

6. AMS Hydrogenation

When AMS is not produced, it must be first converted back to cumene and recycled back to the oxidizers. AMS is hydrogenated over a catalyst in the presence of hydrogen using the Hüls MSHP technology. The process has a very high conversion and yield and operates at relatively low temperatures. The Hüls MSHP process has the advantage of Hydrogen Once-Through (HOT) flow that negates the requirement for a recycle gas compressor.

II. Process Principles

Information from this section will serve as the foundation upon which the rest of this manual is based. This section is written to assist the reader with understanding why this process performs the way it does.

The process principles and chemistry associated with the UOP Phenol Process are complex. A basic understanding, however, of the primary reactions involved and the respective mechanisms is essential to enable operating the phenol plant in an efficient and safe manner.

No attempt is made here to present the mechanisms of the process chemistry to the degree of sophistication acceptable to a chemist specializing in the area. The intention is simply to provide practicing engineers with a basic introduction to the subject. For easy reference, and to become better acquainted with the various organic compounds encountered in the UOP process, a summary of these compounds and their main physical properties is shown in Table II-1 on the next page. The abbreviated symbols assigned to the various chemical species will be referred to throughout this manual. The molecular structure of the most important of these organic compounds is also listed in Figures II-1A, B, C, on the following pages.

Table II-1

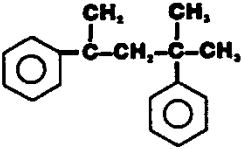
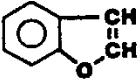
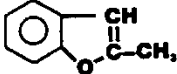
Compounds	Commonly Used Symbol	Chemical Formula	Molecular Weight	Boiling Point, °C	Melting Point, °C	SG @ 20°C	Remarks	UOP Schedule A Symbol
Acetaldehyde	ACA	CH ₃ CHO	44.05	21.0	-123.5	.783	(1)	
Acetone	ACE	(CH ₃) ₂ CO	58.08	56.2	-95.0	.7911	Co-product	
Acetophenone	AP	CH ₃ CO-C ₆ H ₅	120.16	202.0	20.0	1.0281	(2), (4), Oxidation side product	ACPH or ACP
Alphamethylstyrene	AMS	C ₆ H ₅ COCH ₃ :CH ₂	118.17	165.5		.9051	(2), (4), DMPC Dehydration	
Benzene	Bz	C ₆ H ₆	78.11	80.1	5.5	.879	(3)	
Benzofuran (Coumarone)	BF	C ₆ H ₄ OCH:CH	118.13	177.0	-18.0	1.0768	(2), PH+MO side reaction	
Butanal	BCHO	C ₃ H ₇ CHO	72.10	75.7	-99.0	.8039	(1)	
Butylbenzene	BBZ	C ₄ H ₉ C ₆ H ₅	134.21	183.2	-88.0	.8601	(3)	T.Bu.Bz
Crotonaldehyde	CCHO	CH ₃ CH:CHCHO	70.09	102.0	-69.0	.8521	Condensation of ACHO	
Cumene	CUM	C ₆ H ₅ CH(CH ₃) ₂	120.19	152.7	-97.0	.862	Plant Charge	
Cumene hydroperoxide	CHP	C ₆ H ₅ C(CH ₃) ₂ OOH	152.19		-	1.065	Synthesis base product	
Cumylphenol	CP	C ₆ H ₅ C(CH ₃) ₂ -C ₆ H ₄ OH	212.28	315.0	72.0	1.119	(2), (4), DMPC + PH side reaction	CUPH
Diacetone alcohol	DAA	CH ₃ COCH ₂ COH(CH ₃) ₂	116.16	169.0	47.0	.9395	(2), Condensation of AC	
Dimethylphenyl carbinol	DMPC	C ₆ H ₅ (CH ₃) ₂ COH	136.19	202.0	35/57.0	.972	(2), & Oxidation side product	
Formaldehyde	HCHO	HCHO	30.03	-21.0	-92.0	.815	(1)	
Formic Acid	HCOOH	HCOOH	46.03	100.8	8.3	1.2201	Oxidation side product	HCOOH or Formic
Hydroxyacetone	HOA	CH ₃ COCH ₂ OH	74.08	146.0	-17.0	1.002	(2)	Acetol
Isobutanol (sec.)	IBA	C ₂ H ₅ CHOHCH ₃	74.12	99.5	-114.7	.808	(2) PH + MO side reaction	
Isophorone	IPCO	C(O)CHC(CH ₃)CH ₂ C(CH ₃) ₂ :CH ₂	138.20	215.0	-8.0	.9917	(2), (4) AC + MO side reaction	
Isopropanol	IPA	(CH ₃) ₂ CHOH	60.09	82.3	-86.0	.7854	(2)	
Mesityloxide	MO	(CH ₃) ₂ C:CHCOCH ₃	98.14	131.4	-59.0	.8539	(2) Dehydration of AC &/or DAA	MEO or MO
Methyl alcohol, Methanol	MA	CH ₃ OH	32.04	64.5	-97.8	.7928	Oxidation side product	METH
2-methylbenzofuran	2MBF	C ₆ H ₄ OCH ₃ :CH	132.15	193.0		1.037	(2), PH + MO side reaction	
Orthocumylphenol	oCP	C ₆ H ₅ C(CH ₃) ₂ -C ₆ H ₄ OH	212.28				(2), (4), DMPC + PH side reaction	
Paracumylphenol	pCP	C ₆ H ₅ C(CH ₃) ₂ -C ₆ H ₄ OH	212.28				(2), (4), DMPC + PH side reaction	
Phenol	Ph	C ₆ H ₅ OH	94.11	182	42/43	1.072	Main Product	
Phorone	PCO	I(CH ₃) ₂ C:CHI ₂ CO	138.20	198.5	28.0	.885	(2), (4) AC + MO side reaction	
Propanal or Propionaldehyde	PA	CH ₃ CH ₂ CHO	58.08	48.8	-81.0	.807	(1)	
(1) Impurity in Acetone product	(2) Impurity in Phenol product	(3) Impurity in Cumene Charge	(4) Component of residue product					

FIGURE II-1A

<u>CHEMICAL NAME</u>	<u>MOLECULAR STRUCTURE</u>	<u>Mol.Wt.</u>	<u>SYMBOL</u>
1. CUMENE	$\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_3 \\ \\ \text{C}_6\text{H}_5 \end{array}$	120	Cu
2. CUMENE HYDROPEROXIDE	$\begin{array}{c} \text{OOH} \\ \\ \text{CH}_3-\text{C}-\text{CH}_3 \\ \\ \text{C}_6\text{H}_5 \end{array}$	152	CHP
3. DIMETHYL PHENYL CARBINOL	$\begin{array}{c} \text{OH} \\ \\ \text{CH}_3-\text{C}-\text{CH}_3 \\ \\ \text{C}_6\text{H}_5 \end{array}$	136	DMPC
4. ACETOPHENONE	$\begin{array}{c} \text{CH}_3-\text{C}=\text{O} \\ \\ \text{C}_6\text{H}_5 \end{array}$	120	ACP
5. ACETONE	$\begin{array}{c} \text{CH}_3-\text{C}-\text{CH}_3 \\ \\ \text{O} \end{array}$	58	AC
6. PHENOL	$\begin{array}{c} \text{OH} \\ \\ \text{C}_6\text{H}_5 \end{array}$	94	Ph
7. α -METHYL STYRENE	$\begin{array}{c} \text{CH}_3 \\ \\ \text{C}_6\text{H}_5-\text{C}=\text{CH}_2 \end{array}$	118	AMS
8. CUMYL PHENOL	$\begin{array}{c} \text{CH}_3 \\ \\ \text{C}_6\text{H}_5-\text{C}-\text{C}_6\text{H}_4-\text{OH} \\ \\ \text{CH}_3 \end{array}$	212	CPh

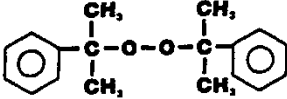
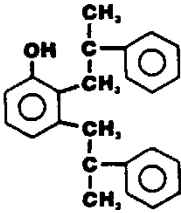
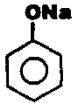
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FIGURE II-1B

<u>CHEMICAL NAME</u>	<u>MOLECULAR STRUCTURE</u>	<u>Mol.Wt.</u>	<u>SYMBOL</u>
9. AMS DIMER		236	AMS DIMER
10. α -HYDROXY ACETONE	$\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2\text{OH}$	74	HDA
11. DIACETONEALCOHOL	$\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2-\text{C}(\text{CH}_3)_2-\text{OH}$	116	DAA
12. MESITYL OXIDE	$\text{CH}_3-\text{C}(\text{CH}_3)=\text{CH}-\text{C}(=\text{O})-\text{CH}_3$	98	MEO
13. PHORONE	$\text{CH}_3-\text{C}(\text{CH}_3)=\text{CH}-\text{C}(\text{CH}_3)=\text{CH}-\text{C}(=\text{O})-\text{CH}_3$	138	PCO
14. BENZOFURAN		118	BF
15. ISOPROPYL ALCOHOL	$\text{CH}_3-\text{CH}(\text{CH}_3)-\text{OH}$	78	IPA
16. ISO BUTYL ALCOHOL	$\text{CH}_3-\text{CH}_2-\text{CH}(\text{OH})-\text{CH}_3$	74	IBA
17. METHYL BENZOFURAN		132	2MBF

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FIGURE II-1C

<u>CHEMICAL NAME</u>	<u>MOLECULAR STRUCTURE</u>	<u>Mol.Wt.</u>	<u>SYMBOL</u>
18. CROTON ALDEHYDE	$\text{CH}_3-\text{CH}=\text{CH}-\underset{\text{O}}{\underset{\parallel}{\text{C}}}\text{H}$	70	CCHO
19. DICUMYL PEROXIDE		270	DCP
20. DICUMYLPHENOL		330	DCPH
21. SODIUM HYDROXODIDE	NaOH	40	NaOH
22. SODIUM PHENATE		116	SPh
23. SULPHURIC ACID	H ₂ SO ₄	98	H ₂ SO ₄
24. ACETALDEHYDE	$\text{CH}_3-\underset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{H}$	44.05	ACHO
25. FORMALDEHYDE	$\text{H}-\underset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{H}$	3.03	HCHO
26. FORMIC ACID	H-C-OOH	46.03	HCOOH
27. METHANOL	OH, OH	32.04	MA

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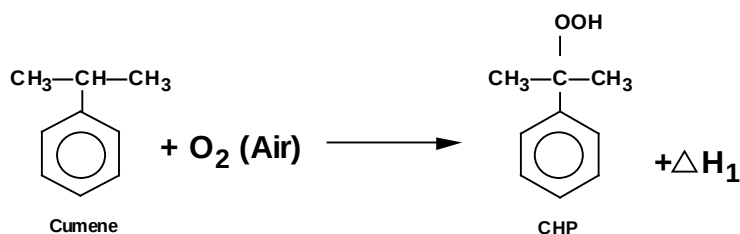
A. Oxidation

The chemistry of cumene oxidation will be familiar to anyone with an understanding of free-radical mechanisms. The objective in oxidation is to maximize the selectivity of the cumene-to-CHP reaction while minimizing by-product formation, all while maintaining the safest possible operating conditions.

The main reaction in the oxidation section and the one to optimize is, of course, the oxidation of cumene to cumene hydroperoxide. The major competing reactions result in the formation of dimethylphenylcarbinol (DMPC) and acetophenone (AP).

1. Cumene Peroxidation

The UOP Phenol Process employs, as the first step of synthesis, the low temperature liquid phase oxidation of cumene to cumene hydroperoxide (CHP). Oxygen is derived from air and the reaction is carried out in a well-mixed reactor. The primary reaction is illustrated below.



$\Delta H_1 \sim -27.7$ Kcal/Mol, Liquid Phase @ 25°C

Cumene peroxidation is a free radical chain reaction and the reaction mechanism involves three sequential steps:

- a. Initiation
- b. Propagation
- c. Termination

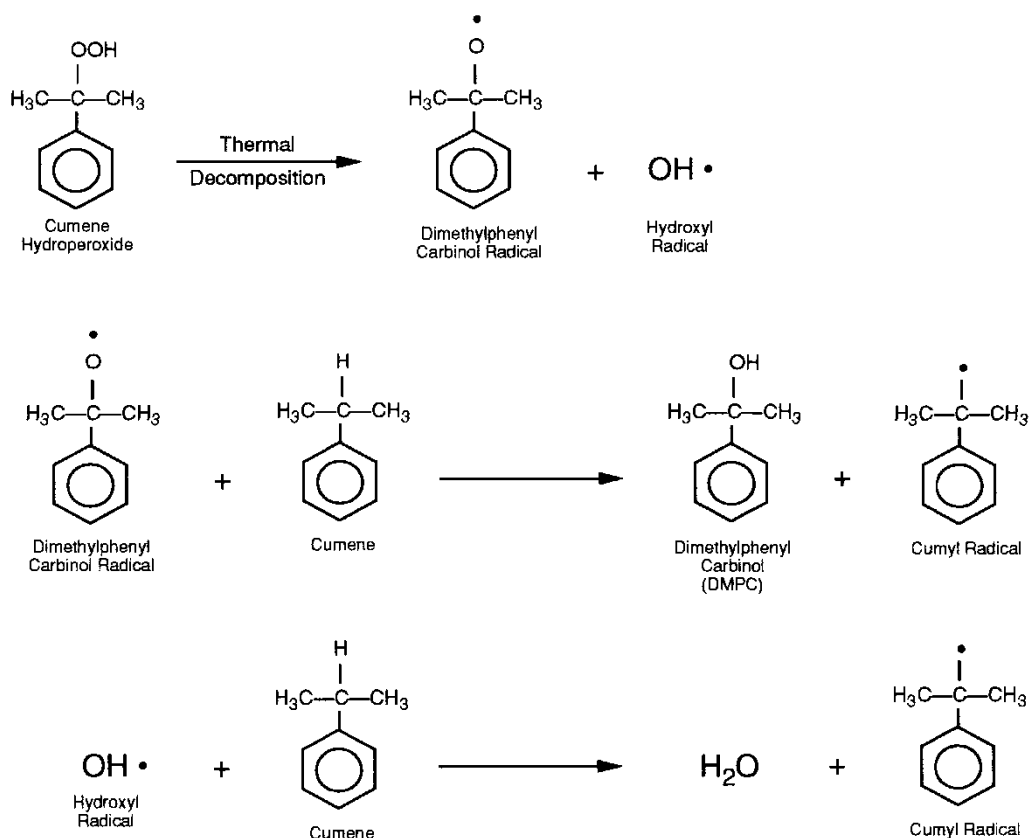
These reaction sequences are illustrated and discussed in this section.

The oxidation reaction of cumene to CHP is highly exothermic, which, if unchecked, could result in the spontaneous decomposition of CHP. In actual practice, this is prevented by close oxidizer temperature control by means of coolers located in circulation loops around the oxidizers. CHP decomposition reactions proceed simultaneously with its formation. At higher concentrations, CHP is inherently unstable. The rate of CHP formation increases with CHP concentration such that in the end, optimum conversion is achieved at that CHP concentration which minimizes by-product formation.

1.1 Initiation

The free radical chain reaction proceeds autocatalytically following initiation. Initiation is the steps that generate the cumyl free radicals. These radicals are generated by the thermal decomposition of CHP and the subsequent reaction of the intermediate radicals formed with cumene.

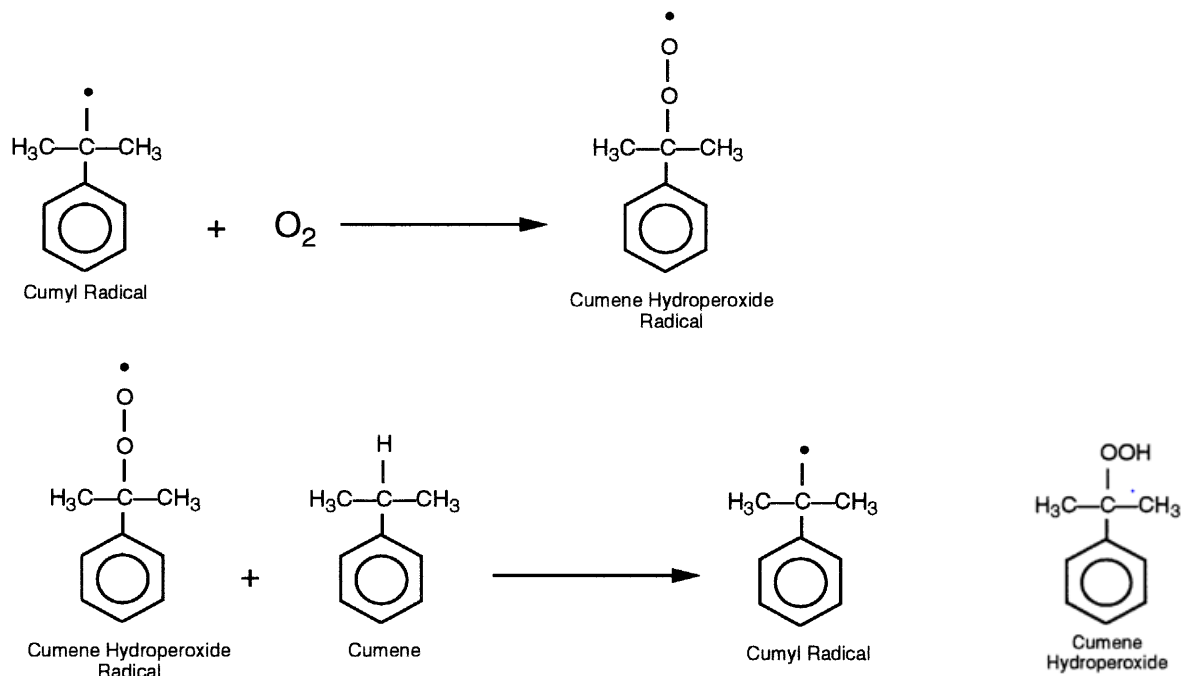
OXIDATION: FREE RADICAL MECHANISM (INITIATION)



1.2 Propagation

The cumyl radicals propagate the free radical chain reaction by reacting with oxygen to form CHP while the cumyl radical itself is regenerated. The second step, propagation, illustrated on the next page, is the rate controlling reaction for cumene oxidation, as it involves both the formation of CHP and the cumyl radical.

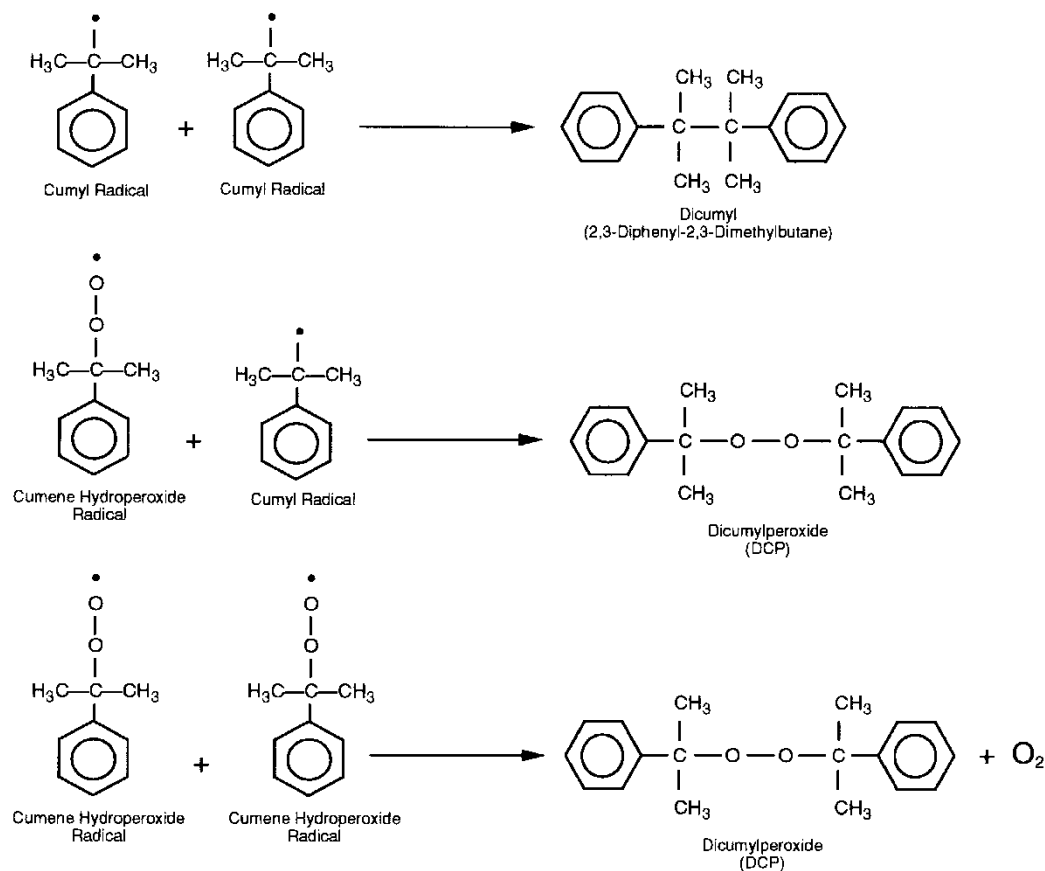
OXIDATION: FREE RADICAL MECHANISM (PROPAGATION)



1.3 Termination

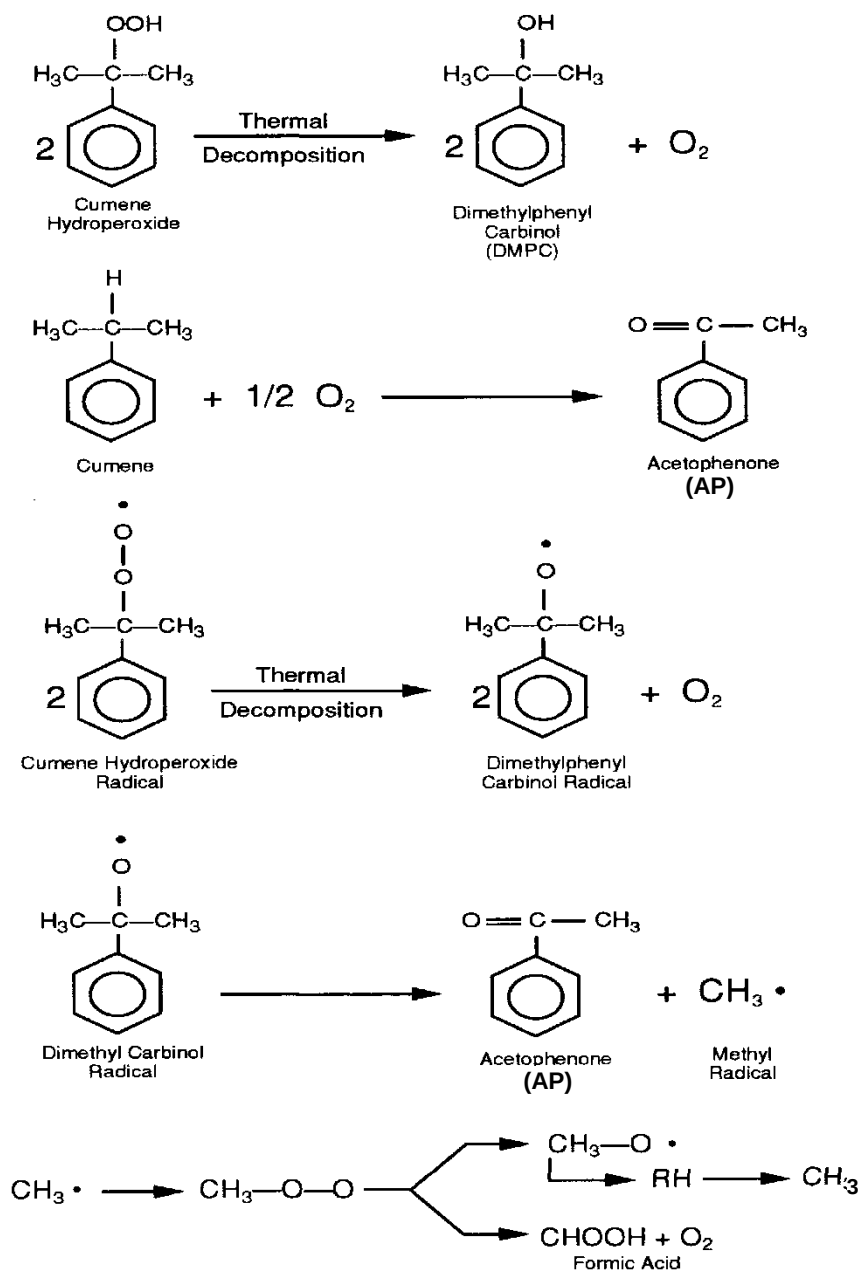
The various termination reactions are shown on the next page. Termination reactions cause the disappearance of radicals and thus stop the chain reaction. Of those reactions illustrated, the third is of more importance when sufficient oxygen is present, as is usually the case.

OXIDATION: FREE RADICAL MECHANISM (TERMINATION)



1.4 Side Reactions

The aim of the oxidation reaction is to produce CHP at a practical rate and the highest efficiency. The rate increases but the efficiency (CHP selectivity) **decreases** as the CHP concentration increases, or as the temperature increases. Oxidation of cumene has to compete with a number of side reactions leading primarily to the formation of dimethylphenylcarbinol (DMPC) and acetophenone (AP). The various oxidation side reactions are illustrated below.



1.5 Rate of Reaction

The generalized rate expressions in the oxidation scheme can be represented as follows:

$$r_{CU} = k_1 C_{CHP}^a C_{CU}^b P_{O_2}^c$$

$$r_{DMPC} = k_2 C_{CHP}^d$$

$$r_{AP} = k_3 C_{CHP}^e$$

r_{CU} represents the rate of cumene disappearance

r_{DMPC} and r_{AP} represent the rates of formation of DMPC and AP

a, b, c, d, e are kinetic exponents obtained by correlating experimental data

P_{O_2} is the partial pressure of oxygen

C_i is the concentration of the component

k_i are reaction rate constants

As is evident from the above equations, the rate of by-product formation can be interpreted in terms of the CHP concentration present in solution. The expression for the rate of cumene disappearance reflects the fact that oxidation of hydrocarbons to their peroxides are catalyzed by the presence of these peroxides in solution, a fact that is obvious upon inspection of the propagation reactions. The rate constants k_1 , k_2 , and k_3 , specific for a given catalytic system, can all be related to temperature through an Arrhenius type expression. The overall CHP formation rate can then be represented by:

$$r_{CHP} = r_{CU} - r_{DMPC} - r_{AP}$$

Besides the by-products mentioned above, trace quantities of other by-products have been found. Dicumylphenol and dicumylperoxide (DCP) are formed from their respective free radicals. Methanol produced with the formation of AP oxidizes to formaldehyde and then to formic acid. The latter has been found to react with CO_2 in the oxidizers to form oxalic acid. These trace amounts of volatile acidic by-products tend to promote CHP decomposition reactions. Caustic washing of recycle cumene from the concentration section neutralizes these acids. Optimum conditions of alkalinity must be maintained for highest oxidation efficiency. Any phenol present in the oxidation system will inhibit the oxidation reaction. The cumene peroxide radical reacts with a phenol molecule to form CHP and a phenyl radical. The phenyl radical does not have sufficient energy to react with cumene to continue the chain reaction and is ultimately removed from the system by reaction with another radical.

2. Charcoal Adsorbers

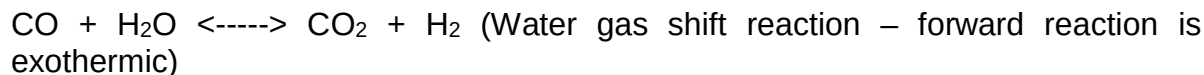
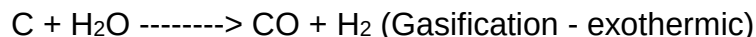
Adsorption of water and organics from the oxidation spent air and other miscellaneous streams is an exothermic process. Additional heat is released as the activated carbon (charcoal) facilitates the reaction of organic compounds (e.g. aldehydes, ketones, peroxides). Charcoal facilitates these reactions by adsorbing and concentrating the compounds so that they come in close proximity and react. Charcoal can act as a catalyst for such reactions.

Only light compounds, cumene and lighter, are easily desorbed during steam regeneration. Higher boiling components (e.g. cumyl hydroperoxide and heavier) tend not to desorb and ultimately lead to fouling of the charcoal. Water (both vapor and liquid) and organic vapors that condense on the charcoal interfere with hydrocarbon adsorption.

At temperatures as low as 80 deg C, carbon, by three classes of reactions, acts to facilitate oxidation of organics (especially acetone) to generate heat, tars and CO/CO₂. These three classes of reactions are:

- a. Oxidation (combustion)
- b. Pyrolysis
- c. "Coking"

Above 200 deg C the following combustion and related reactions can occur:



WITHOUT A FLOW OF GAS OR LIQUID TO REMOVE HEAT, HEAT TRANSFER IN CARBON IS LIMITED. THE RESULT CAN BE THE PROGRESSION OF UNCONTROLLED REACTIONS, INCLUDING COMBUSTION, RESULTING IN A FIRE WITHIN THE BED

The information above indicates that under certain conditions, or sequence of conditions, the potential exists for high temperature ("hot spots") to develop within the activated carbon beds in the charcoal adsorbers. If sufficient oxygen becomes available, the potential exists for runaway combustion reactions leading to a fire within the charcoal adsorbers. The conditions that potentially lead to the formation of hot spots include, but are not limited to, no-flow or low- flow through the charcoal adsorbers while at or near normal operating conditions. The potential for hot spots exists within the charcoal bed both before and after the bed has been stripped with

steam. The conditions under which runaway oxidation reactions can potentially occur within the charcoal adsorbers (with the potential for a fire) include, but are not necessarily limited to, ingress of oxygen while hot spots, elevated temperatures, or conditions otherwise conducive to oxidation reactions exist within the vessel.

To avoid the formation of hot spots within the charcoal bed UOP advises to minimize the amount of time the charcoal adsorbers are under a no-flow or low-flow condition. For a charcoal adsorber that is online, flow can be maintained through the adsorbers by the nitrogen purge used to continually purge the water spray lines at the top of the oxidizer. Additional nitrogen can be provided via the purge line located at the top of each oxidizer.

For a charcoal adsorber that has been taken offline or for which regeneration has been interrupted, UOP recommends the following:

Unless the adsorber has been prepared for unloading, avoid exposing the adsorber to oxygen (air). During a no-flow condition it is recommended that the manways remain closed and to close the valves connecting the adsorber to the vent stack; otherwise, the charcoal adsorber could potentially draw oxygen (air) into the adsorber when steam within the vessel condenses as the adsorber cools.

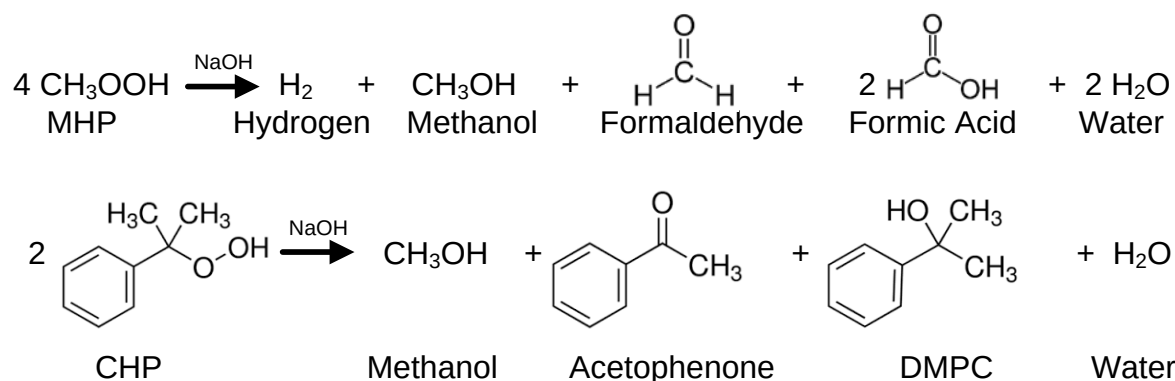
If the normal process flow to the charcoal adsorber cannot be restarted with 8 hours UOP recommends to maintain the adsorber at positive pressure using steam or to flow steam through the adsorber following the flow path used for regeneration.

If steam is not available, after ensuring an adequate exit path for potential vapors formed, water can be added to the charcoal adsorber bed using the same procedure as during charcoal loading to keep the charcoal bed flooded. Water will minimize the contacting of the charcoal with oxygen and will provide a heat sink.

UOP Design Specifications specify installation of thermocouples within the adsorber vessel. If the high temperature alarm is activated UOP recommends to ensure all personnel are a safe distance from the vessel and ensure there is an adequate exit path for the vapors that will be generated when quenching water is added. Any of the following would be sufficient to relieve the vapors formed when the quench is engaged: An open manway, the effluent line from the adsorber to the vent stack, or pressure relief devices on the vessels and sized for this case. After confirming the exit path, UOP recommends starting the water quench to the adsorbers to cool the charcoal bed.

3. Waste Water Peroxide Destruction

The chemistry of destroying or reacting peroxides from the oxidation and concentration waste water streams is shown in the reactions below:

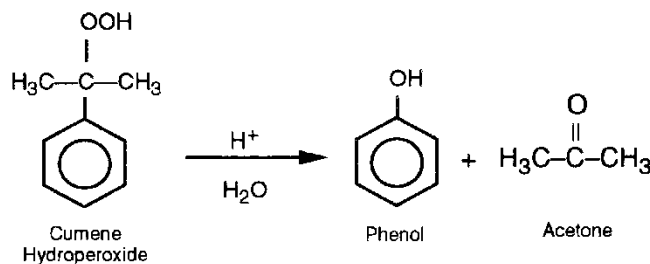


Note the above reactions take place in the liquid phase, in an abundance of water, at 130 deg C and pH = 12. Decomposition/cleavage of CHP is not expected in the peroxide destruction reactor.

B. Decomposition

1. Primary Reactions

The basic chemistry of CHP decomposition is presented in the reaction below.



Selectivity of CHP to phenol and acetone is high, and the main concern is the reduction of by-product formation (typically heavies) to a minimum. This is accomplished by restricting the amount of acid-catalyzed DMPC dehydration in the decomposer loop and leaving the bulk of this reaction to the dehydrator where AMS selectivities are higher, and then forcing the rest of the decomposition thermally.

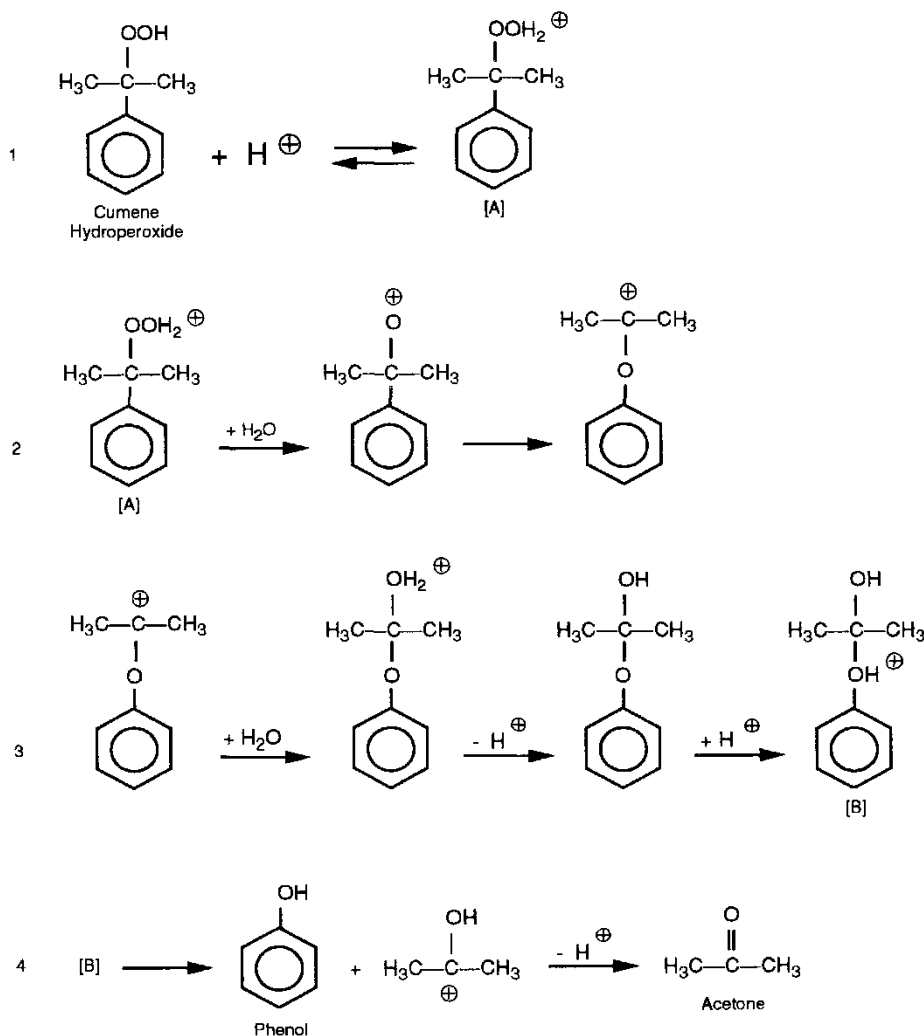
The CHP decomposition reaction is very rapid and strongly exothermic (-60.4 kcal/mol). It is temperature dependent according to an Arrhenius type expression and can be generically represented by the following rate expression:

$$r_{\text{CHP}} = k_4 \cdot C_{\text{CHP}} \cdot C_{\text{H}^+}$$

A postulated mechanism for the acid catalyzed peroxide decomposition is illustrated on the next page. The cleavage mechanism is an example of a 1, 2 shift from carbon to oxygen. It should be noted that such 1, 2 cleavage is highly undesirable

during oxidation because phenols are extremely effective inhibitors of free radical oxidation. The alkaline environment maintained in the oxidation section feed and recycle cumene caustic wash ensures that minimal acidic materials interfere with the oxidation process.

POSTULATED REACTION MECHANISM

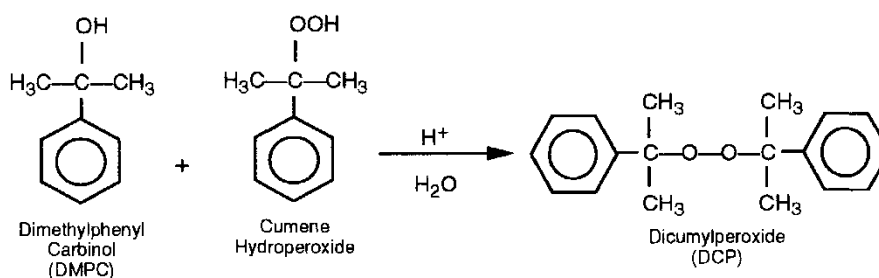


Operating conditions in the decomposition loop favor the suppression of unwanted secondary reactions that produce tars and other impurities. The most important secondary reactions that are present (those that reduce phenol yield) tend to be related to the DMPC concentration in the oxidation product. DMPC readily dehydrates to AMS by both acid catalyzed and thermal mechanisms. The acid catalyzed dehydration of DMPC to AMS has the disadvantage of being too active, which reduces selectivity to AMS and therefore, reduced yields. The AMS that forms from the acid catalyzed reaction continues to react with other components in

the crude product. AMS can react with itself to form AMS dimers and with phenol to form cumylphenol (see side reactions).

The decomposer conditions are controlled to limit the reaction rate of CHP to phenol and acetone. An excess of CHP in the decomposer favors the reaction of the DMPC with CHP to form dicumylperoxide (DCP) over the dehydration reaction. See the reactions below and in Figures II-2 and II-3 on the following pages. The acid catalyzed dehydration of DMPC to AMS results in a poor yield of 50 mol% or less with the by-products going to residue. The thermal dehydration of dicumylperoxide, DCP, yields a much higher yield of 80+ mol% depending on the concentration of CHP circulating in the decomposer loop and DMPC in the decomposer feed. Higher concentrations of residual CHP will improve the yield, while higher concentrations of DMPC in the feed will decrease the yield. The limit for residual CHP in the decomposer is process stability; at high, greater than 2 wt%, residual CHP the decomposer can become unstable under certain conditions.

FORMATION OF DICUMYLPEROXIDE (DCP)



DEHYDRATION OF DICUMYLPEROXIDE (DCP)

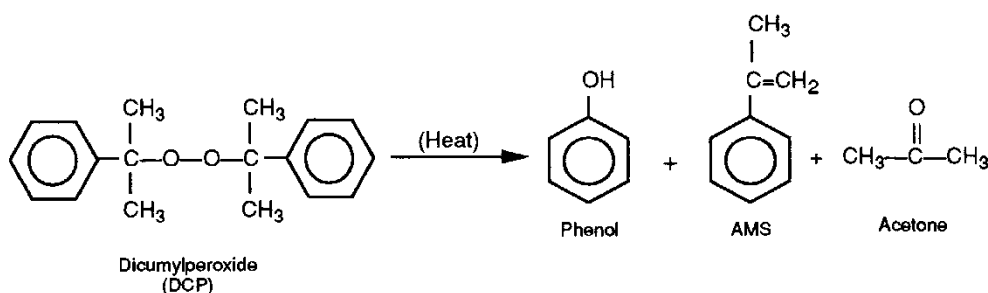


Figure II-2

DICUMYL PEROXIDE CHEMISTRY (IN DECOMPOSER LOOP)

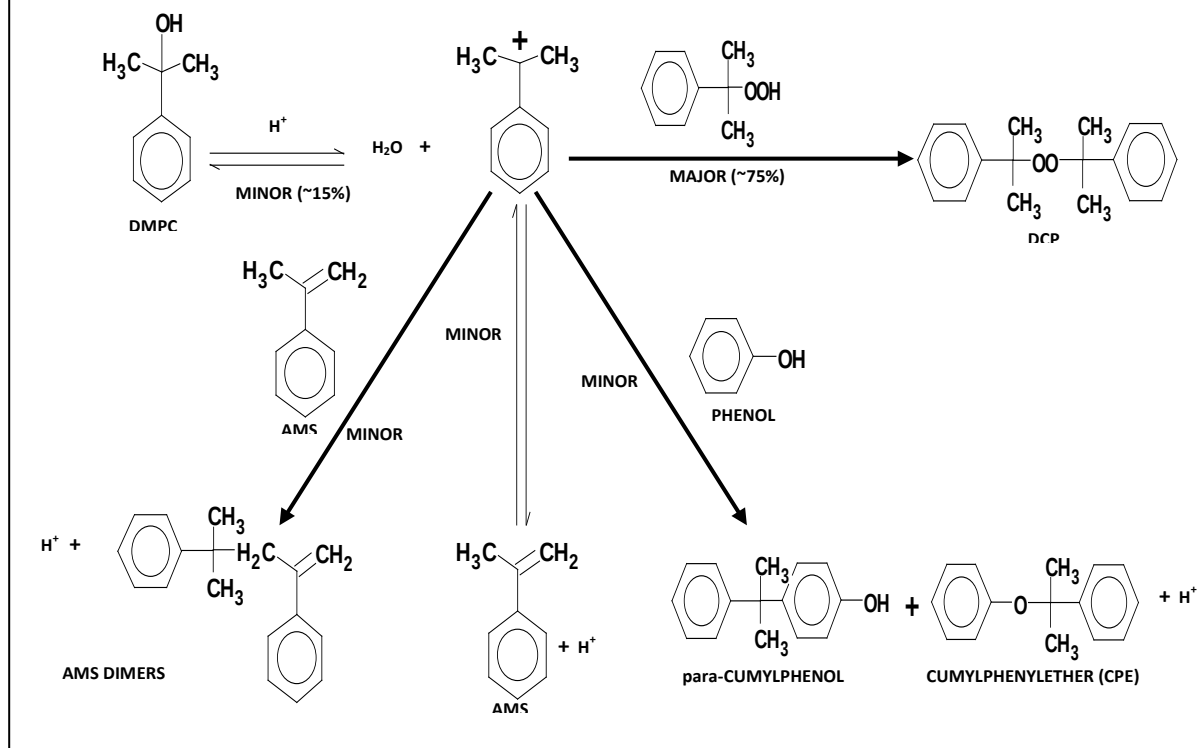
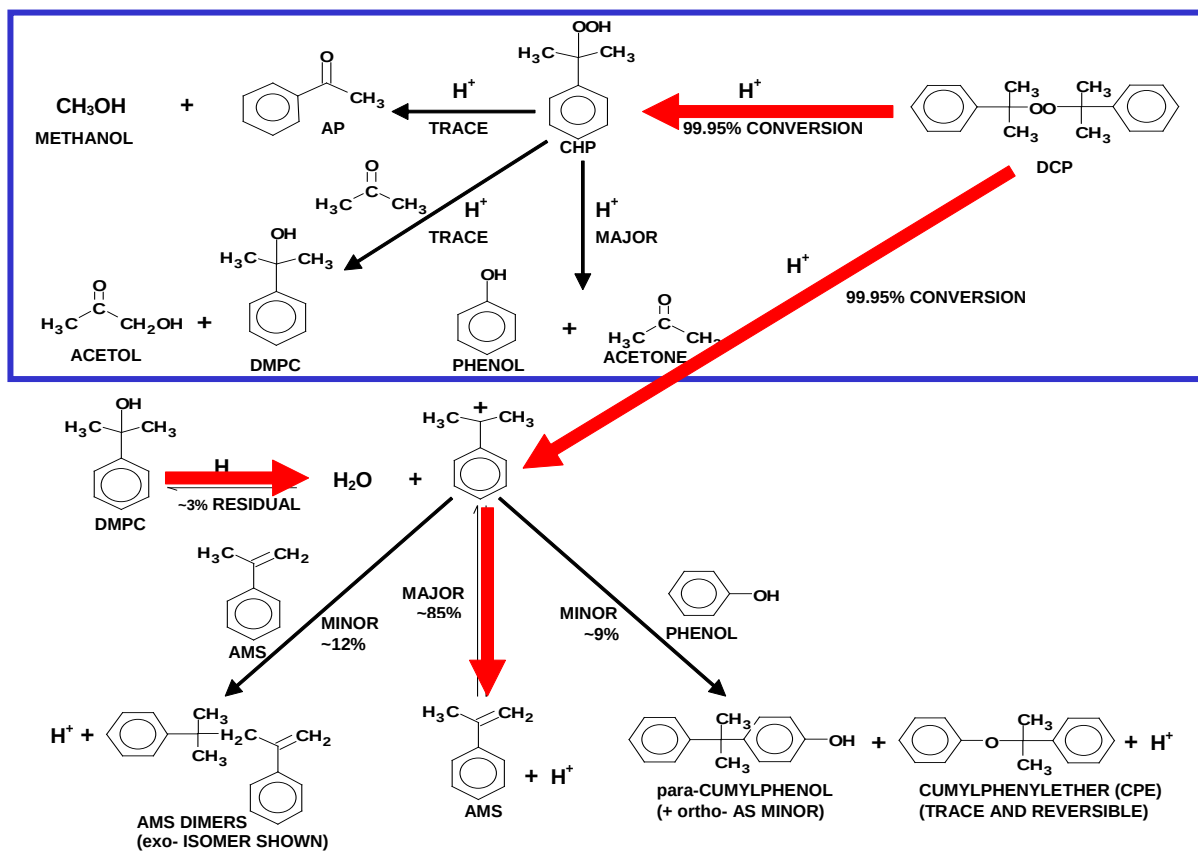
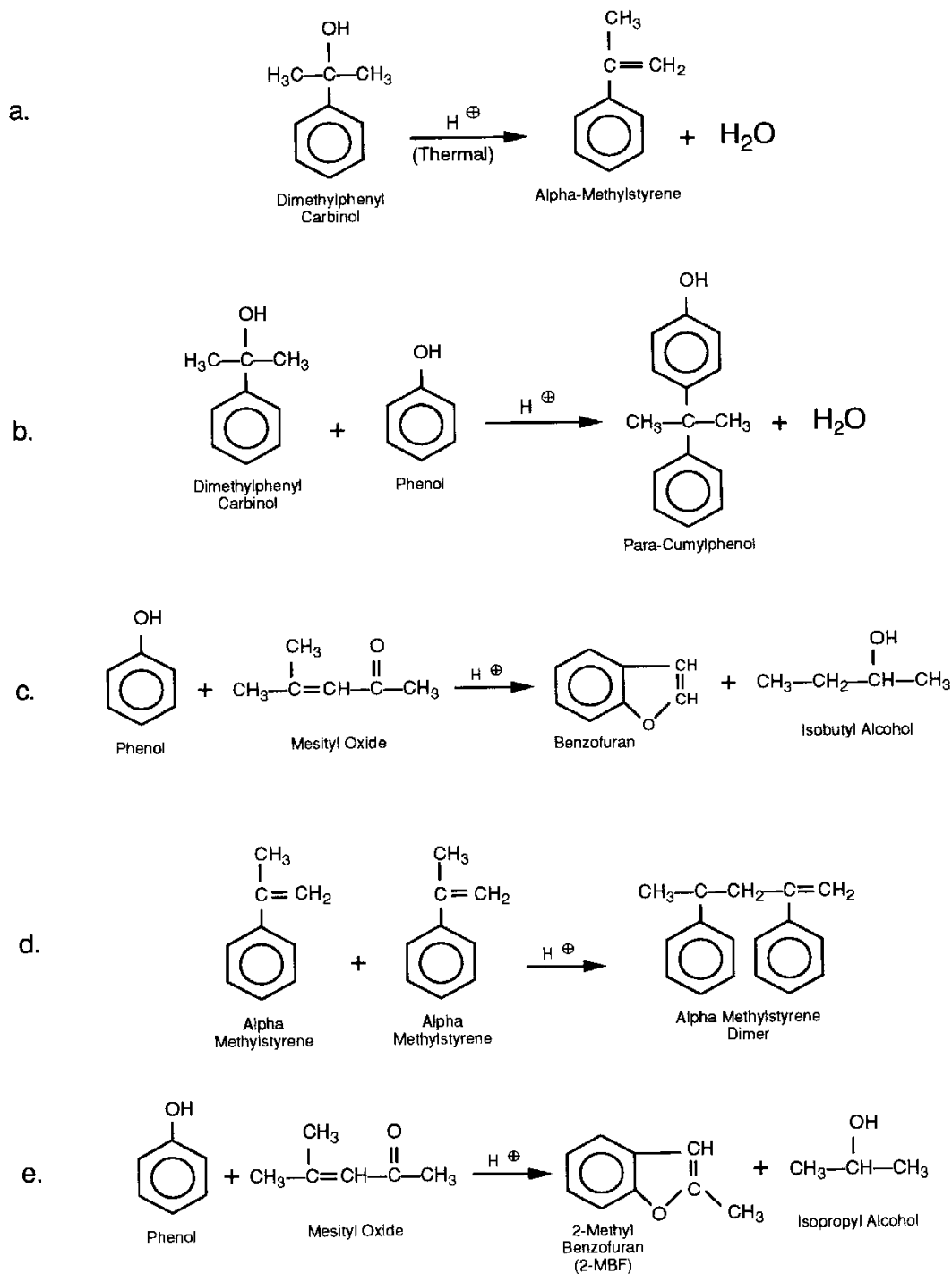


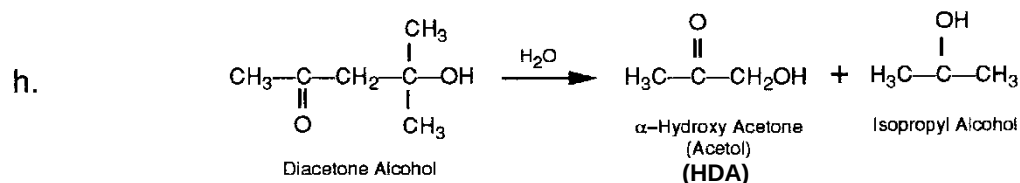
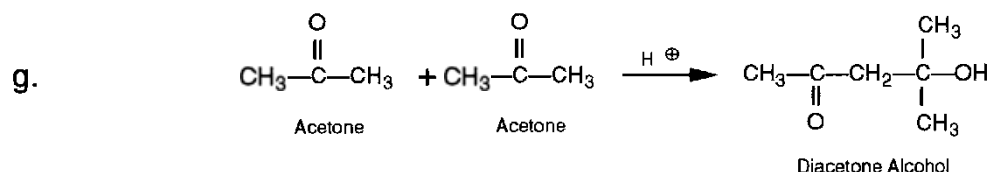
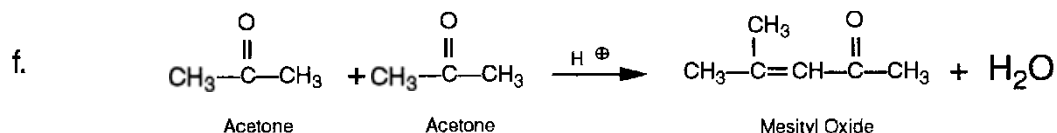
Figure II-3

DEHYDRATOR CHEMISTRY

Acetone also contributes to the production of impurities; it is the precursor to the following compounds: diacetone alcohol (DAA), alpha-hydroxyacetone (HDA or acetol) and mesityl oxide (MO). The various secondary reactions are presented on the next pages.

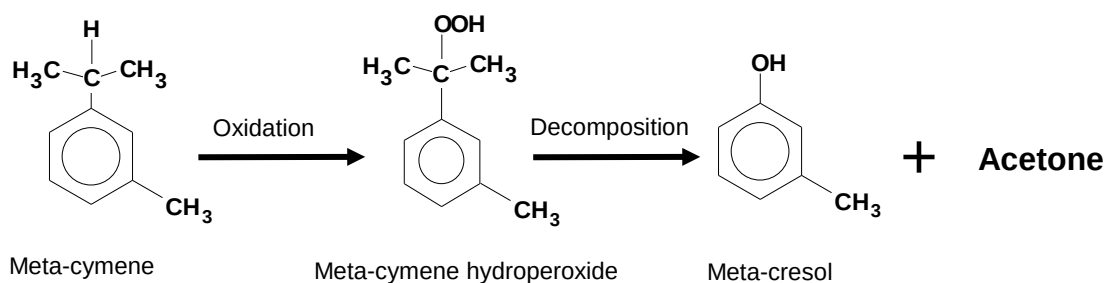
SIDE REACTIONS



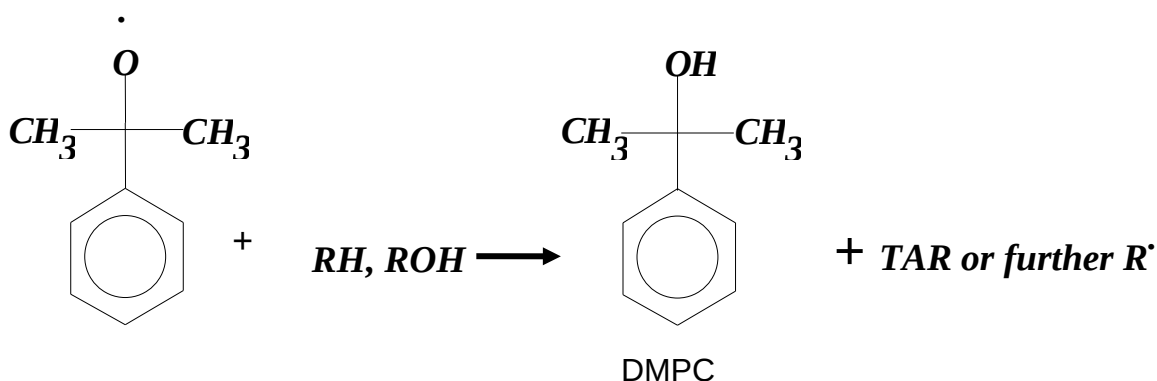
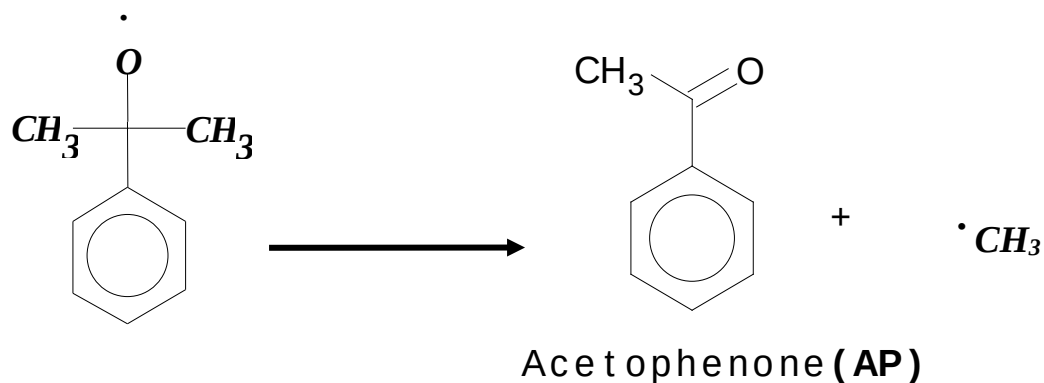


2. Cresol and Other Byproduct Formation

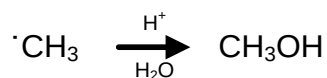
Cresols have become important in recent years, as phenol purity specifications have become tighter. Cresols form due to the normal oxidation and decomposition reaction of cymenes in the cumene feed, specifically, the isomers m-cymene and p-cymene. The isomer o-cymene is typically not found in cumene because of steric hindrance. The cymenes react similarly to the cumene reactions. See the general reaction example below:



Cresols can also form in the decomposer and fractionation section via the dimethylphenylcarbinol (DPMC) radical reactions that form methyl radicals and subsequently react with phenol. The DMPC radical can react to form either acetophenone and a methyl radical or react with another aromatic ring to form DMPC and heavies.



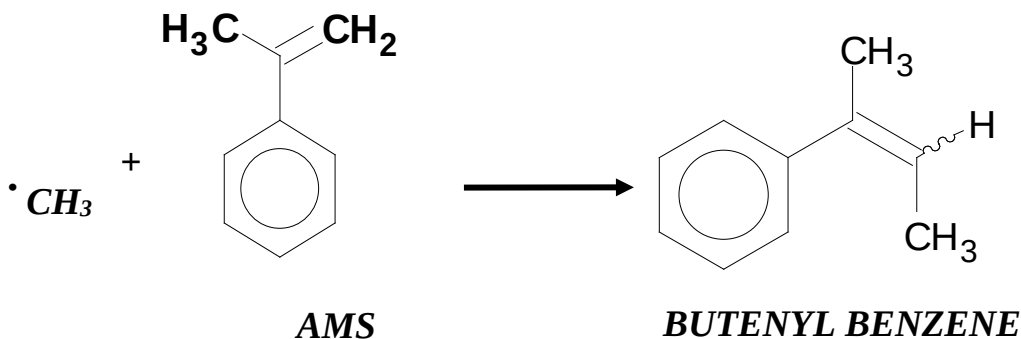
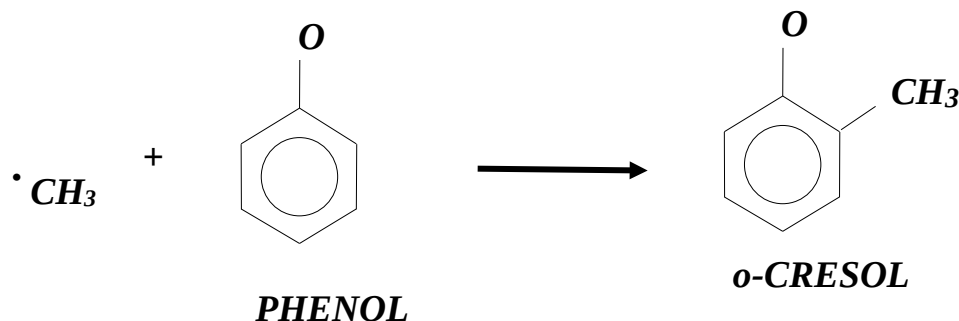
The methyl radical then primarily further reacts in the decomposer to form methanol per the reaction below:



The formation of methanol from a methyl radical is predominant due to the fact that there is hydrogen readily available in the acidic decomposer system. Although not shown above, the DMPC radical can also be cleaved to form benzene and acetone. This reaction is highly unfavored at normal operating conditions, however operating the decomposer in 'start-up mode' (high temperature, high acid and low water) makes the benzene formation pathway more favorable by increasing the concentration of DMPC in high temperature areas such as the dehydrator and the crude acetone column reboiler. Benzene formed in decomposition and fractionation will concentrate in the acetone product.

The reactions shown on the next page occur to a small extent resulting in a cresol concentration of a few ppm in the phenol product, mostly the ortho- and para-isomers. A shift in the reactions will occur if dicumylperoxide (DCP) is allowed to go to the fractionation section. The DMPC radical will form in the fractionation reboilers as the DCP decomposes. In the absence of a strong acid (removed by neutralization section), the methyl radical will tend to react with phenol and other

hydrocarbons instead of hydrating to methanol. This results in an elevated concentration of cresols in the phenol product as cresols are not removed by the phenol purification reactions described in the following section. However cresols can be rejected to the residue stream to a minor degree by fractionation.



C. Crude Product Neutralization

The sulfuric acid is selectively neutralized using the same treating chemical used in phenol purification, HMDA or Dytek. Dytek, or 2-MPMD, is a branched isomer of HMDA. Dytek has the advantage of a low boiling point, but a disadvantage in a generally higher price compared to HMDA. Other triamines or tetramines such as DETA and TETA (but not just DETA) can be used as well. They typically have 2 primary amine groups and 2 secondary amine groups. The primary amine groups will react completely, but the secondary groups do not react 100%. Cost is the major factor in using alternate amines. HMDA is generally the cheapest amine to use, but has the disadvantage that it is solid at ambient temperatures in the recommended 90-100% form and must be melted to move or transfer. Alternatively HMDA can be diluted with water to lower the melting point; however this adds to the

fractionation, PRU and waste water processing and in theory can impact the water concentration in the phenol product.

The simple acid-base chemistry of diamine with sulfuric acid occurs selectively to remove the strong acid. An excess of the stoichiometric quantity is added continuously to the process to ensure all of the sulfuric acid is reacted. The following reaction occurs to form an organic salt.



The organic salt has a low volatility and remains dissolved in the organic phase thus not causing fouling and will go out of the fractionation system into the residue byproduct. The additional organic nitrogen in the residue does not have a measurable impact on the heater spent gas emissions of NO_x when the residue is burned in a furnace or boiler.

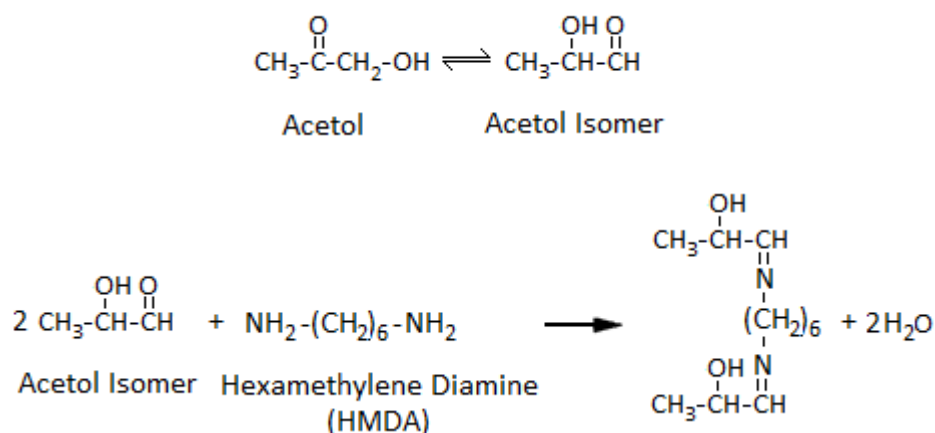
D. Phenol Purification

Most of the by-product impurities such as acetophenone (AP), cumylphenol (CP), and the phenolic tars boil at temperatures higher than phenol and are easily removed by distillation. Some of the acetone derivatives, however, co-boil with phenol and cannot be excluded from the phenol product by distillation alone. To enable production of the highest quality phenol product, crude phenol is chemically treated. The chemical selectively reacts the undesirable impurities into compounds that are subsequently removable by distillation. The treating chemical is a diamine, Hexamethylenediamine (HMDA) or one of its analogs, as described in the crude product neutralization chemistry. In the phenol purification section, pure DETA is acceptable.

Branched analogs of HMDA such as 2-Methylpentanamethylene diamine (MPMD or Dytek) or DETA and/or TETA can be substituted for HMDA and have some desirable properties. The main difference is a lower freezing point (-50 to -60°C) for MPMD versus +30°C for a 90% HMDA solution. MPMD does not then need to be diluted by with water as does HMDA. The one disadvantage of MPMD is there is one supplier, DuPont and the price is generally higher per equivalent quantity of pure HMDA. DETA and TETA, though previously not price advantageous or readily available, are becoming more so in today's market.

The amine group of any one of the above treating chemicals greatly prefers to react with aldehyde groups compared to carbonyl (C=O) or alcohol (-OH) groups. Despite only apparently having carbonyl and alcohol groups, acetol reacts quickly with diamines because the ketone form is in equilibrium with its aldehyde isomer. The product of the diamine reaction with hydroxyacetone (e.g. acetol or HDA) yields a heavy alkylate which is easily removed by rectification of the product. The reactions

below show acetol (HDA) in equilibrium with its isomer and the basic reaction of the isomer with HMDA to form a heavy compound.

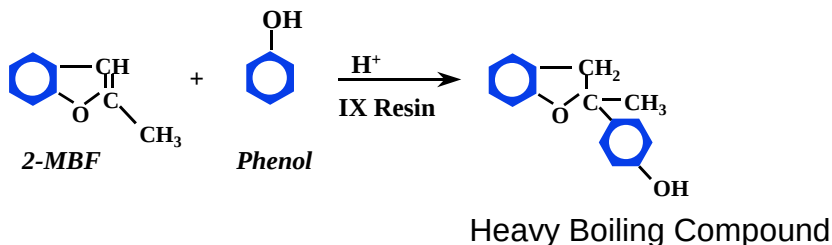


There is one residence time reactor to give the required time for the diamine to react with the aldehydes and ketones. More than the stoichiometric quantity of diamine is required to complete the reaction due to the side reactions of diamine with the organic acids present in the C-AMS column bottoms.

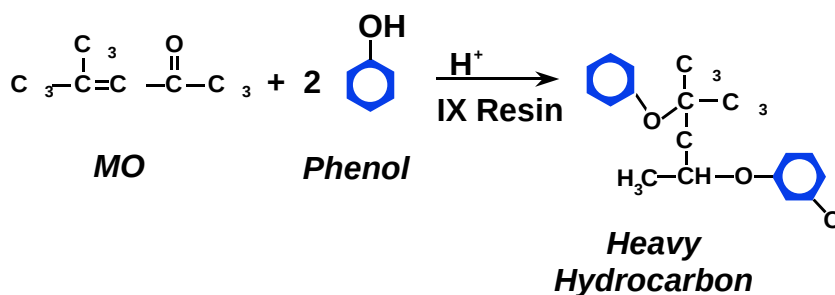
Mesityl oxide (MO) and acetophenone (AP), only having ketone groups, react to a much smaller extent and are not as effectively removed by diamine. MO reacts across the IX resin beds, and is removed via purge cuts in the top of the CPC and PR columns. AP will tend to go out with the residue.

Excessively high residue stripper column reboiler temperatures, greater than 232°C (>450°F), will promote cracking of the heavy alkylate, thus generating carbonyls and fragmented amines back into the rest of the fractionation section. These high temperatures should be avoided to prevent deactivating the downstream ion exchange bed with basic compounds. The column bottoms are designed to operate below 200°C (390°F) to minimize these reactions.

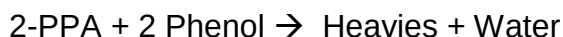
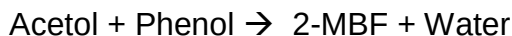
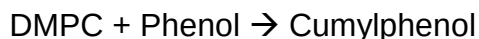
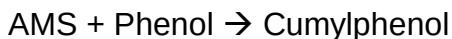
The secondary acid catalyzed reactions, utilizing Amberlyst 36 IX-resin, remove the small quantity, about 25 wt ppm, of 2-methylbenzofuran (2-MBF) that is formed in the decomposer as well as other trace components such as MO and AP. There can also be some small quantity of 2-MBF recycled from the effluent section. Caustic, to a minor degree, catalyzes the reaction of acetol to form 2-MBF. The reaction rate is not high at the near ambient temperature effluent section conditions.



Other compounds such as AMS will react to form heavy compounds such as AMS dimers and cumylphenol similar to the side reactions described previously in decomposition. Any residual carbonyls such as MO and AP will also react to form heavier compounds with phenol similar to the side reactions in the decomposer. MO may also react with two phenols to form a heavy compound as illustrated below:



Other reactions:

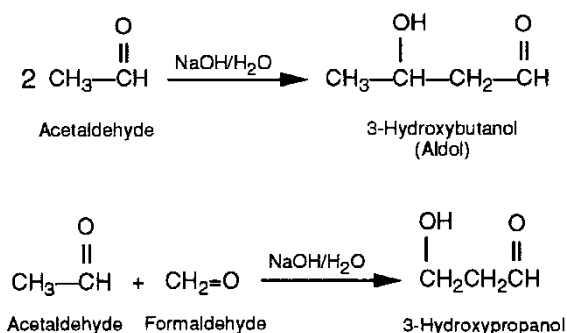


The resin will also adsorb basic compounds such as diamine and amine fragments as well as cationic compounds such as sodium. These compounds will neutralize the resin and make it less catalytic. The resin will eventually need to be replaced when 2-MBF passes through the resin bed unreacted. Typically, the resin will deactivate due to plugging with heavies.

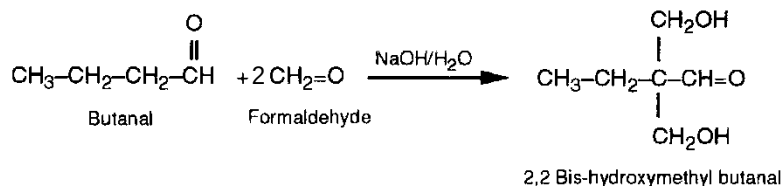
A trickle of sulfuric acid was added to some designs upstream of the IX resin beds with the intent to neutralize amine and amine fragments. However, an unintended consequence occurred when the excess sulfuric acid then catalyzed the formation of 2-MBF downstream of the IX resin treaters. This trickle acid addition is no longer practiced.

E. Acetone Purification

Acetone is purified in the finished acetone column (FAC) where the water and cumene are fractionated. As with phenol, there are impurities present that boil in the same range as acetone; these impurities are primarily aldehydes. Most aldehydes originate from nPB, EB and non-aromatics (propylene trimers) in the cumene feed. Aldehydes in the acetone product can be measured directly by GC or determined indirectly through a test called the permanganate fade time (PFT) test. Generally PFT results greater than 2 hours are sufficient for commercial acetone. However, in addition to aldehydes other compounds commonly found in acetone such as diacetone alcohol (DAA), mesityl oxide (MO), 2,3-butanedione and 2,5-DMF, can influence the PFT test which limits its utility. The formation of DAA is limited by keeping caustic out of the FAC overhead, maintaining sufficient water on the reaction trays and keeping the caustic concentration at or below design. Aldehydes are reacted through a condensation route that is catalyzed by dilute caustic injected into the column. This type of reaction is called an *aldol reaction*, which comes from the reaction product, an aldehyde-alcohol. On the next page are two examples of aldol reactions.



In addition to acetaldehyde, propanol, butanol and formaldehyde can also be present in the FAC feed. Mixed aldol reactions are possible which will form complex product mixtures. Formaldehyde is highly reactive and can form multiple additions to other aldehydes creating heavier compounds. See the reaction on the next page:



The aldol condensation reaction occurs in the organic phase; however the reaction products are water soluble and transition to the aqueous phase. Therefore despite the aldol condensation reaction being equilibrium limited, the fact that the reaction products move to the aqueous phase means that the reaction in the organic phase can continue without limitation. Although in the aqueous phase, the aldol condensation products do not have high water solubility and tend to precipitate in the FAC bottoms and go out with the water phase and get trapped in a filter. The precipitate typically has the appearance and consistency of oatmeal. If loose rust is present in the column bottoms, the precipitate can form into hard agglomerates.

Methanol and other alcohols do not react well in this environment and separate to some extent by fractionation typically trending with water in the acetone product. Water is fractionated out of the acetone to the bottom of the FAC. Of the alcohols, methanol is the largest contaminant for acetone. Methanol formation from decomposition can be minimized by limiting acetophenone formation.

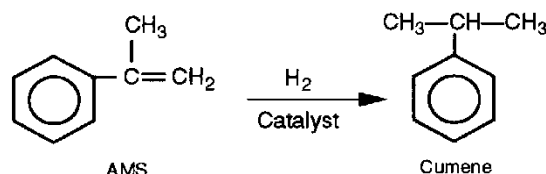
F. Hüls MSHP (AMS Hydrogenation)

The majority of by-products created in the phenol unit are not lost in the process. As stated earlier, DCP readily dehydrates to AMS in the dehydrators. The AMS along with unreacted cumene is distilled from the phenol in the C-AMS column. The trace phenol remaining due to the AMS-phenol azeotrope is then extracted out of the mixture using caustic in the effluent section. After the phenol is removed, AMS is hydrogenated back to cumene using a catalyst and hydrogen in the Huels MSHP™ process. The reaction is carried out over two fixed beds of catalyst. AMS that is not reacted back to cumene will be sent to oxidation where it will react to form organic acids in the oxidizer. Increased concentrations of organic acids in the oxidizer are undesirable as they decompose CHP to phenol which hinders the oxidation reaction.

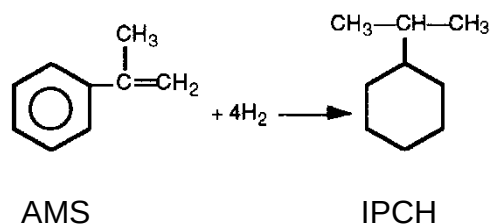
The catalyst is a palladium loaded alumina catalyst with the following properties:

- | | |
|----------------------|--|
| a. Designation | Evonik H14108 |
| b. Particle Size | 1.6 mm (1/16") |
| c. Shape | Extrudate |
| d. Loaded Density | 800 kg/m ³ (53 lb/ft ³) |
| e. Color | Light Gray/Brown |
| f. Palladium Loading | 0.5 wt% |

The catalyst is run under relatively mild conditions to prevent saturation of the benzene ring, which would form isopropylcyclohexane (IPCH). The primary reaction is shown below:



The heat of reaction is $\Delta H_{298^\circ\text{K}} = 26,800$ kcal/kg-mole AMS. If the reactor conditions are allowed to become severe, complete saturation of the AMS to IPCH can occur by the reaction below:



The heat of reaction is higher for complete saturation to IPCH, $\Delta H_{298^\circ\text{K}} = 73,870$ kcal/kg-mole AMS. It also consumes 4 times the hydrogen compared to the desired reaction. The extent of this reaction in the Hüls MSHP process is essentially nil.

G. Theory of Fractionation

For the purpose of this manual, we will restrict ourselves to the two component systems called binary distillations.

1. The Concept of Equilibrium

Equilibrium is a state in which there is no tendency toward change. A vapor and a liquid in a system, for example, containing two components quickly come into equilibrium when mixed or agitated. At this condition of fixed temperature and pressure, molecules of either component in the vapor are condensing at the same rate as molecules of these components in the liquid are evaporating. The composition of each phase is then constant as long as the temperature and pressure do not change. The state of these components in equilibrium can be graphically represented by a line called an equilibrium curve. The curve represents the concentration of one of the components in vapor and liquid phases at constant pressure. We will discuss more about this later.

2. Definitions

Let us define the terms we will be using in describing the mechanism of distillation.

x	=	mol fraction of a component in a liquid phase
y	=	mol fraction of a component in a vapor phase
X	=	mols/hour, liquid phase
Y	=	mols/hour, vapor phase
D	=	mols/hour of distillate
R	=	mols/hour of external reflux
F	=	mols/hour of feed
B	=	mols/hour of bottoms
L	=	mols/hour of internal liquid flow
V	=	mols/hour of internal vapor flow
K	=	vapor/liquid equilibrium ratio of a component = y/x
α	=	relative volatility of two components = K_1/K_2
r	=	ratio of mols of components, light/heavy
π	=	column absolute pressure
p°	=	pure state absolute vapor pressure
T	=	temperature

Subscripts

1	=	the light component (usually unnecessary)
2	=	the heavy component (here $x_2 = 1 - x_1$)
D	=	in the distillate
F	=	in the feed
B	=	in the bottoms

Equilibrium ratio, K, is defined as the ratio of the mol fraction of a component in vapor phase in equilibrium with mol fraction of the same in liquid:

$$K = y/x \quad (1)$$

For a binary distillation $K_1 = y_1/x_1$ and $K_2 = y_2/x_2$. K varies with temperature and pressure. K_2 is always less than 1.0 on all trays, and K_1 is always greater than 1. If $K_1 = K_2$, then the binary is a constant boiling mixture, usually referred to as an azeotropic mixture – the separation of such a mixture is difficult. K values are experimentally determined and are available in tabulated form for different temperatures and pressures for various components.

The principle of fractional distillation is based upon the difference in volatility of the components in a mixture. One term that is used commonly is relative volatility, α , which is a ratio of equilibrium constants, i.e., $\alpha = K_1/K_2$. In effect, this parameter describes the degree of "difficulty" of the separation. Difference in boiling points is not always a good indicator. For instance, if two materials boil close together, the

relative volatility will be much greater if the binary pair are low boiling, compared to a high boiling pair of components. For example, it is much easier to fractionate a pair of components boiling at 30°C (86°F) and 40°C (104°F), respectively, than it is to fractionate a pair of components at 320°C (608°F) and 330°C (626°F). The α for the first pair is approximately 1.27 for the second pair approximately 1.1.

3. The Principles of Distillation

3.1 McCabe-Thiele Diagram

Having defined the necessary terminology, we are now ready to discuss the mechanism of distillation. When the feed is introduced into a column, the more volatile component will be in somewhat higher concentration in the vapor phase above the feed tray. As the vapor rises through the tray above the feed, the heavier component tends to condense first from the vapor being contacted with the colder liquid coming down from above. At the same time, the lighter components are evaporated from the liquid.

When the column overhead vapor is condensed into a receiver, part of this liquid is withdrawn as net overhead product. The remaining portion is pumped back into the top of the column – this stream is called the external reflux. The excess liquid that flows from one tray down to the next is called the internal reflux. The reflux rate is generally related to feed rate as the reflux to feed (R/F) ratio. We will be discussing these variables in later sections. The enrichment of the liquid and vapor phases is directly related to the relative volatility, α , and to the internal reflux ratio. An equivalent amount of enrichment occurs successively on each tray.

Let us now try to relate this physical process graphically. The classical representation of distillation was developed by McCabe and Thiele who developed graphical solutions to the distillation process. McCabe-Thiele diagram (Figure II-4) is an equilibrium plot of x vs. y for a binary distillation. On this diagram, three other lines are also plotted. These are the 45° line where $x = y$, and the two operating lines. The name "operating line" is derived from the fact that the points on this curve represent the actual operating condition. There are two operating lines, one called the "upper operating line" which represents the section of the column above the feed point, and the other "lower operating line" which represents the section of the column below the feed point. The operating lines show the relationship between reflux rate, feed composition, bottom and overhead product quality. The slope of the upper operating line is numerically equal to the L/V ratio in the upper section of the column. Hence, the reflux ratio sets the slope of the operating line, and the feed and distillate compositions set the location of the operating lines. It is customary to express reflux ratio as reflux to distillate, R/D , which is related to L/V by the following equation:

$$R/D = \frac{L/V}{1-L/V} \quad (2)$$

The upper operating line intersects the 45° line ($x = y$) at $x = x_D$. The lower operating line intersects the upper operating line at $x = x_F$, and intersects the 45° line at $x = x_B$.

The limitations in expressing the reflux ratio as R/D must be pointed out here. R/D is highly dependent on the feed composition. For a given degree of separation and a given number of plates, the reflux ratio, R/D , is very nearly inversely proportional to the feed composition, x_F , which is equal to D/F .

$$R/D = \frac{\text{CONSTANT}}{D/F} \quad (3)$$

or

$$\frac{R}{D} \times \frac{D}{F} = \frac{R}{F} = \text{CONSTANT}$$

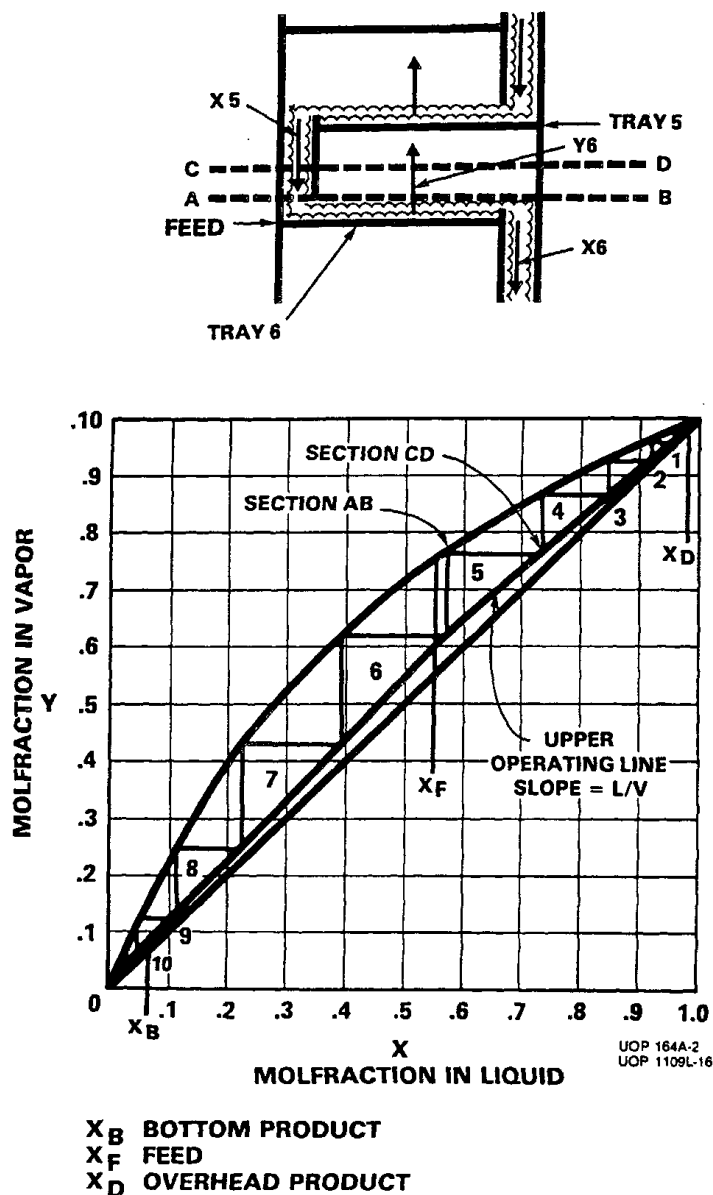
Thus, R/F is usually a constant for a given system. We will use this reflux ratio, R/F , in the development of the equations. R/F is nearly constant over a wide range of variables such as feed composition, pressure and feed temperature. R/D is related to R/F by the equation:

$$R/F = R/D \frac{x_F - x_B}{x_D - x_B} \quad (4)$$

As an example, in the Fractionation Section, the columns may be typically designed with R/F values ranging from 0.3 in the Crude Phenol Column to 2.5 in the Finished Acetone Column.

An example of McCabe-Thiele diagram and the physical significance of this diagram is shown in Figure II-4. As can be seen from this figure, the number of theoretical trays can be obtained by stepping trays off starting from x_D and ending at x_B . In the example considered in Figure II-4, there are 10 theoretical trays.

Figure II-4
McCABE – THIELE DIAGRAM



For a given overhead purity, McCabe-Thiele stipulates two limits. One limit is where the operating line intersects the equilibrium curve at $x = x_F$. The slope of the operating line (L/V) at this condition is related to a term called the minimum reflux ratio $[(R/D)_{\min}]$. At a reflux ratio below this minimum value, it is impossible to achieve the required product purities for a given feed composition. This will be true

regardless of how many trays are installed. $[(R/D)_{\min}]$ can be calculated using one of the following equations:

$$(R/D)_{\min} = \frac{1 x_D}{\alpha - 1 x_F} - \frac{\alpha(1 - x_D)}{(1 - x_F)} \quad (5)$$

or

$$(R/D)_{\min} = \frac{x_D [1 + x_F (\alpha - 1)] - \alpha x_F}{x_F (1 - x_F) (\alpha - 1)} \quad (6)$$

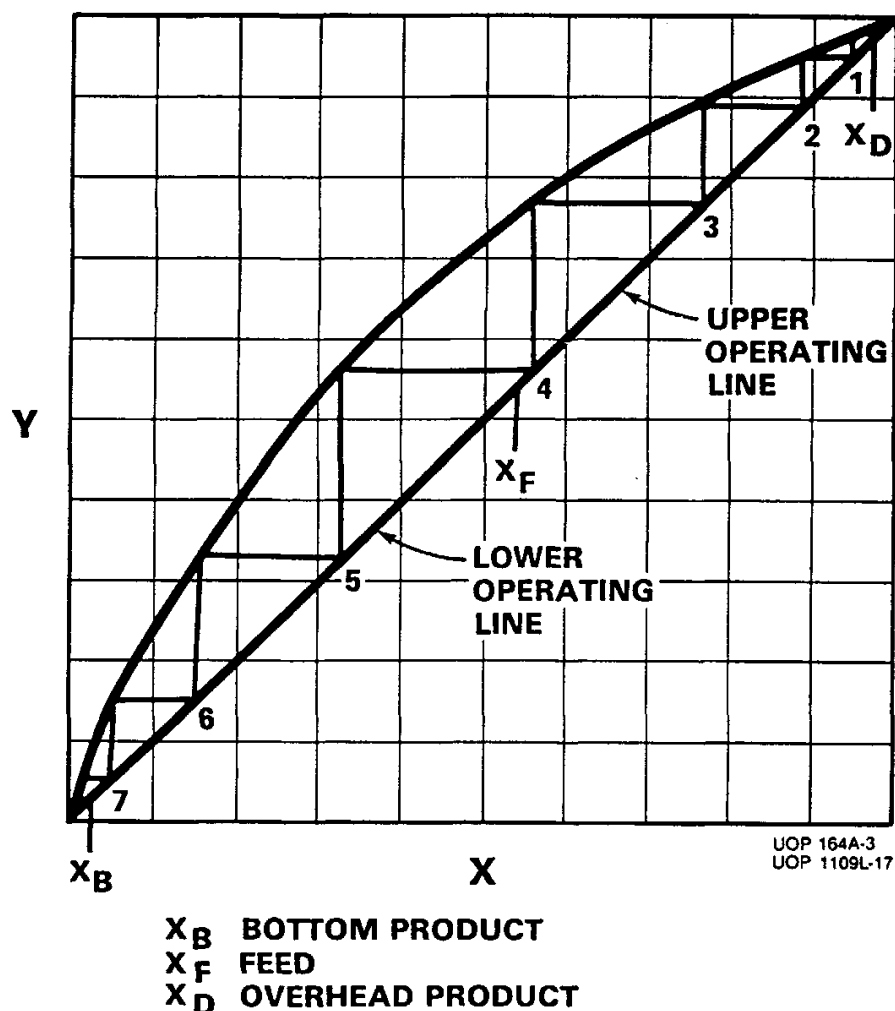
The other limit is when the operating line is equal to the 45° line ($x = y$). This is when all vapors from the top of the column are condensed and returned as reflux. In actual operation, this occurs during the start-up phase of the unit. When the feed to the column is shut off and no net overhead (D) or bottoms (B) are taken off and if all of the material accumulating in the receiver is pumped back as reflux (just enough to heat-balance the reboiler), we have a condition known as "total reflux".

This is graphically represented in Figure II-5. The operating line coincides with the 45° line when all vapors from the top of the column are condensed and returned as reflux. At this condition $L = V$ and hence $L/V = 1$. By stepping off the trays at this condition, the minimum number of theoretical trays (N_M) necessary for the separation can be determined. N_M can also be calculated from the Fenske equation.

$$N_m = \frac{\log ([x_D/(1-x_D)] / [x_B/(1-x_B)])}{\log \alpha} \quad (7)$$

Figure II-5

McCABE THIELE DIAGRAM TOTAL REFLUX CASE



In distillation of a mixture of more than two components in the feed, it is useful to designate two components as "keys". These are usually the ones whose recoveries are specified in advance and are thought of as components where the "split" is made. That is, all components lighter than light key appear mostly in the overhead and all components heavier than heavy key appear mostly in the bottoms. Let us take an example of benzene column feed that contains benzene, toluene, and a mixture of xylenes. In benzene column design, benzene will be taken as the light key and toluene as the heavy key, and their splits in overhead and bottom products

specified. The key components need not be adjacent in terms of order by relative volatility. If there are components between the key components, they are usually called "distributed" components.

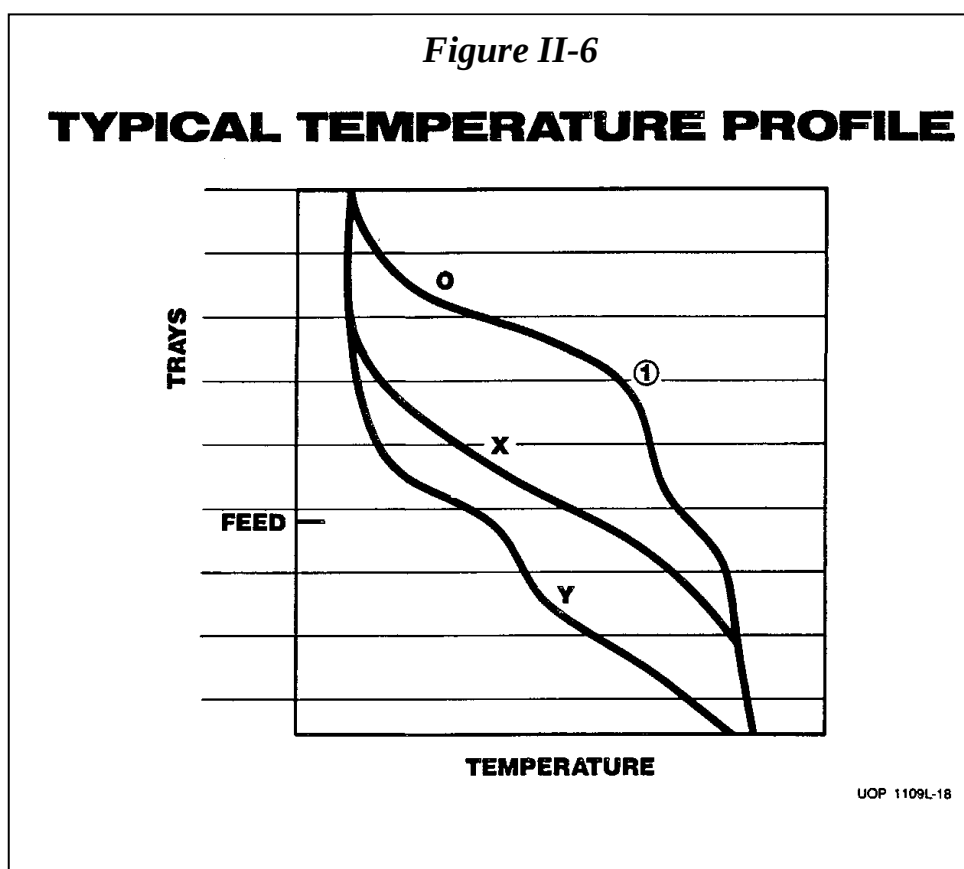
A relationship exists between the composition distribution of the heavy and light keys. The relationship is called Fenske Number:

$$\text{Log } (H_D) (L_B) = \text{Constant} \quad (8)$$

Where H_D = Heavy key concentration in the overhead product

L_B = Light key concentration in the bottoms product

This relationship is plotted in Figure II-6.

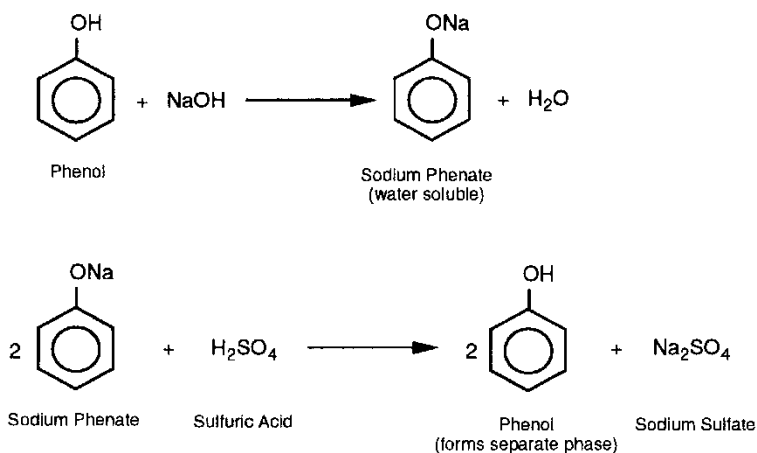


H. Phenol Recovery

The importance of an adequate effluent treatment scheme for a phenol plant cannot be underestimated; first as a means of improving production efficiencies and second, as a criteria to meet strict environmental regulations. An organic stream bearing phenol traces is contacted with sodium hydroxide (NaOH) to form sodium phenate. Sodium phenate is highly soluble in the aqueous phase. The aqueous

sodium phenate streams are then combined and the phenol is "sprung" from the aqueous phase by the addition of sulfuric acid. The organic phase sprung phenol is then sent back to the ion exchange section for recovery.

Phenol can be extracted from water in the same manner. A clean solvent stream (cumene) is contacted with an acidic phenolic water stream to extract phenol into the organic phase. The solvent is then cleaned by contacting with caustic to remove the phenol. By the UOP process, supersaturation of caustic and phenol is achieved resulting in extremely low caustic usage, below the stoichiometric ratio.



III. Process Variables

This section introduces the major process variables (temperature, pressure, flow rate, etc.) and their effect on process performance. The relationship between the process operating conditions, process performance and the process variables is identified. Information is also given about feedstock quality.

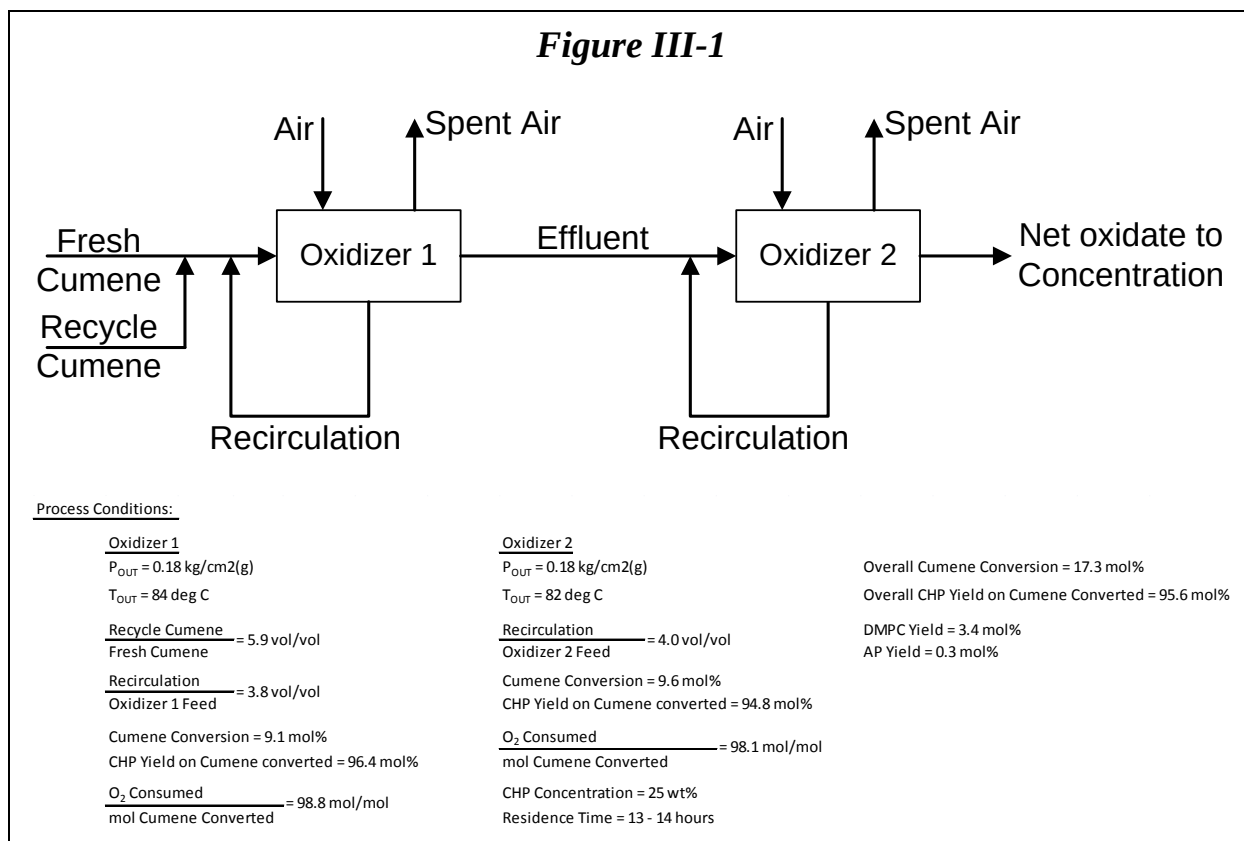
A. Oxidation

1. Oxidizers

The primary process variables involved in the oxidation step are listed below:

- 1) Temperature
- 2) CHP Concentration
- 3) Space Velocity
- 4) Pressure
- 5) Feed Flow Rates
- 6) Spent Air Oxygen Concentration
- 7) Reaction Environment and Inhibitors (Contaminants)
- 8) CHP Yield (Dependent Variable)

See Figure III-1 on the next page for typical oxidizer parameters. A detailed discussion of each of the variables listed above follows:



1.1 Temperature

Oxidizer temperature is one of the variables that control the rate of the oxidation reaction. Conversion rates are increased with an increase in temperature and are decreased with a decrease in temperature. The oxidizer temperature is adjusted to give the desired production rate. There is a minimum start-up value of 80°C (175°F). The effect of temperature is most pronounced in the formation of by-products. In series oxidation, the temperatures are typically cascaded downwards in successive oxidizers, as the effluent CHP concentration from each oxidizer increases. In this manner, the highest CHP yield is realized at the mildest operating temperature possible in the last oxidizer.

During operation, each oxidizer's temperature and air rate should be adjusted to maintain the optimum CHP concentration distribution in the oxidizer at all times. Temperatures above 95°C (203°F) should be avoided to minimize by-product formation. The maximum oxidizer temperature before automatic shutdown of the air is 100°C (212°F). Excessive oxidizer temperature is a major hazard to the safety of operation and must be avoided.

1.2 CHP Concentration

As has been discussed in the preceding sections, the oxidation of cumene to CHP proceeds autocatalytically; i.e. CHP is also the catalyst, as the CHP radical that initiates the free radical reaction scheme is derived by the thermal decomposition of CHP. Thus, the oxidation rate increases simultaneously with the rate of CHP formation. As CHP concentration increases, the rates of by-product formation also increase. It is most important to note, however, that CHP is inherently unstable and simultaneous decomposition reactions take place with its formation. The potential for such decomposition reactions to occur in an uncontrolled and hence hazardous manner is greater at higher CHP concentrations and oxidizer temperatures. Therefore, the optimum concentration of CHP is selected by design that will form the minimum amount of by-products, i.e. optimum selectivity. This entails maintaining an acceptable conversion typically between 20 and 30 wt% and above all, restricting the CHP concentration to a safe value.

Because of the above constraints, it has been found that the most practical approach to oxidation is to stage the oxidation in a series of reactors. The final CHP concentration is increased in stages and with lower temperatures being employed at each consecutive stage. This results in cumene conversion at the highest efficiency.

1.3 Space Velocity

Space velocity is defined as the mass or volumetric rate of feed per unit volume of the oxidizer. A higher space velocity results in a lower residence time and vice-versa. Typically, residence time is used as the standard measure in a continuous stirred tank reactor (CSTR) such as the oxidizer rather than space velocity. As with temperature, residence time mostly impacts byproduct formation. Besides changing production rate, residence time can only be adjusted when at low production rates by decreasing the oxidizer level.

Low residence time does result in the highest CHP selectivity at a fixed oxidizer temperature. However, at that fixed temperature the increase in selectivity is gained at the expense of conversion. Unit optimization entails balancing conversion and selectivity and low conversion is uneconomic as the cumene recycle costs would be high. In practice operating the oxidizers at the highest residence time, highest oxidizer level, is the most economic as it allows operating at reduced oxidizer temperature for a given CHP target (conversion). The gains in selectivity at lower temperature more than offset the reduced selectivity from increased residence time.

To maintain or improve selectivity at lower production rates at fixed residence time requires that the oxidizer temperature be reduced. However, for safety the oxidizer temperature can only be reduced to a minimum of 80 deg C. Below this temperature and there are concerns on high vent gas oxygen due to low reaction rate. To achieve low production rates while keeping within the oxidizer temperature

limitations the residence time must be maintained by reducing the oxidizer level. The minimum oxidizer level for oxidizers designed prior to the year 2000 is 60% of the normal 7.3-7.9 m (24-26 feet) of clear liquid height (e.g. non-bubbling level). 60% is the minimum level that will cover the feed distributor. If the feed distributor is not designed into the unit, then lower levels can be run. The limitation on level is the height necessary to cover the disengaging shroud around the suction nozzle to the oxidizer circulation pumps; this level is typically about 3.6m (12 ft).

1.4 Pressure

The cumene oxidation rate and selectivity is also dependent on the partial pressure of oxygen. The oxidation reaction could be carried out using molecular oxygen, but this approach cannot be justified from an economic and safety standpoint. Using atmospheric air as the source of oxygen is therefore the preferred approach. The system pressure is fixed by design and cannot be changed due to the low maximum pressure rating on the oxidizers. The low pressure rating minimizes the cost of the oxidizer and the air compressor.

1.5 Feed Flow Rates

The fresh cumene feed flow rate is a dependent variable based on the level in the feed surge drum, which is dependent on the combined feed flow rate, which is dependent on the oxidizer level, which is dependent on the flash column bottom level, which is dependent on the decomposer feed flow set point. In a steady state condition, the concentrated CHP flow rate to the decomposer resets the cumene fresh feed flow rate via inventory or level control.

The air flow rate to each oxidizer is independently set by a flow control valve. The flow rate to each oxidizer is based on three other parameters,

- a. CHP flow rate to the decomposer (decomposer feed flow rate * CHP concentration)
- b. Spent air oxygen concentration target
- c. Air distribution between the oxidizers

The CHP production rate is directly proportional to the total net air consumed in the oxidizers at a constant yield. The net air consumed is based on the air flow rate and the spent air oxygen concentration target. The air distribution between the oxidizers will affect the yield observed. Typically, the air is split to be slightly more than 50% going to oxidizer 1 in a two equal oxidizer system. This will give the optimal overall CHP yield.

The net air consumed in the oxidizer must be balanced with the CHP consumed in the decomposer. For example, if insufficient air is sent to the oxidizer relative to the decomposer CHP consumption, then the mass of CHP in the oxidizers will fall given that spent air is kept constant. Consistent oxidizer performance requires that the

CHP concentration, net air consumption, and decomposer CHP consumption all be in balance.

A chart is typically generated with phenol production rate and air flow rate on each axis. A family of lines can be generated at different spent air levels (e.g. 4%, 5%, 6%, 7%). Once a point on the graph is selected, decomposer section flow rate (e.g. production rate) and spent air composition, the air flow rate to each oxidizer is set. The operator then adjusts the temperature of the oxidizer to maintain the CHP concentration to the target value, typically the design value (for example, Ox1 13%CHP, Ox2 25%CHP). If everything is in balance, the spent air should be near the target concentration.

1.6 Spent Air Oxygen Concentration

The oxygen concentration in the spent air will indirectly affect the selectivity to CHP and DMPC. When the spent air oxygen concentration is high, selectivity to CHP is maximized at the expense of DMPC selectivity. If higher DMPC selectivity is desired, for AMS producers, then minimizing the oxygen concentration will increase the DMPC selectivity. The practical range for the excess oxygen concentration is 4-7 mole%. The upper limit is restricted by the lower explosion limit (LEL) for oxygen/cumene mixtures. An automatic shutdown of the air flow to the oxidizer occurs at 9.5 mole% oxygen. The lower limit is restricted by the amount of DMPC that the decomposer section calorimeters will allow.

The spent air oxygen concentration is controlled indirectly by the oxidizer temperature. The oxidizer temperature controls the CHP reaction rate that consumes the oxygen. Other parameters such as CHP concentration, oxidizer residence time, and feed contaminants concentration will also affect the CHP reaction rate.

1.7 Contaminants

Below is a list of contaminants that will impact oxidizer performance. Each one will be discussed in the following paragraphs.

- a. Phenol and Sulfur
- b. AMS and Organic Acids
- c. Benzene and Toluene
- d. Metals
- e. Caustic

1.7.1 Phenol and Sulfur

The most common inhibitor of oxidation is phenol. At concentrations of just 100 wt ppm in the oxidizer feed, the reaction rate can be reduced by as much as 50%.

Other feed contaminants, such as sulfur compounds, aniline, or styrenes all have a similar, though perhaps not as significant, effect on oxidation by tying up the cumene peroxide free radical and thus breaking up the chain reaction.

1.7.2 AMS and Organic Acids

AMS can also adversely impact the oxidation reaction by producing acids and tars. Several hundreds of wt ppm AMS can be tolerated in the combined feed before the unit capacity would have to be reduced. Any acids formed will directionally reduce the oxidizer pH. The more important point to note is that acids may promote decomposition of CHP to generate phenol. If allowed to propagate this highly exothermic reaction may result in a potentially hazardous situation, hence it cannot be overemphasized that it is absolutely important to ensure that acids never be allowed to accumulate in the oxidizer loops or be introduced into the system as a feed contaminant.

The organic acids present in the highest concentration in the oxidizers are formic and oxalic acid. The acid concentration of the oxidizer is measured by regular sampling of both pH and acidity as benzoic acid. The acidity in the oxidizer should never be allowed to increase above 700 wt ppm as benzoic acid. The acidity will normally be 100-250 wt ppm at the terminal (e.g. last) oxidizer.

1.7.3 Benzene and Toluene

Reaction inhibition can also occur due to other feedstock impurities, e.g. benzene, toluene, etc. Such organic impurities are generally not as reactive as cumene, but can form some organic acids and take up volume in the oxidizer.

1.7.4 Metals

Other materials that can catalyze the decomposition of CHP include soluble metal compounds of copper, cobalt, manganese or iron. Some of these accelerate oxidation to such an extent that complete oxidation of cumene can result in liberating carbon dioxide and heat. These contaminants should be kept out at all costs to avoid hazardous conditions developing in the oxidizers. Iron will also catalyze the decomposition of CHP. The metallurgy of the oxidizer is 304SS, so iron or rust will not normally be present. Iron or rust can come in via the air system or the cumene feed system. Filters and caustic scrubbing are used to remove acidity and entrained particles from the air. As well the fresh cumene feed can be caustic washed and has a fine mesh filter to remove particles.

1.7.5 Caustic

Small amounts of caustic carryover to the oxidizer are not a problem as caustic will neutralize some of the organic acids and also react with CHP to form the sodium salt of CHP (Na-CHP) which is effectively removed in the preflash column feed filters. Na-CHP is a yellow viscous material. With significant Na-CHP present there

will be an increase in the rate of delta P increase of the preflash column feed filters and change-out would be required often to avoid them breaking.

High concentrations of caustic in the oxidizer will quickly overwhelm the preflash column feed filter and the will no longer be effectively removed. At that point the sodium salts will exit the flash column bottoms to the decomposer and neutralize the sulfuric acid catalyst in the decomposer resulting in the inability to operate the phenol unit. At this point oxidation, concentration and decomposition must be shut down so the oxidizer can be washed with water to remove the sodium.

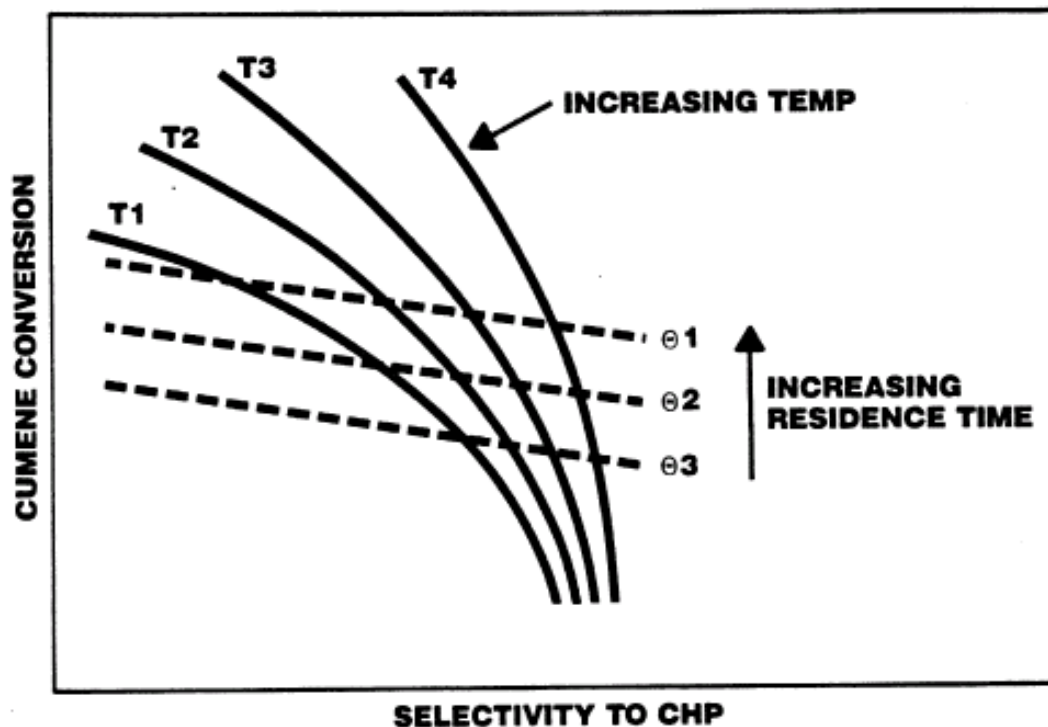
In addition caustic in the oxidizer will increase the rate of oxygen uptake in the oxidizers, forming byproducts such as Acetophenone and DCP. A reduction in the spent air oxygen concentration during a caustic excursion might be observed.

1.8 CHP Yield (Dependent Variable)

All process variable changes to the oxidizer's operating conditions should be made in a cautious and controlled manner. Large step changes should be avoided at all times. Figure III-2, shown on the next page, is a plot that relates cumene conversion and CHP selectivity as a function of temperature. The conversion and temperature ranges indicated are, for all practical purposes, the limiting range to ensure optimum process efficiency and safe operation.

FIGURE III-2

OXIDIZER CONVERSION vs. SELECTIVITY



$T_1 > T_2 > T_3 > T_4$ —OXIDIZER TEMPERATURE, °C

$\theta_1 > \theta_2 > \theta_3$ —RESIDENCE TIME

PRACTICAL OPERATING RANGES:

TEMPERATURE - 80-95 °C

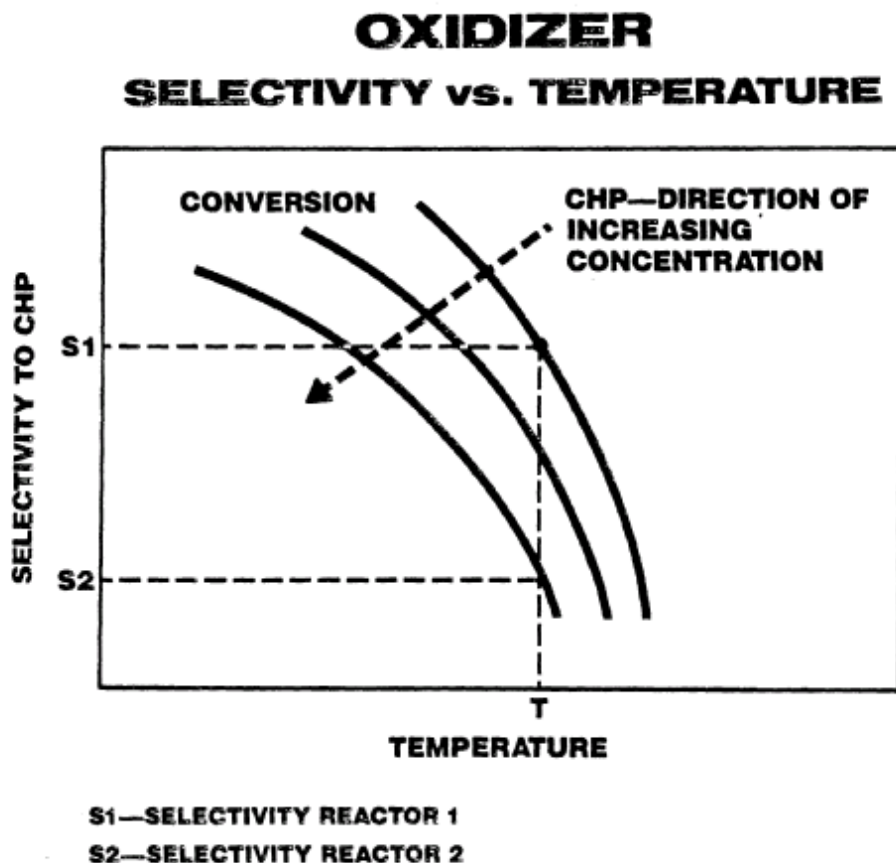
CHP CONCENTRATION - 25-40 WT%

OVERALL CHP CONVERSION - 20-30 WT%

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Figure III-3, shown on the next page, illustrates the basis of stage-wise oxidation at successively lower oxidizer temperatures. If the consecutive oxidizers were operated at the same temperature, then the reaction selectivity would decrease. A small decrease in CHP selectivity is to be expected as the CHP concentration in the oxidizer effluent increases. To minimize the selectivity loss, however, the operating temperatures of consecutive oxidizers are lowered to an optimum value.

FIGURE III-3



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2. Compressed Air Scrubbing

Pretreating the air to the oxidizer is essential to good oxidizer performance. Air potentially has enough acidic compounds to impact oxidizer performance. This potential is dependent on the local environment. Pollution, such as SO_x and NO_x, from nearby manufacturing plants is typically the cause. When this potential exists, an air scrubber is required. A compressed air scrubber will remove acidic compounds. Optimum washing efficiency is realized by manipulating the following variables.

- a. Water Wash Flow Rate
- b. Caustic Concentration

2.1 Wash Water Flow Rate

Appropriate makeup water and caustic rates and dilute caustic recycle rates are employed to provide the necessary water to air ratio to ensure adequate contacting, maintain the target caustic concentration and purge salts and other contaminants removed from the air. Metals such as iron will catalyze CHP decomposition in the oxidizers and are efficiently removed by contacting the air with the recirculating water. When the incoming air is dry, water addition is required to maintain dilute caustic circulation as the air will strip the water from the scrubber. When the incoming air is humid, water must be purged from the scrubber and additional caustic must be fed to maintain the target caustic concentration of 0.1wt%.

2.2 Caustic Concentration

A basic environment is essential to ensure removal of acidic contaminants from the air such as H_2S which forms sulfuric acid when contacted with water. The pH should not be allowed to get too high, however, as this increases the interfacial surface tension and thereby affects phase separation efficiency; resulting in caustic carryover to the oxidizer which is undesirable. The typical circulating caustic concentration is 0.1 wt%.

3. Cumene Feed Wash Column Caustic Scrubbing

Pretreating the cumene feed to the oxidizer is essential to good oxidizer performance. The FWC removes light acids and can remove trace phenol from the recycle cumene during unstable operations. Optimum washing efficiency is realized by manipulating the following variables.

- a. Water Wash Flow Rate
- b. Tray Hole Jetting Velocity
- c. Caustic Concentration

3.1 Wash Water Flow Rate

The appropriate aqueous makeup and dilute caustic recycle rates need to be employed which provide both the necessary aqueous to organic phase ratios and ensures adequate purging of the sodium salts to keep them from accumulating in the aqueous phase. Accumulation of salts lengthens the time required for efficient phase separation.

3.2 Tray Hole Jetting Velocity

The tray hole jetting velocity is defined by dividing the combined cumene flow rate to the feed wash column by tray hole area. The contact efficiency between the aqueous and organic phases is proportional to the jetting velocity. When the jetting velocity is increased, the efficiency goes up. As the jetting velocity goes up the

settling time will go up. At some point increasing the jetting velocity will cause an emulsion to form that will be difficult to break. This point is dependent on the composition and temperature of the phases. A typical range of jetting velocities is 1.2-1.7 m/s (4-5.5 ft/s).

3.3 Caustic Concentration

A basic environment is essential to dissolve the various organic salts into the aqueous phase and neutralize recycled organic acids. The organic salts present may include sodium formate, sodium benzoate, and sodium acetate. The pH should not be allowed to get too high, however, as this induces the formation of emulsions, increases the interfacial surface tension and thereby affects phase separation efficiency.

The sodium salt of CHP will form and leave with the aqueous phase when the caustic concentration approaches 2-3 wt%. This caustic concentration should only be used in an emergency such as an oxidizer acid excursion caused by phenol contamination. At the typical circulating caustic concentration of 0.5-1.0 wt%, phenol is not efficiently scrubbed in this column.

4. Charcoal Adsorbers

Trace quantities of volatile acidic by-products (primarily formic and oxalic acid) are formed during oxidation. These acids are stripped from the oxidizers by the spent air along with some cumene. The spent air is directed to a condenser after which the organic liquid, primarily cumene, is recovered as a separate phase in the decanter. The recycled organic liquid from the decanter is contacted with dilute caustic in the feed wash column to remove the acidic material which is extracted into the aqueous phase. The decanter also removes a separate aqueous phase which can be fed to the PRU, the peroxide destruction reactor or to waste water treatment.

Prior to the charcoal adsorbers a small superheater after the oxidizer chilled vent gas separator increases the temperature of the spent air prior to entering the charcoal adsorbers. Increasing the temperature of the spent air prior to the charcoal adsorbers is another recent innovation to increase the capacity of the charcoal. When the spent air exits the oxidizer chilled vent gas separator it is water-saturated. Increasing the spent air temperature from 5°C to 15-20°C lowers the relative humidity of the spent air. By reducing the likelihood of water condensing on the charcoal the capacity of the charcoal is retained. The life of the charcoal is typically dependent on its hydrocarbon removal efficiency therefore this will directionally increase the charcoal life because the hydrocarbon removal efficiency will be retained longer.

Process variables impacting charcoal adsorber operation are:

- a. Regeneration Cycle Time
- b. Steam Flow per Pound of Charcoal
- c. Cooling Time
- d. Superheat of Incoming Spent Air

4.1 Regeneration Cycle Time

Typically, the regeneration cycle is set at 2-3 hours, with each adsorber on stream for 4-6 hours. More recent designs are for a 2 hour regeneration. The adsorbers are typically operated in series to maximize the adsorption of hydrocarbons and to cool down the regenerated air. Some units operate parallel adsorbers, but this requires a cool down blower. The regeneration cycle time is divided into three periods, steam out, draining, and cooling.

4.2 Steam Flow

The steam out is performed using low pressure steam. Approximately, 0.5 weight of steam is required for each weight of charcoal. This typically corresponds to 1 to 1.5 hours of steaming at design steam rates. The drain time is set to 15-30 minutes with the cool time being set by the rest of the time in the regeneration cycle after steam out.

4.3 Cooling Time

Cooling is typically accomplished by putting the charcoal adsorber in the lag position for series flow operation. Also, cooling can be performed by circulating charcoal-treated-spent air with a blower through the hot charcoal adsorber and to a cooler. A cool down blower is designed for unit with parallel charcoal adsorbers.

4.4 Spent Air Superheat

The spent air from the oxidizer chilled vent gas separator is heated to 15 deg C with low pressure, typically 1 bar(g), steam. Superheat of the spent air avoids condensation of water and hydrocarbons on the charcoal which reduces charcoal capacity therefore influencing emissions and regeneration cycle time. If the superheater outlet temperature is high, greater than 20 deg C, the charcoal bed temperature will be increased which can reduce its adsorption efficiency and capacity.

5. Peroxide Destruction Reactor

A Peroxide Destruction Reactor (PDR) is used to thermally decompose the peroxide compounds in the oxidation waste water streams. Peroxides can have a negative effect on the waste water biological treatment systems. The 2 main peroxide containing water streams are generated by the Feed Wash Column and the Decanter. The streams are preheated to the desired temperature and then sent

through a residence time reactor to allow the reactions to proceed to completion. The reactor is not liquid full allowing any gases generated to be vented. The product is cooled and sent to secondary treatment facilities for further processing.

The main process variables that govern the performance of the peroxide destruction reactor are:

- a. pH
- b. Temperature
- c. Pressure
- d. Residence Time

5.1 pH Adjustment

pH in the Peroxide Destruction Reactor is maintained between 11 and 12 during normal operation. There are two analyzers, one located before the 20% caustic injection point and one located after the injection point, that help monitor the pH of the combined stream entering the PDR. Caustic injection flow is regulated to maintain the pH of combined stream in the required range.

5.2 Temperature

The required temperature to decompose peroxides is 130°C. This temperature is attained first by exchange with the reactor effluent to increase the stream temperature to ~100°C. The temperature is further raised to 130°C by exchange with low pressure steam in the peroxide destruction trim heater. The feed temperature is controlled by a TIC that operates the condensate outlet valve on the trim heater.

5.3 Pressure

Though reactor pressure does not immediately impact the performance of the PDR, sufficient pressure is maintained to keep the water in the liquid phase.

5.4 Residence Time

The residence time of the Peroxide Destruction Reactor is only a function of the liquid flow rate to the vessel. The vessel is conservatively sized to ensure adequate residence time within the vessel during periods of high water production such as shutdown.

B. Concentration

The CHP produced in the oxidation section is concentrated in the Preflash (PFC) and Flash (FC) columns in the concentration section. Concentrating CHP is easily

achieved because cumene is significantly more volatile than CHP. When subjected to excessive temperatures, CHP will undergo rapid decomposition. This is a situation, as must be repeatedly emphasized, which should be avoided. Because of this inherent instability of CHP, the evaporation of cumene from CHP is carried out under very deep vacuum conditions to minimize the temperature required.

The temperature in the Preflash and Flash columns should not exceed about 110-120°C (230-250°F). An automatic shutdown will occur if the temperature is allowed to go this high. The actual shutdown temperature is unit specific and dependent on the normal operating temperature because it is based on the design flash zone temperature (10°C over normal operating temperature). The automatic shutdowns for the PFC can be a little less rigid due to the lower concentration of CHP and lower flash zone temperature. The PFC shutdowns are set to 20°C (36°F) above the normal operating temperature. Some recent designs have increased the pressure and flash zone temperature in the PFC, so in those cases the temperature shutdown guidelines are set to be the same as the FC.

Vapor pressure-temperature relationships (low and high ranges) for cumene, phenol, CHP and other organics present in the evaporator feed is shown on the next pages in Figures III-4, III-5, and III-6.

Figure III-4

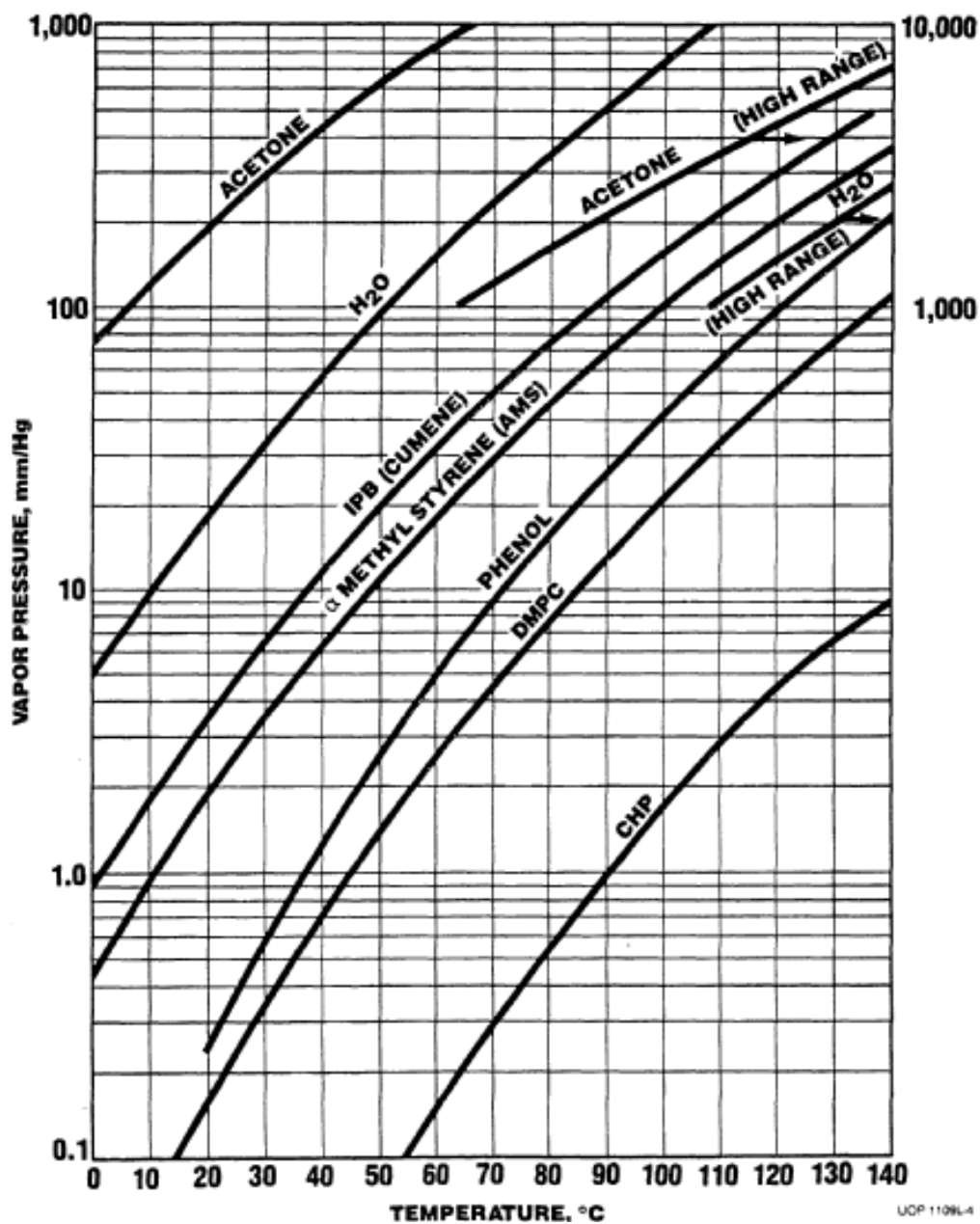
VAPOR PRESSURE vs. TEMPERATURE

FIGURE III-5

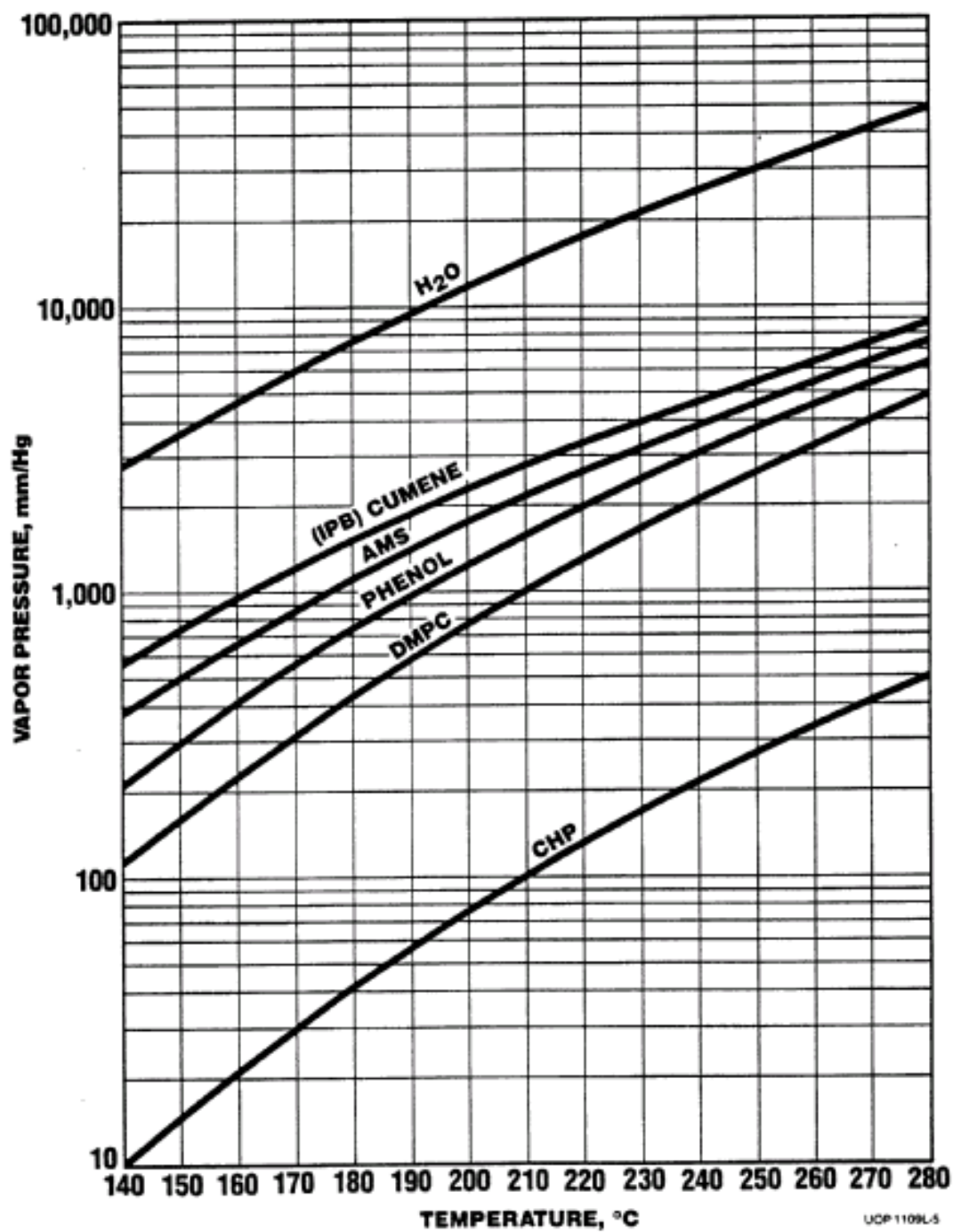
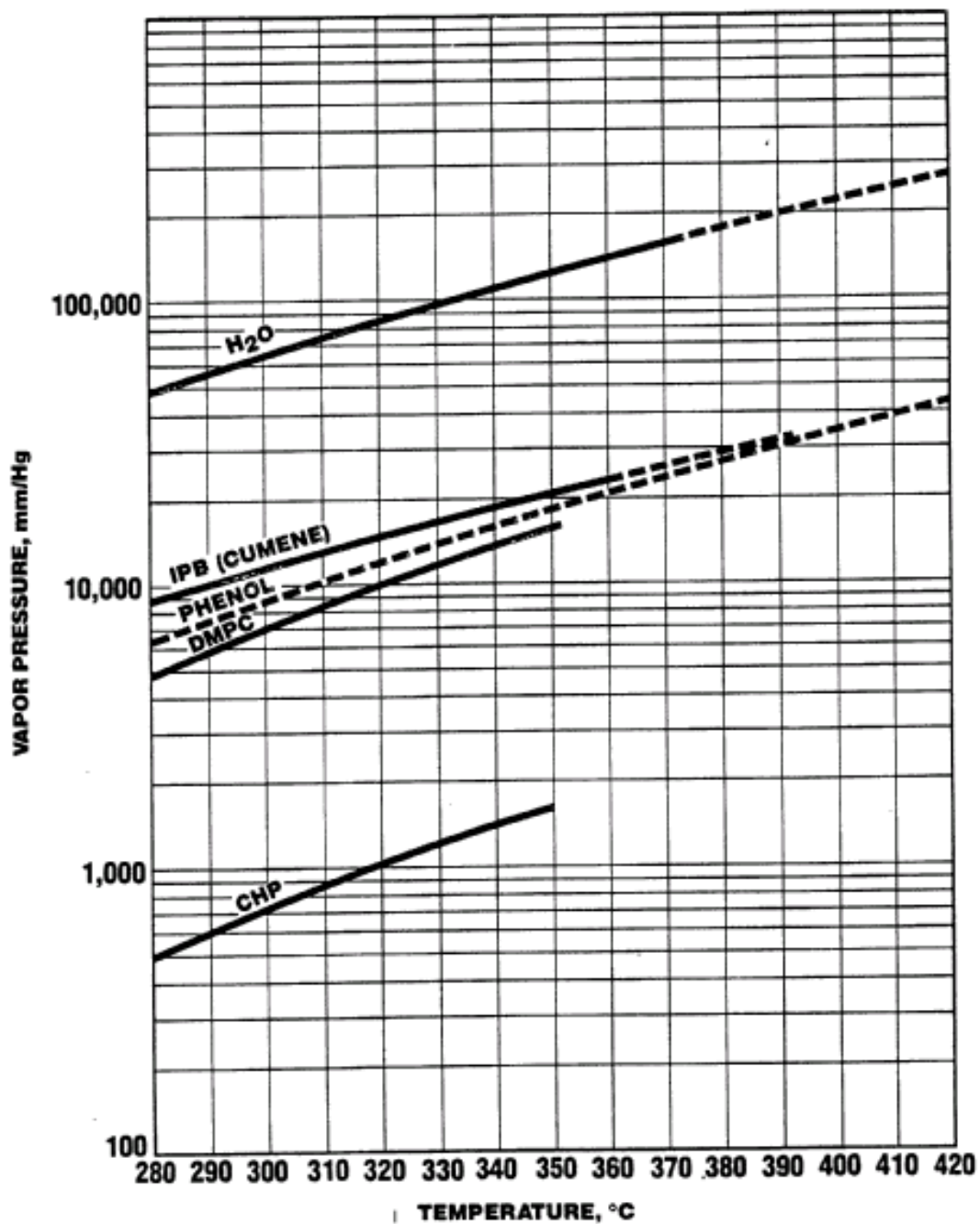
VAPOR PRESSURE vs. TEMPERATURE

Figure III-6

VAPOR PRESSURE vs. TEMPERATURE

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The primary process variables governing the performance of the concentration section are:

- a. Pressure
- b. Flash Zone Temperature
- c. Reflux Flow
- d. Contaminants

1. Pressure

The pressure in the flash zone below the structured packing determines the temperature in this zone. This dependent variable is a function of the overhead pressure, and the pressure drop through the packing. Typically, the flash zone pressure is set at 22-27 mm Hg in the design. Operating at the lowest flash zone pressure will minimize the temperature in the flash zone and increase safety. The vapor velocities in the column will increase with decreasing pressure and may affect the CHP-cumene separation and reflux flow rate required to maintain the separation. The PFC may have a higher flash zone pressure to allow the flash zone temperature to operate in a similar range to the FC flash zone temperature. This will allow the overhead condenser to be smaller, but have the negative effect of increased utilities since the heat exchange with oxidation will no longer be used.

The overhead pressure is controlled by the vacuum system spillback and sets the minimum pressure on both columns. The PFC pressure floats on the FC overhead pressure controller. The vacuum system is typically able to go to a lower pressure than the condensing system can handle. Cumene will boil near ambient temperatures at 10 mm Hg. Excessive quantities of cumene passing the internal condenser may overload the overhead chiller when the pressure is run below 10 mm Hg. Overloading the chilled condenser may cause slugs of cumene to go to the vacuum system, which may cause a pressure upset and automatic shutdown.

2. Flash Zone Temperature

The flash zone temperature is dependent on three other parameters:

- a. Terminal Oxidizer CHP concentration
- b. Vaporizer or Reboiler Outlet Temperature
- c. Flash Zone Pressure

The terminal oxidizer CHP concentration and flash zone pressure are typically fixed by the design. The normal control over this temperature is the vaporizer outlet TIC, which controls the steam flow to the vaporizer. The vaporizers are configured differently for the PFC and FC. The PFC uses a stab-in reboiler in a well, while the flash column employs a vertical shell and tube type. The temperature is varied based on the desired CHP concentration in the bottoms. As the vaporizer

temperature is increased the CHP concentration will increase given that everything else is constant. Variations in the terminal oxidizer CHP concentration may require adjustments to this temperature. Some units have implemented feedback temperature control on the preflash column vapor temperature above the steam heated reboiler to control the steam rate.

3. Reflux Flow

The reflux flow affects the overhead CHP concentration. Increasing the reflux relative to the feed flow rate (reflux/feed ratio) will reduce the CHP in the overhead to an extent. Increasing the reflux will also increase the delta pressure across the structured packing that will increase the flash zone pressure. This is undesirable from a safety aspect. A typical range of reflux to feed ratio is 0.2-0.25 vol/vol for the FC and 0.1-0.15 vol/vol for the PFC. The reflux primarily acts to knock down CHP that is entrained in the cumene vapor.

4. Contaminants

Ordinarily, there are no process contaminants that affect the operation of the flash columns and vaporizers. If some basic compounds are allowed to enter the column through bypassing the feed filters, then the effective surface area of the vaporizer will be reduced through fouling. This could limit the throughput in the column because of high vaporizer temperatures. This would be unusual. Caustic or other salts from oxidation can react with CHP to form Na-CHP which will appear as a yellow slime and rapidly plug the PFC feed filters causing a shutdown due to low feed flow to the concentration section.

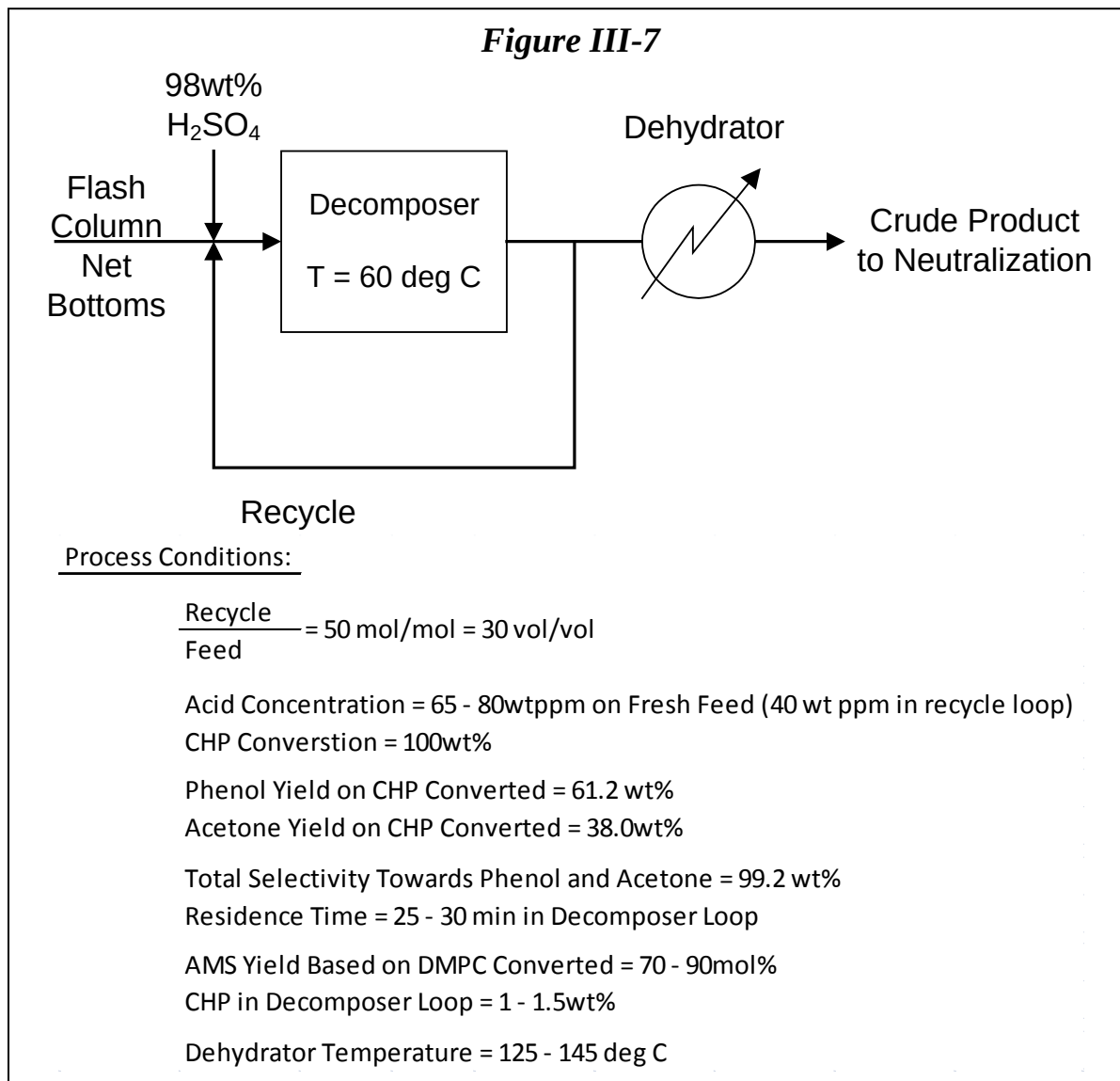
It is also possible, though not probable that an acidity excursion in the oxidization section could cause the sudden decomposition of the concentrated CHP in the bottom of the flash column. The acidity in the oxidizer will concentrate in the bottom of the flash column. To this end, UOP recommends that the terminal oxidizer not be operated over 700 wt ppm acidity as benzoic acid.

C. Decomposition

The concentrated CHP stream is decomposed into phenol and acetone under highly controlled process conditions in the decomposition section. An acid catalyst, concentrated sulfuric acid, is employed to affect the reaction rate. Acid catalyzed decomposition as opposed to thermal cleavage of CHP results in better selectivity to phenol and acetone (i.e. by-product formation is minimized).

The decomposer loop is designed to operate at conditions that provide adequate residence time to convert the majority of CHP. A residual amount of CHP is required in the loop to prevent the acid catalyzed dehydration of DMPC. A simplified block

flow scheme detailing the process and performance parameters of the Decomposer section is shown below in Figure III-7.



The primary process parameters that govern the Decomposer section operation are as follows:

- a. Feed Rate
- b. Sulfuric Acid Concentration
- c. Decomposer Temperature
- d. Water Concentration
- e. CHP Concentration in Decomposer Loop and Calorimeter
- f. Dehydrator Temperature
- g. Decomposer and Dehydrator Piping Volume
- h. Pressure

i. Contaminants

Each of the above parameters is discussed separately.

1. Feed Rate

The decomposer loop is designed such that an adequate residence time based on design fresh feed is available to convert CHP. This time is sufficient from the standpoint of assuring a thorough mixing of the reactants, CHP conversion and removal of the evolved heat. The decomposer feed flow normally sets all the feed flow controllers upstream to the fresh cumene make-up to the oxidizers. If there is an upset in the preflash/flash columns, it is possible that the decomposer feed FIC would be adjusted by the flash column bottoms LIC.

2. Sulfuric Acid Concentration

Sulfuric acid is required to selectively drive the decomposition reaction of CHP to phenol and acetone. The sulfuric acid concentration is adjusted by flow from three separate acid injection pumps. Two sources come from the calorimeters, while the third source is downstream of the decomposer cooler. The typical range is 40-60 wt ppm sulfuric acid in the decomposer loop. Increased acid concentration will increase the reaction rate and lower the residual unreacted CHP in the decomposer loop.

It is possible to operate at a lower sulfuric acid concentration, but the safety margins would be compromised. It would take less basic compounds to neutralize lower acid concentrations. Inadvertently neutralizing the decomposer loop will stop the decomposition reaction and result in the build-up of CHP and a potential uncontrolled decomposition of the CHP.

3. Decomposer Temperature

Temperature also influences the CHP reaction to phenol and acetone with the same relationship as acid concentration. Increasing the temperature will increase the CHP reaction rate and lower the residual CHP in the decomposer loop. The typical operating range is 60-70°C (140-160°F). The range is limited by an automatic shutdown at 57°C (135°F) on the low side and too quick of a reaction rate on the high side. At steady conditions the decomposer TIC is set at 60°C (140°F).

A temperature increase of about 5-7°C (9°F) will be noted between the decomposer and the inlet of the decomposer cooler, depending on the recirculation to feed ratio. A circulation to feed flow volume ratio of 25:1 to 50:1 minimizes the CHP and temperature variation in the piping and maximizes reaction selectivity. Lower ratios can result in an uneven CHP distribution and therefore generate temperature

oscillations in the decomposer circulation loop ultimately resulting in unit shutdown. Low circulation ratio and high process control temperature have the potential of causing vaporization at the decomposer circulation pump suction resulting in cavitation.

4. Water Concentration

Water inhibits the CHP decomposition reaction to phenol and acetone and is used to control the residual CHP concentration in the decomposer loop at 1 – 1.5 wt% which is monitored by measuring the first stage calorimeter delta temperature, discussed further in Section 5 below. The water concentration in the dehydrator cooler product is monitored and targeted between 1 - 2 wt%. Since water is formed by the dehydration of DMPC to AMS, measuring the dehydrator effluent will take this into account. The decomposer loop should also be monitored periodically, especially during the start-up period as too much water at start-up can prevent the initial decomposition of CHP during feed-in and be a safety concern.

5. CHP Concentration in Decomposer Loop and Calorimeter

The CHP concentration in the decomposer loop will directly affect the yield of AMS in the dehydrator. If acid catalyzed dehydration of DMPC were allowed to proceed, the molar yield of AMS would be less than 50%. With 1-1.5 wt% CHP in the decomposer loop, the AMS yield will be between 70-85 mol%. The decomposer temperature, sulfuric acid concentration, water concentration, feed CHP concentration, and residence time all influence the CHP cleavage reaction rate. As the decomposer feed CHP is lowered, the cumene concentration increases, resulting in some inhibition of the CHP decomposition reaction.

As the DMPC concentration in the decomposer feed goes up, then it is more difficult to maintain the AMS yield at a constant CHP concentration. This is due to the fact that the reactant (DMPC) concentration is increasing and the side reactions will become more competitive. The CHP concentration can be increased to some extent to minimize the yield loss. A higher DMPC concentration in the decomposer feed occurs when the terminal oxidizer CHP concentration is increased, for example to 32-40 wt%.

All parameters must be monitored and adjusted to maintain optimal yield and safe operation. The CHP concentration is monitored by the laboratory on a regular basis as well as the first stage delta temperatures in the two calorimeters. The first stage delta temperature results from the residence time induced decomposition of CHP. The calorimeters are spared so on-stream monitoring can be maintained while one calorimeter is periodically cleaned. Approximately 7.2°C (13°F) delta temperature will be measured per weight percent CHP in the stream at the design calorimeter first stage residence time. Approximately 4.1°C (7.4°F) per percent DCP will be measured at the design calorimeter seconds stage residence time.

Sulfuric acid will add an additional 3°C (5.4°F) to the calorimeter delta temperature by the heat of solvation of the acid into the organic stream. Sulfuric acid will react with phenol to form phenol sulfonic acid, which is an exothermic reaction.

The second stage in the calorimeter measures the DCP (major) and DMPC (minor) concentration by the exotherm of the DCP dehydration reaction by acid catalysis. Concentrated sulfuric acid is injected at the beginning of the second stage delta temperature. 4000-6000 wt-ppm sulfuric acid is required to dehydrate all of the DCP. Though the yield from this stream is lower than what is attained in the dehydrator, the flow is very small and has an insignificant effect on the overall yield. The acid catalyzed dehydration tends to cause fouling of the small orifices in the calorimeters resulting in the need for periodic calorimeter cleaning.

6. Dehydrator Temperature

The dehydrator temperature and residence time control the DCP dehydration reaction to form AMS. The dehydrator temperature is considered the fine tuning adjustment, while the dehydrator effluent volume is considered the coarse adjustment. The typical operating range for the calorimeters is 125-145°C (260-295°F), with 135-140°C (275-285°F) being the typical design value. The optimum temperature is attained by monitoring the DCP concentration in the decomposer cooler outlet. The DCP concentration is targeted at 100-300 wt ppm to give the best AMS yield and not contaminate the phenol product downstream. The optimal calculated yield occurs when the residual DCP is 400-900 wt ppm. However, this elevated DCP concentration can cause the phenol to go off-specification on o-cresol and possibly the water lights test (old test to measure phenol soluble hydrocarbons). Operation with DCP greater than 400 wt ppm in the crude product should be monitored carefully.

When the DCP decomposes in the fractionation section, some residual DMPC and AMS will continue downstream. It is then possible that some may react to form AMS dimers and cumylphenols that end up in the phenol product. The IX resin treaters do afford some protection against this as they will form heavies that will easily fractionate out in the phenol rectifier and residue stripper.

Operation of the dehydrator below 120 deg C should be avoided as the dehydration reaction rate decreases significantly below this temperature. It will be difficult to reliably control the DCP in the crude product. Operation at this low dehydrator temperature is typically only encountered at low production rates.

7. Decomposer and Dehydrator Piping Volume

The volume in the decomposer and dehydrator and surrounding piping is restricted to assure that the decomposition and dehydration reactions proceed at the

expected rates. Too large piping in the decomposer will result in the inability to adjust the other operating variables to maintain the proper residual concentration of CHP. Too large piping in the dehydrator effluent will result in additional reaction of AMS to tars. An extra piping loop with isolation valves and drains is provided to remove excessive dehydrator effluent piping volume during turndown. Typical volume restrictions as measured in residence times are listed in the Table III-1 below:

Table III-1: Decomposition Section Residence Times

	Time, Minutes
Decomposer Circulation Loop	15-30
Decomposer to dehydrator Piping	0.2 – 0.35
Dehydrator and Outlet Piping	1.0 – 1.3
Calorimeter First Stage (Piping)	0.6
Calorimeter Second Stage	~1.6

8. Pressure

The system pressure is not a variable. It is set by design to ensure that the reaction environment is in the liquid phase at the design process conditions. A minimum pressure is specified to minimize the pressure control gas usage.

9. Contaminants

The contaminants that are a concern are the ones that will neutralize the sulfuric acid circulating in the decomposer loop. Safe operation of the Decomposition Section is the highest priority. The expected basic compounds are carryover caustic and organic salts from the cumene feed wash column from Oxidation. The Oxidation and Concentration Sections are designed to minimize carryover from the oxidation section during normal operation with safeguards of coalescers in both the feed wash column and cumene feed surge drum, the preflash column feed filter to remove salts of CHP from caustic, feed wash column design and operation and a water boot in the cumene feed surge drum. If aqueous carryover occurs from the feed wash column resulting in caustic or organic salts in the oxidizer, then the preflash column feed filters will remove them. Typically, the caustic and salts

present will form a yellow slime on the concentration section feed filter cartridges. The sodium salt of CHP is yellow in color. The sodium salt of phenol (e.g. sodium phenate) is red.

D. Neutralization

The main process variable in the operation of the crude product neutralization section is the injection rate of diamine. The pH is monitored by the lab and controlled at 2.3-2.7. A sulfuric acid analyzer will detect ppm amounts of free unreacted acid.

Due to the lack of separate aqueous phase, an on-line pH analyzer in this service typically does not read the same pH as the laboratory method, which requires dilution of the sample with water. pH analyzers also tend to foul quickly in hydrocarbon service. No on-line pH analyzers have been found to work.

E. Fractionation

The neutralized crude product is processed through fractionation columns to achieve the final separation of the desired products, phenol and acetone, from the other compounds. The fractionation scheme is straightforward, consisting of a series of fractionating columns that are operated in a conventional manner. To minimize contamination of the final phenol product with co-boiling impurities, the crude phenol is further treated in a reactor and IX resin treaters that are a part of the fractionation section. The immediate discussion that follows is a brief overview of fractionation column variables. This discussion is meant primarily as a review of known concepts that should aid the operations in better understanding of fractionation column operation.

1. Fractionation Process Variables

1.1 Reflux and Flooding

Heat input and reflux are the most important variables in distillation. The reflux is directly related to the heat input because the vapor boiled up to the receiver must be returned to the column if it is not drawn off as overhead product. Most columns are operated with a constant heat input. Figure II-6 in Section II can be used to represent the effect of reflux on product quality. The slope of the upper operating line is L/V in the upper section of the column. If the heat added to the column is reduced, thus reducing the L/V ratio and if all other variables are constant, the operating line will approach the equilibrium curve. The number of trays is fixed by column design. Hence, as reflux is decreased, the overhead purity will decrease since the magnitude of rectification (rectification occurs above the feed tray) per tray will decrease. The same argument holds true for the stripping section (below the

feed tray) resulting in decrease in purity of the bottoms. Conversely, if the heat input and reflux were increased, L/V will increase and both the top and bottom product qualities will increase.

Even though the increase in reflux and heat input to the column increase the product purities, it must be kept in mind that there is a limit. The limit is governed by "flooding" of the column. The condition of flooding is said to have been reached when the pressure drop of vapor across a tray is greater than the liquid head in the downcomer or when liquid traffic through the downcomer is excessive. Some of the vapors travel up the downcomer, when a fractionating column "floods". Under such conditions, proper fractionation of the components will cease and the product purities will be affected. The column then has to be unloaded by lowering the reflux rate and heat input to put the column back into normal operation.

1.2 Feed Temperature

Feed to the fractionating column is generally close to its bubble point temperature. If the feed is at a temperature below its bubble point temperature, then the rectifying section (above feed point) will be helped at the expense of the stripping section (below feed point). This will result in higher overhead purity and lower bottoms purity. Conversely, if the feed is partially vaporized, then the stripping section will be helped at the expense of the rectifying section. This discussion, of course, assumes that other variables are held constant.

1.3 Pressure

In distillation, the column pressure is not an operating variable. This does not imply that the degree of fractionation does not change with pressure. We simply mean that under normal operation, the column pressure is held constant at its design pressure. If the pressure is increased, the equilibrium will change such that it is necessary to add more heat input to do the same fractionation. Column operating pressure and required heat input are not linear however. The control of the column pressure is performed by a separate column overhead pressure control system. Once the required pressure is attained, it should not be changed. Operating pressures can vary anywhere from sub-atmospheric to many atmospheres depending on the specific nature of the fractionation being carried out. For the UOP phenol unit, a number of columns are operated under vacuum primarily to limit the process temperatures required to perform the particular fractionation. A hot reboiler temperature generally causes side reactions to occur, which can reduce the phenol product purity and lower the yield by making additional residue.

1.4 Reflux Temperature and Internal Reflux

In Process Principles, Section II, we defined internal reflux and external reflux. In separations of a mixture of two or more components into high purity products the

internal reflux is a very important control variable. Let us develop an equation to define this variable.

By energy balance on top tray

$$H_{in} = H_{\text{vapor to top tray}} + H_{\text{external reflux}} \quad (1)$$

where H = enthalpy, Kcal/kg

$$H_{out} = H_{\text{vapor to condenser}} + H_{\text{internal reflux}} \quad (2)$$

There is no accumulation across the top tray; therefore, heat in = heat out.

$$H_{\text{vapor to top tray}} - H_{\text{vapor to condenser}} =$$

$$H_{\text{internal reflux}} - H_{\text{external reflux}} \quad (3)$$

$$\Delta H_{\text{vapor}} \text{ across top tray} = \Delta H_{\text{liquid}} \quad (4)$$

$$\Delta H_{\text{vapor}} = (\text{kg of vapor condensed}) \lambda \quad (5)$$

where λ = latent heat of vapor

$$\Delta H_{\text{liquid}} = (R_{\text{ext}}) (C_p) (T_v - T_r) \quad (6)$$

where	R_{ext}	=	external reflux
	C_p	=	specific heat of reflux
	T_v	=	temperature vapor from top tray
	T_r	=	temperature of external reflux

Assuming heat losses are zero, on top tray ΔH_{vapor} must be equal

$$\text{to } \Delta H_{\text{liquid}}, \text{ hence, kgs of vapor condensed} = R_{\text{ext}} \frac{C_p}{\lambda} (T_v - T_r) \quad (7)$$

$$\text{By material balance } R_{\text{int}} = R_{\text{ext}} + \text{kgs of vapor condensed} \quad (8)$$

where R_{int} is internal reflux

Substituting (7) in (8) we get

$$R_{\text{int}} = R_{\text{ext}} \left(1 + \frac{C_p}{\lambda} (T_v - T_r) \right) \quad (9)$$

Equation (9) is a simple equation. In practice, control schemes have been developed based on this equation where reflux FIC controls external reflux to hold internal reflux at the constant set point despite variations in the reflux temperature such as those caused by rain storms or changing fan pitch. Without this control, the composite profile in the column would change before the TIC could compensate. With the internal reflux control, the column never "sees" any change due to variations in the reflux temperature.

2. Fractionation Column Control

If sufficient time is given to come to a steady state all fractionating columns will heat and material balance. The law of conservation of mass and energy requires:

$$\text{Input} - \text{output} = \text{accumulation}$$

For a steady state, there is no accumulation in a given system. This essentially means that what goes in must come out. Both the mass and energy inputs must be equal to outputs. Let us examine this concept with respect to fractionation.

2.1 Material Balance

The instrumentation to a column must be designed such that the column material balances in order to establish a stable operation. To establish an overall material balance, both overhead and bottom draw rates must be properly controlled. The portion of the feed that is not taken out as overhead product must necessarily appear as the bottom product or vice versa. The material balance at the bottom of the column can be satisfied by two different schemes.

2.1.1 Net Bottoms Product on Level Control

The bottoms draw rate is controlled by bottoms level controller. Either the reboiler is controlled to provide a constant heat input or a composition controller located in the lower section resets the reboiler heat input. This type of control is used on most columns with only a few exceptions. In the Fractionation Section, the CAC, FAC, C-AMSC, CPC, and RS all have this type of control scheme, with constant reboiler heat input control.

2.1.2 Net Bottoms Product on Flow Control

The bottoms level controller resets the reboiler heat input. The bottoms draw rate is on flow control, which may be reset by a composition controller located in the stripping section of the column. This control scheme is used when bottoms draw rate is small (less than 20%) compared to the liquid rate down the stripping section of the column. Only the phenol rectifier, in the fractionation section, has this variety of control system. However, the PR bottoms flow controller is not reset by a composition controller due to the high purity of the feed and bottoms. Also, due to

this high purity, the reboiler heat input is set on a feed forward ratio control that has reset by the bottoms level control.

For the overhead system, two control schemes are used to material balance the column.

2.1.3 Net Overhead Product on Level Control

The composition controller resets the reflux flow controller and the overhead product is controlled by the receiver level. The net overhead products from the C-AMSC, CPC, and PR are controlled by the receiver level, with reflux on flow control. The C-AMSC reflux has composition controller reset.

2.1.4 Net Overhead Product on Flow Control

The net overhead product is on flow control, which is reset by a composition controller, and the reflux rate is controlled by the receiver level. This is a widely used control scheme. The overhead products from the CAC, and FAC are on flow control. The CAC flow rate is reset by the feed rate to the column using a feed forward ratio control. Only the FAC, however, has flow control reset from a composition controller. The FAC and PR are a special case in that a net side draw is taken instead of net overhead. There is some possibility that light contaminants will accumulate in the overhead. They can be purged by recycling back using a FIC.

It must be remembered that the material balance control on top and bottom are interdependent. The control scheme should be so chosen so as to avoid any interaction.

2.2 Heat Balance

The heat balance is not independent of the material balance. The rate of vapor flow up the column is a function of heat input at the bottom of the column and at the feed point. Hence, the heat input by the reboiler and the feed stream must be removed in the overhead condenser and net products if the column is to heat balance. Of course, this assumes negligible heat loss from the column.

Another aspect of the heat balance is the natural temperature profile in the column. In order to obtain a good separation, the temperature profile must not be constrained. An example of constraining the temperature profile would be to try to hold a tray at a temperature lower than the boiling point of the material on that tray, such as subcooling the reflux below its bubble point. This is a violation of heat balance, independent of temperature profile.

Another example would be to have the reboiler heat input and the reflux controlled independent of each other. This system would be unstable because the heat is not balanced.

2.3 Controlling Product Purities

2.3.1 Temperature Control (TIC)

The simplest type of composition control is the temperature controller (TIC). This is used where high product purities are not necessary. The temperature sensor for this control method is generally located in the upper (rectifying) section of the column on a tray where the temperature change is greatest for a small change in overhead composition. It is limited for use with low purities, however, because a pressure change on the sensor tray causes temperature change without there actually being a composition change.

As mentioned in the previous discussion on material balance, either the overhead draw or the reflux rate is controlled by the TIC, while the receiver level resets the other, thereby maintaining the material and heat balances. The heat supply to the reboiler is placed on flow control to provide a constant heat input, so that the vapor rate up the column is constant. TIC is also used to reset the reboiler heat input when the bottoms composition is critical such as the residue column.

Three of the columns have bottoms compositional control, the CAC, CPC and RS. The CAC is unique in that the bottoms composition (acetone concentration) is adjusted by the injection of steam or condensate. The water acts to change the partial pressure in the vapor phase, which influences the stripping of acetone from the bottoms. Raising the injection rate will reduce the reboiler outlet temperature and improve the stripping. Acetone lost out of the bottom of the CAC is typically not recovered and can cause problems in the effluent section. The separation in the effluent section is affected by the acetone as it would get into both the organic and aqueous phases and can create emulsions.

The CPC and RS both adjust the reboiler heat input to change the reboiler outlet temperature. The reboiler outlet temperature is directly related to the phenol concentration. A higher temperature translates to a lower phenol concentration.

2.3.2 Differential Temperature Control (TDIC)

A good way to compensate for tray pressure fluctuations is to measure the temperature at two points in the column and hold the temperature difference between these two points constant. The principle is very simple. One of the points is located near the top of the rectifying section where the composition is relatively pure. This is called the reference temperature tray. The other point is located on a lower tray in the rectifying section, the tray on which the temperature change is the greatest for a small change in composition.

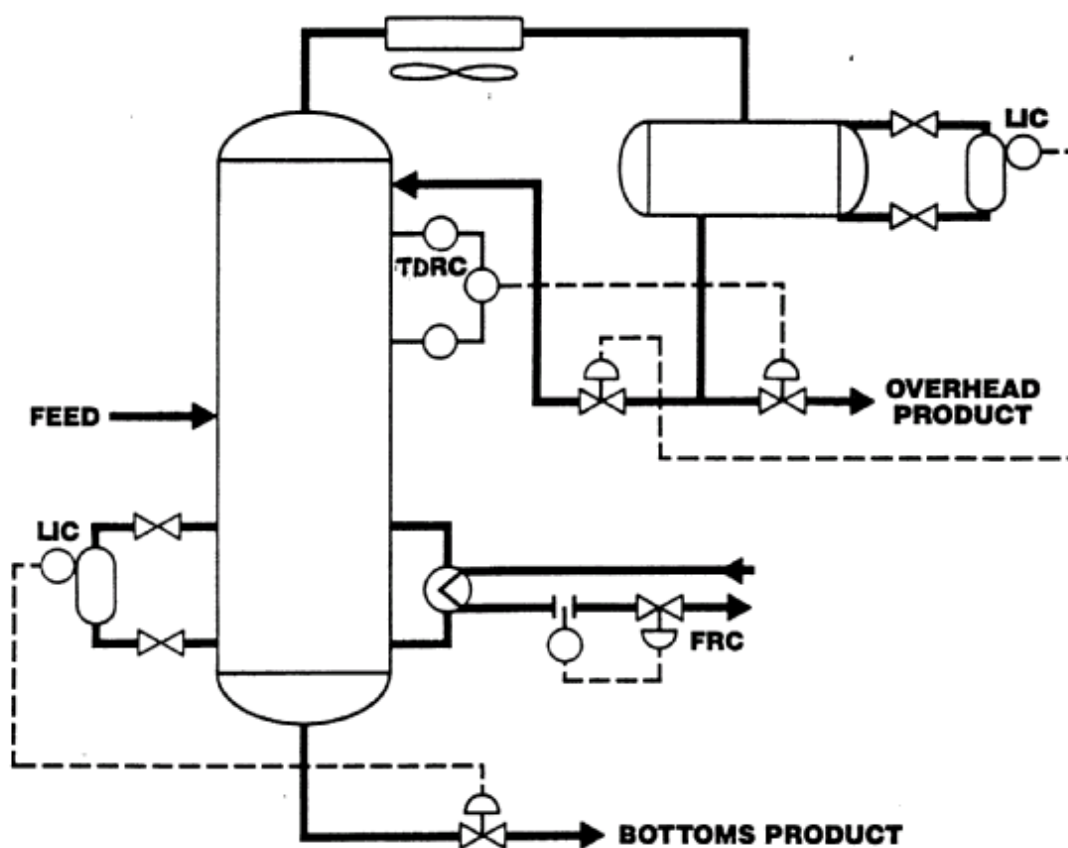
Any pressure fluctuations will affect both tray temperatures equally, thus not affecting the temperature differential. However, if a composition change takes place, the reference temperature will remain the same since this is immersed in pure

material, but the temperature on the lower tray will change according to the amount of composition change. Therefore, by finding a suitable ΔT and holding it constant we can fix the overhead composition.

The TDIC controls overhead draw rate while the reflux is controlled by the receiver level or vice versa. The heat supply to the reboiler is placed on flow control in order to provide a constant heat input. The bottoms draw rate is usually on bottoms level control. This control scheme is illustrated in Figure III-8 below.

FIGURE III-8

DISTILLATION COLUMN



The FAC and C-AMSC are equipped with a variation of this scheme that controls the side-draw flow rate and reflux, respectively.

3. Phenol/AMS Purification with Chemical Treatment

The chemical treatment reactor (CTR) and C-AMS column bottoms provide the environment for the diamine to react with carbonyls that boil in the range of phenol. Most of the variables related to treating are fixed in the design. They are listed below and will be discussed individually.

- a. Diamine Injection Rate
- b. Residence Time
- c. Temperature
- d. Carbonyl Concentration
- e. Contaminants (other reactive species)

3.1 Diamine Injection Rate

The diamine injection rate is based on the phenol product quality requirements. To make ultra-high purity phenol about 1000 wt ppm of pure HMDA per weight phenol is required. This ratio can be adjusted up and down to match the carbonyl concentration observed. Typically, 10-25% in excess of the stoichiometric quantity is needed to react the carbonyls. It is possible to reduce the carbonyl concentration in the phenol to concentrations below the detection limit of the analytical method, but it comes at the expense of excessive diamine usage.

The diamine will also neutralize organic acids in the fractionation system. The source of the organic acids is mainly the oxidation section. In general, the more severe the oxidizer operating conditions, the more organic acids produced. This will tend to increase the diamine consumption at constant phenol product purity.

3.2 Residence Time

The residence time in the CTR is fixed by design at 1.5 hours. The residence time in the Crude Phenol Column and piping give an additional fraction of an hour. At higher than design throughput it will be necessary to compensate for the reduced residence time (e.g. reaction rate) by adding more diamine.

3.3 Temperature

The temperature in the CTR is ~178°C (350°F) and is fixed by the Cumene-AMS Column (C-AMSC) bottoms design temperature and is not an independent variable. The typical column reboiler operating range is sufficient to drive the reaction of carbonyls with diamine to heavies.

3.4 Carbonyl Concentration

The carbonyl concentration in the C-AMSC bottoms is a function of the operations in the oxidation and decomposition sections and the crude acetone column. It

cannot be independently controlled, and should be monitored so the diamine injection rate can be optimized.

3.5 Contaminants (other Reactive Species)

The organic acids in the crude product feed to the fractionation section are formed mostly in the oxidizers. The concentration and type of organic acids will depend on the oxidizer severity and fractionation. The two main acids are formic acid and oxalic acid.

4. Phenol Ion Exchange Resin Treaters

The phenol IX-resin treaters convert 2-MBF to heavies, and carbonyls not converted in the CTR such as acetol (HDA) and MO to heavier compounds that boil outside the range of phenol. The three main process variables are:

- a. LHSV
- b. Temperature
- c. Contaminants

4.1 LHSV

The resin bed is designed on the basis of a specific LHSV = 2.0. Reaction completion is a function of LHSV, temperature and water concentration. At low LHSV's, lower temperatures are required to achieve reaction completion and vice-versa. The LHSV is fixed in the design and cannot be independently controlled since it is a function of the fractionation feed flow rate. The space velocity can be adjusted by changing the loaded quantity of resin. There is sufficient room in the IX resin treaters to load up to 50% extra resin.

4.2 Temperature

As discussed above, the reaction rate is a function of several factors. Due to the small concentration of reactants in the feed, there should be no temperature change across the bed. The resin is very active at the start of operation and a temperature of 80-100°C (176-210°F) should be sufficient. While it is possible to increase the resin temperature to 150°C (300°F) as it deactivates, experience has shown that this tends to make more byproducts and does not greatly increase the activity of the resin. Operating at an excessive temperature may cause the resin to break down. If resin containment is lost, the resin can become reactive and produce additional 2-MBF from the trace carbonyl in the phenol rectifier bottoms/reboiler. Typically, the phenol product is monitored for 2-MBF and the reaction temperatures are adjusted to just high enough to maintain the product spec.

4.3 Contaminants

Normally, the phenol feed should be relatively free of contaminants. However, if any neutralizing compounds such as amine or caustic contact the resin, the resin acidity will be lost resulting in a loss in activity. Regeneration facilities may be available to restore the IX resin acidic activity from a sudden loss in activity.

F. Hüls MSHP (AMS Hydrogenation)

In order to maximize the overall yield of phenol and acetone, AMS produced in the synthesis section is recovered along with cumene in the Fractionation Section. It is caustic scrubbed to remove phenol, and catalytically hydrogenated to cumene. A proprietary catalyst is employed to selectively hydrogenate the AMS. The cumene can then be recycled back to the Oxidation Section.

The primary process variables that govern the operation of this reaction are listed below:

- a. Reactor Temperature
- b. LHSV and SFR
- c. H₂/HC Ratio, H₂ Purity and Rx Outlet Pressure
- d. Contaminants

1. Temperature

Reaction rate is a function of catalyst condition, AMS concentration, LHSV, superficial flow rate (SFR), and the reactor temperature. Given all other variables being constant, higher LHSV's require higher reactor temperatures to achieve reaction completion and vice-versa. The reaction temperature is initially ~60°C (140°F) at the reactor inlet. If possible, the fresh catalyst should be run at a lower temperature for a few days, 50°C, because the fresh catalyst may possess a high initial activity that may result in excessive ring saturation, resulting in the production of isopropylcyclohexane (IPCH). Since hydrogenation is an exothermic reaction, saturation of the rings will generate additional heat. During this initial break-in period, the delta T across the reactor can be lowered by running at a higher recycle to feed ratio.

The primary criteria that sets reactor temperature should be the AMS (unsaturated) concentration of the product which should be at less than 100 Bromine Index (600 wt ppm AMS). It is possible to get below 100 wt ppm AMS in the product. The lead reactor effluent target should be 1 wt% AMS. The IPCH concentration should also be monitored and should be maintained at less than 100 weight ppm.

Bromine Index is a measure of titrated Bromine in a mass of sample. It is defined at the number of milligrams of bromine consumed by a 100 grams of sample under the

conditions specified in laboratory method, ASTM D1492 or D5776. This can be converted to wt ppm AMS with the knowledge that for every two moles of Bromine (Br) reacted, one mole of AMS is consumed. This ratio assumes that there are no other compounds that would react with the bromine; a good assumption for this relatively pure stream. To convert from Bromine Index to wt ppm AMS multiply the BI by MW-AMS/MW-Bromine*10 or $(118.17/79.904*5)$ or 7.394.

Toward the end of run, the reactor temperature should be increased to maintain conversion. Temperatures of up to the limit of 95°C (205°F) at the lead reactor inlet can be used. However, higher reactor temperatures must be accompanied by higher reactor pressure.

A reactor effluent recycle is employed to limit the temperature increase across the lead reactor. Typically, the increase across the lead reactor will be 29°C (52°F) while the increase across the lag or finishing reactor will be 6°C (11°F). Over the catalyst life, the delta temperature can be shifted between the reactors, but the maximum total delta temperature should remain constant at or below 35°C (63°F). Excessive delta temperatures will increase the catalyst deactivation rate as well as by-product ICPH formation.

2. LHSV and Superficial Velocity

The maximum LHSV for the lead recycle reactor is 16 hr⁻¹ at 6 wt% AMS in the combined feed. The maximum LHSV for the lag finishing reactor is 6 hr⁻¹ at 1 wt% AMS. At lower AMS concentrations, a somewhat higher space velocity can be tolerated. Note: the reciprocal value of space velocity is defined as the contact time or residence time in the reactor.

The superficial velocity influences the mass and heat transfer of the AMS hydrogenation at a given LHSV. It also ensures that the hydrogen bubbles stay well distributed in the reactor, ensuring good contact with the catalyst. A higher superficial velocity will reduce the conversion at a given space velocity. The superficial velocity is designed at 40 m/hr liquid and in conjunction with the space velocity sets the reactor bed dimensions.

UOP recommends both reactors be online during all operation to ensure sufficient AMS conversion to not have an impact oxidation acidity.

3. H₂/HC, H₂ Purity and Reactor Outlet Pressure

Hydrogen/Hydrocarbon, Hydrogen purity and reactor outlet pressure are related by the following equation that is the definition of hydrogen partial pressure.

$$P_{H_2} = \frac{P_{rx \text{ outlet}} * (H_2/HC)}{1 + (H_2/HC) * (1/y_{H_2})}$$

where P_{H_2} = hydrogen partial pressure, kg/cm²

H_2/HC = hydrogen to hydrocarbon, molar ratio

y_{H_2} = H_2 purity of the recycle gas, mole%

The design partial pressure should always be maintained in the reaction environment. Reduced partial pressures result in accelerated coke formation and a shortening of catalyst life. When the proper partial pressure environment is maintained by the appropriate combination of H_2/HC ratio and hydrogen purity, catalyst lives are typically very long. As the catalyst ages the process conditions will inevitably have to be adjusted to maintain the desired performance and product quality. In conjunction with the reactor temperature, reactor pressure should also be increased over the life of the catalyst to increase the hydrogen partial pressure up to the limit of the reactor design pressure. Hence, for a given catalyst condition, increasing the partial pressure will increase the catalyst activity.

Typically, the system pressure is increased first as compared to reactor temperature as the catalyst shows signs of deactivation. Increasing pressure has less of an impact on by-product formation compared to increasing temperature.

4. Contaminants

Aside from the variables discussed above, it is also very important to ensure that the quality of the reactor feed be such that it is devoid of potentially harmful contaminants as well as impurities such as caustic, sulfur and halogen containing compounds. Contaminants in the hydrogen such as CO, H_2S , Cl, sodium salts and mercury (Hg) and acetone will decrease the catalyst activity and are therefore limited. Acetone concentrations above 1 wt% will reduce catalyst activity. Various side reactions can otherwise occur which will not only compromise product quality, but also affect the catalyst life.

Recovery from large upsets of sodium salt in the liquid feed have been demonstrated by washing the catalyst with clean water and then drying and re-reducing the catalyst.

In addition, the phenol should be minimized to trace amounts in the feed to the reactor. Mis-operation of the C-AMSCWC is the primary reason for phenol being present in the feed to the hydrogenation reactor. Phenol typically exits the system unreacted. However, as stated earlier it does serve as a severe oxidation reaction inhibitor when recycled to the oxidation section.

G. Phenol Recovery

The discussion of process variables in effluent treatment will be divided into three areas, delay pot/PRU, caustic scrubbing, and oil extraction from water with cumene.

1. Delay Pot/PRU

The delay pot/PRU takes the highly phenolic streams that are mainly from sodium phenate sources and acidifies them to change the solubility of the phenol in the water phase resulting in a separate organic phenol phase. The process variables involved are as follows:

- a. Feed Composition and Cumene Addition
- b. Separator Temperature
- c. pH
- d. Residence Time and Recycle/Feed Ratio

1.1 Feed Composition and Cumene Addition

Figures III-9 and III-10 illustrate the influence of the inorganic salt, sodium sulfate, on the respective solubility of phenol and acetone in water. From a maximum phenol recovery or yield standpoint, the maximum sodium sulfate concentration in the aqueous phase also leads to the minimum phenol and acetone concentration in the aqueous phase. The sodium sulfate concentration is related to the feed sodium phenate composition.

Figure III-9

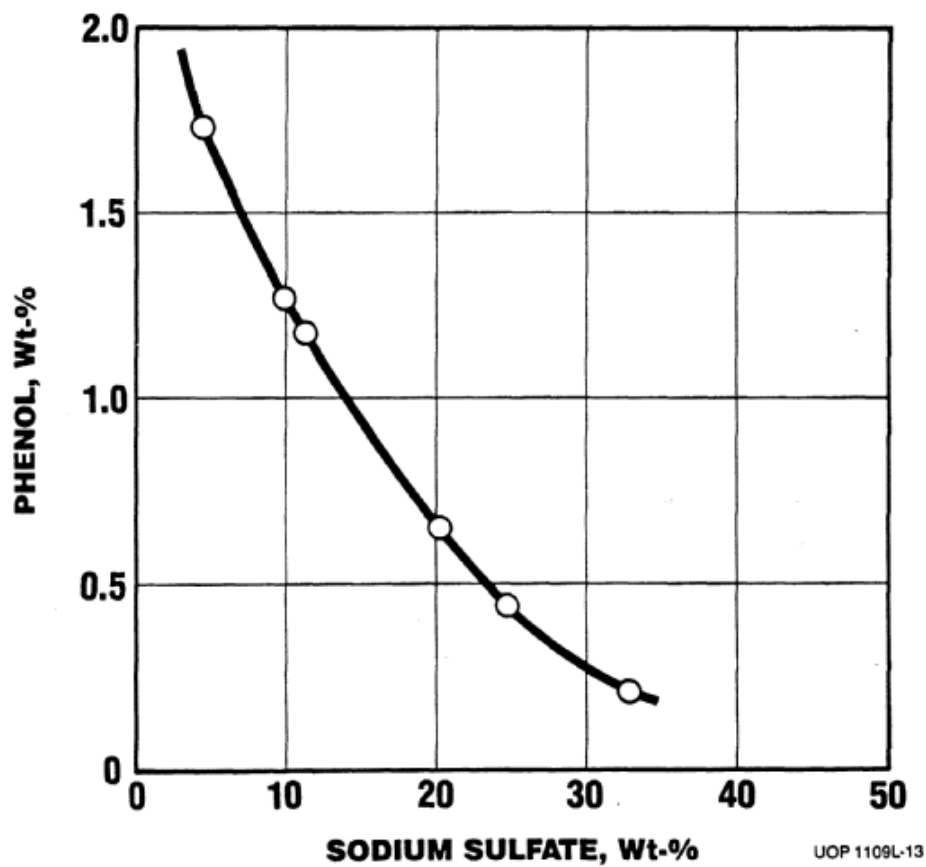
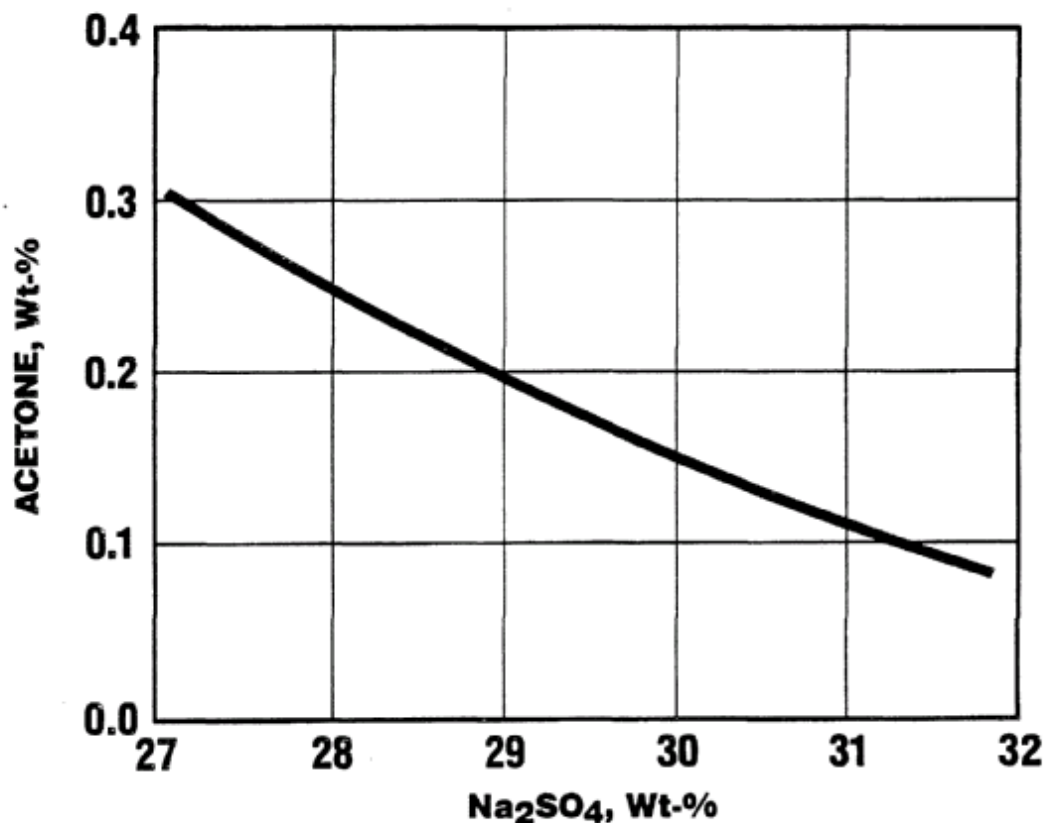
**SOLUBILITY OF PHENOL IN
AQUEOUS SOLUTIONS OF SODIUM
SULFATE AT 25°C**

Figure III-10

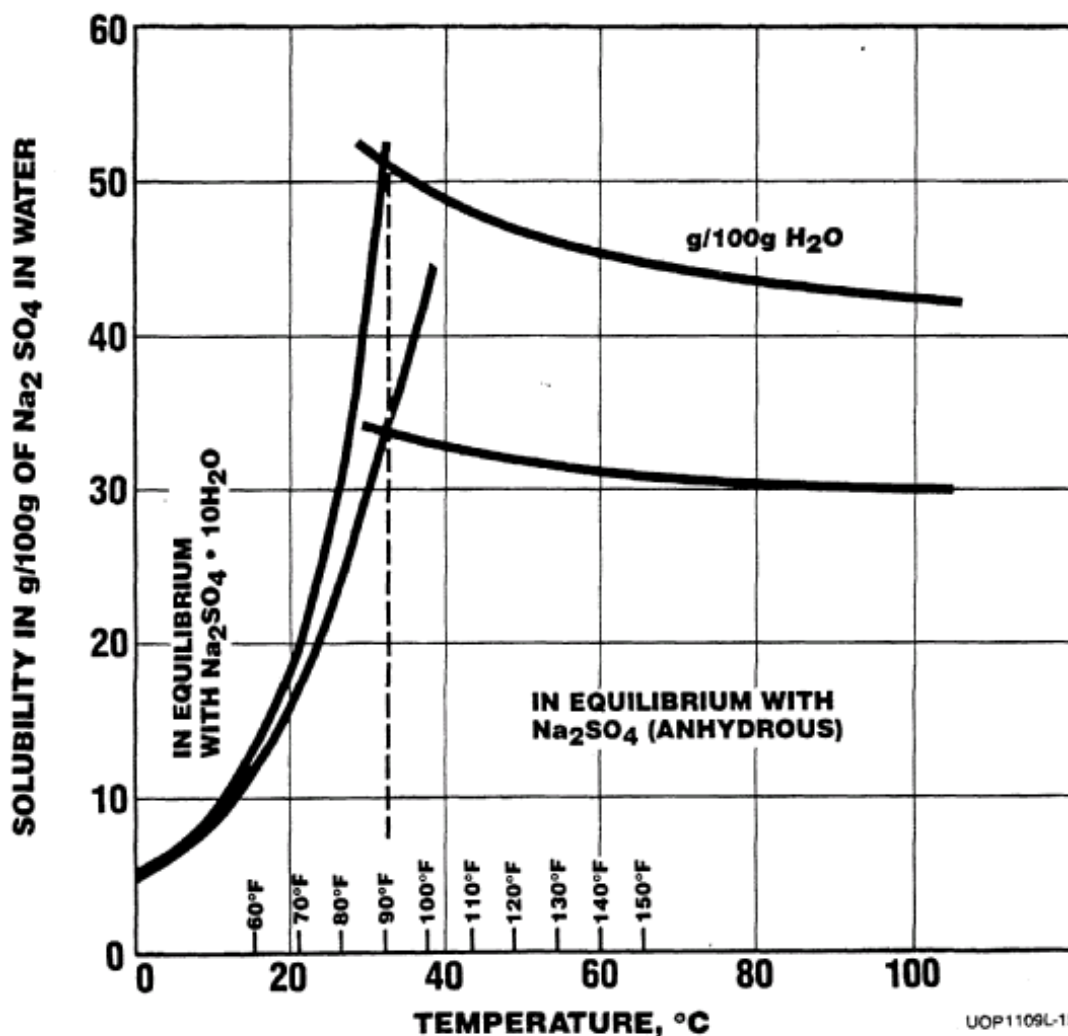
**SOLUBILITY OF ACETONE IN
AQUEOUS SOLUTIONS OF SODIUM
SULFATE AT 43°C**

UOP 1109L-14

Referring to Figure III-11, however, it is evident that there are solubility constraints that must be taken into account. Figure III-11 is a plot of the solubility of sodium sulfate in water as a function of temperature. The two sets of curves are representative of the saturation solubility or equilibrium between an aqueous solution of sodium sulfate and hydrated or anhydrous sodium sulfate crystals. From these curves it is evident that it is necessary not only to operate the PRU separator at a specific temperature so as to stay away from the saturation curves at the left side of the plot, but also to operate at a sodium sulfate concentration below the top saturation curves while staying close to it. The optimum combination of these two aspects is utilized in the process to minimize losses of phenol and acetone to the net aqueous effluent stream from the PRU separator.

Figure III-11

SOLUBILITY OF SODIUM SULFATE IN WATER



The typical sodium sulfate concentration at design conditions results in a specific gravity of 1.1-1.2 in the aqueous phase. The sodium sulfate concentration in the aqueous phase from the PRU is ultimately constrained by the make-up caustic concentration used in the scrubbing columns, 15-20 wt%. Typically, these concentrations lead to having the phenol concentration in the range of 1-2 wt% in the aqueous phase.

Again, as in the discussion immediately preceding, the static mixer combining the sulfuric acid and the sodium phenate stream are designed to take into consideration

the applicable velocity criteria that should be maintained to prevent the formation of an emulsion.

The organic phase sprung phenol will contain a 75%/25% mixture of phenol and water. Sodium sulfate is typically insoluble in organic phases, however this stream is not a typical non-polar organic phase. Phenol is slightly polar and water is very polar. About 2000-3000 wt ppm of sodium sulfate is soluble in the organic phase. By adding 50 vol% cumene to the sprung phenol phase, the organic phase becomes much more non-polar and the water and sodium sulfate concentration in the organic phase decreases to less than 100 wt ppm. See Figures III-12A and B for the relationship of ppm sodium sulfate and the addition of cumene. Additional cumene will reduce the sodium sulfate further, but at the cost of increased load on the fractionation system.

Figure III-12A

PRU Separator Sodium Sulfate in Phenol Phase versus Cumene Added to Phenol

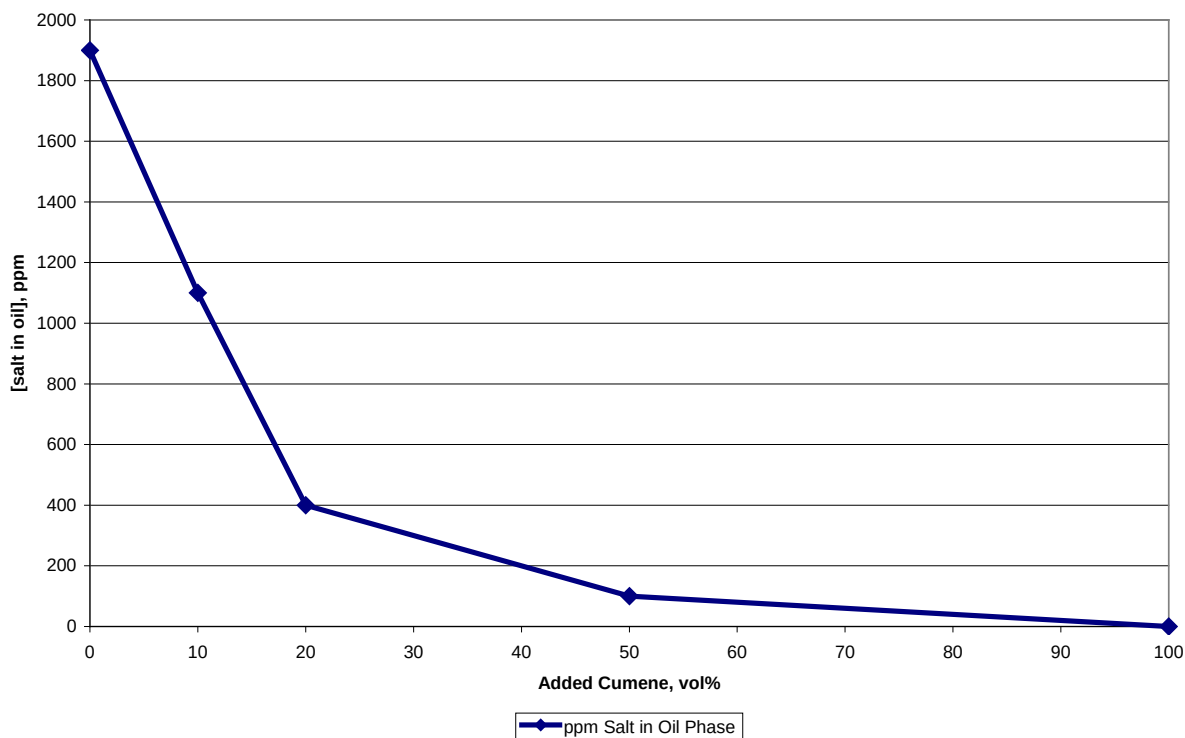
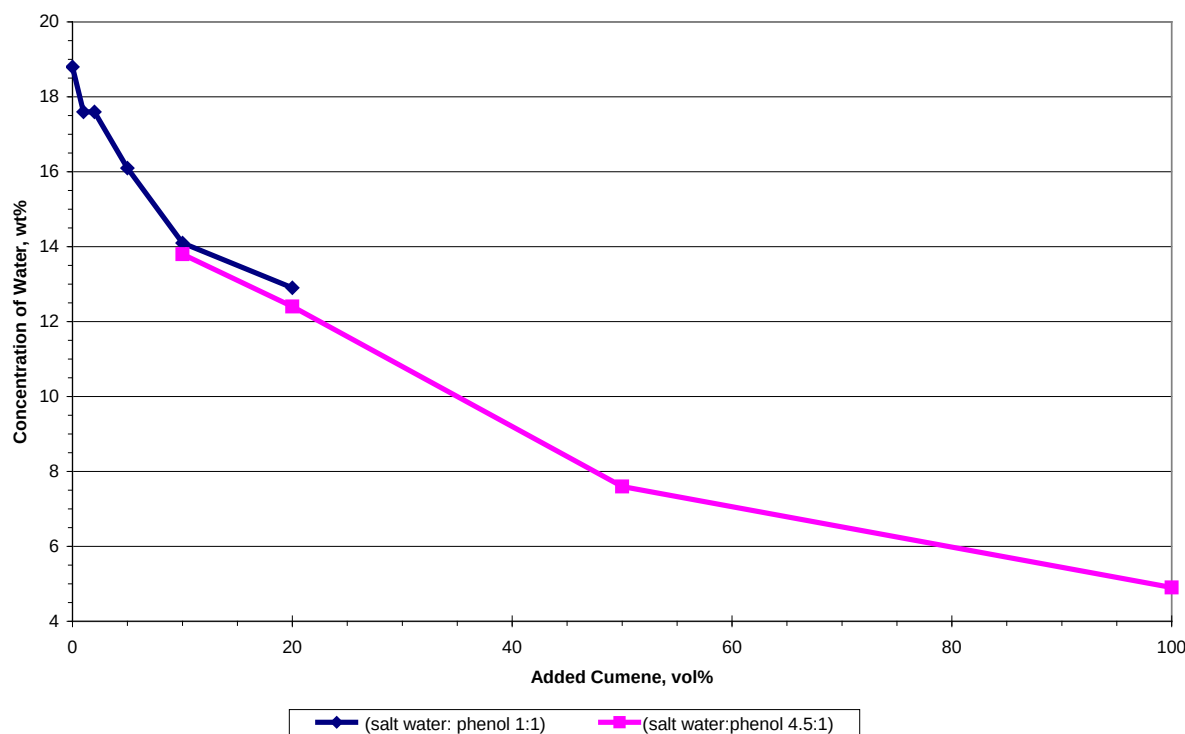


Figure III-12B**PRU Separator Water in Phenol Phase versus Cumene Added to Phenol**

1.2 Separator Temperature

As discussed previously in Figure III-11 the PRU separator temperature will impact the solubility of phenol and sodium sulfate in the aqueous phase. This temperature is controlled by the TIC on the outlet of the PRU cooler. The typical operating temperature range is 35-40°C (95-104°F). If the unit is operated at decreased PRU temperatures then sodium sulfate precipitation in the instrumentation is likely, while operating at elevated temperatures will increase the phenol concentration in the aqueous phase and increase the potential for sodium sulfate in the sprung phenol.

1.3 pH

The pH of the reacted mixture is controlled by the injection of sulfuric acid upstream of the delay pot. For the reaction of sodium phenate to phenol and sodium sulfate to proceed, both time and an acidic environment is required. Phenol has a pH of 5.5-5.8. A pH range of less than 5 is required to spring phenol. A pH below 4 is undesirable from a corrosion standpoint and should be avoided. Some organic salts will spring to organic acids below a pH of 5. The PRU should be operated as close to a pH of 5 while maintaining adequate separation of organic and aqueous phases in the PRU separator.

1.4 Residence Time and Recycle/Feed Ratio

The delay pot is required to allow the reaction of sodium phenate to phenol and sodium sulfate to proceed to completion. The design residence time for the delay pot is fixed by design and is typically greater than 1 hour. The residence time is calculated based on the combined feed (sodium phenate and sulfuric acid) to the delay pot.

The recycle flow from the PRU recycle pumps is independent of the residence time variable. The recycle/feed ratio determines the operating temperature for the delay pot. This temperature is important from a corrosion point of view. Sulfuric acid at a high temperature is very corrosive. Temperatures above 55°C (130°F) should be avoided to maintain long vessel and piping service. A typical design recycle/feed ratio is 3:1.

The PRU separator residence time requirement is dependent on the densities of the two separate phases that are formed. Due to the large ratio of aqueous phase to phenol phase in the receiver, the limiting factor in the separation is removing the aqueous phase from the organic phenol phase. It is important to maintain a low enough interface level to provide adequate residence time for the organic phase. The PRU separator level should be maintained at or below 50%. The interface float level indicator will vary with the densities of the two phases. A periodic check of the level glass should be made to ensure the float is accurate. A density analyzer has been specified to aid in troubleshooting should the density change significantly.

2. Oil Extraction Column

The main variables that determine the scrubbing efficiency and hence recovery of phenol are:

- a. Cumene Solvent/Aqueous Feed Flow Ratio
- b. Lean Solvent (Cumene) Phenol Concentration
- c. Aqueous Stream pH

2.1 Cumene Solvent/Aqueous Feed Flow Ratio

The cumene solvent/aqueous feed flow ratio is one of the variables that affect the extraction efficiency of phenol from the aqueous phase to the solvent, cumene. The flow rate is fixed in the design at approximately 7-9:1. The solvent flow affects the contact efficiency on the OEC trays as well as the tray efficiency in the solvent caustic wash column (SCWC). Related to this flow ratio is the jetting velocity for each tray. The jetting velocity is defined as the solvent flow rate divided by tray hole area. A typical jetting velocity for the OEC is 0.3 m/s (1 ft/s).

2.2 Lean Solvent (Cumene) Phenol Concentration

The phenol concentration in the lean solvent is critical in achieving very low phenol concentrations in the treated waste water. To achieve the target phenol concentration in the treated waste water of 50-100 wt ppm phenol, the phenol concentration of the lean solvent must be kept below 20 wt ppm. A lower lean solvent phenol concentration is possible at the sacrifice of caustic utilization in the SCWC. A lower phenol concentration in the lean solvent will correspond to a lower phenol concentration in the treated waste water. Phenol concentrations as low as 50-75 wt ppm have been achieved with good control of lean solvent phenol concentration and feed pH.

2.3 Aqueous Stream pH

The proper acidic environment must be maintained for the extraction of phenol from the aqueous phase to the organic phase. The pH of phenolic waste water feed must be maintained between 4 to 5, preferably 4 - 4.5. A more acidic pH will cause corrosion problems while more basic pH will reduce the extraction efficiency. Note that the lean solvent from the SCWC has the potential for entrained caustic and can increase the pH at the bottom of the OEC and cause a reduction of efficiency at the bottom of the column. It is possible to extract the phenol from the lean solvent into the aqueous phase if this situation is allowed to occur. A pH meter at the bottom of the column is provided to monitor treated waste water.

3. Caustic Wash Columns

The effluent treatment section includes two liquid full vessels which employ jet deck trays to recover phenol present in primarily cumene streams from both the OEC and the C-AMS column overhead. Phenol is recovered as sodium phenate by contact with 15-20 wt% caustic. To ensure maximum extraction efficiency, the organic phase velocity moving up the tower (jetting velocity) must be maintained at or close to design. The jetting velocity operating range is 1.2 - 1.52 m/s (4 - 5.0 ft/s).

Operation of these two towers is simple, however a stable emulsion can form if the jetting velocity is set too high or the concentration of contaminants such as acetone is significant. Avoiding caustic carryover in the scrubbing columns is one of the top priorities in monitoring.

It is important to maximize caustic utilization without sacrificing scrubber efficiency. Caustic utilization is monitored by measuring the concentration of free caustic in the sodium phenate leaving the bottom of the scrubber. This number is then compared to the fresh caustic entering the scrubber. Typically, caustic utilization should be better than 85 mole%. To that end, an aqueous phase recirculation system and pH monitoring have been added to the C-AMSCWC. The pH is monitored at the middle and bottom to ensure that all of the caustic has been utilized and actively controls how much caustic enters and leaves the column.

IV. Process Flow and Control

The following section will describe the general process flow and control of the unit and its associated equipment.

The following description is typical of the UOP Phenol process design philosophy. The control systems and equipment may vary for individual designs based on the customer's utility and procurement requirements.

A. Oxidation

The oxidation section converts cumene and oxygen to cumene hydroperoxide (CHP) by an auto-catalytic route. In other words, CHP is required to convert cumene and oxygen to CHP. This section includes additional equipment besides the reactor vessels for preparation of the cumene and oxygen (air) feeds, and treatment of the spent air to recover cumene for recycle.

1. Fresh Cumene/Recycle Cumene Feed Circuit

The fresh cumene feed comes from the oxidation day tank (see Figure IV-1). The oxidation day tank serves as a surge tank for both production and safety. It is the source of cumene required for quenching the oxidizers, CHP sump, and the equipment in the Concentration and Decomposition Sections. Fresh cumene is also used as a solvent in the Phenol Recovery Section to extract phenol from phenolic waste water and for the pump flush systems in the Oxidation Section. Due to the critical service for these pumps, UOP specifies these pumps to have a 'Reliable Power' supply. Reliable services are described further below in Section H, Auxiliary Systems.

Fresh cumene is pumped primarily to the combined feed surge drum (CFSD) on split range LIC control reset by the level in the CFSD. The recycle cumene streams from other sections of the complex are sent to the feed wash column (FWC) feed line. An FIC on the FWC feed line controls a spillback from the combined feed pumps to maintain a constant organic feed flow rate to the FWC. A constant organic flow is essential to the tray jetting velocity for optimal extraction of cumene contaminants.

There are two sections to the FWC; a dilute caustic section and a water wash section. Water enters the column at the top tray on FIC control. The dilute circulating caustic, typically 1 – 1.5wt% NaOH, enters the column near the column center. Fresh make-up caustic (15-20 wt%) on FIC control is mixed with the dilute caustic that is recirculated from the bottoms via the caustic wash circulation pumps.

The circulating caustic is controlled by FIC. The spent caustic is removed from the column bottom by interface LIC control to the Peroxide Destruction Reactor.

There are two pH analyzers that continuously monitor the column operation. The pH analyzer on the circulating dilute caustic measures the degree that the caustic is spent and is normally basic, 14. The pH of the aqueous liquid on Tray 2 checks the water for caustic entrainment and is normally neutral, 7-8.

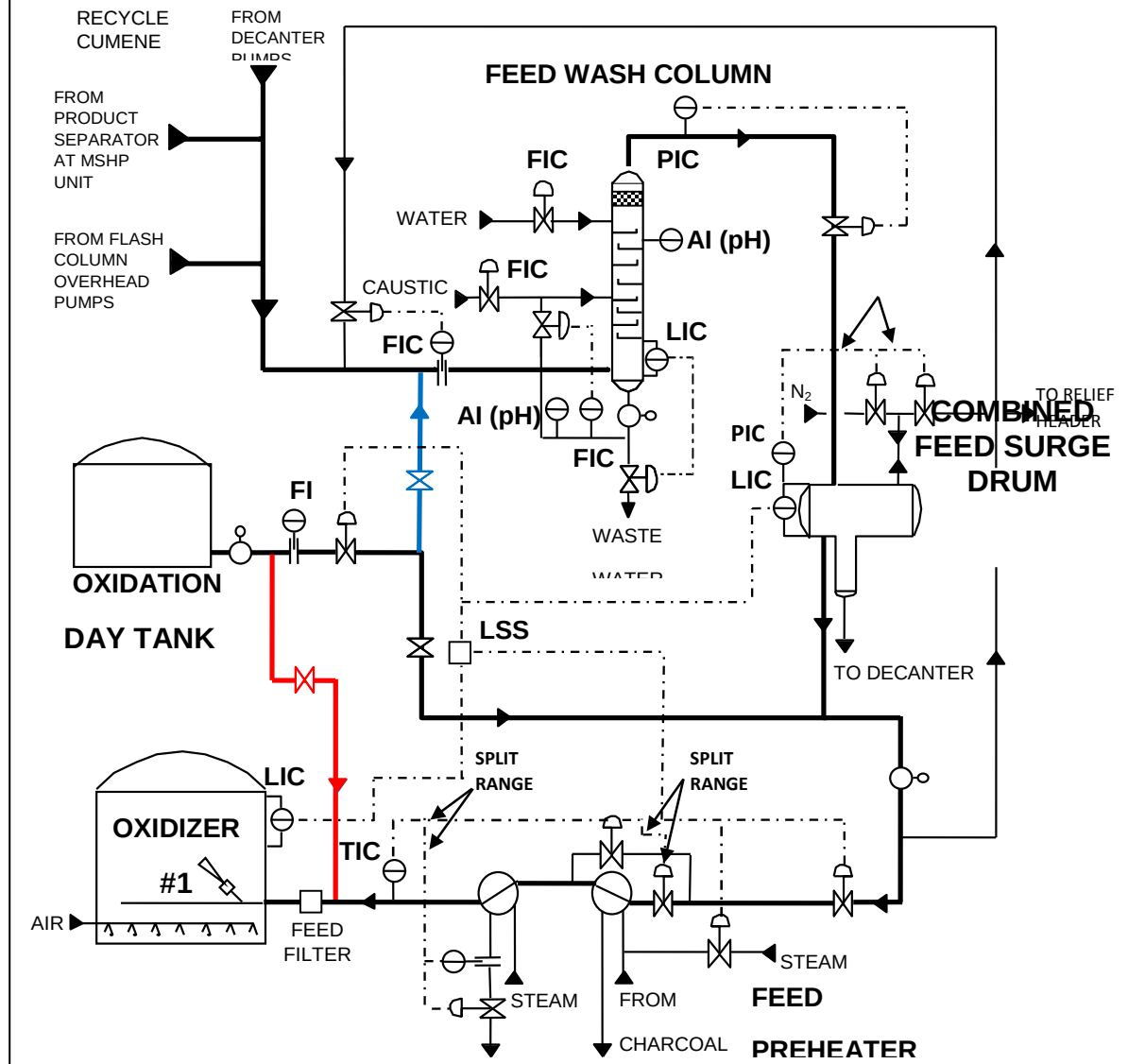
The treated cumene exits the column overhead on PIC control that holds back pressure on the liquid full column. After the pretreatment, the cumene feed is sent to the combined feed surge drum. The pressure on the surge drum is maintained by a split range PIC that controls nitrogen make-up and vent flows to the relief header. This is known as a push-pull control system. A water boot is provided for water removal to the decanter with an LIC controlling the cumene-water interface. The combined feed is pumped to oxidizer #1 by the combined feed pumps.

A low level in the CFSD will open the valve supplying cumene feed to the surge drum and send a signal to a low signal selector on the valve supplying treated cumene to Oxidizer #1. The low signal selector receives inputs from both CFSD level and the Oxidizer #1 level and will adjust its opening according to the LIC signal from the vessel with a level below its setpoint. A high level in the CFSD will do the opposite.

The combined feed is heated by the oxidizer feed preheater and oxidizer feed heater, which uses recovered heat from the charcoal adsorber during regeneration and steam, respectively. A TIC on the oxidizer feed heater outlet controls the heat input by split range controller. During charcoal regeneration excess heat is available so the primary temperature control is the bypass valve around the oxidizer feed preheater (steam to the oxidizer feed heater is closed). When there is no regeneration steam, the bypass around the preheater closes and the steam valve on the feed heater opens.

Finally, a set of filters removes any entrained particles (i.e. rust) that could promote side reactions to organic acids in the oxidizer. A pressure differential indicator (PDI) across the filter monitors the build-up of particles in the filter.

Figure IV-1
Oxidizer Feed Circuit



2. Air Delivery Circuit

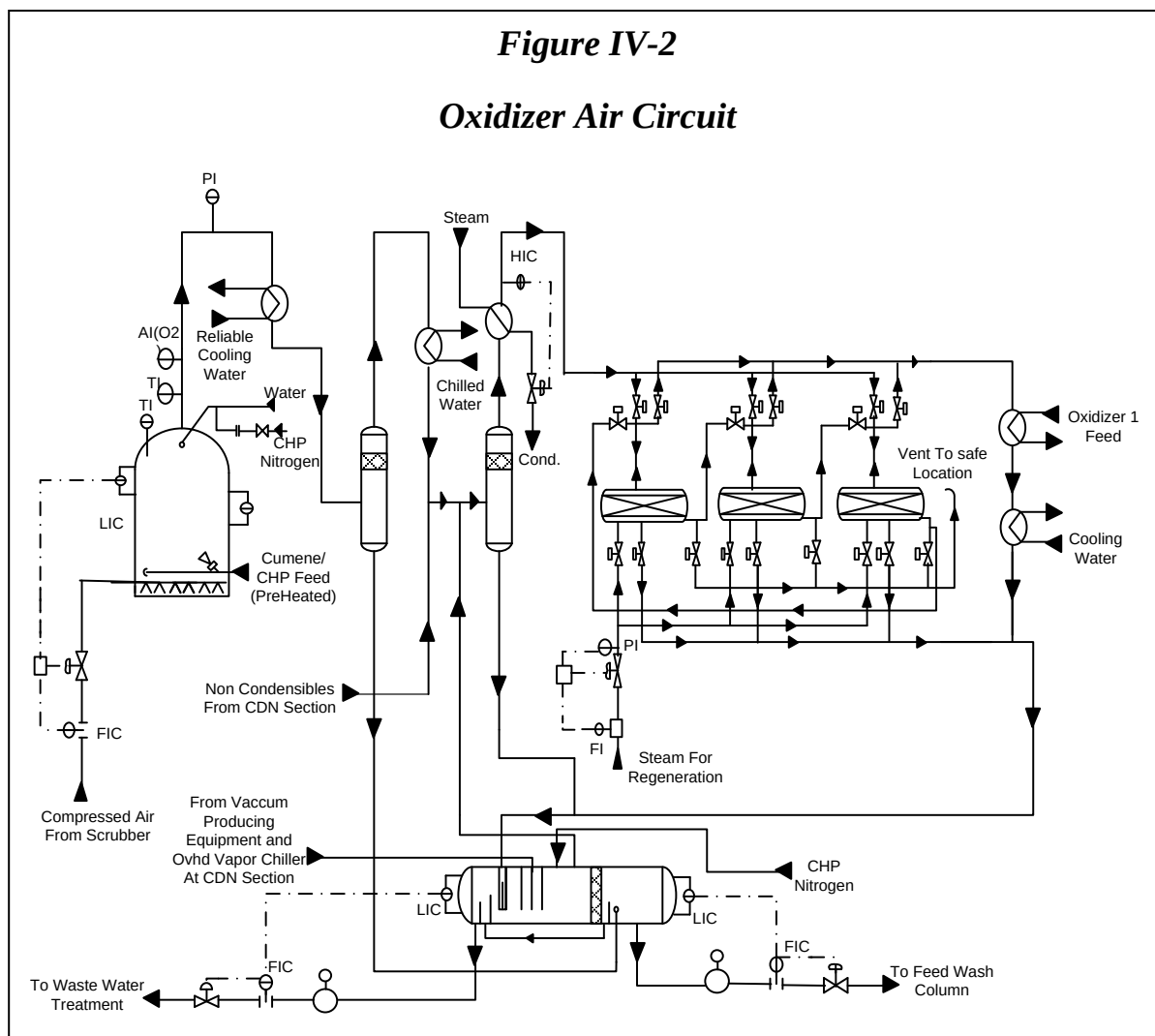
One centrifugal compressor is typically specified to provide the oxygen feed, in the form of air, to the oxidizers (see Figure IV-2). A PIC on the compressed air scrubber overhead along with a compressor load control system protects the compressor from surging. The hot pressurized air is cooled in the air compressor discharge

cooler before being fed to the compressed air scrubber. A bypass around the cooler is used to adjust the temperature.

The compressed air scrubber has four trays to provide contact between the air and weak caustic while minimizing pressure drop. It also has a mesh blanket above tray 1 to prevent liquid entrainment. Basic water is circulated from the bottom of the scrubber to the top tray by the compressed air scrubber pump. A globe valve and local FI are used to control the circulation rate. An LIC is cascaded to an FIC which controls the rate of water from the bottoms to the sewer via high signal selector and a split range controller. If the air is dry, additional water can be added to the air scrubber to maintain the liquid level.

Caustic is added to the column manually with a globe valve and local FI to maintain the bottoms caustic concentration at ~0.1wt% free caustic. The water injection rate will depend on the humidity of the ambient air. Since the air leaving will be saturated with water, dry air entering the compressor will require more make-up water than humid air.

From the scrubber, the compressed air is sent to each oxidizer via independent FIC control. A PDI measures the pressure drop across each air flow control valve. The air FIC, the PDI and a separate automatic valve (UV) are tied into an automatic shutdown system to protect the oxidizers during an emergency situation. The air line to each oxidizer loops up to 4500 mm above the oxidizer top tangent with a check valve at the highest point to prevent the oxidate from back flowing up the air piping. Isolation valves are provided at the oxidizer air inlet for maintenance. The PDI ensures that there is a pressure differential across the air control valve prior to allowing the UV valve to open. The purpose of this is to prevent back flow of hydrocarbon from the oxidizer into the air system.



3. Oxidizers

For new units, two oxidizers in series is optimal (see Figure IV-3). Air enters the oxidizers through spargers at the vessel bottom. Cumene combines with recirculating oxidate and enters the oxidizer through jet eductors. The auto-catalytic exothermic heat of reaction is removed by circulation through external cooler(s) or feed temperature control. On oxidizer #1, the low temperature of the combined feed absorbs the heat generated. The temperature of the oxidate from oxidizer #1 resets the steam flow to the oxidizer feed heater, or bypass control valve around the oxidizer feed preheater, depending on the charcoal adsorber regeneration cycle, on low temperature signal. On high temperature signal the temperature of the oxidate from oxidizer #1 resets the tempered cooling water flow to the oxidizer #1 emergency cooler. The cooling water supply to the oxidizer emergency coolers is off

during normal operation. Oxidizer #2 has two sets of coolers, one for normal operation and a second in series for emergency. The emergency cooler spare is shared with oxidizer #1. The oxidizer #2 cooler for normal operation is the concentration section preflash column (PFC) feed/oxidate exchanger. The temperature of the oxidate from oxidizer #2 resets the bypass valve around the feed/oxidate exchanger on low temperature signal. On high temperature signal the temperature of the oxidate from oxidizer #2 resets the tempered cooling water flow to the oxidizer #2 emergency cooler. The oxidizer pumps serve both to circulate and send the net oxidate to the downstream equipment.

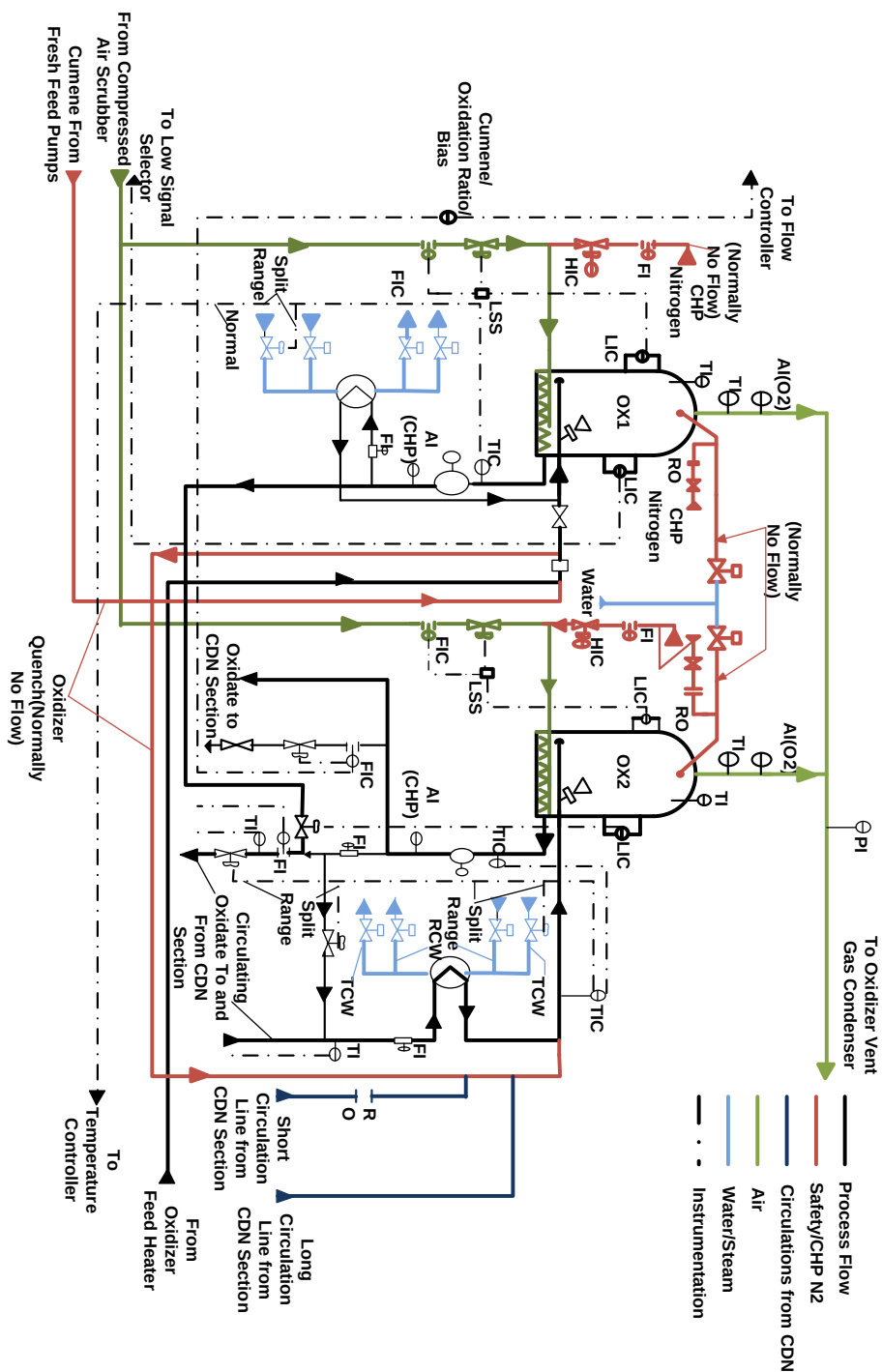
The LIC on oxidizer #1 controls the combined feed flow from the combined feed pumps. The LIC for oxidizer #2 controls the oxidate flow from oxidizer #1. The oxidate from oxidizer #2 (effluent from oxidation section) is sent to the concentration section. The oxidizer #2 net oxidate is normally controlled by the LIC at the flash column bottoms.

An independent LIC at the top of each oxidizer will close the air to the respective oxidizer if a high liquid level is detected. This control serves to protect both the air compressor and from liquid carryover from the top of the oxidizer.

A low delta P across the air control valve, a high concentration of oxygen in the spent air or high pressure in the oxidizer will cause an automatic shutdown of the air to the oxidizer.

A total of 18 temperature sensing points for each oxidizer are tied into the automatic shutdown system. If a high temperature condition persists in either oxidizer, then the emergency cooling system is activated.

Figure IV-3
Oxidation Section



4. Oxidizer Effluent Air Circuit

Spent air from the top of each oxidizer combines into one stream (see Figure IV-2). The spent air in each stream is analyzed for oxygen by redundant continuous oxygen analyzers. The oxygen concentration is limited to keep well below the lower explosion limit (LEL) for cumene in air. The combined stream is then sent through a water cooled oxidizer vent gas condenser. The condensate collected in the oxidizer vent gas separator is mostly cumene and gravity drains to the decanter. A seal loop is provided to hold a liquid level in the separator. A mesh blanket in the oxidizer vent gas separator coalesces small droplets of liquid from the spent air.

The spent air continues to the oxidizer chilled vent gas condenser where the temperature is reduced further and then goes to the oxidizer chilled vent gas separator. A mesh blanket in the oxidizer chilled vent gas separator coalesces small droplets of liquid from the spent air. The condensate collected in the oxidizer chilled vent gas separator is mostly acidic water and gravity drains to the decanter. A seal loop is provided to hold a liquid level in the separator. The spent air from the oxidizer chilled vent gas separator continues to the oxidizer vent gas superheater.

In the oxidizer vent gas superheater the saturated spent air is superheated by exchange with waste steam or other low value utility. The superheated spent air is then sent to the charcoal absorbers. Most (>99%) of the organic components remaining in the spent air are removed by downflow adsorption over activated charcoal. Two charcoal adsorbers are always in service, while one is being regenerated. The treated air from the charcoal adsorbers is vented to atmosphere or sent to an incinerator.

The charcoal adsorber section has the flexibility to be designed a number of different ways depending on local environmental regulations. Some designs have parallel adsorption. The advantage of this is there is little impact on the hydraulics of the air system when an adsorber is brought in and out of service. With parallel adsorbers the regenerated adsorber is cooled by circulating treated spent air from a cool down blower. Cooling the adsorber before putting it back in service reduces the trace organic concentration in the spent air.

Another method used has been to run the adsorbers in series. Field measurements have indicated that the delta pressure of each adsorber is less than $\frac{1}{2}$ of the design value; therefore running the adsorbers in series does not cause a hydraulic constraint. The advantage of series operation is the hot regenerated adsorber is cooled by the treated effluent air of the lead adsorber thus eliminating the need for a cool down blower. The lead adsorber also can adsorb more hydrocarbon compared to the parallel scheme because some breakthrough hydrocarbon can be adsorbed by the lag adsorber.

An automatic adsorber logic system rotates each adsorber through the regeneration cycle. The regeneration cycle lasts 2-3 hours, while each adsorber is online for 4-6 hours. The cycle length is customized for each unit.

The regeneration cycle time is divided into three periods, steam out, draining, and cooling. The steam out is performed using low pressure steam. Approximately, 0.5 weight of steam is required per weight of charcoal. This typically corresponds to 1 to 1.5 hours of steaming at design steam rates. The drain time is set to 15-30 minutes with the cool time being set by the rest of the time in the regeneration cycle after steam out. For units with a UOP Charcoal Adsorber Regeneration Control System (CARCS) the time for each step will be provided in the CARCS EI&DB (Equipment Instructions & Data Book).

The adsorbers are steam regenerated upflow to remove adsorbed hydrocarbons. The spent steam from the outlet of the regenerating charcoal adsorber is condensed by exchange with the combined cumene feed to oxidizer #1 and the regenerant steam condenser. The condensate then gravity flows to the decanter. During steaming and cooling of the oxidizer, condensate collected in the charcoal adsorber gravity drains to the decanter via a seal leg.

5. Decanter

As mentioned above there are numerous liquid streams that enter the decanter. They are:

- a. Mostly organic phase from oxidizer vent gas separator
- b. Mostly aqueous phase from regenerant steam condenser effluent
- c. Mostly aqueous phase from oxidizer chilled vent gas separator
- d. Regenerant liquid from charcoal adsorber during regeneration
- e. Condensate from concentration section ejectors
- f. Seal liquid from concentration section vacuum pumps

The mostly organic streams are sent directly to the organic side of the decanter. Cumene makes up the vast majority of the organic phase. The mixed phase or mostly aqueous streams go into the mixed phase portion of the decanter. A mesh blanket aids in separation of the two phases. A baffle on the aqueous side holds the aqueous level in the decanter and gravity flows to the aqueous closed drain header (CDR) or is pumped to the peroxide destruction reactor. The organic stream is pumped by the decanter pumps to the feed wash column for recycle back to the oxidizers.

Any non-condensable gases from the charcoal adsorber resulting from regeneration along with the gases from the decanter vent to the charcoal adsorbers. Older designs had a separate vessel that was filled water.

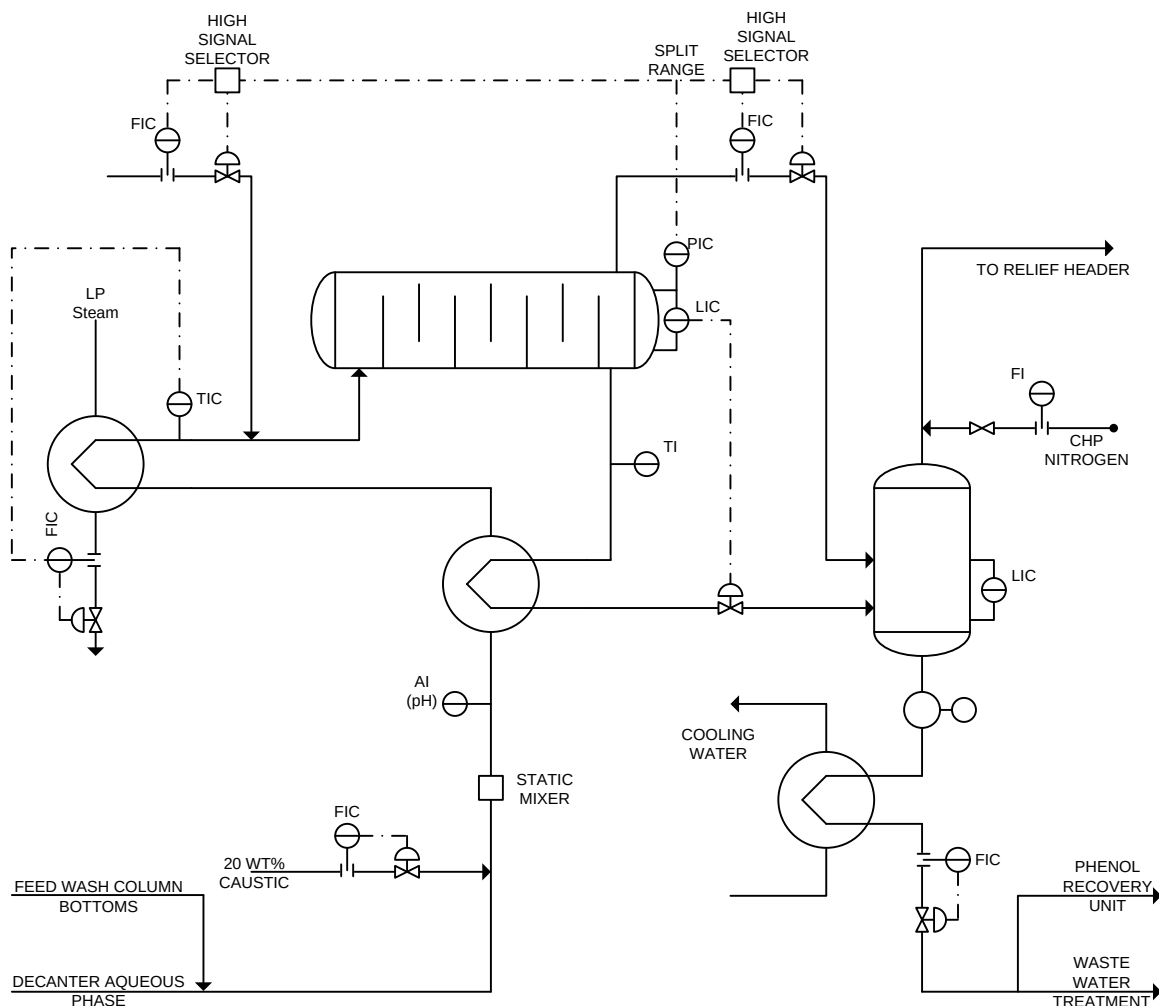
When the spent air from the charcoal adsorbers is sent to an incinerator an automatic shutoff system is required to isolate the system for safety.

6. Peroxide Destruction Reactor

The peroxide destruction reactor (PDR) is a baffled plug flow vessel that provides a 1.7 hour residence time based on the normal flow of net FWC bottoms and the maximum expected water flow from the Decanter for the reaction of peroxides at high pH and 130 deg C.

The PDR feed consists of FWC bottoms, decanter aqueous phase and caustic as needed for pH correction (see Figure IV-4). The caustic injection rate is set by FIC upstream of the PDR from the 20wt% caustic pumps. Nitrogen is added to help strip out light compounds from the reactor liquid. Some light gases may be formed in the reactor. Nitrogen and some lights that are formed in the reactor vent from the top of the reactor and are sent on flow control to the peroxide destruction separator.

Figure IV-4
Peroxide Destruction Reactor



To further remove light components the reactor liquid effluent is cooled and sent to the peroxide destruction reactor at a separate nozzle from the peroxide destruction reactor vent. The offgas from the separator is vented to the relief header. The liquid from the separator is cooled further and is pumped to waste water treatment by the oxidation waste water pumps.

Reactor pressure is controlled by a split range controller. The split range controller adjusts the nitrogen to the reactor via FIC high signal selector. The nitrogen is mixed with the reactor feed to promote disengaging of reaction products as the feed enters the first baffled area. The split range pressure controller also controls the

vent rate from the vessel via high signal selector to the vent FIC. The reactor vents to the product separator which floats on the relief header pressure.

7. CHP Sump

The CHP sump's main purpose is to collect the liquids containing CHP from the CHP closed drain header (CDH). CHP must be handled separately from the other streams to maintain safety. A level of cumene remains in the CHP sump at all times to dilute concentrated CHP drained to the sump. The CHP sump pump then sends the CHP/cumene liquid back to the lead oxidizer or feed wash column automatically by a high level pump start switch. An automatic quench of cumene fresh feed is added on high sump temperature in case of thermal decomposition of the CHP in the sump. In addition cumene is added to the CHP sump as needed to maintain the CHP concentration in the sump below 25wt% at all times. Finally, during the initial unit start-up CHP can be added to the sump for initial seeding of the oxidizers.

B. Concentration

The concentration section quickly, efficiently, and safely increases the oxidate CHP concentration from the terminal (last) oxidizer to a level required for the decomposition to phenol and acetone. Typically, 25-40 wt% CHP concentrates up to 80-84 wt% CHP in this section. The CHP is concentrated by deep vacuum flashing in two columns. This section includes the preflash and flash columns and the feed preparation equipment along with a number of safety related vessels used to quench (dilute) the CHP streams during a shutdown or emergency (see Figure IV-4).

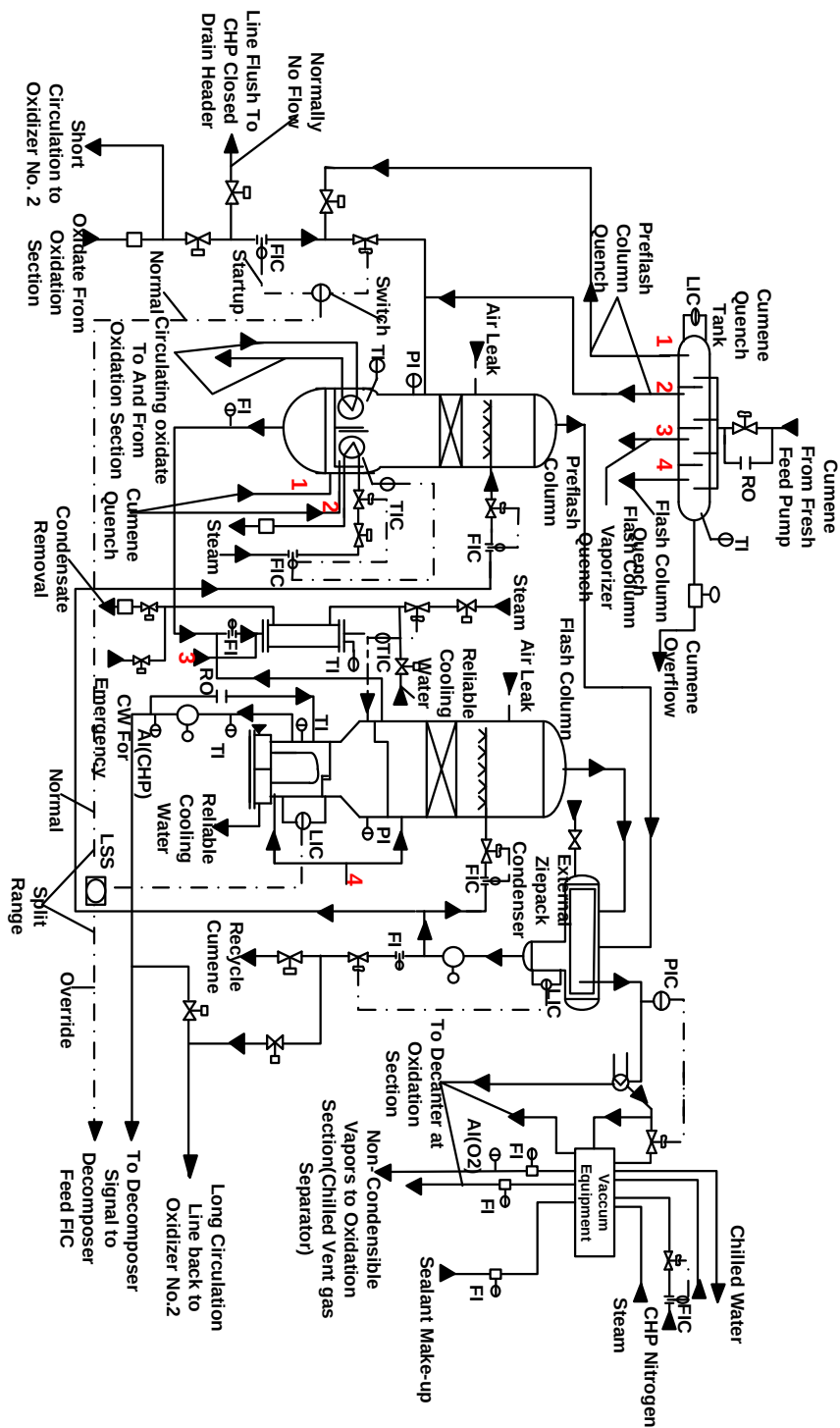
1. Feed Circuit and Pre-Flash Column

The oxidate from the terminal oxidizer is first sent through a set of filters to remove any particles (e.g. organic salts, rust) from the oxidate. Rust particles can be catalytic point sources for the premature uncontrolled decomposition of CHP. Organic salts and other basic compounds can quickly neutralize the sulfuric acid in the decomposer leading to an unsafe situation. The organic salts can also foul the heaters/vaporizers resulting in higher outlet temperatures that can also be unsafe. A spillback is provided after the preflash column feed filters to return the oxidate from the terminal oxidizer back to the same oxidizer. This circulation is termed "short circulation".

The UOP Phenol Process is designed to minimize the stagnant areas where CHP can adiabatically decompose. Automatic shutdown valves with minimum distance requirements are designed to stop the feed flow when required and drain and flush the stagnant piping.

Oxidate flashes across the preflash column feed FIC to a deep vacuum pressure. This auto-cooled material is reheated by cross exchange with the feed-oxidate exchanger, which is stabbed-in type. The material overflows from this tub into the tub containing the stab-in preflash column steam heater. The steam input to the heater is controlled with a total heat input controller based on the circulating oxidate and based on the vapor or liquid temperature above the tub. The liquid temperature at the top of the steam tub can have a small cycle and therefore might not be optimal for feedback control to the steam rate to the exchanger. If the controlling TI cannot be moved to the vapor space then multiple TI's in the liquid across the length of the steam exchanger might provide for more accurate feedback control. Some licensees have cascaded the vapor temperature above the steam tub to the steam FIC.

Figure IV-4
Concentration section Flow Diagram



The flashed cumene and some entrained CHP rises through a packed bed where it contacts recycle cumene on FIC control which knocks the CHP back down the column. The remaining cumene vapor is condensed by an integrally attached or external water-cooled condenser. The condensed liquid falls to a collector plate or surge drum and is drawn off to reflux or net product. The reflux and net product are pumped by the flash column overhead pumps on surge drum level control to the oxidation feed wash column. The non-condensable overhead vapor from the preflash column and flash column combine and go to the concentration section vacuum producing equipment which includes a chilled condenser.

The preflash column bottoms gravity flows to the flash column via the flash column vaporizer. No liquid level is normally observed in the preflash column bottoms. The preflash column bottoms is mixed with some flash column collector tray liquid before entering the flash column vaporizer. This mixture has a lower CHP concentration that requires less over-flash (i.e. temperature) to vaporize the cumene. The less volatile CHP is heated in the vaporizer, which then transfers this heat into the volatile cumene.

Due to the tube wall temperature limit that the CHP can tolerate, a low pressure steam/condensate handling system is required for the steam heater and vaporizer. The handling system equipment consists of either a power trap or condensate pot and condensate pumps that remove the condensate from the vaporizer. The newer systems use a power trap due to reduce cost. The condensate pot has an LIC that controls the flow of condensate to the collection system. An automatic emergency control system will stop the steam; depressure the exchanger; and shutoff the cumene feed flow upon high temperature in and around the exchanger. If the temperature continues to increase, then dedicated cooling water will flush any remaining condensate from the vaporizer and cool the exchanger. There is an additional line for firewater addition to the vaporizer in case the dedicated cooling water system fails. The preflash column steam reboiler only has the steam isolation and venting system due to decreased safety implications at the lower CHP concentration.

2. Flash Column

The flash column separates additional cumene from the oxidate under a deep vacuum. The column can have either four high vapor flow, low pressure drop screen trays or structured packing. An internal flash column condenser minimizes the pressure drop in the overhead system, however an external condenser can also be used. Both the packing and the condenser are designed to minimize pressure drop so the flash zone pressure and temperature are safe.

The partially vaporized feed enters the column flash zone below the collector tray. The cumene vaporizes and goes up the column and the CHP drops to the bottom. The vaporizer outlet temperature is adjusted to give the desired bottoms CHP concentration. The vaporizer outlet TIC directly controls the steam flow to the vaporizer.

The cumene condensed on the internal flash column condenser is collected in an internal accumulator. Reflux enters the column on the packed bed distributor on FIC control similar to the PFC. The non-condensibles from the flash column go to the common concentration vacuum system. The vacuum is generated by a hybrid vacuum pump and ejector system and is controlled by a PIC on the overhead line. The vendor-designed vacuum system typically has both a 100% and 50% high pressure steam driven ejector for the first stage and a single vacuum pump (with back-up) for the second stage. The 100% and 50% ejectors operate on a single inter-condenser. Typically, a spillback of gas from downstream of the ejectors to upstream of the chiller and also from the liquid ring vacuum pump separator overhead to the liquid ring vacuum pump suction controls the vacuum on the preflash and flash columns. Varying the upstream chilled water rate to the flash column overhead vapor chiller has been previously used for control. The condensed liquid from the chiller flows by gravity to the decanter.

The flash column bottoms LIC primarily controls the level by adjusting the feed to the preflash column under normal operations. When the preflash column feed FIC is at 100% of scale, then the LIC begins to close the decomposer feed flow control valve directly.

An internal flash column bottoms cooler minimizes the quantity of hot, concentrated CHP for safety and cools the concentrated CHP before pumping it to the decomposer. A small flow of flash column bottoms is recirculated back to the flash column bottoms cooler, controlled by an RO, to maintain duty on the exchanger and ensure the CHP in the flash column bottoms cooler and subsequent piping is cold and does not stagnate.

3. Emergency Equipment

Concentrated CHP and/or CHP at elevated temperatures is considered unstable over time. Even though the UOP design minimizes the time that CHP is at elevated temperature, there is some possibility that decomposition could take place in the preflash and flash columns. A cumene quench drum is provided in this section for emergency quenching of the concentration section equipment to cool the column contents and dilute the CHP.

The cumene quench drum has automatic filling via the fresh feed pumps and an LIC on the drum level. The drum is elevated above the equipment it's protecting, so it does not have to rely on a pump during an emergency. The quench cumene flows by gravity from the quench drum to the column. The drum is divided into four

sections. One section quenches the tubs in the preflash column. Another section quenches the preflash column bottoms and preflash feed line upstream of the feed FLOW CONTROL VALVE. The other two sections quench the flash column vaporizer, collector tray, and the concentrated CHP in the flash column bottoms section.

Low feed flow, high temperature, high pressure, and high bottoms level will all shutdown the decomposer and flash column vaporizer and put the flash column overhead and bottoms on long and short circulation back to the terminal (last) oxidizer, simultaneously, the preflash column feed FLOW CONTROL VALVE closes, and the feed is rerouted back to the terminal oxidizer on short circulation.

In addition, a high temperature in a specific area of the flash column will result in cumene quench being sent to that area. Automatic line-size valves will open upon high temperature in the area that the quench line is dedicated. Typically, the flushing is stopped after the emergency situation is over, to prevent too much cumene being sent to the oxidizer, which will reduce the CHP concentration.

There is also emergency purging of condensate on the steam side of the flash column vaporizer with dedicated cooling water as a result of a high temperature within and at the outlet of the vaporizer. There are two switches that will cause the cooling water to start, a high temperature with a time delay and a higher temperature without a time delay. The two switches are added to minimize the chance of dedicated cooling water being mixed with condensate unless a high temperature persisted or was excessively high.

C. Decomposition

The decomposer and dehydrator produce the high yield, low residue phenol and acetone products by acid catalyzing the cleavage of CHP to phenol and acetone, and selectively dehydrating the intermediate product dicumylperoxide (DCP) to AMS, phenol, and acetone. Due to the large exothermic heat of reaction from the cleavage reaction, a large cooling loop and safety related automatic shutdown systems are required in this section.

1. Decomposer Feed

Concentrated CHP from the flash column bottoms is pumped via the flash column bottoms pumps to the decomposer (see Figure IV-5). Water is added from the process water break drum in the phenol recovery unit and process water pumps with FIC to set the flow rate of water to the decomposer feed to attenuate the cleavage reaction. An FIC sets the flow rate for the feed to the decomposer. During normal operation, all upstream flows are reset by LIC to maintain this flow. A signal

selector from the flash column bottoms LIC will override the decomposer feed FIC signal and reduce flow when:

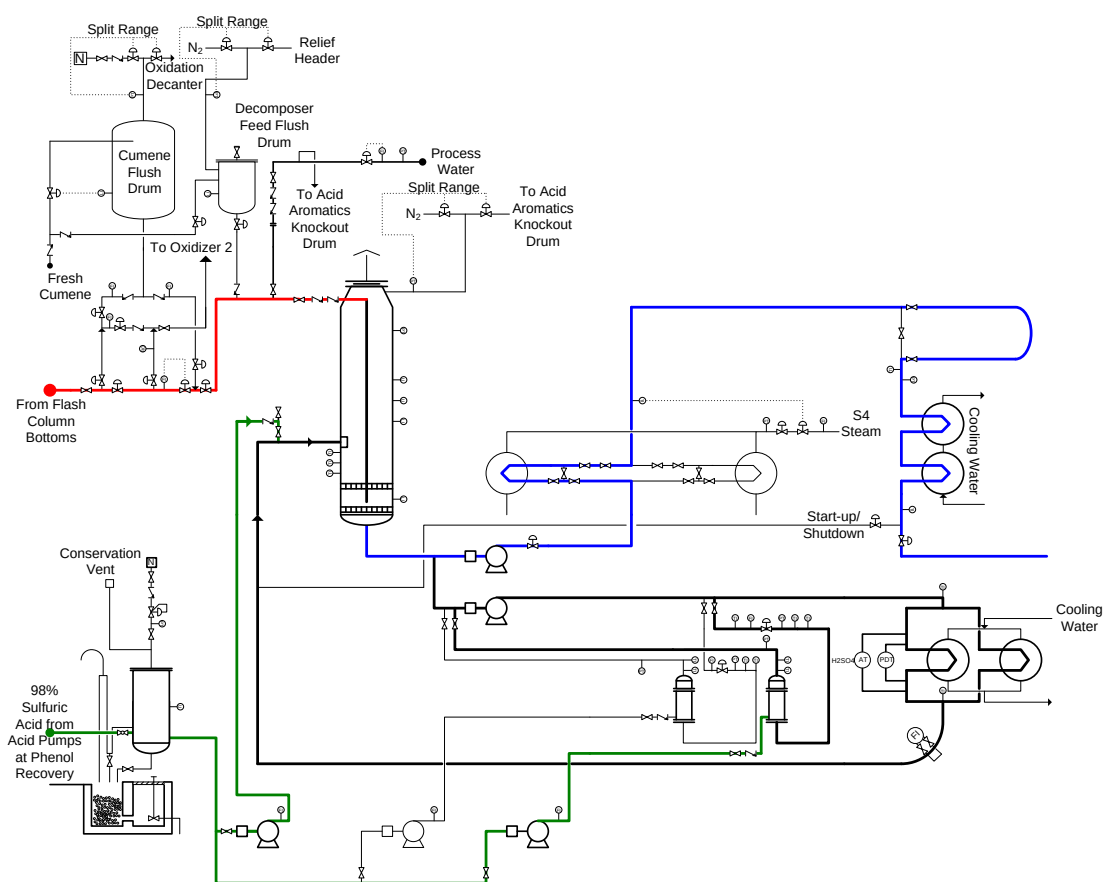
- The flash column bottoms level is below set point and
- The preflash column feed FIC is at 100% of scale.

A signal selector from the decomposer LIC will override the decomposer feed FIC and reduce flow when:

- The decomposer level is above set point and
- The decomposer product FIC is at 100% of scale.

Figure IV-5

Decomposition Section



2. Decomposer

The concentrated CHP enters the decomposer through a dip tube distributor to the vessel bottom between two sets of mixing gratings. The concentrated CHP mixes with the circulated partially decomposed product. A large ratio of circulated product

to decomposer feed (25:1 to 100:1) mixes with the feed and minimizes the temperature increase. Operating below a recycle to feed ratio of 25 can make unstable temperatures and requires a reduction in feed rate to maintain the minimum ratio until the circulation pump can be revamped. The recycle to feed ratio has been reduced in recent designs. The decomposer is a nitrogen blanketed vessel with a push-pull pressure control system similar to a storage tank. The decomposer pressure is controlled at 0.70 kg/cm²(g) and vents to the acid aromatics knockout drum.

Most of the bottoms from the decomposer flows to the decomposer circulation pumps and then to the decomposer cooler. The design volume of the circulating loop is restricted to a maximum to allow adjustment of the cleavage reaction rate within the limits of the variables available. The circulating flow rate is monitored indirectly with a PDI across the decomposer cooler and an elbow tap flow meter. A low elbow tap FI reading will activate the spare circulation pump. A very low PDI reading across the decomposer cooler will shut down the decomposer feed FLOW CONTROL VALVE, stop heat to the dehydrator and put the decomposer/dehydrator on internal circulation.

The decomposer temperature TIC directly controls the decomposer cooler dedicated cooling water flow rate. The cooling water flow rate is monitored with an FI and return water TI. A number of automatic shutdowns are tied into the decomposer temperature. The decomposer temperature is one of the variables that affect decomposition reaction rate.

3. Sulfuric Acid Injection and the Calorimeters

The sulfuric acid concentration in the decomposer is another variable that affects the cleavage reaction rate and yields. Sulfuric acid is injected via injection pumps into the circulation loop to adjust the reaction rate for the cleavage of CHP to phenol and acetone. The acid is injected into three places, the two calorimeters, and the decomposer circulation line downstream of the decomposer cooler. Each injection point has its own set of pumps. Each injection point also has an FIC.

The system is designed to control the total ppm of sulfuric acid injected based upon the decomposer feed flow and the total acid flow rate. The acid flow rates to the calorimeters are set at the minimum required to decompose the DCP (4000-6000 wt ppm H₂SO₄). The control system will adjust the direct injection flow rate first and then adjust the calorimeter injection.

The calorimeters serve to measure the concentration of CHP, DCP and dimethylphenylcarbinol (DMPC) in the decomposer recirculation loop. At a constant flow rate (i.e. constant residence time), the temperature increase in the calorimeter

is directly proportional to the concentration of these compounds. The calorimeters are redundant so one can be taken offline for cleaning while operating the other. The calorimeters are designed to be adiabatic and have a specified volume. They are well insulated and can be put in a separate shelter that is temperature controlled.

The calorimeters measure the exothermic temperature increase in two stages. The first stage temperature increase is proportional to the concentration of CHP entering the calorimeter. During normal operation, the CHP is nearly consumed when the stream enters the second stage. In the second stage, the stream is mixed with sulfuric acid to a 4000-6000 wt ppm acid concentration. The acid catalyzes the dehydration of DCP to AMS, phenol, and acetone, and any remaining DMPC to AMS and water. Therefore, the second stage is a measurement of the DCP, DMPC, and the heat of solvation of the acid. Automatic decomposer shutdowns are tied into the calorimeters' high delta T switches.

4. Dehydrators

The effluent from the decomposer is pumped by the decomposer product pumps to the dehydrator on FIC control reset by the decomposer LIC. The volume of the piping from the decomposer to the dehydrators is limited to minimize the acid catalyzed dehydration reaction of DCP in the piping and to allow the decomposition of any remaining CHP. The effluent is heated in the dehydrator to a temperature where DCP and the remaining DMPC are selectively decomposed to phenol, acetone, and AMS. This results in higher phenol and AMS yield, lower cumene consumption and less residue formation.

The dehydrator is spared so one can be cleaned while the other is on-line. The dehydrator outlet TIC controls the steam flow to the dehydrator. The piping volume downstream of the dehydrator to the crude product cooler is controlled to limit the residence time for the exothermic reaction. The residence time is a "coarse control" variable for the yield. The pipe volume (e.g. residence time) can be varied by opening and closing valves to flow through or bypass extra pipe volume. The temperature of the stream would be considered the "fine control" variable.

The crude product is cooled in the crude product cooler with cooling water before going to the neutralization section.

5. Emergency Equipment and Shutdowns

A set of two automatic shutdown valves in series isolate the concentrated CHP feed from the decomposer downstream of the decomposer feed flow control valve. The concentrated CHP will circulate back to the lag oxidizer from directly upstream of the decomposer shutdown UV valves during a shutdown. All other stagnant decomposer feed piping is flushed with fresh cumene during a shutdown by the

cumene flush drum and the decomposer feed flush drum. Each of the cumene flush drums provide a cumene surge volume which is always available with gravity flow. Safety is further enhanced by maintaining these two systems separate to avoid cross contamination of CHP and the acid environment.

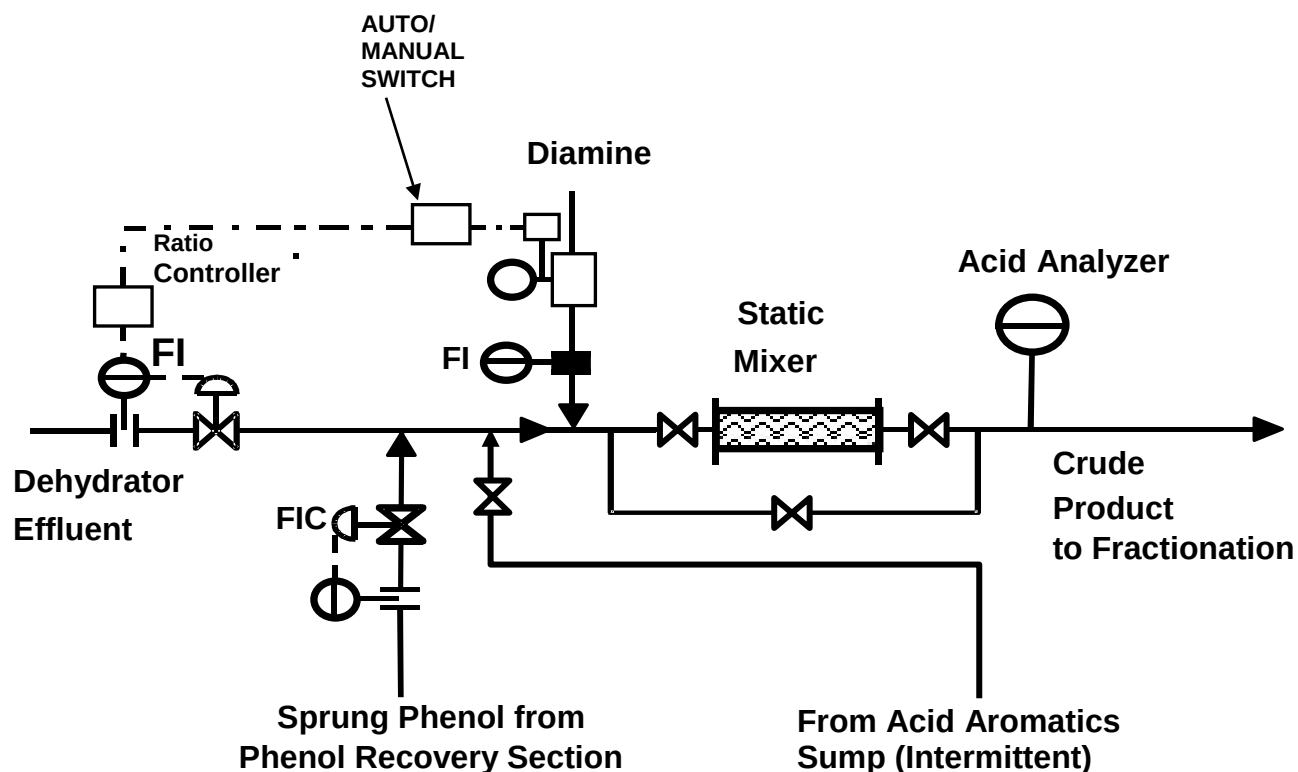
The decomposer can be shut down by high and low decomposer temperature, high decomposer level, high calorimeter temperature, and high dehydrator outlet temperature. When the decomposer is shutdown, the water flow control valve closes to prevent a build-up of water in the decomposer. Too much water will completely inhibit the cleavage reaction.

D. Neutralization

In this section, the decomposition section sulfuric acid catalyst is removed by reaction from the dehydrator effluent to prevent yield loss and minimize downstream corrosion in the fractionation section. The sulfuric acid in the decomposer effluent and sprung phenol from the effluent section is reacted selectively with a diamine to form an organic sulfate salt. The diamine injection is on ratio control to the decomposer net product rate. The ratio is adjusted based on pH readings from the laboratory and on-line sulfuric acid analyzer.

The flow from neutralization to the fractionation feed tank is controlled by a PIC upstream of the crude product cooler since this is the point of highest vapor pressure.

Figure IV-6
Direct Neutralization Section Flow



E. Fractionation

The fractionation section separates and purifies the co-products phenol and acetone. The columns also separate the cumene with AMS for hydrogenation of the AMS for recycle back to the oxidizers from the small quantity of residue that forms in the decomposer. The phenol is purified by reacting the carbonyls to compounds that boil at a different temperature compared to phenol, followed by distillation of light and heavy compounds (see Figure IV-7).

1. Fractionation Feed Tank

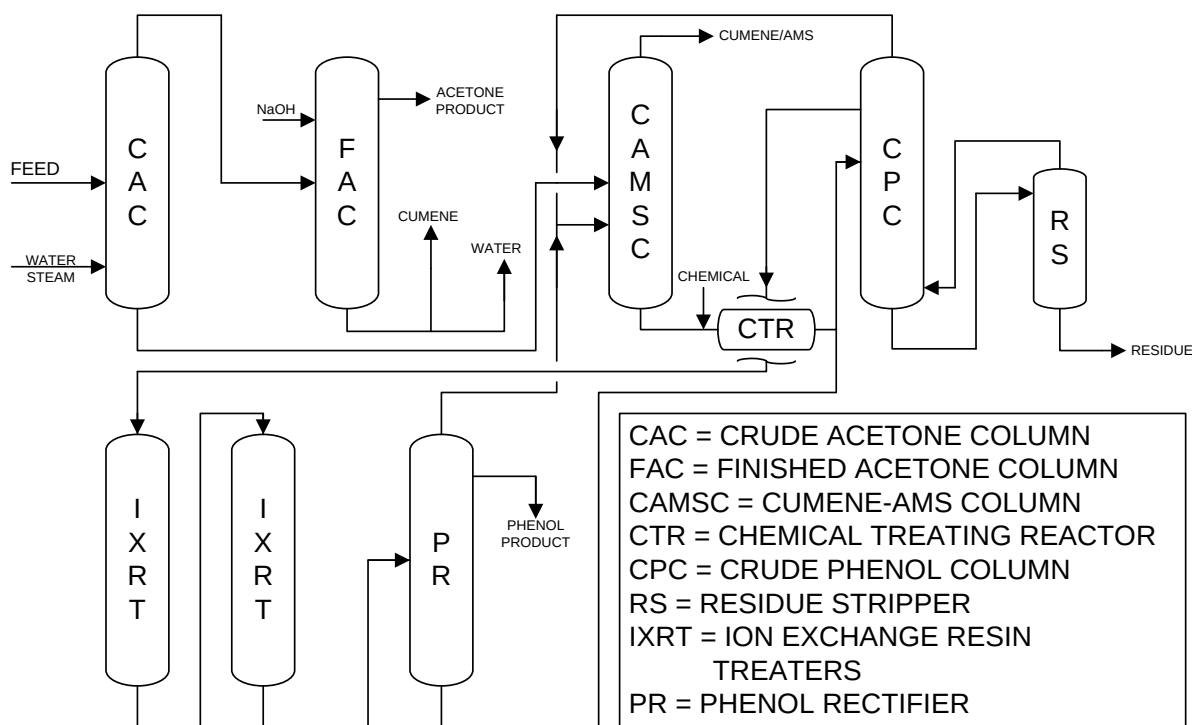
The fractionation feed tank(s) is the surge tank between the phenol/acetone synthesis sections and the fractionation section. The surge time enables one section to run while the other is shut down for a short period. The surge time and number of tanks is dependent on the customer needs. The number of tanks is typically one or two. If the tank is run at 50% level, the surge time can vary between 1-2 days depending on the design.

Due to the presence of acetone in the crude product, a feed tank vent condenser using chilled water can be added to minimize the loss of acetone to the atmosphere if a cone roof tank is used. A floating roof tank does not require any additional equipment. The tank(s) are steam heated to prevent solidifying phenol during re-processing of off-spec. Mixing of the tank is required to prevent stratification of the crude product and to provide a constant temperature feed to the crude acetone column.

The crude acetone column charge pumps, located at the fractionation feed tank, pump the crude product through a filter. The filters remove heavy residue in the feed as well as entrained rust particles from the tank that might be present.

Figure IV-7

Fractionation Section



2. Crude Acetone Column (CAC)

The objective of the CAC is to reduce the acetone concentration in the crude product to a low value (e.g. <0.1 wt%). At the same time, the CAC must keep most of the phenol and AMS out of the overhead. Any acetone that goes out the bottom of the CAC is considered unrecoverable because it will either go out to the phenol fractionation vacuum system or out with the cumene-AMS which contaminates the

product. If significant acetone were recycled, then a higher feed point would be used.

The CAC is reboiled by one or two vertical thermosiphon reboilers. A spare is sometimes provided to allow for greater fractionation section on-stream efficiency. The reboiler(s) may eventually foul with water soluble salts that carryover from the effluent section during upsets. With a spare reboiler, the fouled reboiler can be isolated and water washed while the column is operating. Without a spare, the feed to the fractionation section must be stopped and accumulated in the feed tank during the boil off. Salts can be cleaned from the reboiler by injection and boiling of process water at the bottom of the reboiler. Steam and non-condensables are vented from the CAC receiver during the boil-off. The recent innovations of adding cumene to the PRU and amine to neutralize the crude product has eliminated the need for a second reboiler as regular boil outs are no longer necessary.

Water or steam is injected to the bottom of the CAC during normal operation to increase the relative volatility of the acetone and reduce the bottom temperature. A low bottoms temperature is important to maintain a low reaction rate for the formation of phenol formaldehyde resin, a foulant for the CAC reboiler. Formaldehyde is created in the decomposer which reacts with phenol in the dehydrator to form phenol formaldehyde resin – this is one of the main dehydrator foulants. However it does not all react there, some formaldehyde stays in solution until it contacts the crude acetone column and then, at high temperature, reacts with phenol, resulting in fouling. Operating the CAC at low bottoms temperature, which is driven by live steam injection, significantly decreases the reaction of formaldehyde and phenol to form phenol formaldehyde resin. The steam FIC to the blind tray above the reboiler is reset by a TIC on the reboiler outlet. In stable operations, an increase in reboiler temperature will result in an increase in the live steam injection rate as increased reboiler outlet temperature indicates a 'dry' reboiler and risk of fouling conditions. The CAC bottoms LIC resets the FIC on the discharge of the CAC bottoms pumps controlling the flow to the C-AMSC. If and when water is added to the CAC bottoms it is added as available from the fractionation vacuum system which is typically a liquid ring vacuum pump.

Some of the overhead vapor of the CAC, consisting of acetone, water and some cumene, is sent to the finished acetone column (FAC). 20-40% of the cumene in the crude product is taken overhead. The amount of cumene is dependent on feed flow rate and feed tray. The net overhead vapor draw FIC is reset by a feed forward ratio controller from the CAC feed FIC. The set point for the ratio controller is determined by composition of the crude product, which when operations are stable and there is no re-processing of off-spec, should not change. Note that during re-processing of off-specs the ratio will need to be changed accordingly with the feed composition. The benefit of the ratio controller is operations should not need to make adjustments to the vapor feed flow rate to the FAC when CAC feed rates are

changed. The remaining CAC overhead vapor is condensed in the CAC condenser and then drains to the CAC receiver. The liquid is then pumped by the CAC overhead pumps back the top tray on LIC control.

Because the mass balance of the column is being set by the ratio controller from the feed, a TIC is not required within the column. With an accurate ratio setpoint and stable column control (levels and pressure) then the column splits will be made to the required precision. The crude acetone column is what the name implies, a crude split. The live steam injection increases the volatility of the acetone sufficiently high that even without 'precise' control, the bottoms acetone specification can be met reliably. The flexibility in the column comes from how much cumene is taken overhead; more or less can be taken overhead so long as the bottoms acetone concentration is met.

The CAC overhead pressure is controlled by a vent valve on the overhead receiver and is set at +0.70kg/cm²(g). The increased pressure is to ensure sufficient driving force of the net overhead to the FAC and to increase the CAC overhead temperature to allow reasonable design of the air-cooled condenser. This vent should normally be lined up to the FAC feed, however during start-up operations should be lined up to the flare while excess nitrogen is vented. This recent change eliminated the need for a nitrogen push-pull system and a vent condenser.

3. Finished Acetone Column (FAC)

The finished acetone column purifies the acetone through distillation and reaction. The cumene, water, and C₈-C₉ saturate compounds are separated by distillation. The acetone boiling range aldehydes and organic acids are reacted with caustic to form heavy compounds that are separated by fractionation.

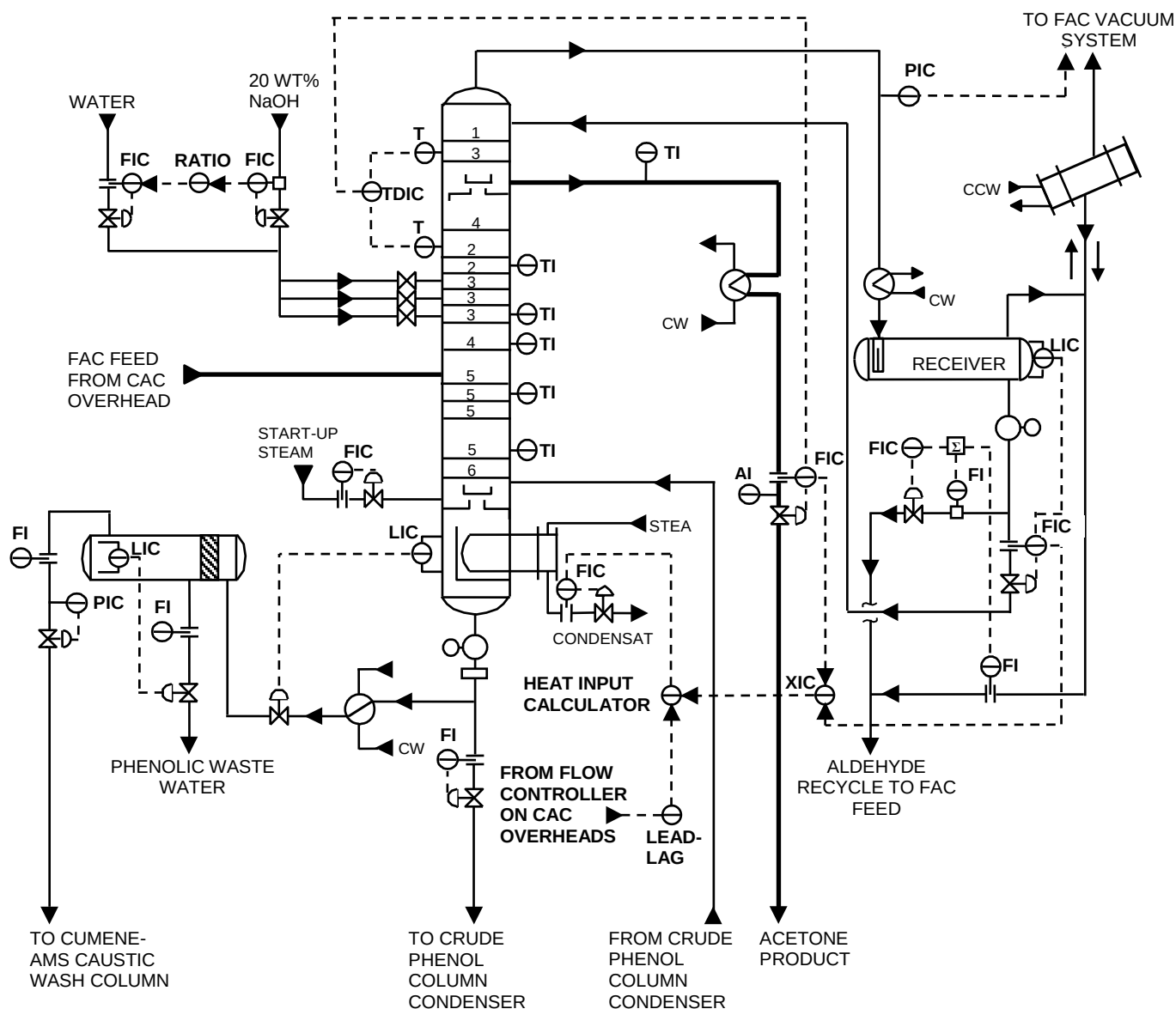
The main feed to the FAC is vapor. It enters the column about 10 trays from the bottom part of the column. A dilute caustic stream is normally fed to the column at one of three locations (see Figure IV-9). Typically, the top point is used to maximize the contact time of caustic with aldehydes.

Caustic is injected into the FAC to ensure that aldehydes present are reacted and rejected to the bottom of the column. Two flow controllers, one on the water and one on the 20wt% caustic, are used to dilute the caustic down to approximately 1 wt% before flowing into the column. There are three feed points with open end distributors available in the middle third of the column to optimize the caustic injection point versus acetone product quality.

The FAC is reboiled using several heat sources. Greater than ½ of the heat is supplied by a pump-around circuit to the crude phenol column (CPC) condenser-FAC reboiler. The FAC bottoms pumps circulate through an FIC to the CPC condenser-FAC reboiler returning back to the column below the bottom tray. The secondary reboiler is typically a steam-heated stab-in type. It supplies less than ½

of the heat requirement. The condensate flow from the reboiler is controlled by an FIC, which is reset by a heat input calculator which receives inputs from both a feed forward controller on the vapor feed to the column (as the vapor feed flow increases the condensate flow increases) and a FAC reflux/FAC product ratio controller (as the ratio of reflux/product increases the condensate flow decreases). The water analyzer on the acetone product side-cut provides information for the operator or a direct controller to adjust the side-cut flow rate to control the TDIC. Finally, there is also a provision to add heat by injecting live steam to the bottom of the FAC. This steam is useful during start-up, or when the other reboilers are not performing sufficiently.

Figure IV-9
Finished Acetone Column



The acetone product is drawn off an accumulator tray below tray 3. The acetone product is pumped through the acetone product cooler, the FIC and then to storage. The FIC is reset by a TDIC that measures the differential temperature between the nearly pure acetone on Tray 3 and the acetone/water mixture approximately 20 trays below. Several other temperature indicators (TI) are located on the tower to monitor the operation. An increase in the TDIC is an indication that too much water

is coming up the column and the acetone will go off-spec for high water. The response of the TDIC controller to high TDIC is to reduce the product rate to allow more acetone back down the column to push the water lower in the column and reduce the TDIC back to set-point.

The overhead vapor from the column goes through the FAC condenser where most of the organic vapors are condensed and then drain to the receiver. The non-condensable vapor separates in the receiver and goes to a chilled water vent condenser and the FAC vacuum system. The FAC column operates at a pressure of $-0.37 \text{ kg/cm}^2(\text{g})$. The vent condenser condensate and liquid ring vacuum pump seal water purge are recycled back to the feed line of the FAC to minimize acetone losses and further react the aldehydes by putting them below the reaction zone again. The aldehyde recycle purges contaminants that accumulate in the upper part of the column. The reflux pumps acetone from the receiver back to the column on FIC control cascaded to the receiver LIC.

The net bottom stream of the FAC is pumped to either the FAC bottoms drum or the pump-around reboiler. The flow to the FAC bottoms drum is first cooled by the FAC bottoms cooler. The FAC bottoms drum has a packed section of anthracite coal that aids in separating the cumene and water phases. The water coming off the drum bottom is sent to the oily water separator, or to the phenol recovery section, depending on its phenol concentration. The cumene from the top of the liquid full drum is ultimately sent to the phenol recovery section so trace phenol can be extracted or can be sent back to the fractionation feed tank for start-up.

Due to the two phase liquid in the FAC bottoms, the level control scheme is more complicated than typically observed. The level in the bottom of the column is controlled by LIC that resets the two phase flow to the bottoms drum. The bottoms drum interface LIC controls the water flow rate out of the drum bottom. The cumene flow from the top of the drum is controlled by a back pressure controller.

4. Cumene-AMS Column (C-AMSC)

The primary objective of the cumene-AMS column (C-AMSC) is to separate the cumene and AMS from the higher boiling phenol. Any AMS left in the phenol will react in the ion exchange resin treater (IXRT) to form a heavy compound, which will lower the phenol yield and foul the resin. The phenol concentration in the overhead is at the azeotropic condition with AMS.

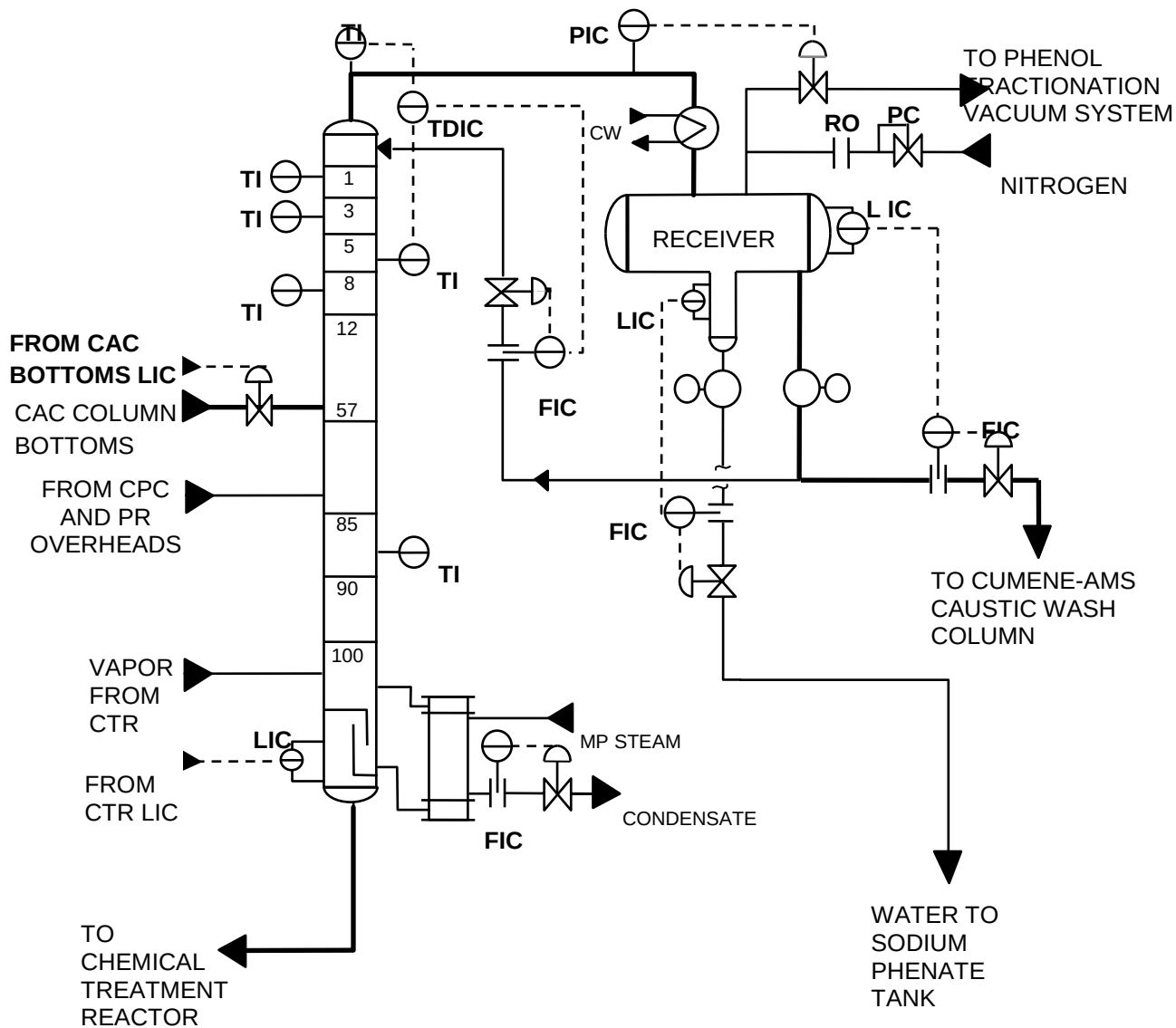
The acetone-free crude product from the CAC bottoms enters the feed point located in the middle of the C-AMSC. Recycle phenol from downstream columns are fed to a separate, lower, feed point (see Figure IV-10).

The overhead vapor is cooled in the C-AMSC condenser and then free drains into the C-AMSC receiver. The column runs at a relatively high vacuum. The PIC on the overhead vapor controls a valve on the vent line from the C-AMSC receiver. Nitrogen can be injected upstream of the valve to maintain good pressure control.

Two phases of liquid, aqueous and organic will be collected in the C-AMSC receiver. A boot on the receiver accumulates the aqueous phase. An interface LIC controls the flow of the aqueous from the boot. A pump can be provided depending on the destination pressure of the water. The water, containing a few percent phenol, is sent to the sodium phenate tank in the phenol recovery section for treatment. The organic phase from the receiver is pumped by the C-AMSC overhead pumps as reflux and net product. The reflux is pumped back to the top tray on FIC control reset by a TIC in the upper part of the column. The net overhead cumene/AMS containing 5-10 wt% phenol is pumped to the C-AMSCWC in the phenol recovery section on FIC reset by the receiver LIC.

The CAMSC bottoms liquid is vaporized by a vertical thermosiphon reboiler. The heat source is medium pressure steam with the appropriate flow control system on the heating medium. A feed forward ratio control of column feed to the reboiler heat source flow rate may be added to improve operation stability. The bottoms level is controlled at the chemical treating reactor by the LIC, which resets the FIC for the flow to the crude phenol column.

Figure IV-10
Cumene-AMS Column

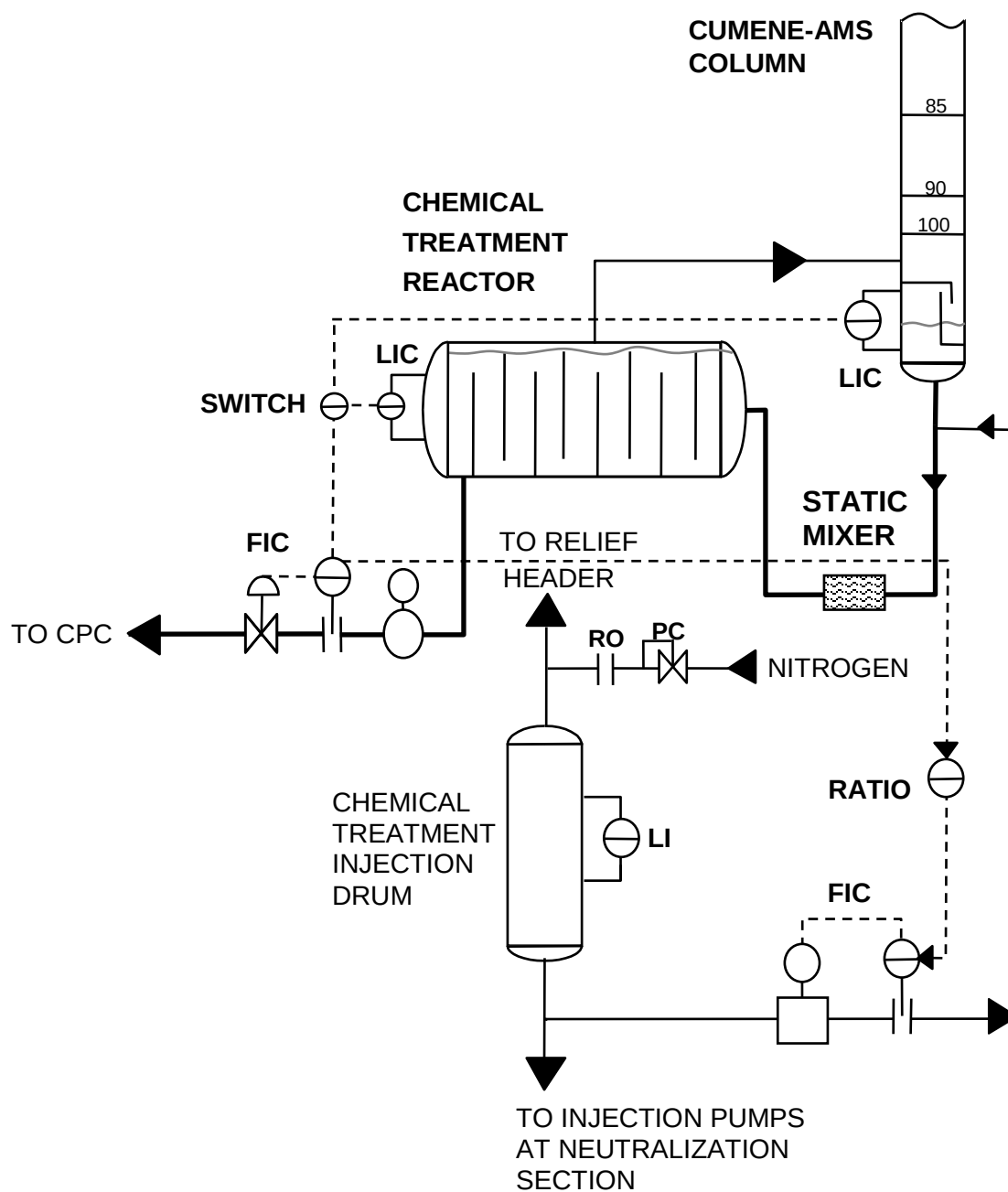


5. Chemical Treatment Reactor (CTR)

The chemical treatment reactor is a baffled, plug flow vessel that provides a 1.5 hour residence time for the reaction of diamine, or equivalent amine, with carbonyls to occur.

The chemical treatment reactor (CTR) feed consists of C-AMSC bottoms and diamine (see Figure IV-11). The diamine is injected upstream of the CTR to a static mixer from the chemical injection pumps from a drum rack or surge tank. The rate of diamine injection is controlled by a ratio controller to the CTR effluent rate.

Some light gases may be formed in the reactor therefore it is not run liquid full. A vent from the top of the reactor is connected to the vapor space of the C-AMSC above the reboiler return nozzle. The liquid from the reactor is pumped to the crude phenol column feed by the chemical treatment reactor effluent pumps. The CTR LIC is cascaded to the product FIC which controls the flow rate to the CPC.

Figure IV-11**Chemical Treatment Reactor**

6. Crude Phenol Column (CPC) / Residue Stripper (RS)

The crude phenol column (CPC) and residue stripper (RS) are designed to remove the heavy components from acetone free crude product. The separation objective is to minimize the acetophenone in the overhead (<0.05 wt%) while minimizing the phenol lost in the bottoms (target <10%, preferably <5 wt%). The two objectives are not mutually exclusive. If the phenol is reduced to a very low value, then acetophenone will be higher in the overhead (see Figure IV-12).

The two columns can either be stacked or side-by-side. The stacked arrangement saves on one set of pumps and plot space, but requires a somewhat higher total column length and the associated costs that go with it. The crude phenol column and residue stripper are heated by reboilers with MP steam as the heat source. No high pressure steam is required. A spare reboiler is sometimes provided to the residue stripper to minimize the fractionation section downtime for cleaning. The recent innovations of adding cumene to the PRU and amine to neutralize the crude product has eliminated the need for a second reboiler as regular cleaning is no longer necessary. A TIC on the outlet of each reboiler controls the heat input. A net product stream from the residue stripper bottoms pumps is controlled by the column bottoms LIC. The TIC is set to give about 60% phenol in the CPC bottoms. The residue bottoms stream is cooled using tempered water cooler before going to the residue tank.

The overhead vapors are condensed by the CPC condenser-FAC reboiler. The condensed liquid collects in a receiver where non-condensable gases are vented to the fractionation vacuum system. The reflux is controlled by FIC cascaded to the receiver LIC. The net liquid to the IX resin treaters from the side-draw is controlled by FIC, which is reset by a ratio control based on the reflux flow. The reflux sets the separation of heavy and light compounds in the overhead. The reflux rate is dependent on the heat input to the reboiler.

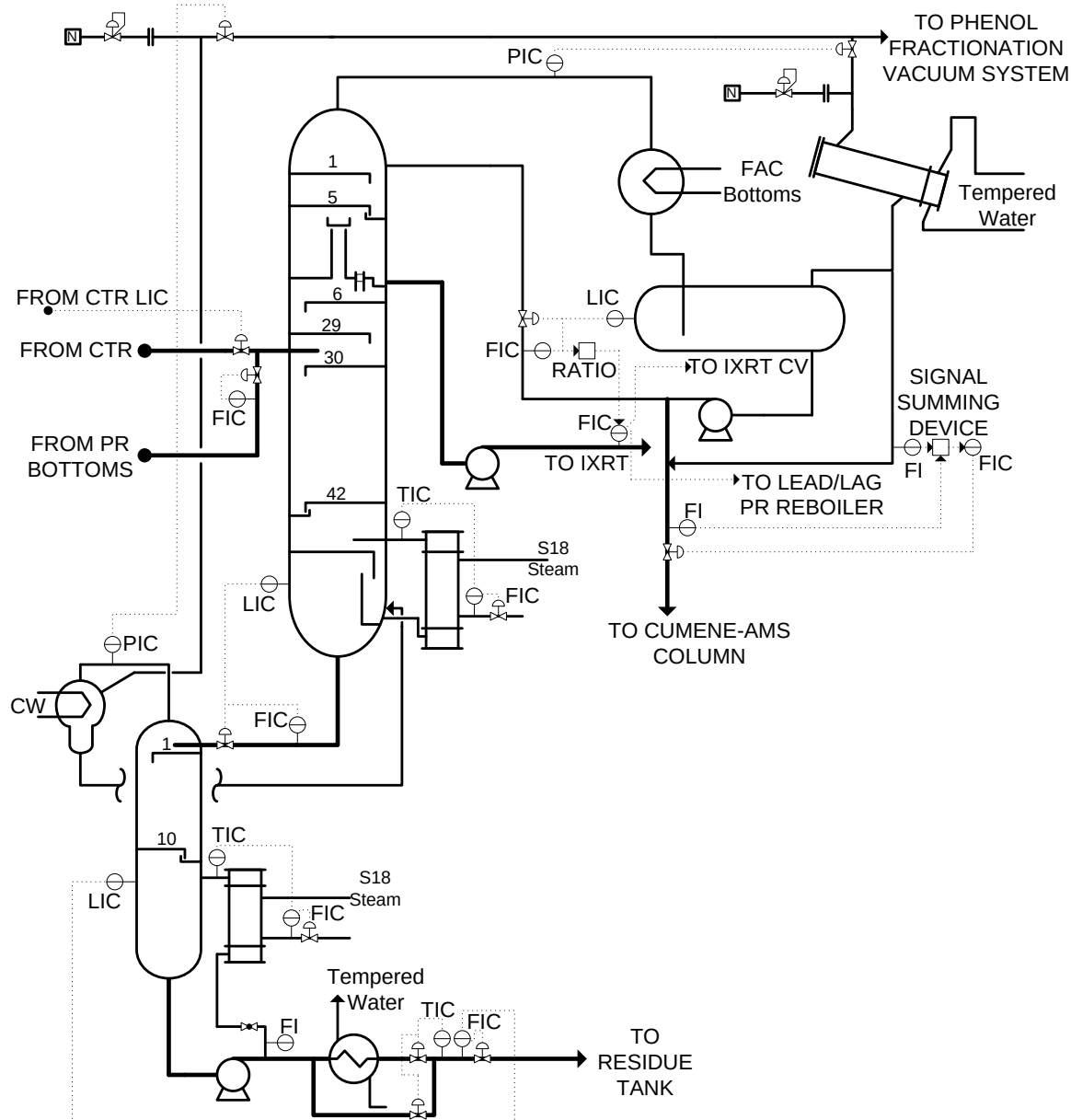
CPC net overhead product quality and RS bottoms phenol concentration are adjusted through the CPC reboiler condensate FIC. The condensate sets the heat input to the CPC and provides the stripping in the bottom of the column to reduce the phenol concentration. The vapors from the top of the residue stripper are condensed and returned to the CPC reboiler inlet by gravity through a seal leg. This flow scheme allows vapors a chance to be flashed once before being refluxed back to the RS.

The CPC column overhead pressure (vacuum) is controlled by PIC. The PIC controls the non-condensable flow to the vacuum system. Nitrogen may be added to help control the column pressure and a manual valve will also aid in adding pressure drop between the header and the column. The residue stripper pressure is also controlled by an independent PIC on the non-condensable vent line to the

vacuum system. Seal legs are required between the columns in a stacked arrangement because the top of each column run at the same vacuum pressure.

Figure IV-12

Crude Phenol Column and Residue Stripper

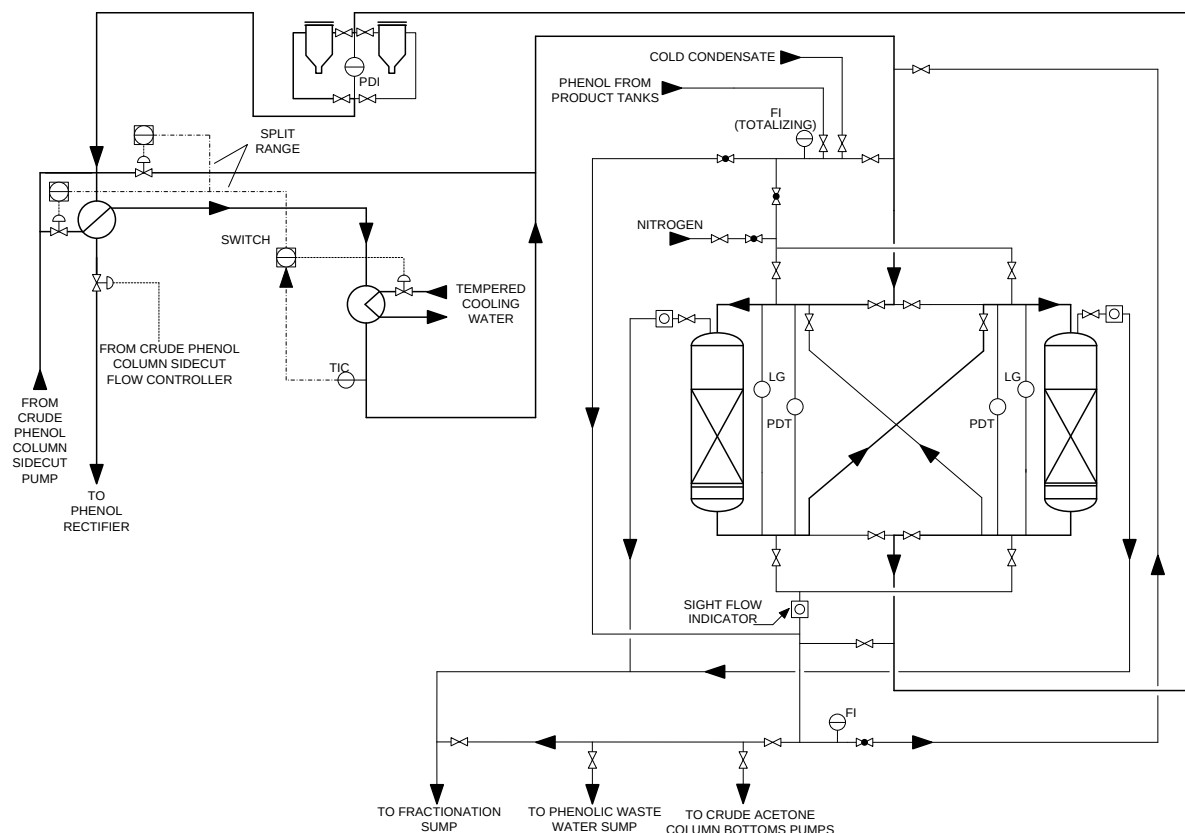


7. Ion Exchange Resin Treaters (IXRT)

The IXRT finish the purification of the phenol product (Figure IV-13). The crude phenol column side-draw product is cooled by heat exchange with the treater effluent. A trim cooler/heater controls the final temperature to 60 -150 °C, depending on resin type and life. The treaters are arranged in a lead-lag position with piping available to put either treater in the lead position.

Figure IV-13

Ion Exchange Resin Treatment System



8. Phenol Rectifier (PR)

The objective of the phenol rectifier (PR) is to take a heart-cut of phenol and recycle the light and heavy cuts back to upstream fractionation.

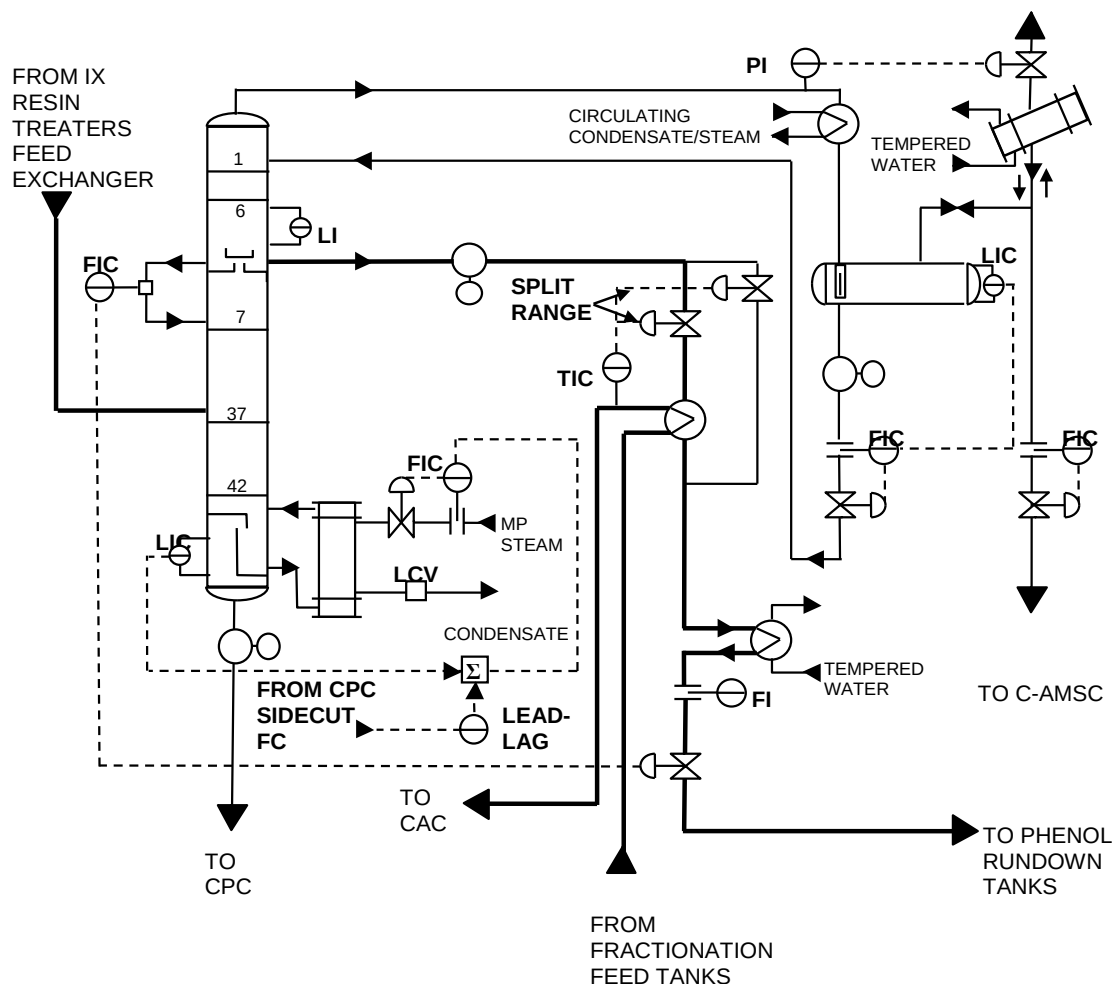
The IXRT effluent flows to the PR (see Figure IV-14) via the CPC side-cut pumps. The PR feed enters the column typically about 5 trays from the bottom. The net phenol is drawn from the column as a liquid side-cut typically 6 trays from the top of the column.

The PR reboiler is a thermosiphon type. The flow of MP steam is controlled by FIC that is reset by the bottom liquid LIC. A feed forward controller based on column feed aids the level controller. A net bottoms liquid (heavy cut) is pumped on FIC control by the PR bottoms pumps to the CPC feed. Typically, the flow is 5-10% of the total feed flow to the column. The heavies produced from the IXRT concentrate in the PR bottoms.

The overhead pressure is controlled at a vacuum to minimize the bottoms temperature. High temperatures can accelerate color formation in the phenol product. The overhead vapor PIC controls the venting rate with a control valve. Nitrogen can be added as necessary to maintain the valve in a controllable range.

Overhead vapor is condensed in an internal or external PR condenser by steam generation. If an internal condenser is utilized the reflux gravity flows down the column and is measured at the side-draw accumulator tray. The reflux flow controls the net overhead valve, which maintains a constant level on the tray. If an external condenser is utilized reflux is pumped to tray 1 on FIC control and cascaded to the receiver level. The net overhead flows by gravity on FIC to the C-AMSC feed. Vapors from the condenser/receiver flow to a tempered water cooled vent condenser. The vent condenser removes any condensable vapor from the vent gas. Water is the predominant light end that is concentrated in the overhead of this column.

The phenol product is taken as a side-cut and is either pumped or flow by gravity to a product cooler and to storage. The flow rate of the phenol product is controlled by FIC controller and is cascaded to the internal reflux flow meter from the accumulator tray for the phenol product.

Figure IV-14**Phenol Rectifier**

9. Miscellaneous Fractionation Section Equipment

A number of systems in the fractionation section that have a common use are the vacuum system, sumps, tempered water system and product tankage.

9.1 Vacuum System

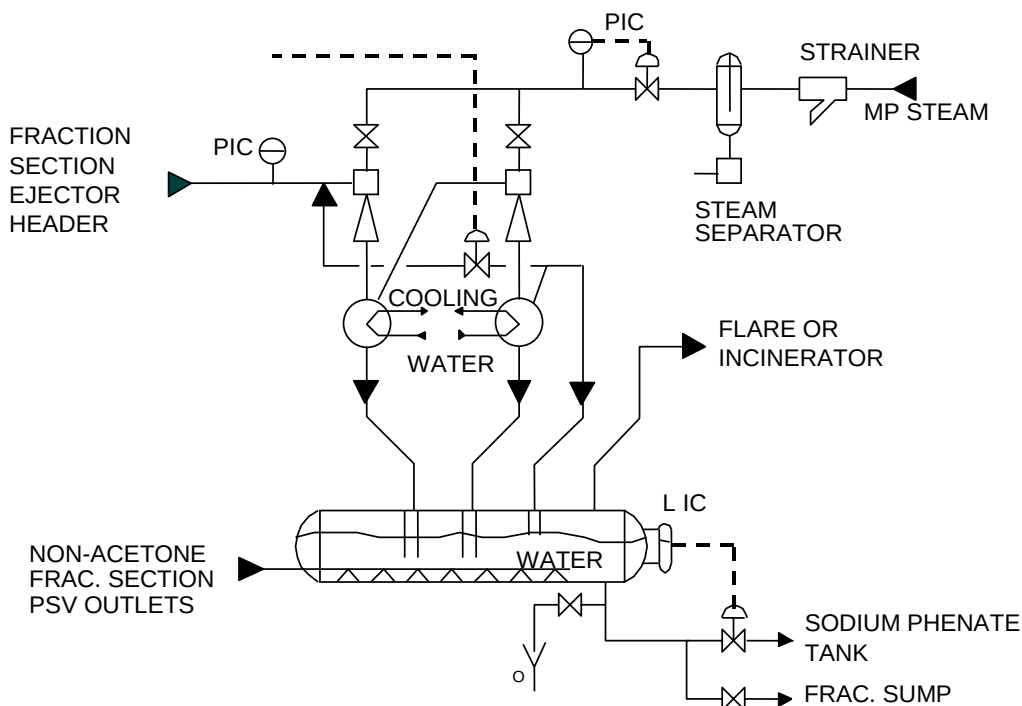
Four columns, C-AMSC, CPC, RS, and PR are tied to a common vacuum system (see Figure IV-15). The non-condensable gases proceed to the fractionation vacuum producing equipment. If ejectors are used, typically a two stage ejector system is required to produce sufficient vacuum for the fractionation section. A PIC

on the vacuum header controls the spillback of gases from the last stage to the first stage. If vacuum pumps are used single stage vacuum pump (with backup) is sufficient. A PIC on the vacuum header controls the spillback of gases from the vacuum pump separator overhead to the suction of the vacuum pump.

If an ejector system is used the ejector steam and any condensable liquids drain from the ejector after-condensers to the fractionation ejector condensate drum. The drum is larger than typical ejector drums because it also serves as a receiver for flow from the fractionation section pressure safety valves (PSV's). It is impractical and undesirable to heat trace the flare header downstream of the PSV's all the way to the flare. Any relief flows containing phenol are retained within the battery limits by sending them to the condensate drum. The phenolic streams are diluted in the drum with water to minimize the heat tracing requirements. The drum is then vented to flare.

The excess aqueous phase is normally pumped on LIC control to the phenolic waste water tank or sewer depending on its composition. An option to recycle the water back to the bottoms of the CAC is provided to conserve on water. This can be employed as long as there is no build-up of light water soluble hydrocarbons that contaminate the products. If there is a PSV relief of a significant quantity of phenol, then the aqueous stream with high phenol concentration is sent to the sodium phenate tank, manually. If an organic layer forms in the drum, then it can be skimmed by draining to the fractionation closed drain header. There is also the option of pumping the drum contents back to the fractionation return header.

If vacuum pumps are utilized the vessel otherwise named the 'fractionation ejector condensate drum' no longer has condensate constantly flowing into it from the ejectors. In this case there is normally no flow to the vessel which is now named the Phenolic Relief Drum. All of the phenolic reliefs are combined into two headers at the bottom of the vessel and submerged below a level of water. The water in the vessel acts to condense phenol from the vapor reliefs limiting how much phenol enters the relief header. Additional water can be added manually with a globe valve. During or following a relief event the contents of the drum can be drained to the fractionation closed drain header for reprocessing.

Figure IV-15**Fractionation Section Vacuum System with Ejectors****9.2 Fractionation Sump**

The fractionation sump collects the liquid drained from any equipment in the fractionation section via the fractionation closed drain header. The fractionation sump contents are pumped through to the fractionation return header cooler to the fractionation feed tank(s). The pump is automatically started and stopped by high and low level switches. A number of start-up lines join the fractionation sump pump discharge before going to the return header cooler. The sump is vented to the relief header. The sump has a nitrogen purge to prevent backflow of gases from the relief header.

9.3 Product Tankage

The product tanks may or may not be included in the UOP design package depending on the customer requirements. Rundown tanks are required for the phenol and acetone products to verify the product quality before sending the contents to long term storage. Typically, three day tanks are used for phenol and acetone each. The third tank is used to recycle off-spec material. Some units have only 2 run down tanks. In this case, it is recommended to have two fractionation

feed tanks, so one can be used as an off-spec tank. The phenol tanks are nitrogen blanketed and constructed of 304 SS or phenolic resin lined carbon steel to maintain product quality. The tanks are also heated to keep the phenol from freezing. The acetone rundown tanks have a common chilled water condenser on the tank vent to minimize losses of product to atmosphere or are a floating roof type. Only one residue tank is required because the residue product specifications are not rigid.

F. Hüls MSHP (AMS Hydrogenation)

The AMS hydrogenation section reacts α -methyl styrene (AMS) with hydrogen to produce cumene. If AMS were allowed to go to the oxidizers, an increase in oxidizer acidity would result impacting yield and requiring a capacity reduction.

After the AMS is scrubbed with caustic to remove phenol in the effluent section, the treated AMS is sent to the hydrogenation feed tank. The tank is a multi-day surge volume that allows the AMS to accumulate while the rest of the phenol complex is operating during a hydrogenation catalyst change-out.

The fresh AMS feed to MSHP is fed to the unit on local FIC. A combined feed FIC controls the reactor #1 effluent recycle to maintain the velocity and composition through the reactor. The combined reactor liquid is then mixed with hydrogen (see Figure IV-17). The rate of fresh hydrogen is controlled by local FIC cascaded to a ratio controller on the vent hydrogen from the reactor separator; this ensures the correct amount of excess hydrogen is fed to the reactor.

The reactor #1 hydrocarbon recycle controls both the combined feed AMS concentration (reactor delta T) and the reactor inlet temperature. A portion of the reactor #1 effluent recycle is either cooled (normal) or heated (start-up) in the reactor recycle cooler. A reactor #1 inlet temperature controller resets the bypass control valve around the reactor recycle cooler and the control valve from the recycle cooler on split range.

Reactor #1 has two outlets, a liquid phase exiting the vessel bottom to the recycle pump, and a two phase side draw that sends the net effluent to reactor #2. Reactor #2 is the finishing reactor; it completes the saturation of AMS in a once-through flow reactor (e.g. without recycle hydrocarbon dilution). It operates in a downflow configuration like reactor #1. A portion of cooled reactor #1 effluent can be bed to reactor #2 to maintain a low temperature in reactor #2 to extend catalyst life.

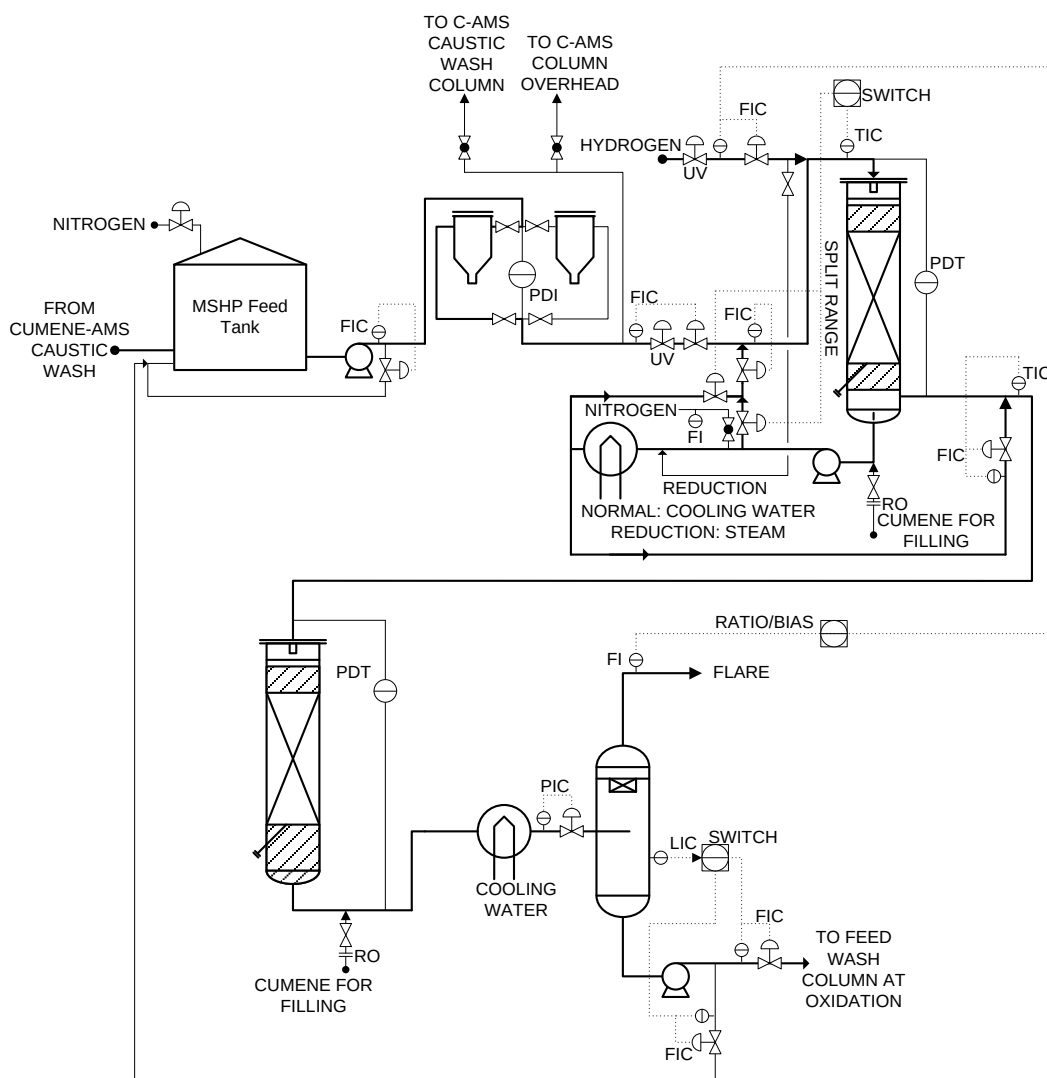
The reactor #2 effluent is cooled and then flashed across a pressure control valve to a low pressure separator where the residual hydrogen is vented to the relief header. The reactor back pressure control valve maintains the reactors at the desired

setpoint. The liquid from the product separator is pumped via LIC to the oxidation section feed wash column.

The excess hydrogen from the product separator is vented to the relief header. The FI measuring the vent rate resets the make-up hydrogen FIC to maintain an excess of hydrogen in the reactors.

Figure IV-17

MSHP Unit



G. Phenol Recovery

The section accomplishes multiple objectives (see Figure IV-18):

- a. Recovery of phenol from sodium phenate
- b. Removal of phenol from phenolic waste water
- c. Removal of phenol from cumene and AMS before hydrogenation and recycling to the oxidation section
- d. Make-up and storage of sulfuric acid and caustic

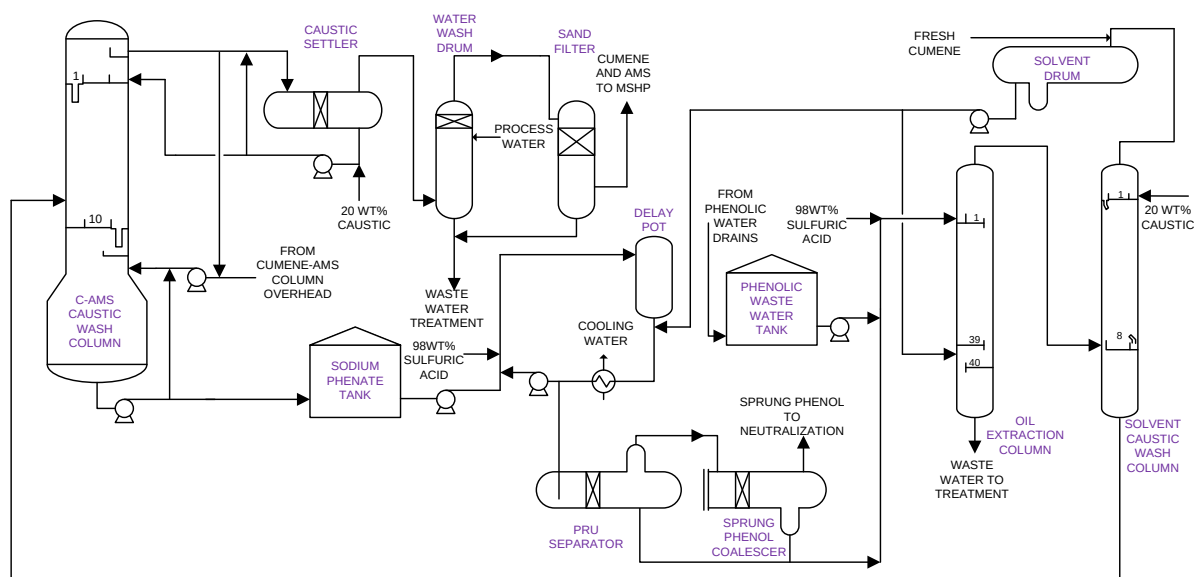
In this section there are four tanks: a phenolic waste water tank containing phenolic waste waters from various sections of the plant, a sodium phenate tank containing sodium phenate (from phenol and caustic reaction) from other sections of the plant, a sulfuric acid tank and a caustic tank.

1. Sulfuric Acid Tank

There is typically one main storage tank for concentrated sulfuric acid. Carbon steel can be used to store 98 wt% sulfuric acid with minimal corrosion, however an upgrade to stainless steel may be justified if erosion corrosion is observed due to high velocities at nozzles or other locations. The acid circulation pumps transfer the acid to small vessels in the decomposition and phenol recovery sections. Stainless steel piping and vessels are required downstream of the sulfuric acid filters to minimize the chance that iron sulfide will foul the proportioning pumps downstream.

2. Caustic Make-up System

Two caustic storage tanks are typically required, one concentrated (45 wt%) and one diluted (20 wt%). The concentrated caustic is used to make-up more dilute solutions in the 20 wt% caustic tank. The dilute caustic (20 wt% NaOH) is used in the oxidation, fractionation and phenol recovery sections. An on-line density analyzer controls the concentrated caustic FIC to the 20 wt% caustic tank. The 20 wt% caustic tank LIC controls the flow of water added to the tank. The water and concentrated caustic are mixed in a static mixer before entering the tank.

Figure IV-18**Phenol Recovery Section****3. Phenol Recovery Section**

The feeds to the sodium phenate tank are mainly sodium phenate from the C-AMS caustic wash column and the C-AMSC receiver aqueous phase. Other highly phenolic waters, such as the acid aromatics sump and the phenolic relief drum pumps (or fractionation ejector condensate pumps) when a phenolic column PSV relief occurs, are also sent to the sodium phenate tank. There is also the option of sending the C-AMSC receiver aqueous phase to the phenolic waste water tank if the phenol concentration is less than 2 wt%. The sodium phenate and phenolic water tanks have skimming connections that are tied into a common tank skimming pump. Organic or rag layers formed in the tanks are drained to a separation vessel and then pumped/metered to the delay pot to recovery organics and reject salts.

The sodium phenate tank contents are pumped by the sodium phenate pumps, mixed with skimmed hydrocarbons, delay pot recycle, and sulfuric acid, and fed to the delay pot static mixer (see Figure IV-19). A minimum flow spillback is normally provided for sodium phenate pump protection. The feed rate to the delay pot is set by local FIC at the sodium phenate pump discharge. The flow rate of 98wt% sulfuric acid to the delay pot feed is controlled by a ratio/bias controller with inputs from the delay pot FIC and the pH meter on the PRU recycle. The flow rate of sulfuric acid is set based on a ratio of the sodium phenate feed rate which can be biased up or down based on the pH of the delay pot effluent. The two-phase pH-adjusted feed undergoes an acid-base reaction in the delay pot. The delay pot has a 30 minute residence time to allow the exothermic reaction to go to completion.

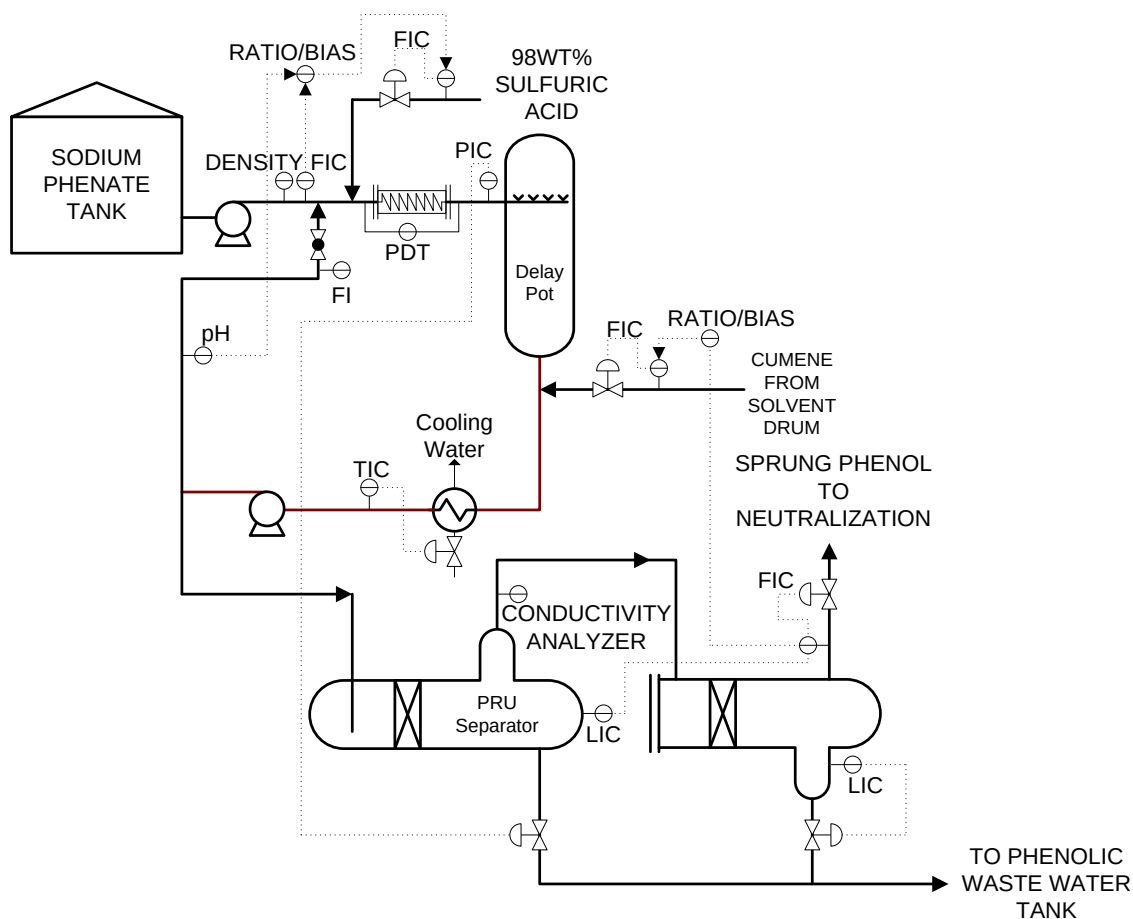
Cumene is added to the two phase stream exiting the bottom of the delay pot. The cumene is added on ratio control to the sprung phenol rate at a rate to target 50 vol% cumene in the organic phase from the PRU separator overhead. The additional cumene minimizes the entrained sodium sulfate salts and water in the organic phase, and minimizes the organics in the aqueous phase.

The mixture is cooled through the PRU cooler. The PRU effluent is temperature controlled by adjusting the cooling water rate to the exchanger. A process side bypass is not used around the PRU cooler because this could result in over-cooling in the exchanger resulting in salts precipitating and fouling the cooler. The cooled mixture is then fed to the PRU recycle pumps.

The PRU recycle pump discharge then splits with the recycle to the feed controlled by a globe valve and local flow indicator. Most of the cooled delay pot effluent is recycled back to the delay pot inlet to cool the feed and maintain the design residence time in the delay pot. The balance of the liquid flows to the PRU separator.

The two phase stream containing the "sprung" phenol and the aqueous phase is settled in the PRU separator. The overhead of the PRU separator flows to a coalescer for further salt/aqueous phase separation. The interface LIC in the PRU separator controls the FIC regulating the flow of sprung phenol leaving from the top of the coalescer. A conductivity analyzer on the overhead liquid detects if high salts are sent overhead and issues an alarm to the operator. The PRU separator overhead acidic phenol stream containing about 8% soluble water is sent to the neutralization section to be neutralized.

The aqueous phase, containing 1-2 wt% phenol, from the PRU separator bottoms and any additional liquid collected from the coalescer, are pressured directly to the oil extraction column or the phenolic waste water tank after pH correction. In this liquid full system, the PRU separator bottoms flow is the balance of the liquid to mass balance the unit and maintain a constant pressure at the delay pot inlet. The delay pot inlet is hydraulically the highest pressure point in the system and maintaining constant pressure here ensures there is sufficient pressure to keep the reacted material in the liquid phase and pressure the net organic phase to the neutralization section and net aqueous phase to the OEC.

Figure IV-19**Phenol Recovery Section Flow****4. Phenolic Waste Water Treating**

The water from the phenolic waste water tank (PWWT) is combined with sulfuric acid and fed to a static mixer and then to the oil extraction column (OEC) mixing pot (see Figure IV-20). A pH analyzer on the outlet of the mixing pot controls the sulfuric acid injection rate. The total flow FIC to the OEC controls the net flow from the phenol waste water tank pumps. The total flow is fed to the top OEC tray and flows by gravity down the column. The OEC has multiple reverse rain deck trays (e.g. weirs on the top and bottom of the one pass sieve tray). An LIC above the top tray controls the treated water flow out the column bottoms. A lean solvent (mostly cumene and essentially free of phenol) enters the column bottom distributor and flows up the column on FIC control.

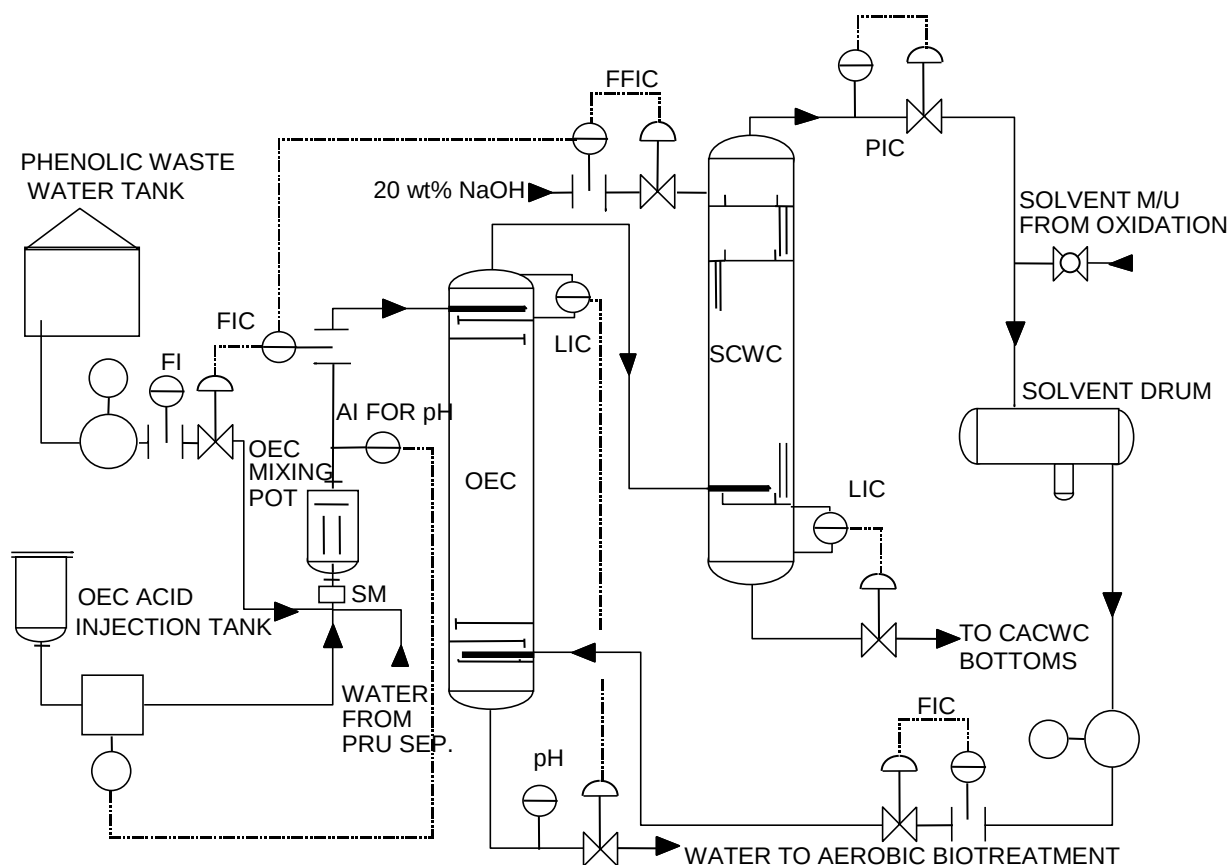
The rich solvent with phenol leaving the OEC top flows to the bottom tray of the solvent caustic wash column (SCWC). Phenol is extracted from the solvent with 20

wt% caustic by jetting up through the multi-tray tower. Caustic gravity flows down the tower. The fresh caustic FIC is reset by the total waste water flow to the OEC through a ratio controller. The column bottom interface LIC controls the spent caustic and sodium phenate flow rate to the C-AMSCWC, tray 7, to recover the phenol. The lean solvent exiting the top of the column goes through a back pressure PIC.

The solvent flows to a nitrogen purged surge drum named the solvent drum where any residual aqueous phase droplets are separated. Fresh cumene from the oxidation section can be added periodically to the drum to make-up for losses. The lean solvent is circulated with the solvent pumps back to the OEC to maintain the minimum jetting velocities in the OEC and SCWC. The cumene source for the PRU is the solvent drum which is fed from the oxidation day tank. If a build-up of undesirable compounds in the solvent occurs (e.g. acetone, AMS), then solvent can be pumped back to the fractionation feed tank, or fed at a higher rate to the PRU.

Figure IV-20

Phenolic Waste Water Treatment Flow

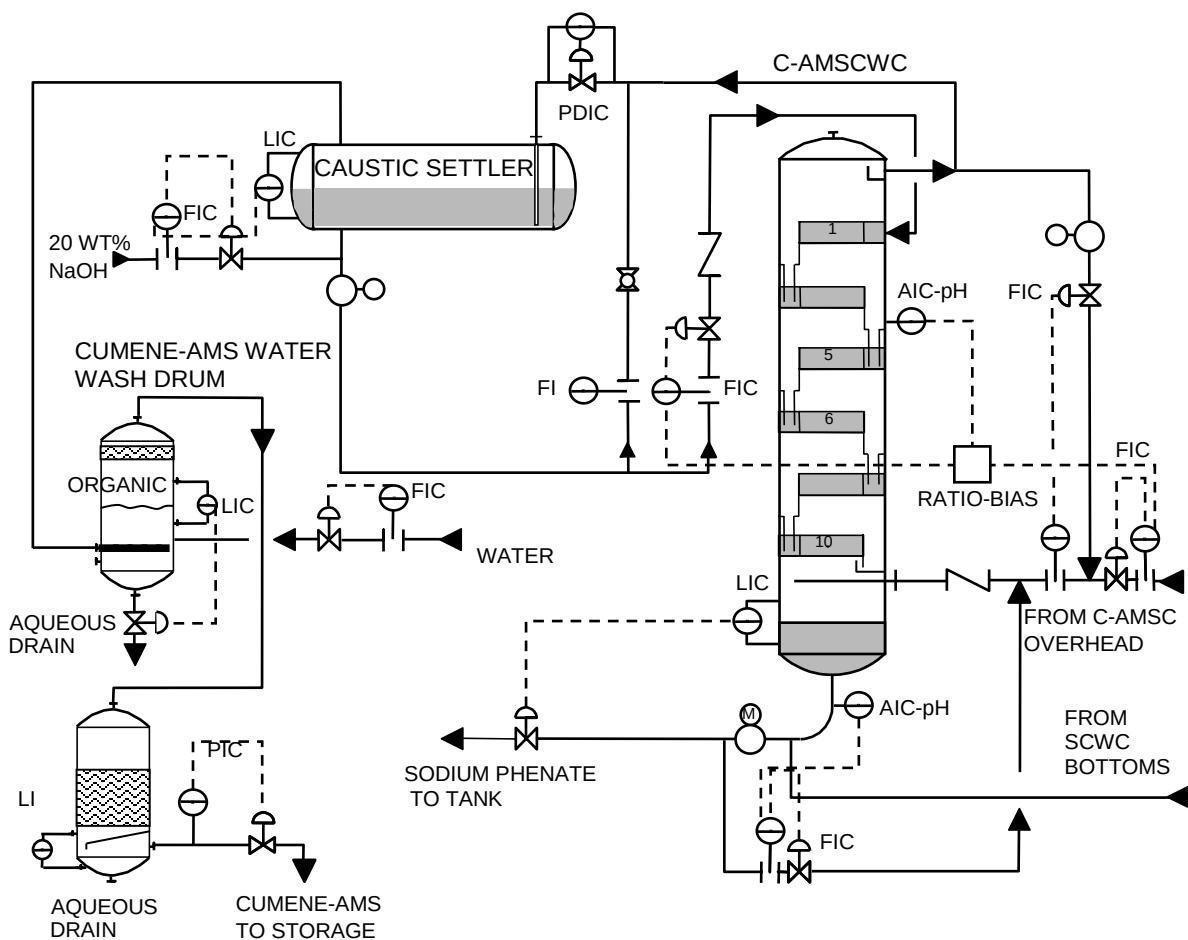


5. Cumene and AMS Treating

The organic phase from the C-AMS column overhead contains cumene and AMS with 4-10 wt% phenol. This stream is washed counter-currently by 20 wt% caustic in the Cumene-AMS Caustic Wash Column (C-AMSCWC) to remove the phenol. The fresh caustic enters the column top and gravity flows down (see Figure IV-21). The fresh caustic FIC is reset by a ratio control to the net C-AMSCWC feed FIC in the fractionation section. The caustic solution containing sodium phenate is sent to the sodium phenate tank on bottoms LIC.

The cumene jetting velocity through the trays is maintained by C-AMSCWC pumps. The treated cumene/AMS is circulated back to the column inlet controlled by an FIC on the total flow to the C-AMSCWC. The treated cumene/AMS is mixed with a small fresh caustic flow and then sent through an adjustable PDIC. The cumene and caustic then separate in the caustic settler. The treated organic phase (cumene & AMS) is then sent through a water wash column and sand filter to remove any entrained caustic. The effluent cumene/AMS from the sand filter is sent to the Hydrogenation Feed Tank. A back pressure PIC maintains pressure on the system.

The caustic utilization is a measure of the caustic used per phenol recovered. This utilization can increase above 100mol% as some phenol is naturally soluble in sodium phenate streams. A recycle system of sodium phenate is added to the C-AMSCWC because of its relatively high caustic load compared to the SCWC. The pH analyzers balance the flow of fresh caustic to the top of the column and the flow of mostly spent bottoms stream back to the combined feed point at the bottom of the column to yield a high caustic utilization.

Figure IV-21**Cumene-AMS Caustic Wash Column Flow****H. Auxiliary Systems****1. Reliable Cooling Water****1.1 Definition of Reliability**

For all pumping services designated as critical, it must be possible upon initial failure to reestablish the pumping service within 20 hours with 99.99% certainty. To meet this criterion typically requires multiple sources of power. Options for multiple independent power sources can include, but are not necessarily limited to, an emergency electric power generator, steam-driven turbine, a direct drive engine, or

multiple independent external electric power supplies. While the first three options are less reliable than normal electrical power, the combination of multiple power sources provides a more robust system than a typical single external electrical power supply. For example, if a diesel-powered emergency electric power generator is utilized, a probability of 97% is typical. Emergency generators are less reliable than normal electrical power because of the probabilities associated with failure to start, failure to run, and unavailability due to testing and maintenance. Quantitative risk assessment is typically performed to verify the reliability of such systems.

Below are the pumping services that would significantly impact on the safety aspects of the UOP Phenol Process Unit and are thus considered critical pump services:

- a. Fresh Feed Pumps
- b. Oxidizer #1 Circulation Pumps
- c. Oxidizer #2 Circulation Pumps
- d. CHP Sump Pump
- e. Flash Column Bottoms Pumps

Note: Equipment outside of UOP scope that is typically considered critical includes the Reliable Cooling Water Supply pumps, supply pump for quench water to oxidizers, and cooling water fan. (See below)

1.2 Cooling Services

For all cooling services designated as critical, it must be possible upon initial disruption to reestablish cooling within 20 hours with 99.99% certainty. This availability criterion typically requires a cooling water supply system, referred to throughout the process documents as Reliable Cooling Water, with a minimum of three pumps, with 2 normally operating and the third in standby; and multiple independent power sources for the pumps and cooling tower fan. Below are the critical coolers:

- a. Oxidizer #1 Emergency Cooler
- b. Oxidizer #2 Emergency Cooler
- c. Flash Column Bottoms Cooler
- d. Flash Column Vaporizer
- e. Oxidizer Vent Gas Condenser

Note: Reliable Cooling Water is used during normal operations as source of cooling water for the Decomposer cooler in the Decomposition section of the Phenol unit. During emergency conditions the Decomposition section is automatically shut down and the Reliable Cooling Water normally used at the Decomposer

Cooler becomes available for use at the Oxidizer #1 and #2 Emergency Coolers.

2. Dedicated Cooling Tower

The dedicated cooling tower provides normal cooling for the oxidizers and decomposer and emergency cooling for the oxidizers and flash column vaporizer. A pump control system regulates the number of pumps running. High and low flow conditions as indicated by the common pump discharge FI sends a signal to the control system to start the spare pumps. A high oxidizer temperature will also activate the spare pumps after a time delay. High temperatures will also shutdown the decomposer feed FIC and steam to the preflash/flash column vaporizers. The dedicated cooling tower falls under the definition of 'reliability,' discussed above. Biocide, sulfuric acid for pH adjustment and corrosion inhibitor injection facilities are added to maintain good cooling tower operation.

3. Chilled Water

A chilled water supply system provides 4-5°C (39-41°F) glycol/water mixture for use in the Oxidation, Concentration, and Fractionation sections. There are three major sets of equipment in this section

- a. Chilled Water Storage Tank
- b. Chilled Water Circulation Pumps (2)
- c. Refrigeration unit (2)

The chilled water storage tank is the surge tank for the system. The circulation flow normally bypasses this tank because of heat adsorbed by the tank from the environment during warm periods would increase the system duty. If the tank were included in the normal circulation, then the refrigeration equipment would have to be significantly larger. A line is always open from the tank to the chilled water circulation pumps.

The chilled water circulation pumps circulate the warmer propylene glycol water from the exchangers back to the refrigeration equipment. The refrigeration equipment is provided as a vendor package. The package typically includes a compressor and exchangers and is Freon based. Both the pumps and refrigeration equipment are spared to maintain on-stream efficiency. The chilled water quality (e.g. pH) is monitored and adjusted with anti-corrosion additives to minimize corrosion.

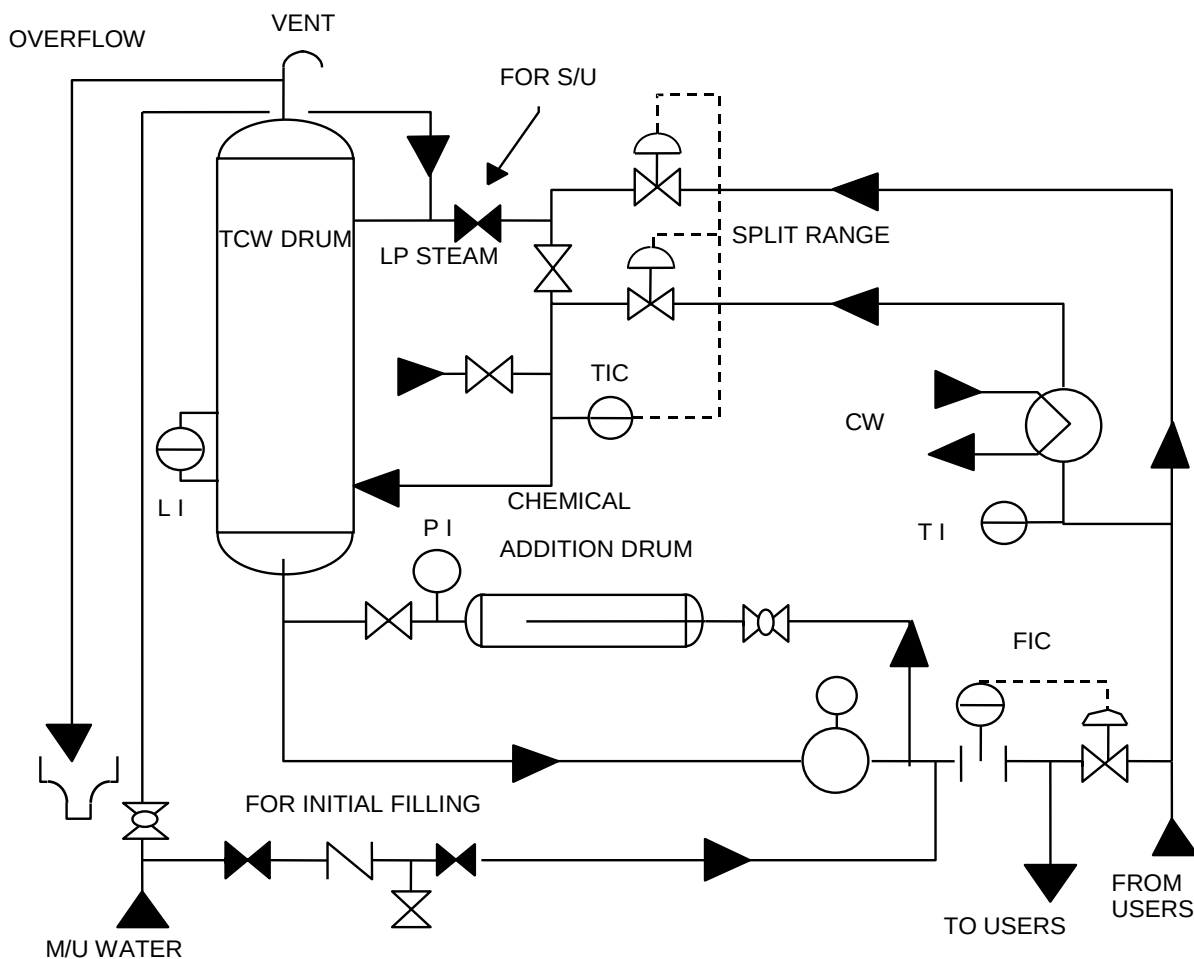
4. Tempered Water

A tempered water section is required to provide water for cooling phenolic streams that have the potential of freezing if normal cooling water is used (see Figure IV-16 on the next page). The tempered water system is comprised of the following:

- a. Surge tank
- b. Tempered cooling water drum
- c. Tempered cooling water pumps
- d. Tempered cooling water cooler
- e. Chemical addition tank
- f. Temperature controller

The tempered water from the drum is pumped to the users (e.g. heat exchangers). A line from the pump discharge is also sent to the chemical addition drum. The flow to the drum is restricted with a globe valve so the corrosion inhibitor can be metered into the main stream at the suction side of the pumps. The pumps are protected with a FIC that maintains a minimum flow through the pumps. The FIC controls a bypass valve around the tempered water header. A low flow switch automatically starts the spare pump. The pumps typically have emergency power.

Figure IV-16
Tempered Cooling Water System



The water temperature is maintained at 50°C (120°F) with a TIC that controls the amount of water bypassing the cooler, which is located on the return line from the tempered water header. Steam can be added to the tempered water circuit as needed to maintain the control temperature.

5. CHP Nitrogen

For the oxidation and concentration sections the introduction of acidic material could result in the unintended and rapid decomposition of cumene hydroperoxide (CHP). For these sections, to avoid the back-flow of acidic material from acid-containing systems to CHP-containing systems, a separate CHP nitrogen header is specified. The nitrogen to the CHP nitrogen header must come from a system that minimizes the opportunity for nitrogen contamination from other plant users such as a separate source of nitrogen or separate headers of nitrogen from a common liquid tank. The following CHP nitrogen header options may be considered:

5.1 CHP Nitrogen Header

CHP nitrogen can come from an independent nitrogen source which is not connected to any other users. This can be accomplished by a separate storage vessel which is refilled batch-wise but otherwise isolated from its source.

5.2 Liquid Nitrogen Storage

If the plant nitrogen supply is from liquid nitrogen storage, the CHP nitrogen header may come from the plant nitrogen supply through a dedicated vaporizer and supply line. The plant nitrogen supply must come from a point downstream of the instrumented back-flow prevention system.

5.3 Pipeline Nitrogen

If the plant nitrogen supply is from pipeline nitrogen, the CHP nitrogen header may come directly from the nitrogen pipeline through a dedicated supply line. The plant nitrogen supply must come from a point downstream of the instrumented back-flow prevention system.

V. Pre-commissioning

The pre-commissioning procedures ensure by thorough inspection and testing that the equipment is safe, operable and constructed as specified. The material in this section gives general guidelines for preparation before start-up. Should a certain section not apply to a particular process unit, that section should be ignored. For example, if this unit does not have fired heaters, ignore the section regarding fired heaters.

As the construction of the unit nears completion, a large amount of work must begin in order to prepare it for start-up. These pre-commissioning activities have three main purposes: 1) to ensure, by thorough inspection and testing, that the unit is safe, operable, and constructed as specified; 2) to prepare equipment for operation by flushing, running in, etc.; and 3) to acquaint the operators with the unit. The importance of these activities cannot be overemphasized. No matter how well a unit is designed, if the equipment is not as specified, not properly brought on stream, or not understood by the operators, it will not perform as expected.

All of the following activities are required to properly pre-commission the unit. However, the exact order of presentation need not be strictly obeyed. Depending on the progress of construction, certain procedures may be required earlier or later than suggested here. A thorough knowledge of the entire pre-commissioning operation will allow the plant personnel to schedule activities in the most time-saving and labor-efficient way. These are the necessary pre-commissioning activities:

- a. Reactor Inspection
- b. Vessel Inspection
- c. Inspection of Other Major Equipment
- d. Piping and Instrument Check
- e. Hydrostatic Testing
- f. Line Flushing
- g. Run-In of Pumps and Drivers
- h. Run-In of Compressors
- i. Servicing and Calibration of Instruments
- j. Commissioning of Plant Services
- k. Availability Check of Chemicals and Other Materials
- l. Plant Pressure Test
- m. Air-Freeing
- n. Commissioning of Additional Plant Services
- o. Charge Heater Refractory Dry-Out
- p. Reactor Loading

A. Reactor Inspection

The inspection of the reactor is very important due to the high operating temperature and hydrogen partial pressure. The reactor vessel nameplate, vendor drawings and specification sheets should be checked against the UOP project specifications. Any deviations should be documented and resolved before placing the reactor into operation. Particular attention should be given to the vessel temperature rating, pressure rating, vacuum rating, shell thickness, metallurgy, vessel elevation, support and construction of internals.

B. Vessel Inspection

Inspection of all vessels, separators, columns, receivers, storage tanks, etc., should be made at an early stage of construction. The actual installation should be compared to the vessel drawings in the Project Specifications. Particular attention should be given to the following details:

- a. Vessel temperature, pressure and vacuum rating
- b. Vessel wall thickness and metallurgy
- c. Vessel elevation and support.
- d. Vessel cladding: thickness and metallurgy
- e. Trays: number, type, spacing, levelness, liquid seal, weir height, valve number and type, sieve hole diameter and pitch, metallurgy, weep holes, supports
- f. Partition walls: location, height, seal
- g. Distributors: type, levelness, orientation, size, number and size of openings
- h. Nozzles: location, size and levelness
- i. Vortex breakers, baffles, wear plates
- j. Demisters and coalescing screens: thickness, materials, support and tightness
- k. Thermowells: location, length, pressure rating and metallurgy
- l. Level instruments: type, length, location of nozzles and metallurgy
- m. Insulation
- n. Fireproofing
- o. Inspection plates in place
- p. Cleanliness

CAUTION: Before entering a vessel, the refinery's safety precautions should be observed. These usually include the following: sampling the vessel for toxic vapors and oxygen concentration, wearing a safety harness, and having an attendant outside the vessel. AN UNATTENDED VESSEL SHOULD NEVER BE ENTERED.

C. Inspection of Other Major Equipment

When inspecting the following equipment, special attention should be given to the associated items:

Fired Heaters: metallurgies, tubes, burners, refractory, insulation, baffles, stack dampers, instrument connections, steam connections, explosion doors, access doors, observation ports, stack arrangement, expansion provisions.

Heat Exchangers: refractory lining, metallurgies, tubes, grounding wires, expansion provisions; manufacturer's ratings for temperature, pressure and pressure drop; nozzles, flanges.

- a. Shell and Tube: elevation, insulation, tubesheets, baffles, channel covers, differential pressure rating.
- b. Fin-Fan: header boxes, louvers, tube fins, fans, motor, bearings, motor grounding wire, fan pitch control, vibration switch, belt.

Pumps: metallurgy, suction and discharge block valves, suction strainer, discharge pressure gauge, check valves, lubrication and cooling systems, piping expansion provisions, drains, steam tracing and insulation of pump and associated lines; vendors ratings for head, capacity, temperature and pressure.

- a. Centrifugal: NPSH and speed rating, balancing line, warm-up lines, seal flush, bearings.
- b. Positive Displacement: discharge relief valve, pulsation dampeners, packing, speed and stroke rating.

Compressors: metallurgy, lube oil system, suction and discharge block valves, suction strainer, instruments, insulation, number of stages, piping supports; vendor ratings for capacity, pressure, temperature, specific gravity and compression ratio.

- a. Centrifugal: seal oil system, vibration monitor.
- b. Reciprocating: cooling water system, packing vents, suction valve unloaders, steam tracing, pulsation dampeners, distance pieces.

D. Piping and Instrument Check

The unit should be checked to ensure that it conforms to the UOP piping and instrument diagram. In addition to verification of the correct process flow, the inspection should include the following:

- a. Line sizes and metallurgy
- b. Flanges and gaskets

- c. Bolting
- d. Drains and vents
- e. Relief valve settings
- f. Piping supports and expansion provisions
- g. Equipment accessibility
- h. Utility lines
- i. Tie-ins with other units/storage
- j. General safety requirements

E. Strength and Leak Testing

Hydrostatic tests are made on new or repaired equipment to prove strength of materials and welds. Following the completion of construction the unit must be strength tested to make certain that no leaks exist in any joints or welds. Any equipment, whether piping, vessels, or exchangers, which has been modified, repaired, or had connections made to it by welding during a downtime or construction period preceding a start-up should be strength tested before any hydrocarbon is admitted to the unit. This strength testing is normally done by hydrostatic pressure. This test is made by completely filling the equipment with water and increasing the pressure. Normally, a portable positive displacement pump is used to raise the pressure. This test should not be confused with other less severe tests generally carried out before a start-up to check the tightness of connections.

NOTE: Chloride deposits from plant water that contains high chloride concentrations can cause stress corrosion cracking in austenitic steels. In areas of the plant that contain austenitic steel, serious consideration should be given to using treated water for hydrostatic testing and flushing. Initially, however, the plant water should be analyzed for chloride concentration. Additional analyses, as hydrostatic testing continues, should be performed to ensure the continued suitability of the plant water. If analyses ever indicate high chloride concentrations in the plant water, treated water should be used exclusively in the austenitic steel equipment.

The contractor typically carries out these tests with the assistance of operation personnel. When hydrostatic pressure testing is performed in the field, operating personnel assigned to witness and record the test should determine that blinds have been properly placed to isolate the equipment to be tested from all other equipment which has a lower design pressure. The recommended procedure is to divide the unit into groups of similar test pressures and isolate them by installing suitable blinds. If for any reason it should become necessary for the operators to carry out the test, these suggestions are extended:

An Air Pressure Test using plant air can be placed on the sections of the unit prior to the hydrostatic test so that any open lines or flanges may be discovered and taken care of before water is admitted. However, a water hydrostatic test is considered necessary for final testing and to determine the location of all leaks.

Care should be taken that the temperature is not too low, causing metal to become brittle. Vessel and water temperature should be above the minimum allowable working temperature to eliminate the possibility of cold-fracturing the metal. The vessel manufacturer should be consulted for the minimum allowable temperature for pressure testing.

Test pressure is normally 1.5 times the design rated pressure of the piping and 1.3 times the design rated pressure of the equipment .

All items under test must be liquid full, with all air vented from high point legs and other pockets.

Relief valves or rupture discs must be removed, blinded, or gagged so that they will not open during the testing.

Internals, such as level ball floats or other instruments that are not designed for the test pressure, must be removed.

Remove all orifice plates in lines to be tested.

It is not practical to test all of the equipment together. Thus the unit will be divided into sections as governed by the location of the various items of equipment and the test pressures to which each item will be subjected. Suitable blinds must be made up for insertion in nozzles and between flanges to isolate the various sections of equipment as required by different pressure ratings. Normally the exchangers, receivers, etc., for the various towers will be tested together with the piping.

Items which are capable of withstanding the test, such as piping connected to a vessel under test, may be subjected to the test. Ensure that all valves within the test system are open.

The refinery piping code ANSI B31.3 must always be followed when testing process piping.

Items which are not capable of withstanding the test, such as vessels connected to piping under test, must be isolated by blinding.

Test Gauges must be placed at the low point of a large line vessel so that the liquid head will be taken into account.

Fine Mesh Screens should be placed in the suction lines of all pumps before any liquid or gas is allowed to pass through them from newly constructed or dirty lines. If this is not done, scale, weld metal, slag, stones, etc., may freeze or damage the impellers of the pumps. The screens also act to trap such dirt in a place where it can be conveniently removed. Suction screens are normally placed in a flange or T-type strainer (for larger lines) between the suction valve and the pump. The screens should be made of fine screen retained by a heavy gauge, punched hole, cone-type strainer. These should be placed in the lines before the unit is pressure tested so that the test liquid can be pumped through the lines for the purpose of washing them. The flow through the screen should preferably be downward or horizontal. Precautions should be taken to place screen in a location where the dirt particles will not drop into an inaccessible place in the line when the flow through the pump stops. If this should happen, it may not be possible to remove the dirt upon removal of the screen. See the section on Line Flushing.

Special attention should be given to shell-and-tube type heat exchangers. If there is a possibility that an exchange medium such as hot oil or cooling water, can enter the process side of the exchanger during normal operation, the tube bundles and tube sheets should be given a very careful hydrostatic test.

NOTE: Check exchanger vendor name plate test pressure regulations in cases where differential design is used on the tube bundles.

All relief valves (PSV's) are required to be bench tested and set before final installation. This testing and setting may be done while hydrostatic testing is in progress. Do not return relief valves to service until after the following line flushing procedure has been completed.

CAUTION: A water filled system requires venting when being emptied in order to avoid pulling a vacuum and possibly collapsing equipment.

F. Line Flushing

All piping must be thoroughly cleaned of debris and scale. This may be done after hydrostatic testing, before the test water is drained. Generally, liquid lines are flushed with water and thoroughly drained. Gas lines may be either water flushed or air blown, but water should be blown from gas lines if water flushed. Gas lines to compressors must be free of water.

The following items are suggested as a guide for line flushing:

- 1) Where practical, clean water should be supplied to the vessels, and contiguous lines should be flushed away from the vessel. Never flush into equipment.

- 2) No matter what the flushing medium – steam, air, or water – maximum volume and velocity should be used for thorough cleaning.
- 3) Orifice plates must not be in place while flushing.
- 4) Control valves should be removed. It is strongly recommended that line flushing should not be conducted through the control valves. If construction trash or debris becomes trapped in the valve and/or valve seat, the valve may be damaged. This is especially true of on/off valves that require a tight shut off. Refer to vendor recommendations for specialized valves. Additionally, refer to UOP Project Specifications 616 sheets for further notes on control valves.
- 5) Instrument lines should be flushed with the transmitter removed.
- 6) Flexitallic gaskets and other final gaskets cannot be reused so they should not be put in until final installation. Temporary gaskets can be used for initial installation.
- 7) At pumps:
 - a) Disconnect suction and discharge piping and flush lines.
 - b) Install temporary screens in pump suction strainers.
 - c) Reconnect lines for circulating water.
 - d) See further discussion under "Run-in of Pumps and Drivers".
- 8) Regulate the flushing medium at its source. As examples, water from a tower should be regulated at the tower; and steam, at the valve in the supply to the line being blown.
- 9) Flush through all vents and drains.
- 10) Where possible, flush downward or horizontally.
- 11) Flush or blow:
 - a) The main header, from source to end, then
 - b) each lateral header, from the main to end, and
 - c) each branch line, from the lateral header to end.
- 12) Always flush through a piece of equipment's bypass to an open end before flushing through the equipment.
- 13) Disconnect lines at exchangers and flush to the openings.
- 14) For steam systems, flush well through dirt leg drains and steam trap bypasses before placing the traps in service.
- 15) Relief valves should be blinded if they have been returned to service following hydrostatic testing. (Set pressure can be checked at this time in the shop.)
- 16) In all cleaning by flushing, whether with water, steam or air, maximum safe volume and velocity should be used for thorough cleaning.
- 17) Water flushed lines which do not drain freely should be blown clean with air.
- 18) Gas-handling piping may be either water flushed or blown clear with air. Water should be blown from the gas lines on completion of flushing. Gas piping to compressors must be free of water.

Upon completion of line flushing of any system, carefully check that all temporary breaks are reconnected, control valves are replaced, and pump alignments are normal. Also, see that all possible water has been drained.

G. Run-In of Pumps and Drivers

Proper installation and operation of pumps and drivers is essential for trouble-free performance. The pumps and drivers should receive careful handling during initial run-in. The initial run-in of pumps is generally done by circulating water through the new equipment. Temporary strainers are installed in the suction line of the pumps, conveniently located for removal and cleaning. The screens also should be positioned so that dirt particles will not gravitate to inaccessible places when flow is stopped. During run-in of pumps, the strainers may cause some restriction of flow. As debris collects in the strainers, flow to the pumps will fall off. When this happens to a centrifugal pump, it is necessary to throttle the pump discharge by partially closing the discharge valve. This prevents the pump from cavitating, which can cause damage to the pump. However, also avoid restricting the pump discharge to the extent that it causes internal slippage and excessive heat generation.

In starting a turbine-driven centrifugal pump, the rotation should be brought up to speed as rapidly as possible. Normal operating speeds are usually attained rapidly and automatically with motor-driven pumps, assuming proper motor starting.

The development of discharge pressure is essential to flush and lubricate the wearing rings. After initially starting the pump, close the discharge valve for a short time. Subsequently, it is always advisable, where practical, to close the pump discharge valve immediately prior to shutting down a centrifugal pump. However, discharge valves on operating positive displacement pumps should never be closed. These pumps can often overpressure themselves and downstream lines and equipment. The following items are suggested for checking prior to run-in:

- 1) The manufacturer's operating instructions for any specific precautions that should be observed.
- 2) Completion of overall installation.
- 3) Alignment of pump and driver for cold operation. No undue strain by the piping on the pump or driver is allowable.
- 4) Cooling fluid piping and seal or gland oil piping:
 - a) Conventionally packed pumps in hot service are generally furnished with gland oil. Verify that this installation is correct and complete.
 - b) For pumps with mechanical seals, verify that all of the components of the flushing system (such as strainers, separators, restriction orifices and coolers) have been correctly installed and are clean. Loss of flush or dirty flush may cause the failure of seals. Process flushing to mechanical seals and to stuffing box jackets for some hot pumps has been specified for this

unit. In addition, cooled product flushing material is specified for stuffing box jackets with return to pump suction or to mechanical seals.

- 5) Packing or seals are installed.
- 6) Bearings and shafts have been cleaned prior to final lubrication.
- 7) Pump and driver are lubricated according to lubrication instructions.
- 8) Rotation of electric motor drivers uncoupled from the pump. Run-in uncoupled for a minimum of four hours, verifying good motor operation.
- 9) During run-in, many pumps are delivering a higher density liquid (water) than the normal process fluid. But, the pumps' drivers are sized for the normal pumping fluid. Consequently, there is potential for the overload of many electric motors. To avoid overloading the motor of a centrifugal pump, the flow must be limited by throttling the pump discharge valve. When doing so, if possible, check the amperage usage against design.
- 10) The following procedure is suggested for pump run-in:
 - a) Rotate pump and driver by hand, verifying that they roll freely.
 - b) Check that run-in water circulation is lined up.
 - c) Open suction valves fully, venting air from piping and pumps, completely filling with liquid.
 - d) Establish flow of cooling fluid, where required.
 - e) Check that lubrication is satisfactory.
 - f) Make sure electric power is available from the switch gear to the starter of the electric motor driver.
 - g) Barely open the discharge valve on the centrifugal pump.
 - h) Start the pump; if the pressure does not build immediately, stop and resolve problem.
 - i) When the discharge pressure has increased satisfactorily after starting, gradually open the discharge valve to obtain the desired flow rate.
 - j) In the event of unusual noise, vibration, overheating, or other abnormal conditions, shut the pump down immediately. Correct the cause before resuming use of the pump. Continue to check for abnormal conditions as these may occur after prolonged operation.
 - k) Check shaft sealing; mechanical seals should show no leakage. Conventionally packed stuffing boxes must always be permitted to leak slightly to provide some lubrication and to prevent overheating. Stuffing box gland nuts are generally only finger tight. A leaking mechanical seal will show some leakage on start-up. However, after the pump is started and stopped a few times, the leakage may stop.
 - l) Operate the pump, directing flows through all suction and discharge piping circuits.
 - m) Inspect and clean screens as required.

- n) Recheck and realign if required, after any disturbance of piping, such as required for suction screen inspection if pipe flanges have to be parted for screen removal.
- o) When shutting down, close the discharge valve first, thereby maintaining discharge pressure while the pump rolls to a stop. This will also help protect against the pump rolling backwards should the discharge check valve leak, and gives the wearing rings a quick flush.
- p) After all lines available to a pump have had suitable flushing, the temporary screen may be removed, but only after it has been free of debris on two successive examinations. The permanent strainers may then be installed where required.

H. Run-In of Compressors

The run-in procedures for centrifugal and reciprocating compressors are somewhat different. Therefore, following some general compressor run-in guidelines, the two run-in procedures are presented separately.

Maintaining the recycle compressor is essential to continuous, safe and efficient unit operation. Therefore, it is very important that all operating personnel be familiar with start-up and operation of the compressors. The run-in of these compressors is a good opportunity for the operating personnel to learn the compressors and the proper start-up and operating procedures. The compressors should be started and operated according to the manufacturer's instructions.

In general, manufacturer's representatives will pre-check the compressors, drivers and the lube oil systems prior to initial machine run-in. Some of their responsibilities are:

- 1) Checking drive motor for proper rotational direction by pushing, once, the compressor starter switch.
- 2) Checking compressor-driver alignment, both hot and cold.
- 3) Checking for proper base grouting and equipment support.
- 4) Removing slushing oils from bearings and seals.
- 5) Checking all seal and bearing clearances.
- 6) Evaluating suction and discharge piping for proper vibrational support.
- 7) Acidizing and flushing lube oil systems.
- 8) Run-in of main and auxiliary lube oil pumps.
- 9) Establishing normal oil levels in all lube oil systems.

Once the pre-check is completed, the compressor should be ready for operation. Since these machines are important and expensive pieces of process equipment, their proper operation cannot be overemphasized. Manufacturer's recommendations should be consulted for a more thorough discussion of any item mentioned here.

1. Centrifugal Compressor Run-in

The following general run-in guidelines should be used for a centrifugal compressor in conjunction with the manufacturer's instructions.

- 1) Because of the high speed involved with this machine's operation, it may be desirable to leave a coarse suction screen in the machine for the duration of the first run to guard against the possibility of miscellaneous construction debris entering the compressor.
- 2) Check the rotating element of the compressor for free rotation by turning slowly by hand. If the rotor rubs, rotation should be stopped immediately and the difficulty located and repaired.
- 3) Line up and start the lube oil circulation system, including filters and coolers, to make certain that the bearings of the compressor and driver are well lubricated. The manufacturer's instructions will give the proper values for the lube oil pressure, as well as other necessary settings. Generally, the minimum inlet lube oil pressure to each bearing should be about 0.7 kg/cm² (10 psig).
- 4) Remove any suction or discharge line blinds.
- 5) Unblock the compressor suction and discharge.
- 6) Put the stand-by lube oil pump on auto-start.
- 7) Drain the compressor casing, before starting the compressor.
- 8) If the compressor driver is a condensing steam turbine, commission the surface condenser system and warm up the turbine. Warm-up should be extended evenly over a thirty minute period. During warm-up, the high pressure steam should be checked to ensure that all condensate is out of the system.
- 9) Start the compressor rotating using the inlet steam block valve for control (start-up is best done on manual control) and bring the machine's speed to 500 rpm. If the compressor is motor driven, start the motor. Note any excessive vibration or noise. If either occurs, shut down immediately and locate the problem.

NOTE: With a flexible shaft, multistage turbine driver, the start-up acceleration places considerable torque strain on the coupling and, with start-up heating effects, can warp the shaft. To minimize this possibility, the machine should be brought up to operating speed in several steps.

- 10) Allow the compressor to run at the specified rpm until the lube oil temperatures increase to near normal operating temperature, usually about 120°F (50°C). Start cooling water flows to the oil cooler to maintain lube oil temperature at the desired level. Do not allow lube oil temperatures to exceed 175°F (80°C).
- 11) If the machine is turbine driven, increase the compressor speed stepwise to above the critical speeds, being sure to increase the speed smoothly and rapidly

through the critical speeds. Put the compressor on governor control and establish the desired gas flow rate.

- 12) While operating, inspect the compressor carefully and record your observation. This information may prove very useful should the compressor subsequently fail.
- 13) Make any necessary adjustments, as required, in the manufacturer's instructions for continuous operation.

2. Reciprocating Compressor Run-In

The following general procedure can be used, along with the manufacturer's instructions, to run-in a reciprocating recycle gas or booster gas compressor:

- 1) By hand, bar or jack turn over the machine to make certain it is free to move.
- 2) Commission the compressor cooling water and lube oil systems. Check temperature, flow, and integrity of systems.
- 3) Air free the circuit with nitrogen and leave the system at nitrogen header pressure.
- 4) Purge the compressor with nitrogen from the purge line on the compressor suction. Vent to flare or atmosphere.
- 5) Roll the machine over to ensure complete purging.
- 6) Open the compressor suction block valves to pressure up the machine to reactor system pressure (also, nitrogen header pressure). Open the compressor discharge block valves.
- 7) Drain the suction line and snubbers of any liquid. Liquid is extremely harmful to reciprocating compressors, if allowed into the cylinders.
- 8) Start steam to the steam tracing on the suction line of the compressor. This steam tracing prevents vapor condensation between the compressor suction drum and the compressor. It serves no purpose with only nitrogen in the system, but should be commissioned to ensure that it functions.
- 9) Make sure that there are no restrictions to gas flow in the reactor recycle circuit.
- 10) Start the machine as follows:
 - a) Open all of the suction unloading valves. Close the bypass-vent line, if open.
 - b) Check the compressor's lubricating oil level in the crank case or reservoir.
 - c) Check the cooling water temperature and flow through the cylinder jacket.
 - d) Start the compressor and check the oil pressure.
 - e) Let the compressor idle for a few minutes while closely watching the suction temperature.
 - f) Put the machine on line by closing the suction unloading valves. Follow the manufacturer's loading sequence, if one has been provided.
- 11) Start cooling water to the lube oil cooler, when the lube oil temperature reaches a normal operating temperature (50°C).
- 12) While operating, inspect the compressor carefully and record your observations. This information may prove very useful in the event the compressor fails.

13) Follow the manufacturer's suggestions regarding maintenance of the compressor.

I. Servicing and Calibration of Instruments

Preparation of the plant instrumentation for start-up and operation should include the following inspections and tests:

- a. Visual inspection of instruments to verify compliance with specifications.
- b. Visual inspection of instrument installation to verify compliance with installation specifications in the instrument instruction manual.
- c. Pressure testing of instrument piping
- d. Testing and calibration of instruments
- e. Testing of instrument air signal lines
- f. Testing of thermocouple wiring
- g. Loop checking
- h. Testing of safety relief valves, if not done during hydrostatic testing
- i. Final preparation.

In inspecting instruments for compliance with specification, attention should be given to instrument range, pressure rating, connections, and materials of construction, size, electrical characteristics and special features. Special attention should be paid to notes in the specification. Orifice plates should be inspected as soon as possible so they will be ready for installation when line flushing is complete; inspection should include size, materials of construction, location of weep hole, if any, counter bore, overall condition, measurement of orifice bore, and verification of information stamped on tab.

Inspection of installations should include attention to location of transmitters with respect to the connection point; to piping for proper drainage, and for proper valves and gaskets; to transmitters and control valves to ensure that they are not installed backwards; and to special treatments such as winterizing for proper coverage and insulation. Orifice installations should be checked as soon as possible to ensure that straight run requirements are met, so that required piping changes can be completed before start-up.

Instrument piping should be inspected visually for completeness and condition of welds and connections, and tested by pressuring with instrument air and brushing joints with soap solution. During hydrostatic testing of the plant, control valves should be isolated from the test system and adjacent drain valves should be opened to relieve any pressure. Control valve assemblies may be included in the hydrostatic test, provided that it is understood by all personnel that any leaks at control valve

stem packing will be corrected by a qualified instrument technician. It is generally safer to exclude the control valve assemblies from the hydrostatic test.

Testing and calibration of instruments should include calibration of transmitters (in place, if possible), control board equipment and local controllers. Control valves and positioners should be adjusted. Special attention should be given to instrumentation used in critical services (such as temperature control) and accounting service (such as charge and product meters). Temperature transmitters should be checked for proper thermocouple burnout protection. Meter factor tags and valve position indication on controller output indicators should be installed at this time, and control actions should be set.

Instrument air signal lines should be checked for leakage, kinked tubing, and for connections to the proper port of shutdown solenoid valves and controllers. Thermocouple wiring should be inspected for proper polarity at the thermocouple head. Point by point verification of the switch identification at the temperature indicator in the control room can be accomplished by selecting a point to be checked and opening the circuit at the thermocouple head. With proper identification, the corresponding temperature indicator will drive upscale or otherwise indicate an open circuit. The same applies for TRC instruments on the control board. Loop checking involved checking inputs and outputs to ensure that piping and wiring has been properly installed. A signal should be simulated at the transmitter, and the control should be watched for response. The controller output should be adjusted manually, and the control valve watched for response. Proper response of control valves to air failure should be verified. Special attention should be given to the complicated trip systems in the Reactor Section of the unit.

All safety relief valves should be bench tested and set, if this was not done previously during the hydrostatic testing.

Final preparation includes checking instrument piping to verify that valves are in the proper position, making preliminary control mode adjustments, filling seal pots and seal legs, and making sure that instrument air and electrical power are supplied to all instruments. It may also be necessary to have the steam tracing operable at this time.

Commissioning of instruments will be carried out as the plant comes on stream. In commissioning of instruments, most attention will be directed toward liquid flow measurements. Differential pressure instrument piping must be drained to eliminate accumulated dirt and water, and the instrument zero must be adjusted to obtain a correct zero reading on the indicator scale on the control board. Controllers must be tuned, interface level transmitters must be zero checked. Scale, weld slag, and trash must be removed from sticking control valves, and other malfunctions must be corrected. Inspection and commissioning of analyzers should include attention to sample line lengths, analyzer enclosures, auxiliary equipment (such as carrier gas

for chromatographs) and other items as indicated in the instruction manual for the particular analyzer.

In general, analyzers should be inspected, calibrated, and commissioned in accordance with procedures and instructions in the instruction manual for the analyzer. The specifications for the chromatographs include the services of a competent service engineer to review installation and supervise calibration and initial start-up. Chromatographs should be inspected to see that all external piping and wiring is complete, and that no parts have been lost or damaged in shipment. No other action should be taken until the arrival of the service engineer.

J. Commissioning of Plant Services

For ease of operation, the plant's utility systems should be placed into service as soon as possible. Lines should be flushed and leak tested. Steam lines should be warmed up slowly to prevent damage from water hammer. All steam traps and control valves are to be placed into service and tested. The following list of systems should be commissioned:

- a. Plant water and treated water systems. These systems should already have been commissioned prior to the line flushing procedure.
- b. All electrical and light systems, including emergency power backups.
- c. All plant and instrument air systems.
- d. Nitrogen system.
- e. Fuel oil and fuel gas systems should be flushed and leak tested, but not commissioned within the units battery limits until it is needed, as for heater dry-outs.
- f. Steam and condensate systems, excluding steam tracing for the moment.
- g. Tempered water and refrigerated water systems should be flushed and leak tested, but not commissioned until start-up is imminent.
- h. All drains and effluent systems
 - a. Cumene hydroperoxide drains and pumps
 - b. Acid aromatics drains and sump
 - c. Fractionation Section drains and sump
 - d. Sulfuric acid drains and neutralizing pit
 - e. Caustic drains and dilution pit
 - f. Secondary water treatment facilities

A leak test of all drain and sump systems should be conducted to ensure that the systems are tight. Also, it should be verified that all drains lead to the proper sumps, pits or other collection facilities. This leak test can be effected by blinding or blocking all drain lines at the collection vessel and filling the drains to the brim of the drain funnels. A level loss indicates the existence, but not the exact location, of the

leak(s). To verify correct drainage piping, the blinds and blocks should be removed. By water hosing each drain point individually and watching the collection vessel for the appearance of water, the correct drainage line-ups can be confirmed.

All storage tanks must be thoroughly flushed, leak tested, dried, and perhaps air-freed if the tank is in hydrocarbon service. Those tanks that have been air-freed should be left under slight positive nitrogen pressure. All lines to and from tankage must be flushed, blown dry with nitrogen, and pressure tested.

K. Availability Check of Required Materials

A check of chemicals, catalyst, ceramic balls, sand and other materials should be made to assure no delays due to shortages. The UOP Schedule "A" will list the quantities of these materials required in the 106 specification.

L. Plant Pressure Test

The purpose of the plant pressure test is to check the piping and equipment for tightness of flanges, connections and fittings. These tests should not be confused with the strength tests made during construction. Pressure tests are made near, or at, the design rated pressure of the piping and equipment. Generally, compressed air is used on new equipment to pressure test as it is less expensive than nitrogen, but either is acceptable in most cases.

When pressure testing, particularly in cold weather, care should be taken that the testing is not carried out at a temperature so low that the metal becomes brittle. As metal temperature decreases, brittleness increases. The vessel manufacturer should be consulted on the minimum temperature allowable before the vessel can be brought up to operating pressure. This is especially critical for the reactor, charge heater, combined feed exchangers and steam generation equipment.

As the utility air or utility nitrogen systems are at a lower pressure than is necessary to properly test certain sections of the plant, a source of higher pressure air or nitrogen is required. Generally, a small, portable compressor is convenient as it can be transported around the plant to the various sections being pressure tested.

It is not possible to test all of the piping and equipment at once. Therefore, the unit should be tested in sections according to location and necessary test pressure, as in hydrotesting. Isolation blinds should be used to segregate the sections. During testing, all safety relief valves and rupture discs must be removed or blinded, because the normal relieving pressure may be exceeded.

The following steps are essential in pressure testing a section of the plant:

- 1) Install all necessary blinds. Open all valves within the pressure test system.

- 2) Connect source of high pressure air or nitrogen to a drain or some other convenient point within the system.
- 3) Install reliable pressure gauges at several remote points.
- 4) Pressure up the system to its design pressure.
- 5) Soap test all flanges and screwed connections for leaks.
- 6) Repair leaks as they are located. Stubborn flange leaks may often be stopped by simply unbolting and rebolting the flanges.
- 7) Re-pressurize the system as necessary to maintain the design pressure.
- 8) When all connections have been soap tested and all leaks repaired, the pressure test is complete. As added insurance, the pressure gauges should be monitored for a period of eight hours to verify no pressure loss. Consider the effect of ambient temperature on system pressure. Rain or a decrease in ambient temperature can reduce the pressure.
- 9) Depressurize the system through various low point drains to help remove any remaining flush water.
- 10) If the pressure test was made with nitrogen, do not depressurize completely. By leaving a slight positive nitrogen pressure, the system will remain partially air-free, thereby simplifying the next operation.

M. Air-Freeing

It is necessary to completely air-free all piping and equipment in hydrogen or hydrocarbon service by purging with nitrogen. At this time, all free water should also be removed from the plant. As before, it is easiest to perform the air-freeing operation one section at a time.

Evacuation is the most efficient and effective way of removing air from the system. It is extremely critical that all equipment and instruments in the unit that are not capable of withstanding a vacuum be isolated.

To air-free by evacuation, simply commission the evacuation apparatus and draw the system down to the minimum pressure possible. Stop evacuation and introduce nitrogen. Increase the pressure up to the nitrogen header pressure. Vent through various low point drains to help remove the water from the system. Evacuate a second time and break with nitrogen. Samples should be taken from several remote points in the system and analyzed for oxygen. If the oxygen concentration is less than 0.5 volume percent, this section of the plant may be considered air-free. Vent through various low point drains to help remove the water from the system. Leave a slight positive nitrogen pressure to ensure exclusion of air.

If it is not possible to pull a vacuum on the equipment, one or two additional cycles of pressurization and venting with nitrogen may be necessary to achieve less than 0.5 volume percent oxygen.

The target of 0.5 vol% oxygen is historical and other values such as 0.3 vol% or mol% are acceptable as long as process materials can be introduced without the risk of spontaneous combustion. Lower values may be required in certain instances if required for process reasons.

Alternatively, steam can be used to air-free some sections. If steam is used, nitrogen will still be required for drying and blanketing these sections of the plant.

CAUTION: Until now, all activities have been carried out with piping and equipment containing air. With a nitrogen environment now existing in most equipment, a work and entry permit system should be established. A permit system will protect personnel by restricting work in unsafe areas. This system should be maintained for the entire life of the plant.

N. Commissioning of Additional Plant Services

At this time, the following plant services should be put into operation:

- a. Plant cooling water system.
- b. Plant steam tracing.
- c. Plant flare header, flare drum and flare stack.

The plant cooling water system should be flushed, drained and leak tested. When flushing, the cooling water lines should be disconnected at exchangers and the exchanger flanges blinded.

The flare system should be flushed and drained. Also, the flare header must be purged of oxygen down to an acceptable concentration, generally less than 1.0%.

If desired, the storage tank blanketing header can be commissioned at this time.

The steam tracing system should be flushed with steam to verify free flow of steam to all users. After commissioning, the tracing should be checked to verify that it is hot.

O. Commissioning of Ejector Systems

The concentration and, if applicable, fractionation ejector systems cannot be commissioned until the steam system is operating. However, they may be desired very soon after steam is available to assist in the air-freeing operation. If the ejector systems are used in air-freeing, the vacuum leak test may be done as part of or immediately following the air-freeing operation.

For this discussion of the start-up of an ejector system, it will be assumed that the equipment from which suction will be taken is under a slight positive pressure of nitrogen. It will also be assumed that the valves around the ejectors are closed. The normal sequence of operations would be as follows:

- a. Open cooling water fully to each of the inter-condensers and after-condensers.
- b. Break the ejector steam line as near as possible to the ejector and blow steam through to clean the lines. This need be done only on initial commissioning of new installations, but its importance cannot be overestimated. The jets on the ejectors are typically quite small, sometimes just a few mm or less in diameter. If the steam lines to the ejector system are not clean, there will be many reductions in capacity if not outright shutdowns during the initial commissioning of the plant while the steam system cleans itself out.
- c. Check that the after-condenser air vent is open and free to discharge to the condensate separator. Check that the separator vent to flare (or blowdown sump condenser) is open.
- d. Open the inlet valve of the smaller ejector of the first and subsequent stages.
- e. Open steam to the last stage ejector. Next, open steam to the preceding stage and so on until all stages are in operation.
- f. Check that the steam is dry and adjust the steam pressure to that given on the equipment name plate. As soon as a level appears in the condensate separator, an event that can be speeded by introducing treated water, start the condensate pump and place the level on control. Put the condensate to drain until the downstream equipment is ready to receive it.
- g. Allow the ejectors to run until a constant vacuum is obtained.

Observe the interstage pressures that give the most stable operation on the different sets of ejectors. If the vacuum decreases or becomes sensitive to process capacities, additional ejectors should be added. If an interstage pressure rises, i.e., the vacuum decreases, add the parallel ejector of the downstream stage. If the interstage pressure is unchanged but the equipment pressure is rising or not sufficiently low, add the parallel jet of the stage taking suction on the equipment. This may in turn render necessary the addition of the downstream parallel ejectors should the load prove too great for only one. To put the parallel ejector in service, do as follows:

- a. Open steam to the nozzle.
- b. Check that the nozzle steam pressure is still under control.
- c. Open the ejector suction valve.

Once the ejector operation has been stabilized, put the respective pressure controllers in service. Vent the 2nd or 3rd stage non condensables to the first stage inlet as necessary to maintain pressure control.

Occasionally, ejectors will fail to pull an adequate vacuum or will perform erratically. This can be the result of any of a large number of problems, a few of which are listed below for checking.

- a. Air leaking into the system. Tighten all bolts on flanges and manways. Tighten and oil all valve packing glands. Plug all vent and drain valves and tighten all screwed connections. Check that pump gland flushing is adjusted to maintain a positive pressure on the gland.
- b. Air leaking into the interstage. This will be confirmed by an increase in interstage pressure. Tighten all flanges and screwed connections.
- c. Certain styles of ejectors can leak air or steam through a leak in the point where the nozzle is attached to the body. Check this.
- d. Check that the steam pressure corresponds to that stated on the ejector name plate data.
- e. Check the performance of steam traps, since the pressure of excessive steam due to wet steam will cause back pressure at the jet.
- f. Check that the diffuser nozzle has not been worn due to use of wet steam.
- g. Check that the main supply steam separator is not clogged.
- h. High condensate temperature can be the result of insufficient cooling water flow or fouling of either tube or shell sides of the condensers.
- i. Failure to properly align gaskets and similar details that are normally insignificant can trap condensate pockets or cause turbulence that can affect the performance of vacuum equipment. If an ejector has been dismantled, each nozzle must be re-installed to the correct diffuser.
- j. Backpressure can be due to deposits in the condensers, plugged vents, condensate pockets, or other obstructions.
- k. Attempts to obtain an absolute pressure lower than the vapor pressure of a liquid in the system is not possible.

Having commissioned and tuned the three ejector systems, the vacuum equipment in the concentration and fractionation sections should be vacuum leak tested.

P. Vacuum Testing

During normal operation, most of the equipment in the concentration and fractionation sections is under a vacuum, drawn by the respective ejector and/or liquid ring vacuum pump systems. Should there be leaks at flanges, fittings, or other connections around this equipment, air could enter the system. At the very least, this would cause reduced capability and efficiency in the vacuum systems. At worst, a very serious situation could develop with air and hydrocarbon combining in the equipment. To eliminate vacuum leaks, all vessels and piping that normally operate

under a vacuum should be leak tested under vacuum before introducing hydrocarbon. Vacuum testing entails the following steps:

- a. As in the plant pressure test, smaller test systems should be created within the entire testing area. Install blinds where necessary to isolate these test systems from each other and from the greater system. Blinds will ensure that leaks through block valves will neither frustrate the leak test nor draw a vacuum on equipment that is not designed to withstand it.
- b. Put masking tape around all flanges, fittings, and other connections. Make sure that the tape forms complete seal.
- c. Open one test system to the vacuum system and pull the maximum vacuum possible on the equipment.
- d. Begin checking for leaks. A leak can be detected by poking a small hole in the masking tape on a flange, fitting, etc. and then covering the hole with any tell-tale substance (shaving cream works well or a soap solution). If a leak exists, the substance will be visibly drawn into the hole.
- e. When all connections have been leak-checked and all leaks have been repaired, isolate the test system from the vacuum system. With a vacuum gauge located in a remote part of the test system, monitor the gauge reading. The pressure should not change appreciably over the period of one hour. If a dramatic change occurs in ambient conditions during the one hour, perform the test again. This gauge test will ensure that no major leaks have been overlooked.
- f. Break the test system vacuum with nitrogen and leave the equipment with a slight positive nitrogen pressure. Continue the above procedure with other test systems.

When all vacuum testing is completed, the vacuum systems can be shut down until start-up to conserve steam.

VI. Start-up

The various aspects associated with start-up of the UOP Phenol process, both initial start-up, and typical start-ups following the initial start-up are included in this section. The purpose of start-up procedures is to provide a safe and reliable way to bring the process on stream. All procedures will be included in a general format which must be modified to be used for a specific operation. This section will address special process considerations, if any, specific to start-up conditions which may be required to assure optimum performance.

Though the instructions are given in serial form, obviously many of the procedures can (or must) be carried out concurrently. It is the responsibility of the start-up team to organize the necessary jobs in such a manner as to ensure that the unit start-up proceeds in a safe and orderly manner and that all auxiliary equipment is brought up as necessary.

A. Initial Start-up

Being titled "Initial Start-up", this section contains instructions for the start-up of a new plant or of one that has been completely shut down and de-inventoried. It is quite possible to extract from the following two subsections those steps necessary for a start-up, which follows a "normal shutdown" or an "emergency shutdown". There are situations in which production has been stopped, and most of all of the plant is left inventoried.

It is further advised that due to the complexity of this unit and the hazardous nature of some of the chemicals involved, that these instructions be read fully and understood before the actual start-up activities commence. This will facilitate both safe and optimum operation.

1. Inventorying of Tankage

NOTE: For revamp units, the following will most likely not apply. The reason for phenol and acetone on site is to synthesize a fractionation feedstock, which can be used to inventory, commission and line out the fractionation section (and Hüls MSHP section, if applicable) prior to the production of crude feedstock from the new sections. In the case of a revamp, it is generally expected that utility chemicals (caustic, acid, etc.) and fractionation feedstock (real, not synthesized) will be made

available for start-up. This will make the rest of this section superfluous. The text is provided for the sake of completeness.

Before production can begin, quantities of feedstock, products, and other process chemicals need to be on site. Sufficient acetone and phenol should be brought in to fill their respective product rundown tanks. The large cumene product tank should be partially filled with cumene. In addition, the sulfuric acid tank and concentrated caustic tank should be fully inventoried.

In general, these materials will be brought in by road tankers. The following guidelines are recommended for off-loading a road tanker.

- a. Direct the tanker to the off-loading zone. Shut off the engine and allow it to cool. Have the driver leave the cab of the truck, entrusting the vehicle's keys to the off-loading supervisor.
- b. Connect a grounding cable from the tanker to the off-loading connection and check that it is secure.
- c. Open a vent and/or sample tap and extract a sample of the tanker material for laboratory analyses. Do not connect the tanker to the off-loading connection until confirmation has been received that the material is correct as specified.
- d. Connect nitrogen to the tanker to provide an inert gas blanket when hydrocarbon is to be discharged.
- e. Check that the off-loading line is isolated to the tank and that all valves and drains are closed. Check the tank level and confirm that the tanker contents will not overflow the tank.
- f. Confirm that the off-loading hose to be used between the tanker and the off-loading station is clean, and then make the connection. Open the tanker to the station and check that no leaks are present. Line up from the station to the tank and begin the transfer.
- g. When the transfer has been completed, check the tank level to confirm that the correct amount has been off-loaded. Isolate the off-loading station from the tank and disconnect the tanker from the station. Clean the hoses and off-loading area of any spills.

Before inventorying the tanks, all support services should be available; e.g., blanketing gas, heating steam, steam tracing, drains, etc.

1.1 Sulfuric Acid Tank

For the initial filling of this tank, it and its associated equipment and piping must be totally drained of water and blown dry with air. Avoid any localized water pockets to prevent corrosion. In the drying operation, include the sulfuric acid pumps, the acid filters, and the acid lines to cleavage and effluent recovery. After commissioning the high level trip on the off-loading pump, fill the tank with acid. Line up the acid pumps to circulate the tank, and remain isolated from the filters and supply lines. Utilize all

the pumps and spillback lines, and then check for any leaks or signs of corrosion such as local hot spots around drains or pump casings. Continue to circulate the tank until acid is required. At that time, open the filters and fill the supply lines, maintaining spillback from the pump to the tank.

1.2 Concentrated Caustic Tank

Use the caustic pump to transfer caustic from the truck to the tank. Fill the tank. Line up the caustic pump to circulate the tank and remain isolated to the 20 wt% caustic tanks until they are ready.

1.3 Phenol Rundown Tanks

These tanks are typically either made of stainless steel, or carbon steel coated on the inside with a material which is phenol resistant. Depending on the manufacturer's recommendations this coating may need to be cured by contact with phenol prior to use. This first batch of phenol will generally be off-spec for high purity uses, but may be adequate for resin grade material. Of course, if the contaminated phenol is reprocessed, it may easily meet any spec.

For each of the tanks, check that steam tracing on the vents is working properly and heating coil steam is available. Commission these heating systems before filling with phenol. After filling, test the motorized mixers for a short period, and then stop them. When the curing is completed, most of this phenol will be used to prepare a synthetic feed mixture in (one of) the fractionation feed tank(s) and for inventorying the crude phenol tank.

1.4 Acetone Rundown Tanks

For the initial start-up, these tanks will be dried with acetone. If applicable, before filling, confirm that the tanks' vent condensers are receiving chilled refrigerant. Inventory the tanks and circulate to absorb as much water as possible. Test the motorized mixers, if provided. When the dry-out is completed, the acetone will be used to prepare a synthetic feed mixture in one fractionation feed tank.

1.5 Cumene Product Tank

Most of the UOP phenol process unit will be filled with cumene from this tank. A full tank is probably not necessary for commissioning purposes, however, so the actual volume required should be determined prior to off-loading the cumene. Remain isolated to the day tank until the oxidation section is ready to be inventoried.

When off-loading is completed, some additional tanks can be inventoried with material from the off-loaded tanks.

1.6 20 wt% Caustic Tank

When satisfied that the concentrated caustic is circulating properly, prepare a batch of 20 wt% caustic. Open concentrated caustic and treated water to the 20 wt% caustic tank and commission the on-stream specific gravity controller. Balance the caustic and water flows to give the correct specific gravity. Line up the 20 wt% caustic pumps to circulate the tank, while remaining isolated to the effluent treatment section supply lines. Start the tank circulation and check for leaks. Raise the level to 50% and commission the tank level controller to hold this level. Stop the concentrated caustic to the 20 wt% caustic tank until 20 wt% caustic usage begins.

1.7 Fractionation Feed Tanks

When satisfied that the phenolic resin lining in the phenol rundown tanks are fully cured (if required) and that the acetone rundown tanks are fully dried, begin preparation of a fractionation feedstock in one of the fractionation feed tanks. Phenol, acetone, cumene, and water are required, in the following volumetric proportions:

Phenol	21.2
Acetone	16
Cumene	11.2
Water	1

One-quarter to one-third of one fractionation feed tank should be sufficient to inventory and commission the fractionation columns. Special blends need not be made up. One can simply wait for the crude product to be manufactured from the decomposer. After inventorying the tank, start the motorized mixer or eductor spillback to verify its operability. Stop the mixer until commissioning and start-up operations begin.

All pre-commissioning activities are complete. The UOP phenol process unit is now ready for start-up. Before beginning the following start-up procedures, however, a final check of the unit should be made. Ensure that all pancake blinds have been removed, all spectacle blinds have been turned to their proper position, all safety relief valve gags have been removed, and all block valves on pressure relief lines are locked open. Also, notify all other affected process areas UOP phenol process unit start-up operation is beginning.

2. Passivation of Oxidizers, Flash Columns, and Related Piping

2.1 Passivation of Oxidizer Upper Section

Before passivation is started, the associated equipment to the oxidizers and flash columns, such as pumps, piping, instrumentation, utilities, air injection system, feed

wash system, charcoal adsorbers, CHP sump and the CHP closed drain header needs to be mechanically complete.

The oxidizer internals are passivated so their exposure to CHP will not cause decomposition from any acidic material remaining from welding, fabrication or from other residues left behind. Passivation is completed by spraying a basic 0.5% water solution of soda ash (Na_2CO_3) onto the walls and upper sections of the vessel to neutralize acids. By this method, we will not have to fill the entire vessel with a soda ash solution. High-pressure water jets capable of reaching all locations within the oxidizer are recommended. Passivation of equipment and piping downstream of the oxidizers will require partially filling the oxidizers with soda ash solution. The air spargers and other internal and external piping and equipment will be passivated as part of the lower section passivation.

Passivation can be completed immediately after completion of construction and hydrostatic testing of the oxidizers. Only steam, a clean water supply and the soda ash are required for passivation. The water low chloride concentration must be less than 100 wt ppm chlorides, preferably less than 50 wt ppm. It might be necessary to erect scaffolding within the oxidizers to adequately passivate the upper section of the walls and roof.

After passivation with soda ash the oxidizer upper walls, roof and piping must be rinsed with clean water to rinse away the remaining soda ash.

All equipment must be installed within the oxidizers including the large thermowell in the center of each oxidizer. Any equipment installed after passivation must be individually passivated by hand.

- 1) Vacuum away any dust and welding grinds, slag, and debris that remains in the vessels.
- 2) Thoroughly inspect and remove foreign debris that may be “hiding” in eductors, pipe supports, stilling chambers, or any other location.
- 3) Set up the high-pressure spray equipment and prepare the 0.5% soda ash in water solution.
- 4) Personnel who will be working inside the oxidizers are to wear the appropriate personal protective (PPE) for soda ash solution exposure.
- 5) Enter the oxidizers and begin spraying the upper walls and roof. Attempt to spray the outlet vapor line up to its high point. It may be necessary to erect a scaffold within the vessel to properly reach all surfaces with the spray.
- 6) Rinse the soda ash from the upper walls using clean water. Rinsing of the spargers and lower sections will be done separately.
- 7) Remove the scaffolding and spray the lower walls and oxidate circulation distributor and air distributor in the same manner as the upper walls. Ensure all of the welds are sprayed or hand cleaned with soda ash solution.

- 8) Remove all mechanical equipment, tools, and debris used for the spraying work from the oxidizers. It is advisable to remove all residual water from the bottom of the vessels to better assure no debris remains.
- 9) Close the bottom manways and install all valves, thermowells, and any other items that to make the vessels mechanically complete.

2.2 Passivation of Oxidizer Lower Section

The oxidizer will be filled to a 50% level with clean water and sufficient soda ash to make a 0.5% solution. With this solution the remaining wall surfaces and equipment inside of the oxidizer will be passivate; this includes the air spargers, circulation manifold and eductors. The external circulation piping, pumps, valves and heat exchangers will also be passivated in this procedure. The soda ash solution is heated to 50 deg C during circulation.

Utilizing the main air compressor, air will be purged through the air distributors in oxidizer 1 and 2 while there is water in the respective oxidizer.

To optimize the use of the soda ash solution and minimize waste handling the soda ash solution in oxidizer 1 is then transferred to oxidizer 2 for its passivation. The soda ash solution will be warmed and used to passivate the PFC and FC. The CHP sump is passivated at this time as well.

The soda ash is cooled prior draining it to the sewer. Arrangements must be made to provide for proper draining and disposal of this large quantity of soda ash solution.

The same steps are repeated with clean water to wash out the soda ash so that none remains in the equipment in preparation for startup.

It is important to note that a CHP compatibility test will be performed after cumene is inventoried to the oxidizer to test whether the passivation was adequate. The entire procedure will have to be repeated if this test fails. It is very important to do this procedure properly the first time.

2.2.1 Water Inventory to Oxidizer #1

- 1) Charge oxidizer #1 with clean water, having less than 100 wt ppm chloride, to 50 % level as read on the total span instrument. Visual level indication can be utilized by looking and measuring through the manway at the top of vessel.
- 2) Flush the suction line with water from oxidizer #1 through the suction strainer of the oxidizer circulation pumps.
- 3) Start oxidizer #1 circulation and flow through each emergency cooler separately.
- 4) Flush the density analyzer sample lines and all drains to the CHP closed drain header to the CHP sump.
- 5) Flush forward to oxidizer #2 through the level control valve and its bypass.

2.2.2 Passivate Oxidizer #1

- 1) Add soda ash through the top manway of oxidizer #1 to give 0.5% solution. Add clean water as needed to return the level to 50%. A large amount of soda ash will be required. Plan in advance to do it efficiently. For example, a loading hopper and crane can be used to add the soda ash to the top of the oxidizer.
- 2) Circulate the soda ash solution through oxidizer #1 and its heat exchangers, flushing through all drains, sample loops, and alternate piping. Check all piping compared to P&ID to assure no section is missed.
- 3) Drain any water from the shell side of the emergency cooler to prepare to start steam to the exchanger to heat oxidizer #1. Open a high point vent valve to allow air into the shell.
- 4) Commission steam to and condensate from the exchanger. Drain accumulated condensate that is in the S1.5 header from ahead of the steam block valve before slowly opening steam to the exchanger to avoid water hammer.
- 5) The condensate leaving the emergency cooler is to be drained to the ground and not returned to the condensate header as there is not expected to be enough pressure for the condensate to be returned.
- 6) Heat the soda ash solution in oxidizer #1 to 50 deg C. Do not allow the temperature increase across the heat exchanger to exceed 10 deg C.
- 7) Stop steam.
- 8) Open the two air inlet block valves at the bottom of oxidizer #1 to allow the soda ash solution to flow backwards through the air spargers into and up the air feed line so that the levels equalize. Let the valves remain open for the subsequent clean water flush.
- 9) Start cooling water to the air compressor after-cooler.
- 10) Start the oxidizer air compressor with all flow venting to atmosphere.
- 11) Slowly divert air from the vent to oxidizer #1 using the compressor control system. It will be necessary to reset the air shutdown valve.
- 12) Send air to oxidizer #1 for a minimum of 20 minutes. The oxidizer top manway or the charcoal adsorbers must be available for the exiting air.
- 13) Shut down the compressor until ready for the next step.
- 14) Allow the soda ash to circulate for a total of four hours to assure good contact and neutralization.

2.2.3 Passivate Oxidizer #2

- 1) Pump the soda ash solution from oxidizer #1 to oxidizer #2. Use the oxidizer #1 pump-out nozzle for this transfer to remove the maximum amount of liquid.
- 2) Once started it might be required to slowly close the pump discharge to maintain flow when the oxidizer level becomes very low. Do not operate the circulation pumps dry. Stop the circulation pump at the first sign of loss of suction as indicated by a large variation of the discharge pressure or unusual sounds coming from the pump.

- 3) Add clean water as needed to raise the level in oxidizer #2 to 50%. Add more soda ash through the top manway to give 0.5 wt% solution.
- 4) Start oxidizer #2 circulation towards the PFC feed-oxidate exchangers. Then bypass the feed-oxidate exchanger to send all of the circulation flow to the oxidizer #2 emergency coolers. Flush through all drains, sample loops, and alternate piping in this circuit.
- 5) Following the same instructions as for oxidizer #1, heat oxidizer #2 to 50 deg C with steam to the emergency coolers.
- 6) Following the same instructions as for oxidizer #1, purge air to oxidizer #2 for at least 20 minutes.
- 7) Shut down the air compressor.
- 8) Circulate for at least four hours to thoroughly neutralize any acidity.

Go to Section 2.3: Passivation of Pre-Flash Column and Flash Column. Return to this procedure when the Preflash and Flash Column step 2.3.4.5 is complete.

- 9) Start cooling water to the shell sides of the emergency coolers and cool oxidizer #2 to cooling water temperature.
- 10) Stop circulation pumps.
- 11) Drain the soda ash solution from the oxidizers and all process equipment and piping to minimize residual soda ash remaining after final flush to prevent filter blockage problems when the oxidizers are operated.
- 12) There may be 20-30 metric tons of soda ash contained in the solution. Make the necessary arrangements to allow this amount of basic material to be properly disposed.

2.2.4 Water Flush of Oxidizers

- 1) Repeat steps 2.2.1.1 to 2.2.2.14 to dilute the remaining soda ash solution in oxidizer #1.
- 2) Repeat steps 2.2.3.1 to 2.2.3.8 in this procedure to transfer the water in oxidizer #1 to oxidizer #2.

Go to Section 2.3.5: Water Flush of PFC and FC for continuation of the water flush procedure. Return to this procedure when the PFC and FC water flush steps of section 2.3.5 are complete.

- 3) Empty oxidizer #2. Determine the appropriate disposal for this water.
- 4) Open the bottom manways of both oxidizers.
- 5) Introduce low-pressure steam to each oxidizer using pipe or hoses through the manways. The process water to the oxidizer top spray nozzles should be evaluated as a method for getting steam in to the oxidizers if the line is designed for the temperature increase and resulting expansion. The steam will condense on the oxidizer walls and provide the final flush of the oxidizer.
- 6) Shut off steam after 24 hours of steaming.
- 7) Start process water through the oxidizer top spray nozzles to cool the oxidizers.

- 8) Prepare oxidizers for entry, cleanout and drying.
 - a) Look in stilling wells and eductors for any soda ash residue
 - b) Clean all surfaces to remove any residual soda ash

2.3 Passivation of Pre-Flash Column and Flash Column

The PFC and FC are passivated to prevent the unwanted decomposition of CHP during normal operation caused by its contact with acidic materials remaining from construction. The PFC and FC are passivated using the soda ash solution remaining from passivation of oxidizer #2.

After initial clean water flushes the warm soda ash solution in oxidizer #2 is fed to the PFC feed-oxidate exchangers as well as the insides of the PFC and Flash Column FC.

The soda ash solution will also flush and neutralize the associated piping of the PFC and FC up to the main decomposer feed isolation valve using the short and long circulation circuits returning to oxidizer #2.

After passivation the soda ash solution is drained from all equipment. Oxidizer #1 is then re-filled with clean water to begin the final flush procedure to remove any residual soda ash solution from all equipment. Water is the drained from all parts of the unit and any remaining water is removed by blowing with air or inert gas.

2.3.1 Flush the Cumene Quench Drum

- 1) Open the vent at the top of the PFC and FC to prevent pressurizing the columns.
- 2) Fill the cumene quench drum with clean water (no soda ash) and flush the drain lines from each compartment. Assure there is water in each of the four sections of the drum.
 - a) Allow some of this flush water to go to the CHP Sump once it flows clean and clear of debris at the low point drain.
 - b) Once the water starts draining from the low-point drain, open the drain valve(s) at the bottom of each heat exchanger tub in the PFC to clear that piping, and then allow them to remain open for ten (10) minutes then close both valves.

2.3.2 Flush the Cumene Flush Drum and Decomposer Feed Flush Drum

- 1) Charge water to the cumene flush drum and the decomposer feed flush drum from a convenient location.

- 2) Stop adding water when the flow coming out the top of the vessels is clean and clear. Allow the two drums to empty.
- 3) Refill the drums and flush water from each drum to the FC bottom line (decomposer feed) between the isolation valves, but not into the decomposer, for 30 minutes each.
- 4) Flush water from the decomposer feed flush drum into the decomposer.

2.3.3 Passivate the PFC

- 1) Bypass the PFC feed low flow shutdown and the flash column bottoms high level shutdown by activating the 'Feed to Preflash Column Feed Start' hand switch. This is a timed bypass. The feed flow valve will close when the timer expires if flows or level are not in acceptable ranges. The hand switch must be reactivated to resume flow for another time period.
- 2) Begin pumping soda ash solution from oxidizer #2 through the PFC feed filter. Drain water from available drains.
- 3) Flush the short circulation line returning to oxidizer #2, leaving the valve open at the filter outlet and at the inlet to oxidizer #2. Remove the RO from the short circulation line.
- 4) Flush both feed filter drain lines to the CHP Sump.
- 5) Allow the heat exchanger tubs to fill and overflow to the FC. Stop the transfer of soda ash to the PFC when water comes out from the FC bottom pump suction.

2.3.4 Passivation of Long Circulation

- 1) Drain the soda ash solution from the FC bottom pumps' suction strainers until the drained water is clear.
- 2) Start the FC bottoms pump and establish circulation through the density analyzer, sample station and the re-circulation line returning to the FC bottom.
- 3) Resume feeding the PFC. Note that the transfer rate from oxidizer #2 must be controlled to avoid over-filling the bottom of the FC.
- 4) Passivate the PFC tub drains for 10 minutes by first opening, and then closing the valves at the end of this time. Open the valve from the PFC bottoms to the CHP sump during this time.
- 5) Passivate the Long Circulation piping
 - a) Flow must travel from FC bottoms pumps back to oxidizer #2.
 - b) Do not allow soda ash solution to enter the decomposer.
 - c) Continue flushing for one hour.
 - d) Stop feed from oxidizer #2 and shut down all pumps that may be running.
 - e) Drain all soda ash solution from all equipment.

Return to Passivation of Oxidizer Lower Section 2.2.4 for preparation to rinse all equipment with clean water.

2.3.5 Water Flush of PFC and PFC

- 1) Repeat the previous steps of 2.3.3 and 2.3.4 in this section using clean water from oxidizer #2.
- 2) Drain all lines of water when the clean water flushing is done. It may be necessary to clear lines using nitrogen or air.
- 3) Proceed to vacuum test the PFC and FC. It is important to remove as much water as possible in order to be able to pull vacuum on these columns.

3. Passivation of the CHP Sump

The CHP sump and CHP closed drain header can be cleaned separately or in conjunction with oxidizer #2 and the PFC and FC. Temporary piping for the discharge of the sump can be used to circulate the cleaning solution through the header and back to the sump. Alternatively the soda ash solution can be drained from the many drains points during passivation to clean the header. The soda ash solution collected in the CHP sump can then be pumped to oxidizer #2 during the appropriate step via the oxidation feed filters. Note the filter elements should not be installed during passivation. Think of the sump as a mini-oxidizer.

4. Tempered Cooling Water System

To conserve utilities, the tempered cooling water system (TCW) should be commissioned as near to start-up as possible. Once the TCWS is inventoried and circulating, it should always remain inventoried and circulating, excepting perhaps during extended shutdowns of the UOP phenol unit. The following guidelines should be used when commissioning the TCWS:

- 1) Check that all pumps and coolers are ready for service.
- 2) Using treated water, inventory the TCW drum with a 70% level. Monitor the tank's level and hold the level by periodically adding water.
- 3) Line up a circulating pump to the TCW air cooler, through the cooler bypass leaving the TCW cooler isolated, and back to the return header and the tank. Start the pump and circulate.
- 4) Commission the TCW header FIC and continue to circulate. Open the valves to one user heat exchanger at a time, circulating through the exchanger bypass first, if one exists, and then through the exchanger. And so on, through each exchanger in turn until they have all been commissioned.
- 5) Introduce steam to the TCW return line to the drum and increase the water temperature to 50°C (120°F).
- 6) Commission the TCW cooler cooling water if not done already and put flow to the cooler.

- 7) Commission the system temperature controller, which at this time will send most of the flow through the tempered cooling water cooler bypass.
- 8) Add the corrosion inhibitor to the chemical addition drum and open the outlet gate valve while controlling the inlet globe valve to allow water to circulate the inhibitor into the TCW system.

Continue to circulate the TCW, leaving the heat exchangers and the other users isolated until needed. Drain water from the system if the tank level rises due to the condensate from the heating steam. When users come into service and heat needs to be removed from the water, start air cooler fans to maintain control of the water temperature. Shut down the TCW drum steam when it is no longer needed to hold the water temperature at 50°C (120°F).

5. Chilled Water System

As with the tempered water system, the chilled water system (CWS) should be commissioned as near to start-up as possible to conserve utilities. Once the CWS is commissioned and circulating, it should always remain so, as long as there are vents from tanks and equipment that require chilling for environmental or safety reasons. Chilled water may be used in a number of tank vent condensers to prevent acetone losses; it is necessary to minimize acetone evaporation losses while maintaining nominal tank pressure and safe oxygen concentrations in the tank vapor. The following general procedure should be used in commissioning the CWS:

- 1) Check that pumps and the refrigeration unit are ready for service. Consult the manufacturer's recommendations for the start-up of the refrigeration unit.
- 2) Using cooling water and ethylene glycol, inventory the chilled water storage tank with the appropriate mixture of glycol and water and inhibitor to a 70% level.
- 3) Line up the chilled water circulation pump through the refrigeration unit back to the tank, via the recycle flow controller. Start the circulation pump and circulate through the tank.
- 4) Continue to circulate and, one by one, open each exchanger (user) of the CWS into the circuit. Circulate through the exchanger bypass first, if one exists, and then through the exchanger. Maintain the storage tank level by bringing in treated water.

Continue to circulate the system, leaving the heat exchangers in the circuit. Commission the refrigeration unit and reduce the water temperature to design, usually about 4°C (39°F). Controlling this temperature may be difficult while the heat load is reduced prior to start-up. However, it is important that the temperature be kept low for the acetone rundown tanks vent condenser after these tanks are inventoried with acetone. Once the load on this system increases, lock open the storage tank bypass and close the start-up circulation line to the tank.

6. Oxidation

6.1 Pre-commissioning Status

Verify that all pre-commissioning procedures, including passivation, have been completed and that all trips and bypasses are working and are in their normal operating position. Cooling water should be in service to all exchangers.

The main concern during start-up of the oxidation section is safety. While it will be desired that the start-up proceed as quickly as possible, it is critical to note that the oxidation system needs to be started up in a careful and controlled manner, due to the danger associated with the handling of concentrated CHP solutions. Always keep this in mind when making changes to oxidation section process variables or starting the section up from a prior shutdown.

6.2 Feed Wash Column and Cumene Feed Surge Drum

- 1) Inventory the CHP process water break drum with water.
- 2) Start the CHP process water pumps on total spillback
- 3) Set the FWC overhead pressure controller to the CFSD at the design value and put in auto.
- 4) Inventory water from the CHP process water break drum to the FWC to a bottoms level of 50%. Drain the initial water from the FWC bottom until it is clear.
- 5) In preparation to inventory cumene to the FWC and CFSD, first isolate the other cumene users on the cumene feed line to avoid sending cumene to other locations. For safety reasons there is no valve to the concentration section cumene quench drum and so it will be commissioned at this time also.
- 6) Start the cumene fresh feed pumps and start feed cumene to the cumene quench drum. Inventory the cumene quench drum to an 80% level.

Flow to the cumene quench drum will continue through the RO at all times to ensure the drum is always full. An overflow line back to the oxidation day tank or a pump is provided to spillback this continuous flow. Ensure this flow is active at this time.

- 7) Fill the CFSD to a 20% level by back filling through the outlet line from the oxidation day tank.
- 8) Inventory cumene to the cumene flush header.
- 9) Inventory cumene to the FWC. As the FWC fills nitrogen is displaced on auto control to the CFSD. When the CFSD is full liquid will automatically start to flow to the CFSD. Continue filling the FWC to a 50% level in the CFSD.
- 10) Start cumene circulation from the combined feed pump back to the FWC setting the FWC flow controller at the design value. The control valve bypass will likely need to be opened to achieve the design flow since flow from the flash columns is not yet available. This is normal and expected for start-up. Achieving the

design jetting velocity with the design FWC feed rate is critical to the proper caustic scrubbing the separation in the FWC.

- 11) Start the flow of water from the bottom of the FWC and set the LIC at 50%
- 12) Resume CHP process water to the FWC at the design rate.
- 13) Observe if water accumulates in the CFSD water boot, if so, this is evidence the FWC is not functioning properly. Check the jetting velocity and ensure proper organic-aqueous separation before continuing.
- 14) Establish flow to the tray 2 pH meter by starting the sample pump and set the rate at the design value. The normal pH will be 7.0.
- 15) Start 20 wt% caustic to the FWC at 50% of the design rate. Target a sufficient caustic rate to achieve 0.7 – 1.0wt% free caustic in the FWC bottoms.
- 16) Commission the pH analyzer at the FWC bottoms. The normal pH will be 12.5.

6.3 Oxidizer Inventory and CHP Compatibility Testing

Inventory the oxidizers by bringing in fresh cumene from outside cumene tankage, to the oxidation day tank, through the feed wash column to oxidizer #1. If the cumene comes from a UOP Q-Max unit and meets all specifications, then it is not required to be washed in the FWC.

The overall procedure involves adding some cumene to oxidizer #1 and #2 for an initial test of CHP compatibility. This initial cumene is added using the normal feed piping through the oxidizer preheater via the combined feed pumps. Additional cumene is added to be able to circulate the first oxidizer followed by heating the cumene to 75 deg C. Nitrogen is bubbled through the cumene to increase mixing. The cumene is then cooled and analyzed for CHP compatibility. Cumene is then transferred to oxidizer #2 to allow circulation. During circulation of oxidizer #2 cumene is also circulated through the feed-oxidate exchanger, the PFC and FC. The cumene in oxidizer #2 is heated, purged with nitrogen for mixing and cooled to analyze for CHP compatibility similar to oxidizer #1.

A positive (failure) CHP compatibility result will require emptying the oxidizers and treating the cumene in the FWC. The oxidizers and PFC and FC will need to be cleaned and passivated again. Thorough passivation with soda ash is important, since failure of this step will result in significant additional work and project delay.

Note: The preflash column feed filters will tend to plug up frequently during start-up with scale and scum from washing. A thorough water wash will minimize the plugging. The filters may have to be changed every couple of hours for the first 2-3 days. Sufficient cartridges should be on hand or available at the local distributor to replace them with fresh ones. The cumene quench drum safety system can be checked at this time. See Initial start-up - Concentration Section procedure.

- 1) Assure the charcoal absorbers are online and each has vessel has been regenerated and cooled at least one time. Cooling water and chilled water should be commissioned to all appropriate exchangers.
- 2) Air-free the oxidizers to less than 1 mol% O₂ by opening the nitrogen lines to each of the main air feed lines. The oxygen concentration can be actively monitored with the oxygen analyzers already present at the top of each oxidizer. Confirm the online analyzer results with lab samples.
- 3) Commission the nitrogen purge through the water spray lines. This nitrogen will ensure CHP does not build-up in the water spray lines.
- 4) Inventory oxidizer #1 and #2 each with 1000 mm of cumene via the oxidation feed filters.
- 5) Sample each oxidizer and complete a CHP Compatibility Test on each of these samples using Honeywell V-180-FR.
- 6) If the test is negative, indicating CHP compatibility, add additional cumene increase the level in oxidizer #1 to 5000 mm, or enough to be able to start circulation using the normal draw-off and cover the feed sparger.
- 7) Start the oxidizer #1 pumps and circulate cumene for two hours, circulating cumene through each pump and to each of the emergency coolers.
- 8) Sample oxidizer #1 and perform a CHP compatibility test. If the test results are negative, continue with procedure.
- 9) Heat oxidizer #1 to 75 deg C with steam to the emergency cooler with a maximum temperature increase of 10 deg C across the heater on the cumene side.
- 10) Increase the nitrogen flow to the oxidizer #1 feed line to 2,000 Nm³/hr and hold for one hour. Increasing the nitrogen feed will improve the mixing so that a more representative sample can be taken.
- 11) Reduce the nitrogen purge to 500 Nm³/hr.
- 12) Sample oxidizer #1 and perform a CHP compatibility test. If the test results are negative, continue with procedure.
- 13) Remove steam from the emergency cooler and drain all condensate. Start cooling water at the maximum rate and cool oxidizer #1 to cooling water temperature.
- 14) Assure that nitrogen continues to flow to the oxidizer #2 feed line at 500 Nm³/hr.
- 15) Open the low draw-off valve (For Pumpout) and close the high draw-off valve from the oxidizer #1 circulation pump.
- 16) Transfer cumene from oxidizer #1 to oxidizer #2 until the level in oxidizer #2 decreases to 30% on the full range transmitter.
- 17) Stop the oxidizer #1 circulation pump.
- 18) Add fresh cumene to oxidizer #2 to increase the level to 50%.
- 19) Start the oxidizer #2 circulation pumps and circulate oxidizer #2. Run both pumps separately and circulate through both emergency coolers separately.
- 20) Commission the cooling water to the PFC and FC condensers.

- 21) Ensure the PFC feed filters are lined-up with filter elements installed and liquid full.
- 22) Open the drain valve(s) on the PFC heat exchanger tubs to allow draining to the bottom of the column.

The cumene quench drum be used to put an initial level of cumene into the PFC bottoms line, FC vaporizer and FC to enable starting the FC bottoms pumps.

- 23) Push the concentration section emergency shutdown button to start the flow of cumene to the PFC and FC.
- 24) Push the 'Cumene Quench Reset' button to stop the flow of cumene to the PFC and FC when the FC bottoms level is 50%.
- 25) Commission the FC bottoms pumps on total spillback to the FC bottoms cooler.
- 26) Pushing the 'Feed to PFC Feed Start' button and within 30 seconds increase the flow rate from oxidizer #2 to the PFC to above the shutdown value. It might be necessary to bypass the low-flow shutdown switches to achieve this flow rate without activating the low-flow to PFC shutdown. Increase the feed flow to above the low-flow shutdown amount if bypassing of this low-flow switch is not an option.
- 27) Ensure the Short Circulation line from the discharge of the PFC feed filters back to oxidizer #2 is locked open.
- 28) Open the Long Circulation flow control valve from the FC back to oxidizer #2. Continue circulating cumene for two hours, operating both FC bottoms pumps.
- 29) Sample the FC bottoms and analyze this cumene for CHP compatibility. If the test is negative, proceed with the procedure.
- 30) Stop cumene to the PFC and close the valve(s) draining the PFC heat exchanger tubs.
- 31) Stop the FC bottoms pumps before the FC bottoms level decreases to zero.
- 32) If all CHP compatibility tests have passed (negative result), proceed to heat oxidizer #2 to 75 deg C with steam to the emergency coolers with no more than a 10 deg C temperature increase across the exchanger.
- 33) Circulate oxidizer #2 with both pumps separately and flow through each of the emergency coolers separately.
- 34) Increase the nitrogen flow to the feed line of oxidizer #2 to 2,000 Nm³/hr and hold for two hours. Then reduce the nitrogen purge to 500 Nm³/hr.
- 35) Cool oxidizer #2 to cooling water temperature.
- 36) Sample the cumene in oxidizer #2 and perform the CHP compatibility test. If the CHP compatibility result is positive, the cumene must be removed and the oxidizers, PFC, FC and all associated equipment must be cleaned and passivated again.
- 37) Stop oxidizer #2 circulation.

6.4 Oxidizer Seeding

At least a 5 wt% CHP solution is required to start CHP production in a quick, efficient and low risk manner. The lower the CHP concentration for start-up, the

higher the risk that undesirable byproduct (e.g. phenol) will be formed that will poison the reaction.

If CHP is not available, an alternate chemical for start-up that is commercially available is dicumylperoxide (DCP). A 0.2 - 0.5 wt% DCP solution has been demonstrated to be sufficient to initiate the peroxidation reaction.

A survey done to locate potential licensors for technology to produce dicumylperoxide (DCP) resulted in UOP contacting the following suppliers, which may be useful to obtain price information for DCP to use in seeding the oxidizers:

Akzo Nobel Chemical
Website: www.akzonobel.com
Note: Offer two grades: "R" and "free-flowing"

Concord Chemical Industrial Co., Ltd., Taiwan
Tel. (02) 391-1388

The following commissioning procedure is not specific to any unit. In a revamp, start-up procedures assume that a fair quantity of CHP is already available on site, and that fractionation feedstock is available for the start-up of the decomposition and fractionation sections. This would not normally be the case for new units, and both of these materials would normally have to be brought in from the outside in order to commission the various sections. However, once the materials are on site, the procedures for start-up are unchanged.

NOTE: The following scenario describes one particular means by which the oxidation section of a typical unit can be safely started up. There are many possible scenarios, with the differences mostly due to how much CHP is available on site. If sufficient CHP is available (a revamp in which a new unit is being constructed alongside an existing unit for example) the quickest and possibly safest way to commission the new unit would be to bring in sufficient CHP from the existing unit to commission the new unit with typical CHP concentrations. On the other side, CHP may have to be imported in drums or special tank cars, which is a completely different start-up.

Note: Dicumylperoxide (DCP) is an acceptable substitute for CHP. The general principles given below will allow procedures to be written based on the situation at any particular plant site.

When a large inventory of CHP for start-up is available, a "seeding" operation (typically from tank cars or drums) will not be necessary. The inventory in oxidizer #1 should be sufficient to start the oxidation reaction there at a rate almost equal to normal. Therefore, this oxidizer can supply CHP to the rest of the oxidizers as they are brought on line. This makes the start-up of the oxidation section a relatively quick activity compared to having to synthesize the entire CHP inventory from a few tons of seed material. If sufficient CHP is available from existing plant capacity, or other sources, then all oxidizers can be inventoried with sufficient CHP to enable the reaction to start at a reasonable rate (5 wt% CHP should be sufficient, at a physical level of at least 50% in each oxidizer). This requirement is based on a safe and efficient mixing of the oxidizer contents during the start-up. If a cumene feed or oxidizer feed distributor is provided, then the level must cover this distributor after warm up, at least a 60% level.

CHP is typically inventoried through the CHP sump with the connection provided. If the CHP must be washed due to the presence of a neutralizing agent, then the feed wash column can be used to water wash the CHP. If this is required, then the decanter, FWC and CFSD must be closely inspected for cleanliness prior to start-up. The CHP will likely have to be diluted with cumene to accommodate the hydraulics and inventory in this section.

If DCP is used, it typically comes in the form of a powder in bags. A manway is available at the top of the oxidizer to add DCP powder. Circulation should be established to mix the powder as it is being dumped into the oxidizer. A small crane is likely necessary to lift the bags to the top of the oxidizer.

To add DCP powder to the top of either one of the oxidizers, do the following:

- 1) If bags of DCP will be dumped into the oxidizer individually, begin to place the bags of DCP on top of the oxidizer. The bags will not all fit on the top of the oxidizer at the same time.
- 2) If a hopper and crane will be used to lift the DCP to the top of the oxidizer, then construct a loading platform at grade, outside of the oxidizer dike area.
- 3) Increase the level of cumene in the oxidizer to 6650mm total cumene level using the cumene feed line to the feed wash column to ensure all cumene feed has been processed through the feed wash column.
- 4) Stop nitrogen purges to the air feed line of both oxidizers.
- 5) Maintain an inert atmosphere in the oxidizers by continuing the nitrogen purge through the water spray nozzles at the top of the oxidizers.
- 6) Remove the top manway from oxidizer #1. Protect the surface flange of the manway as appropriate.
- 7) Install a course grating over the manway opening. This grating is to prevent bags of DCP or personnel from falling into the oxidizer during DCP loading. A chain or grounding wire shall be connected from the grating into the oxidizer, suspended over the cumene to avoid a build-up of static due to the falling DCP. The grating shall be grounded to the oxidizer, which is itself grounded.

- 8) Due to the presence of a nitrogen atmosphere, along with cumene inventory inside the oxidizers, Self-Contained Breathing Apparatus (SCBA) should be used by all workers on top of the oxidizer. Proper emergency evacuation and rescue plans shall be in place prior to the commencement of all work. See the Safety Caution below:

SAFETY CAUTION:

The small flow of nitrogen flowing to the oxidizer vessels will cause nitrogen and cumene vapors to come out of the top manway. To minimized exposure, personnel are recommended to work upwind of the manway to minimize exposure to these vapors.

Exposure to DCP dust can cause health problems. Personnel handling DCP should wear proper personal protective equipment (PPE) recommended by the DCP SDS.

Dropping powdered DCP from the top manway to the surface of the cumene liquid should be performed using a grounded conductive sock to minimize the length of free-fall and the potential for a static electric charge being generated.

Review the dicumylperoxide SDS for complete handling instructions.

- 9) Empty enough DCP into the oxidizer to yield a final solution of at least 0.25wt% DCP in cumene by weight. Calculate the amount of DCP required based on the actual cumene inventory. There is not a further need for DCP after this step. All of the purchased DCP should be used at this time to avoid further storage.
- 10) Replace the top manway.
- 11) Resume nitrogen purges to both oxidizers via the air lines at 500 Nm³/hr each.

6.5 Oxidizer Heat-up and CHP Increase

When the oxidizers have sufficient inventory, oxidizer #1 heat-up can begin. Keep oxidizer #2 at ambient temperatures until the cold circulation of the concentration section is complete and that section is mechanically sound. Then the heat-up of the terminal oxidizer can begin and some flashing of the oxidizer contents can begin as the concentration section is brought in line.

When there is an adequate level in oxidizer #1, it can be heated up using the steam connection to the oxidizer #1 cooler. Block in the cooling water and route steam to

the shell side of the exchanger, with circulation established at the normal rate. Note: the hydrocarbon rate is hydraulically controlled. Open the steam to the exchanger after lining up the condensate to the condensate return, and begin heating up oxidizer #1 to 90°C (195°F) at a oxidizer cooler delta T of no more than 10°C (18°F). 90°C (195°F) is chosen to have sufficient reaction rate at the low CHP or DCP concentration even though this is not the optimum operating temperature.

As the oxidizer is heating up, ensure that the charcoal adsorption system is ready to receive spent air, the feed system is ready to provide feed, and the remaining oxidizer(s) are being heated up using the same heat-up criterion. If the cold circulation of cumene from the terminal oxidizer to the concentration section has proven successful as measured by the CHP compatibility test, then that oxidizer as well can be heated to 80°C (175°F).

Remember that the heat-up of the oxidizers must be watched carefully. As the material heats up, the level will begin to increase. If it rises above 80%, send some oxidizer #1 material to the next oxidizer to keep the level in range. Remember as the air is injected to oxidizer #1, more material from oxidizer #1 may have to be exported to the downstream oxidizer to make room for the (now expanded) material. Since the CHP concentration in the next oxidizer will not typically be above 10%, there will be no hazard.

Once the CHP concentration of oxidizer #2 is greater than 5wt%, then it can also be heated up and the air injection started similar to oxidizer #1. The big difference between the two oxidizers during start-up is the method of cooling. Oxidizer #1 is cooled with a cooling water cooler and fresh feed temperature while oxidizer #2 has the choice of cooling using the emergency cooler or the preflash column feed-oxidate exchanger.

Long circulation must be established to use the feed-oxidate exchanger. A concentration vacuum must also be established to get the heat consuming flash of the warm feed to the PFC. Essentially, the concentration section must be ready to start-up. If the concentration section is not ready, then all of the circulating oxidate must bypass the feed-oxidate exchanger.

While heat-up of the oxidation section is taking place, other activities in the synthesis section and in the fractionation section must take place as well.

7. Concentration

7.1 Pre-commissioning Status

Verify that all pre-commissioning procedures have been completed and that all trips and bypasses are working and are in their normal operating position. Cooling water should be started to all exchangers in the section.

The main concern during start-up of the concentration section is safety. The heating of concentrated solutions of CHP should not be undertaken lightly. All safety systems should be double checked, all trips verified by simulating the associated emergency condition and verifying that the system responds as designed, and finally, operations personnel should be fully trained and aware of their responsibilities in dealing with the heating of CHP containing streams.

Line-up cumene quench to the quench drum and fill it on automatic level control. This level control valve should have an extremely quick action, virtually on/off, in order to refill the drum as quickly as possible. Verify correct valve action and controller operation.

7.2 Cumene Quench System Check

The cumene quench system provides an additional degree of safety to the concentration section. With the cumene quench drum is inventoried, it is necessary to ensure that the system is working properly. Keep in mind that all of the cumene dumped into the flash columns will recycle back to oxidizer #2, adding to its inventory.

- 1) Check that the cumene quench lines are operational:
 - a) Open by trip simulation each quench valve (one at a time). Check that the valve opens completely, without binding or other problems.
 - b) For each valve opened, check that quench is flowing down the line by observing the level in the vessel into which the quench flows.
 - c) Close the quench valve when satisfied everything is in order.
 - d) Confirm that the cumene quench drum refills on level control.
- 2) After each quench valve is checked, commission the respective emergency quench circuitry.

7.3 Concentration Vacuum System Preparation

At the end of a successful pressure/vacuum test in the concentration section, the section should have been left under a vacuum with the vacuum system working per normal. If this is not the case, verify that steam pressure is adequate, blow down the steam lines to ensure dry steam to the ejectors, and then pull a vacuum on the concentration section again. Put the pressure controller in service.

With vacuum stable in the concentration section, proceed with the commissioning.

7.4 Concentration Circulation

With no heat to the PFC steam heater and FC vaporizer, press the 'Cumene Quench Reset' switch and then the 'Feed to Preflash Column Start' switch, and 'Oxidate Feed Reset' button and begin circulating the terminal oxidizer effluent through the PFC, then to the FC vaporizer, to the FC and from there to the FC bottoms. There are some time delays (30 seconds to 2 minutes) bypassing some shutdowns to allow circulation to be established, such as the low feed flow shutdown and the flash column bottom high level shutdown. These delays are relatively short in nature, so one must be well prepared. This might already have been completed as part of the pre-commissioning procedure: Passivation of Oxidizers, Flash Columns and Related Piping.

When a level appears in the flash column bottoms, start a flash column bottoms pump and route flash column bottoms liquid to the decomposition section, and from there through the long circulation line joining with the flash column net overhead material (zero to small flow at present) and back to terminal oxidizer. Note: the feed flow to the flash column will be limited by the flash column bottoms pump capacity.

If the effluent of terminal oxidizer is pure cumene at ambient temperature, there will be very little flashing of cumene in the feed to the concentration section until that oxidizer is heated up. Use this time to verify correct operation of the flow meters, pumps (bottoms pump) and temperature indicators during the cold cumene circulation, before CHP (even though it is dilute). Ensure that there is sufficient flow through the restriction orifice assembly routing cooled flash column bottoms back to flash column bottoms cooler. Although it would be quite difficult, due to the fact that there is only one substance (cumene) in the feed to the concentration section, the heating system (PFC steam heater/FC vaporizer) can be tested at this time as well. Their operation will be much smoother when there is a wider boiling range of components in the flash column feed.

There will be some flashing of cumene in the concentration section during the above procedure, since the pressure is very low in the concentration section. If this flashing results in enough liquid to establish control of the overhead system, recycling material back to the oxidation section, put the reflux flow FIC into automatic at 50% of design. Otherwise, the collector (inside the flash column) can just spill the condensed cumene to the column, and the combined material can flow from the bottom of the flash column to the oxidation section by way of the long circulation line.

7.5 Concentration Section Heat-up

Long circulation is now established, with material from the terminal oxidizer being routed to the PFC and FC, and this material being returned to oxidation via the long circulation line returning to the terminal oxidizer. The concentration section is now ready for heat-up and introduction of CHP from the terminal oxidizer, when that

vessel's inventory contains the desired CHP concentration. When this is the case, the flash column feed vaporizer can be tested much more easily. Also, the online density analyzer for determining the CHP concentration at the flash column bottoms can be calibrated.

When the terminal oxidizer is up to the desired concentration, if long circulation is stopped it can be re-established with the intent to concentrate the CHP. The FIC on the decomposer feed bypass line will set the circulation rate. Put this FIC in automatic/cascade and set to 70-100% of design. Begin heating up the feed to the preflash column using the steam reboiler and the flash column vaporizer. Slowly increase the temperature to about 10°C (18°F) below design. It is likely that the columns will have a lower flash zone pressure during start-up and be more efficient. Establish the design reflux/feed ratio when enough cumene is flashed overhead.

8. Decomposition

Verify that all pre-commissioning procedures have been completed and that all trips and bypasses are working and are in their normal operating position. Cooling water should be in service to all exchangers.

The main concern during start-up of the decomposition section is safety. All trips in the decomposition section must be verified as working correctly and all instrumentation must be calibrated and in perfect working order in order to be able to proceed with the commissioning and start-up. Fill the flush drums with cumene.

8.1 Decomposer Inventory

The decomposition section must be inventoried with crude product (if available) or an equal-molar mixture of phenol and acetone. Neither pure acetone, cumene, phenol nor any other fluid is suitable for this purpose. It must be 50 mol% acetone and 50 mol% phenol.

Feed crude product to the decomposer from the fractionation feed tank via the line provided for this purpose. When the level in the decomposer approaches 80%, bump (e.g. momentarily start and stop) the circulation pump(s) and check the operation of each of the pumps. Simulate the trips for low cooling flow, and the auto-start mechanism. As soon as these are proven, keep one of pumps running, keep filling the circuit as required, and establish circulation through the calorimeters and the dehydrator going back to the decomposer.

8.2 Decomposer Heat-up

Line up steam to the dehydrator and circulate through to heat-up the decomposer. Simulate the trips by allowing the temperature in the drum to fluctuate to the trip

points and verifying that the system acts as designed. Simulate low cooling water flows, high temperatures in the system, and check that the decomposer TIC works correctly by putting it in auto after the trip simulation above and watching it control the drum temperature. After operations is satisfied that the system is working correctly, cut the steam to the dehydrator and cool off the circulating material to cooling water temperature.

The decomposition section is now ready for final commissioning. These steps are typically taken just prior to start-up and include the following:

- a. Sulfuric Acid Injection and Verification of Decomposition Material Concentration (H_2O , CHP and sulfuric acid) by the Laboratory
- b. Heat-up of the System
- c. Flushing of the Decomposer Feed Line
- d. Feed-In

9. Neutralization

Verify that all pre-commissioning procedures have been completed.

Inventory the amine drum/tank and prime the injection pumps. Calibrate the injection pump stroke versus flow rate using the sight glass. Check that flow meter corresponds correctly with sight glass measurements.

10. Fractionation

Proceed with the commissioning of the Crude Acetone Column (CAC), Finished Acetone Column (FAC), Cumene-AMS Column (C-AMSC), Crude Phenol Column (CPC), Residue Stripper (RS), and Phenol Rectifier (PR). For the initial start-up, stored crude product is typically not available. To expedite the UOP phenol plant commissioning, a synthetic mixture of phenol, acetone, cumene, and water to roughly their normal operating concentrations can be prepared or the synthesis section can be started to fill the fresh feed tank. When the mixture is ready, start the crude acetone column feed pump and circulate the fractionation feed tank per the instructions under step "4" below. Keep the tank circulation in service at all times to prevent material stratification.

For initial commissioning, water running the fractionation section is an easy way to clean debris out of the system and verify the operations of the control systems. Construction debris and dirt can cause problems during normal operation such as plugging up pump suction and control valves to causing the phenol and acetone products to go off-spec. "Water running" is a term used to describe the operation of the columns using water. The utilities are commissioned to the columns and water is added, heated and condense. The vacuum system should not need to be run, as the water will boil at atmospheric pressure. Vents to atmosphere after the

condensers may be opened to ensure that no relief occurs during the water running. Water should be drained from the columns and receivers at appropriate points to check for cleanliness.

10.1 Pre-commissioning Status

- 1) All columns and vessels should have been pressure tested and those in vacuum services, vacuum tested. A slight positive pressure with nitrogen should exist on the CAC.
- 2) All equipment and instrumentation is ready to be put in service.
- 3) All drainage and heat tracing systems are in service. The spectacle blinds in the lines to the Fractionation Closed Drain Header (FCDH) should be turned.
- 4) The fractionation feed tank and the 20 wt% caustic tank should already be inventoried. The acetone rundown tanks, the phenol rundown tank, the hydrogenation feed, and the residue tank should be ready to receive organic feed. Commission the fractionation feed pumps to mix the tank, if not already done.
- 5) Confirm all lines to the FCDH are closed, double blocked and bled. Check that the Fractionation Section sump is isolated from the FCDH. Monitor the sump level throughout the Fractionation I and II procedures.
- 6) Cooling water should be started to the following equipment:
 - a) CAC Condenser
 - b) FAC Condenser
 - c) Acetone Product Cooler
 - d) FAC Bottoms Cooler
 - e) C-AMSC Condenser
 - f) Ejector System Inter-condenser and After-condenser
 - g) Tempered cooling water should be started to the following equipment:
 - h) Fractionation Return Header Cooler (FCDH)
 - i) RS Bottoms Cooler
 - j) PR Condenser (may be steam generator)
 - k) Ion Exchange Resin Treater Cooler
 - l) PR product cooler
- 7) Refrigerated water should be in service to the following equipment:
 - a) FAC Vent Gas Condenser
- 8) The following column reboilers should be ready for steam without circulation going through them.
 - a) CAC Reboiler
 - b) FAC Reboiler

- c) C-AMSC Reboiler
 - d) CPC Reboiler
 - e) RS Reboiler
 - f) PR Reboiler
- 9) Proceed with the start-up when satisfied that these and all pre-commissioning steps have been completed.
- 10) The IX resin treaters should be loaded with resin. See Section XII, Special Procedures.

10.2 Crude Acetone Column

The commissioning of the crude and finished acetone columns is typically done concurrently. CAC bottoms should be routed to the fractionation feed tank, not the C-AMSC until the CAC bottoms is on specification and the C-AMSC is ready for commissioning.

Bring crude feedstock into the CAC from storage through the feed preheater (if provided), set to give a temperature at the outlet of 50°C (120°F). Route the feed to the middle column feed inlet point. If the CAC has been under N₂ positive pressure, the introduction of feed will cause vaporization of acetone in the feed, leading to possible pressure surges. Feed introduction and heat-up should not be done so quickly as to cause the pressure of the column to exceed 1 kg/cm²g (15 psig). At a pressure of 1.12 kg/cm²g (17 psig) heat input to the column from the reboiler (not in service at this time) would be cut-off to prevent over pressuring. Increase the feed temperature to design. Open the non-condensable vent on the CAC condenser to improve the condensation during start-up. This valve may be partially open during normal operation to ensure no build-up of non-condensables in the condenser.

The CAC overhead pressure controller should be put in line at this time, controlling pressure at 0.7 kg/cm²g (10 psig). The control valve can either be lined up to flare or sent to the FAC column feed, depending on the status of the FAC vacuum system start-up.

When a level appears in the CAC bottoms, slowly start steam to the reboiler on local manual. If the heat input is too fast, it is possible that the column overhead pressure will increase above the trip setting. If this happens, block in the steam to the reboiler, then reset the solenoid and try again slowly.

As a level appears in the receiver, start the reflux pump and put reflux back to the column on local auto, increasing the reflux rate as possible. When reflux is established, non-condensables in the column will be essentially removed, and the condenser vent can be nearly closed.

With reflux established, heat can be applied to the column more quickly, and the column can be brought up easily. Line up overhead vapor from the CAC to the FAC

and ensure that column is ready to receive vapor feed to the bottom feed inlet point. Put the water flow on cascade controlling the reboiler outlet temperature. Bring up the reboiler outlet to temperature 155°C (310°F) with a combination of reboiler steam and stripping steam to get a R/F weight ratio of approximately 0.6.

As operations are stabilizing, the FAC can be brought on line concurrently. It should be under a slight vacuum with all instrumentation ready for operation. The FAC bottoms should be filled with water and circulation established with the FAC bottoms pump. The circulation should be established to the Crude Phenol Column (CPC) condenser, though no heat is available from that source. Heat can be added via the tub reboiler and live steam. Temperatures should be increased to 90+°C (195°F).

When the FAC is ready to accept CAC overhead vapor feed open the vapor feed line and start putting CAC overhead vapor to the FAC on local remote (FIC) control.

The vapor flow rate to the FAC is best found by experimentation. Specifications for the CAC operation are that the acetone concentration of the CAC bottoms should be minimized (typically less than 0.2% should be easily attainable) and that the overhead material should contain little or no AMS (<0.1 wt%) and no phenol.

Begin taking vapor feed to the FAC on straight FIC (CAC overhead vapor) control, watching the upper tray and the bottoms temperature of the CAC. When the bottoms temperature reaches about 155°C (310°F) with reboiler water injection, the bottoms should be relatively acetone free. Hold the overhead vapor draw rate at this, and take analyses of the CAC bottoms and the CAC reflux (this is the same material as the overhead vapor). If the acetone concentration of the bottoms is below 0.2%, and the AMS concentration of the overhead material is less than 0.1%, the ratio of vapor flow rate to CAC feed flow rate should be calculated and used in the feed forward FIC. The vapor draw to feed weight ratio is typically about 0.4.

These columns are designed to split the total cumene recycle to the oxidation section (via the C-ACWC) 1:3 to 1:2, i.e. at design rates and CAC feed compositions, approximately 1 part of FAC bottoms organic material is produced, and approximately 2 parts of Cumene-AMS Column overhead hydrocarbon is produced. This rough ratio should be maintained during normal operations regardless of the turndown. If a lower ratio is attainable, say 1:3 or higher, then the FAC will have less load and the acetone product specifications will be easier to meet.

The CAC and C-AMSC are also designed to allow the cumene from the FAC bottoms to be recycled back to the fractionation feed tank. This allows for upsets in the FAC where the acetone concentration of the bottoms is high as it is undesirable to send this material to the effluent section. This extra capacity avoids the need to turn down the fractionation section when this upset occurs.

10.3 Finished Acetone Column

The FAC should be under a slight vacuum with all instrumentation ready and lined up. Feed to the FAC will be overhead vapor from the CAC on FIC control. The bottom of the column should be inventoried with water as mentioned previously.

As soon as hot vapor feed from the CAC is available, route it to the bottom feed inlet nozzle and begin taking vapor to the FAC on CAC control. Line up bottoms heat, bottoms reboiler pump to provide cooling to the crude phenol column overhead.

Before feed is started to the FAC, several things must be done.

- 1) Start one bottoms pump to put bottoms to the CPC overhead at the design rate. When heat is put to the CPC, the heat input to the FAC can be adjusted to give the correct reflux/feed ratio. Establish this flow now, however because the pump has to run above its minimum flow. The CPC condenser will cool the FAC bottoms until the metal has heated up inside of the exchanger.
- 2) Line up and start steam to the stabbed-in reboiler at the bottom of the column to start driving vapor overhead, if this hasn't already occurred. Direct steam injection can also be added to compensate for the lack of input from the CPC. Since the majority of this material is water during start-up (it is typically half cumene and half water during normal operation) heat must be added gradually, if the level is to be maintained. Water may need to be added. Raise the bottoms temperature to 90+°C (195+°F). When vapor feed is added to the FAC, acetone will begin to accumulate in the overhead receiver.
- 3) When a level is established in the reflux, start the reflux pump and adjust the flow. Cascade the flow to the receiver level as the level becomes stable. Heat input adjustments will then be reflected in higher reflux rates, all other things being equal. Start a small drag of reflux material from the top tray to bottom of the column via the aldehyde recycle line provided. Leave this drag (used to give aldehydes which may collect in the overheads another chance to condense via contact with NaOH) at the design rate (or greater, if lab analysis justify it) at all times.
- 4) Put the moisture analyzer on the net draw from accumulator tray in service. When the column has lined out, take a sample of acetone to the laboratory, and use the results to calibrate the moisture analyzer. Start a side-draw of acetone back to the fractionation feed tank.
- 5) After the moisture analyzer has been calibrated and necessary adjustments have been made to put the acetone on-spec with respect to moisture, start diluted NaOH injection to the highest feed point, at the design rate relative to the charge rate (~1 kg dry caustic/1000 kg acetone product). The KMnO₄ fading time should be on-spec within an hour or two. Don't bother taking the sample to the lab for at least two hours after NaOH injection has been started.
- 6) The bottoms system should be relatively acetone free by the time that reflux has been established in the column. Bottoms flow to the crude phenol column

overhead condenser should be established (whether or not the crude phenol column has been started yet). The bottoms pump should be started if this has not been done already, and material pumped through the cooler to the bottoms receiver. The receiver overhead material should be routed at this time to the fractionation feed tank, in case there is too much phenol in the bottoms hydrocarbon. The receiver bottoms aqueous phase should be sent to either secondary treatment or the phenolic waste water tank. The switch between the two systems should be the concentration of phenol that can be attained in the OEC. If less than 150 wt ppm phenol, then send to secondary treatment.

- 7) When the CAC vapor draw is lined out, the FIC can be adjusted to take only cumene overhead, and not more than ppm concentrations of AMS, then the cumene can be routed to the effluent section
- 8) When the column is lined out, adjust reflux/feed to obtain the desired moisture concentration, and the NaOH injection rate to obtain a salable product with regard to KMnO₄ fading time (typically >2 hours is required for sale). If the rest of the fractionation section front end (crude phenol and Cumene-AMS Column) are not yet commissioned, then the product acetone should be routed to the fractionation feed tank in order to maintain a consistent CAC feed composition. When the fractionation section is completely lined out, products can be routed to final destinations.

10.4 Cumene-AMS Column

When the CAC bottoms material composition is satisfactory and operations are stable at design flow, proceed with the start-up of the C-AMSC, as follows:

- 1) Slowly open up the C-AMSC receiver to the evacuation header and commission PIC to 140 mm HgA.
- 2) Line up the C-AMSC bottoms around the Chemical Treating Reactor through the CTR effluent pumps to the FCDH. The Chemical Treating reactor will initially be bypassed until the bottoms are on specification due to its long residence time.
- 3) Line up the C-AMSC overhead pump discharge for flow to the C-AMSC. As the level in the C-AMSC receiver increases with the C-AMSC heat-up, commission the C-AMSC receiver level controller, open the control valve, and start to reflux the C-AMSC.

NOTE: Remember to balance the feed to the C-AMSC with the C-AMSC bottoms rate, gradually reducing the CAC bottoms flow into the FCDH.

- 4) As a level of 20-30% is reached in the bottom of the C-AMSC, commission the C-AMSC bottoms level controller and set it to maintain a 50% level. Once at this level, start the CTR effluent pump, sending material on level control to the

FCDH. Establish steam to the reboiler. Be careful not to heat quickly because there will be some water in the columns bottoms initially. A sudden flash of water in the column could cause a pressure spike and a high upward velocity, resulting in tray damage.

- 5) Complete the heat-up and line-out of the C-AMSC. Slowly divert all of the CAC bottoms from the FCDH to the C-AMSC, while maintaining C-AMSC conditions. With the CAC bottoms no longer going to the FCDH and the C-AMSC operation steady, complete increasing column temperatures to make a good separation.

NOTE: Small amounts of acetone will initially enter the C-AMSC from the CAC bottoms. Open the non-condensable vent on the condenser frequently until the CAC operation is stable.

- 6) Line up the C-AMSC overhead organic pump discharge to the FCDH, keeping it isolated from the cumene-AMS caustic wash column. Line up the C-AMSC overhead aqueous boot level control to the sodium phenate tank. Continually check the water boot to ensure that only water is being sent. Commission the water boot interface level controller, and set it to hold a 50% level.
- 7) Slowly raise the temperature of the bottoms to around 170-180°C (340-360°F) and adjust the reflux to give 110-120°C (230-250°F) on the control tray. Design reflux:feed is 1.3 wt:wt. Commission the column TDIC controller and use it to reset the reflux flow controller.
- 8) Take samples of the overhead organic phase and bottoms product, to determine the concentration of phenol and AMS, respectively. Generally, for the overhead organic phase, phenol should be 4-10%. The bottoms should have only trace amounts of cumene and AMS (<0.1 wt% by analysis). If detailed analyses is performed, less than less than 10 wt ppm cumene and less than 100 wt ppm AMS should be observed.

10.5 Crude Phenol Column & Residue Stripper

With the Cumene-AMS Column heat-up in progress, proceed to the commissioning of the CPC/RS, as follows: (Continue to balance flow forward in the C-AMSC/CPC and through the fractionation return header cooler via fractionation closed drain header (FCDH) until both columns are stable):

- 1) Slowly open the CPC vacuum line to the fractionation vacuum system and commission PIC. Set to the design pressure, 130 mm HgA. Commission the RS pressure control as well, 105 mm HgA. The butterfly valve in the vent line from the CPC to the vacuum system may have to be partially closed to force the fractionation vacuum system to pull a deeper vacuum.
- 2) Commission and open up the C-AMSC bottoms level control valve and begin to slowly inventory the CPC. The majority of the C-AMSC bottoms should still be going to the FCDH. Line up the column bottoms material to the residue column (RC). The RC bottoms should be lined up to the FCDH.

- 3) With a 50% bottoms level established, begin the heat up of the both columns. Commission the steam and condensate flow to and from the CPC and RS reboilers. Raise the reboiler vapor returns to 165-180°C (330-355°F) temperature at about 30°C/hr (50°F/hr). Be careful when first heating the column. It is possible that some residual water will vaporize uncontrollably. Quick heating may cause high vapor velocities up the column, resulting in tray damage. The RS reboiler may have some difficulty in getting the phenol down in the beginning because the inventory of heavies in the column bottoms may take some time to accumulate.
- 4) Line up the net overhead flow rate to the C-AMSC inlet. Establish a flow at 5% column feed flow.
- 5) When a level appears in the CPC receiver, commission the reflux pumps and reflux everything back to the CPC. Commission cascade to level control as receiver stabilizes. Continue to heat up and establish the design reflux-to-feed ratio, 1.3 wt:wt.
- 6) Establish a side-cut flow rate to the FCDH. Proceed with the Phenol Rectifier (PR) start-up.
- 7) Route RS bottoms back to the FCDH until phenol concentration is on-spec. After the RS bottoms phenol concentration is 4-10 wt% begin sending the residue to the residue tank.

10.6 Phenol Rectifier

The Ion Exchange Resin Treaters (IXRT) will be initially bypassed. The phenol for start-up of IXRT must first be relatively dry with low levels of AMS and carbonyls. The resin must first be dried out so it becomes catalytic to react the 2-MBF in the phenol. High concentrations of AMS will cause rapid residue formation. High concentrations of carbonyls will cause rapid formation of the impurity 2-methylbenzofuran (2-MBF).

Line up the PR bottoms pump discharge to the CPC, the overheads to the C-AMSC, and the PR side-draw pump discharge to the fractionation sump. Proceed with the PR commissioning.

- 1) Commission the PR split-range pressure controller, and slowly open up the PR vent condenser and associated overhead system to vacuum. Set the controller to maintain the design pressure (280 mm HgA).
- 2) Commission the CPC accumulator level control valve. Open this valve and start to build a level in the bottom of the PR. Be aware that it will be easy to overfill the PR bottoms until the reboiler is lined out.
- 3) With a 70-80% bottoms level established, commission the PR bottoms flow control valve, start the bottoms pump, and circulate the material in the bottom of the column. Slowly close the control valve, and begin a flow of bottoms liquid to

the FCDH. Set the bottoms flow controller to maintain approximately a 50% of the design value.

- 4) Commission the steam generator/condenser water flow.
- 5) Commission the PR Reboiler condensate flow controller. Open the control valve and slowly heat up the column.
- 6) Commission the reflux flow controller when it begins to flow and use it to adjust net product to FCDH according to the reboiler heat input. Design reflux:feed is 0.72 wt:wt.
- 7) Commission the overhead draw flow back to C-AMSC feed.
- 8) Commission the bottoms level controller and use it to reset the condensate flow controller.
- 9) With the PR operation satisfactory, slowly increase the feed to design. Maintain this condition and check out equipment and instrumentation.

11. Phenol Purification Reactors

The chemical treating reactor (CTR) and ion exchange resin treaters (IXRT) must have close to on-specification feed related to the fractionated impurities prior to commissioning. If not, then additional circulation time will be added to turn over the inventory in the reactor and treaters. The C-AMSC bottoms must be free of AMS (<0.1 wt%) prior to starting the CTR. The IXRT must have low carbonyls prior to commissioning.

- 1) Open the chemical treating reactor vent valve a small amount to pull a vacuum on the vessel. The vessel may presently have some non-condensable gases in it. Do not open the valve so far that the fractionation vacuum system cannot control the pressure.
- 2) Once the CTR pressure has equalized to the C-AMSC pressure, slowly fill the reactor with C-AMSC bottoms. Be careful not to upset the downstream CPC column operation.
- 3) When the CTR is inventoried, line up the reactor for full flow and close the bypass.
- 4) The ion exchange resin treaters must now be inventoried with phenol and dried out if not done previously. The treaters are initially filled upflow from the rundown tanks followed by a downflow flush to remove water. The phenol temperature should be minimized (80°C) to minimize the reaction of carbonyl to 2-MBF during the drying phase.
 - a) The upflow phenol filling velocity is set to target 50% fluidization of the resin bed. For phenol at 60 deg C the target velocity is 2.6 m/hr. For filling with phenol from the process at 90 deg C the target filling velocity is 5 m/hr. The flow rate is dependent on the IXRT bed diameter.
 - b) The initial downflow flush target flow rate is set based on a LHSV = 2 hr⁻¹ and should be returned back to the FCDH rather than the phenol rectifier due to the high water concentration. 3 - 4 bed volumes of phenol should be passed

over the bed prior to line up with the phenol rectifier. In general the IXRT can be lined up to the PR when the water in the effluent is less than 400 wt ppm.

- 5) Soon after the IXRT are lined up in a lead-lag configuration and begin to dry, line up the chemical treatment injection pump and tank to upstream of the CTR. If there is unexpected long delay between drying the IXRT and the start of chemical injection, a significant quantity of 2-MBF will be formed from the carbonyls across the IXRT. If the delay is more than one hour, then the IXRT should be bypassed and kept liquid full until the chemical injection is ready.

When the fractionation section is commissioned, the status of each of the columns should be as follows:

1) Crude Acetone Column

- a) Feed at design flow
- b) Reflux on receiver level control
- c) Bottoms free of acetone and being sent to the C-AMSC
- d) Overhead vapor on FIC control cascaded to feed forward ratio control
- e) Column pressure at 0.7 kg/cm²g (10 psig)

2) Finished Acetone Column

- a) Feed on CAC FIC control
- b) Reflux is adjusted to give desired acetone water concentration
- c) Bottoms free of acetone; all organic bottoms being sent from FAC bottoms drum, to the fractionation feed tank. Water should be sent to sewer
- d) Side-draw material sent to fractionation feed tank
- e) Reboiler heat input on automatic control
- f) Controlled vacuum with PIC at 465 mm HgA
- g) Caustic injection in service

3) Cumene-AMS Column

- a) Feed on CAC receiver level control
- b) Reflux on temperature control to give a control tray temperature of 110-120°C (230-250°F)
- c) Reboiler heat input is on flow control, adjusted to keep reflux rate at R/F = 1.3 wt:wt
- d) Bottoms material should be AMS free and sent to the fractionation feed tank.
- e) Overhead aqueous phase to sodium phenate tank
- f) Column vacuum on PIC control at 140 mm HgA

4) Crude Phenol Column / Residue Stripper

- a) Feed on C-AMSC bottoms level control.
- b) Reflux on flow control, R/F equal to about 1.3 wt:wt
- c) CPC reboiler heat input on flow control cascaded to reboiler outlet TIC at 165-180°C (330-355°F) corresponding to about 60% phenol. The RS reboiler vapor temperature should be lower than normal operation because low residue present in the bottoms. If residues are present, and the phenol concentration is within spec at 4-10 wt%, then the residue can be routed to the residue tank.
- d) Side-draw flow rate is ratio controlled to the reflux flow. The net overhead material is sent on flow control back to the C-AMSC
- e) CPC column vacuum on PIC control at 130 mm HgA. RS at 105 mm HgA

5) Phenol Rectifier Column

- a) Feed on CPC side-draw flow control
- b) Reflux on flow control cascaded to receiver level control at design R/F ratio of 0.72 wt:wt
- c) Bottoms flow to CPC column on flow control set to 5% of product flow rate
- d) Net overhead on flow control set to 5% of the product flow rate going to the C-AMSC

12. Cumene-AMS Caustic Wash Column

Although not specifically a part of the fractionation section, the cumene-AMS caustic wash column may be put into operation at this time, as its feed is C-AMSC overhead organic phase. Place the C-AMSCWC into operation, as follows:

- 1) Line up the C-AMSCWC overhead off-spec line to the fractionation feed tank. Commission the overhead pressure controller and slightly open the back pressure control valve.
- 2) Line up 20 wt% caustic flow to the caustic settler and column inlet. Commission the caustic flow controller and open the valve. Commission the bottoms level controller, leaving it on "manual", and establish a level in the base of the column and the caustic settler. Open the bottoms level control valve and send the caustic to the sodium phenate tank to be sure that this line is clear. Stop the caustic import and export and maintain the base level.
- 3) Slowly open the C-AMSC receiver organic phase to the wash column while slowly closing the line to the FCDH.
- 4) Fill the column and take the hydrocarbon overhead through the back pressure control valve to the hydrogenation feed tank. Set the pressure controller to maintain 3.1 kg/cm²(g) (45 psig).
- 5) Commission the flow controller on the AMS Hydrogenation Section feed line and establish flow through the start-up line back to the C-AMSC receiver.
- 6) Start one C-ACWC recycle pump and set the flow controller to maintain design flow to the column. This ensures consistent, optimum column operation.

- 7) Open the caustic to the column again. Set the caustic rate at 50% of design. Set the bottoms level controller to maintain a 50% level.
- 8) Sample the overhead cumene for phenol. Adjust the caustic rate to give less than 20 wt-ppm phenol in the cumene.
- 9) Commission the spent caustic circulation pump and pH analyzers. It may take some time to establish the expected caustic utilization due to the inventories of caustic in the system.

When satisfied with all column operations continue to circulate the fractionation system. Alternatively, the FAC can be shut down and the CAC, CPC/RS, C-AMSC, and PR left on total reflux. In the latter case, isolate the cumene-AMS caustic wash column by sending C-AMSC overhead organic phase to the FCDH and discontinuing the import of 20 wt% caustic.

13. Hüls MSHP (AMS Hydrogenation)

13.1 Pre-commissioning Status

The following procedures are assumed complete:

- a. The unit has been pressure tested and dried is currently under a slight positive pressure of nitrogen.
- b. The make-up gas compressor (if applicable) has been run in and is available for service. It should currently be nitrogen blanketed and isolated from the unit.
- c. The hydrogen supply line is isolated from the unit with a blind.
- d. The reactors are loaded with catalyst and its inlet and outlet blinds are removed.
- e. The steam header is commissioned.
- f. Cooling water is circulating through the hydrogenation product cooler.
- g. The necessary hydrogen cylinders are available.

If these steps have not been completed, they should be finished before proceeding further with the commissioning of this section.

13.2 MSHP Unit Pressure Test

A pressure test with hydrogen in stages, first at 3.5 kg/ cm²g (50 psig) and then 10.0 kg/ cm²g (150 psig). Confirm that the AMS feed line is isolated at the battery limit block valves. Confirm that the product separator (PS) level control valve is closed and isolated with the “E” assembly block valves.

- 1) Line up reactors to allow nitrogen to flow once-through to the high pressure separator and the flash drum.

- 2) Open the nitrogen supply at the M/U hydrogen supply line and raise the unit pressure to header pressure, generally 7-8 kg/cm²(g) (100-115 psig).
- 3) Depressure the unit to flare through the flash drum. Purge out dead leg lines such as the hydrocarbon feed line and the flash drum from the bottom.
- 4) Repeat steps 2 and 3 twice.
- 5) Repressure and take a sample of the unit gas and confirm the oxygen concentration is less than 0.3 mol %. If it is not, repeat steps (1) and (2).
- 6) Isolate the nitrogen supply with a blind when finished lowering the unit's oxygen concentration.
- 7) Depressure the unit to the flare leaving a slight positive pressure.
- 8) Remove the hydrogen supply blind. Line up hydrogen to the unit. Bypass the automatic shutdown of the hydrogen supply. Commission the pressure controller and open the control valve to introduce hydrogen. Raise the pressure to 3.5 kg/cm²(g) (50 psig), then close the pressure control valve and isolate. Pressure test for 1 hour.
- 9) After a successful low pressure test increase pressure to 10 kg/cm²(g) (150 psig) and pressure test again for 1 hour.

13.3 MSHP Catalyst Hydrogen Reduction for New Catalyst Load

When satisfied with the pressure test, reduce the unit pressure to 3.9 kg/cm²g (55 psig) and prepare to reduce the catalyst, as follows:

- 1) Start the make-up gas compressor (if provided). See manufacturer's recommended procedures for starting the compressor. Start flow at 80 Nm³/hr per m³ catalyst (1.6 SCF per pound catalyst) by the start-up line to the reactor recycle cooler
- 2) Set the pressure controller to maintain 3.9 kg/cm²g (55 psig). The bypass around the product separator vent FIC will have to be opened to allow the large hydrogen flow to pass.
- 3) Commission the steam to the reactor recycle cooler and heat the make-up hydrogen to 50-60°C (120-140°F). The reactor temperature target is 50°C (120°F). Do not allow the bed temperatures to increase above 70°C (160°F).
- 4) After the hydrogenation reactor #2 outlet reaches 50°C (120°F), begin a 2 hour hold. The water produced should be drained at the separator.
- 5) After the 2-hour period, if feed is not available, cool reactor and cut out hydrogen and maintain system pressure.

With the catalyst activated, the unit is ready to receive feed. Note that if the hydrogenation feed tank becomes contaminated (>20 wt ppm phenol), it cannot be used as feed. The tank material would then be reprocessed by sending it back to the fractionation feed tank or by sending it through the C-AMSCWC, depending upon the degree and type of contamination.

14. Phenol Recovery

14.1 Pre-commissioning Status

Verify that all pre-commissioning procedures have been completed and that all trips and bypasses are working and are in their normal operating position. Cooling water should be in service to all exchangers.

The commissioning steps that need to be taken care of at this time in this section are minimal. 20 wt% NaOH solution should already be available; if not, make it now.

Inventory the bottom of the C-AMSCWC, and the SCWC with 20 wt% NaOH. Start a flow of caustic to each of these columns, until a 10% level is observed in the sight glass in the field (the board level indicators will not function correctly until there is hydrocarbon in the vessel as well). When a level is observed in the sight glass, stop caustic injection to the columns and await feed. The caustic in the trays will fall down to the bottom when the flow has stopped. Inventory the caustic settler to a 50% level with 20 wt% caustic and inventory the C-AMS water wash drum to a 50% level with water.

Inventory the solvent drum to a 20% level. Solvent circulation can begin as soon as the delay pot and PRU are ready to be put on line. Since the amount of solvent necessary to inventory the system is difficult to calculate beforehand, initially add just enough solvent to get the system to work with a low level in the solvent drum. Too much solvent in the system requires later dragging to the PRU or fractionation feed tank. Since dragging cumene is undesirable and unnecessary, the start-up of the OEC and SCWC are left until the sodium phenate tank and phenolic waste water tank level are increasing and the delay pot and PRU are put on line.

B. Normal Start-up

There are situations in which production has been stopped, and most or all of the plant is left inventoried, such as when an automatic emergency shutdown occurs. There are also situations when parts of the plant are de-inventoried to prepare for maintenance and must be re-inventoried to restart. Procedures for all of these situations are covered by Initial start-up Section A. The portion of the initial start-up procedures that apply will depend on what has been de-inventoried from the individual sections.

Note that it is very unusual that all of the CHP in the oxidizer section will be de-inventoried due to the complexity and time required to re-seed the oxidizers. The UOP design provides for running a single oxidizer at a time and the ability

maintenance a single oxidizer while the other oxidizer(s) has CHP in it. Therefore, the seeding part of the initial start-up procedure will likely only be used once.

1. Oxidation

The entire synthesis section (e.g. oxidation, concentration, decomposition and neutralization) has to be brought into production at the same time. The actual start of production is when feed to the decomposer is started, and crude product flows to the neutralization section and then to the fractionation feed tank.

If production has not started by the time the oxidation section has lined out and reached the correct CHP concentrations in each of the oxidizers, operations has probably cooled them off somewhat and reduced the air flow in order to minimize the formation of excess CHP, and keep the concentration of CHP in the oxidizers in normal ranges. If this is the case, and the concentration section is operating normally and recycling 80-84wt% CHP from the flash column bottoms, proceed to the decomposition section start of production. When production of crude product has begun and has lined out, return to this section to adjust oxidation parameters.

2. Concentration

When the terminal oxidizer is up to the desired concentration, if long circulation is stopped it can be re-established with the intent to concentrate the CHP. The FIC on the decomposer feed bypass line will set the circulation rate. Put this FIC in automatic/cascade and set to 70-100% of design. Begin heating up the feed to the preflash column using the steam reboiler and the flash column vaporizer. Slowly increase the temperature to about 10°C below design. It is likely that the columns will have a lower flash zone pressure during start-up and be more efficient. Establish the design reflux/feed ratio when enough cumene is flashed overhead.

The concentration section at this time (just prior to start of production of crude product) should be producing concentrated 80-84 wt% CHP at the bottom of the flash column (verified by the laboratory) and all temperatures, flows etc. should be in an acceptable range. Pressure control especially should be functioning correctly holding the correct pressure at the top of the flash column before the decomposition section is brought on line.

During initial operation, the preflash column and flash column do not need to be optimized perfectly. What is important is the both columns operate nominally with reasonable flash zone temperature and flows. A target of 60 wt% CHP should be made for the preflash column. It can be expected that preflash column feed filters will plug repeatedly during initial operation after cleaning the oxidizers.

Except for the re-routing of the concentrated CHP to the Decomposer Section, and the routing of the recycle cumene through the feed pre-treatment system (both done

at about the same time) the current operation, typically referred to as long circulation, is the same as normal operation/production.

When the decomposition section is ready to be brought on line, and the concentration section is lined out correctly, the start-up can proceed.

3. Decomposition

The decomposition section start of production is the start of production for the entire unit. It is here that the crude product is formed from the 80-84 wt% CHP. Putting this section on line is a complicated process and **safety** is the key word.

There are four things needed before the decomposition section can be brought on line:

- a. Sulfuric Acid Injection
- b. Verify Decomposition Section Composition by Laboratory
- c. Heat-up of the System
- d. Flushing of the Decomposition Feed Line

3.1 Sulfuric Acid Injection

Sulfuric acid is brought into the decomposition section in three normal routes. Two of these are to the calorimeters, and the third is to the circulating liquid effluent from the decomposition cooler. The sulfuric acid flow to all three of these points should be verified, and then sulfuric acid should be brought into all three points at the normal rate (i.e. 40 wt ppm for the anticipated feed rate) by putting the pumps in service. The objective is to increase the sulfuric acid concentration of the decomposer liquid to 300 wt ppm (as verified by the laboratory). Since the acid pumps in the unit are so small, this may take a few hours. Overshooting the acid concentration can also cause problems with the start-up as the CHP will react too fast and cause high temperatures at the circulation pump suction. This could cause the pumps to cavitate, which would then cause a decomposer shutdown. The section will be heated up while the sulfuric acid concentration is being increased.

Verify that the flow of decomposer liquid to the calorimeters responds to the flow controllers and set the flow to the calorimeters to design. The volume in the piping and calorimeters in this section has been specified to such tolerances that the flow rates to the calorimeters must be maintained to get 36 seconds residence time, if they are to function correctly.

Verify decomposition section composition by laboratory. Verify that the material in the circulation loop meets the specs for start-up, with less than 2 wt% H₂O and ~300 wt ppm sulfuric acid.

3.2 Decomposer Heat-up

The decomposer is heated using the dehydrator, recycling material from the decomposer, through the dehydrator, and returning the material to the decomposer. The decomposer low temperature bypass switch and the dehydrator 'Steam Enable' switch must be depressed to get steam to flow to the dehydrator.

If the above circulation route has not been established yet, do so now. Open the product FIC (not on cascade to the decomposer level at this time) and set the dehydrator feed flow to approximately 25% of scale. Put the decomposer TIC in auto at 70°C (160°F), controlling the cooling water to the decomposer cooler. Verify that the cooling water control valve closes (since the decomposer temperature is now less than 70°C, 160°F). Target a dehydrator outlet temperature of 135°C (275°F). The decomposer temperature should start to increase.

When the decomposer temperature reaches 70°C (160°F), increase the flow to the dehydrator to 70% of design while keeping the dehydrator outlet temperature at 135°C (275°F). Verify that the decomposer cooler is capable of maintaining the decomposer temperature by temporarily decreasing the decomposer temperature controller setpoint to 60°C (140°F) and observing the process variable. It should decrease quickly, and may overshoot, since the current heat load on the cooler is quite small compared to that of the decomposition of the concentrated CHP. Still, the controller action should be correct. Verify that this is so. Good control is required because of the low temperature shutdown at 55°C. This automatic shutdown is the most likely to save the system upon loss of CHP decomposition reaction since the optimal temperature is 60°C.

After being satisfied that the decomposer TIC is operating correctly, put the setpoint back to 70°C (160°F) for feed in. Verify that the temperature controller operates correctly.

ENSURE THAT NEUTRALIZATION IS READY FOR CRUDE PRODUCT BEFORE PROCEEDING.

When the decomposer temperature reaches 70°C (160°F), check the sulfuric acid concentration in the circulating liquid by lab analysis. Continue to inject sulfuric acid until the concentration is at least 300 wt ppm and then stop acid injection.

Verify that conditions in the concentration section are stable.

The unit is ready for start of production.

3.3 Flushing of the Decomposer Feed Line

The decomposer feed line should be flushed with cumene to ensure that no acid, or other foreign material that might promote cleavage, remains in the feed line. This step is particularly important for restarts after a shutdown that has been waiting for hours or days to restart. It is automatic, upon shutdown, so if a restart will occur quickly, say within a few hours, this flushing does not need to be restarted. It is the last step prior to introduction of concentrated CHP to the decomposer. The button is called the “Cumene Flush to Decomposer Feed Line Start”.

Flush the feed line from the cumene flush point to the decomposer for a time and rate so as to flush the volume of affected piping twice, in one minute. This volume should be calculated and then translated to the delta level in the flush drum. The feed piping “free drains” by gravity into the distributor in the decomposer. The liquid occupying this volume during a shutdown case is equivalent to the normal liquid level in the decomposer down to the end of the distributor. It is not necessary to assume the whole pipe is liquid full and must be flushed twice, but rather this standing liquid at the end of the distributor that must be flushed twice. The whole piping volume should be used for the normal shutdown flushing when refilling the flush drum

3.4 Decomposer Feed-in (Once-through Production)

When operators are confident that the necessary conditions exist in the decomposer section for a safe start-up of this section, the concentrated CHP can be introduced. As always, **SAFETY** is the key word in this procedure. If the instructions are followed closely and attention paid to detail during the start-up, the procedure is safe. Close attention is the key.

The concentration section flash column bottoms (80-84 wt% CHP) should be bypassing the decomposition section at this time, joining with the net flash column overhead and returning to oxidation. This flow is routed through a bypass flow controller. Both automatic block valves are closed on the feed line, and the automatic drain valve open to the sump. The next steps commission the decomposition section.

The FIC to the decomposer should be set in automatic/cascade with a setpoint equal to the FIC on the decomposer long circulation line.

Open the two automatic block valves and close the automatic bleeder and cumene flush valves. This will require pressing the "Decomposition Feed Start" button. By pressing this button, the two block valves in the feed line will open and the flush and bleeder valves will close. The dehydrator effluent valve to the neutralization section will open and the valve recirculating back to the decomposer will close.

A ramp function has been added to the long circulation FIC to allow this controller to automatically close over a short time (< 5 minutes) at start-up. As the long circulation flow control valve is closed, the decomposer feed FIC will automatically open the control valve to the decomposer to maintain the total FC bottoms flow constant. The CHP must be pushed forward quickly enough to generate the reaction in the decomposer otherwise the decomposer temperature will decrease quickly and shutdown the decomposer on TSL at 55°C (130°F).

Verify:

- a. Circulating decomposer inventory is at 70°C (160°F)
 - b. Circulating decomposer inventory contains ~300 wt ppm sulfuric acid
 - c. Water concentration of the decomposer inventory is less than 2 wt%
 - d. Current sulfuric acid injection rate is equal to 40 wt ppm of the feed rate that will be in effect when the entire flash column bottoms is routed to the decomposer
 - e. Dehydrator outlet temperature should be near the design value, 135°C (275°F)
 - f. Cooling water to the dehydrator product cooler is online and all instrumentation is ready
 - g. Neutralization section is ready for crude product
 - h. The flash column bottoms CHP concentration is at least 80 wt%
- 1) Begin the decomposer feed ramp function immediately.
 - 2) The board operator should see a response by the system immediately (it does take a few seconds for the feed to get through the pipe into the decomposer). The first response may be an increase in the decomposer temperature, followed by an increase in the cooling water flow to the decomposer cooler as the controller maintains the 70°C (160°F) temperature in the decomposer. If the response is not quick, due to the feed not going forward, the system will shut down on low temperature because of the large mass of cooling water in the decomposer cooler. So it is possible the first response may be a decrease in temperature followed by an increase in temperature, if the feed comes in slowly.
 - 3) Put the decomposer level controller on cascade if not done already, sending the decomposed crude product to the dehydrator and forward. The dehydrator and dehydrator product cooler should still have their instrumentation on automatic and should require minimal operations input.
 - 4) At the decomposer start-up conditions (high temperature and high acid concentration) the reaction of the concentrated CHP should be instantaneous. If the operator is certain that there is some flow into the decomposer, and yet there is no indication from the instrumentation (i.e. TI's and cooling water flow control) that an exothermic reaction is occurring, stop feed to the decomposer, flush the feed line, and sample the decomposer contents in the laboratory to ensure that the specifications for start-up are met. The most typical reasons for lack of or slow reaction are:

- a. Insufficient Temperature
- b. Insufficient Acid Concentration
- c. Low CHP Concentration in the Feed
- d. High Cumene Concentration in the Decomposer Circulating Liquid

Check all of the above to ensure that there is no problem with the circulating liquid.

ABOVE ALL, IF THERE IS NO REACTION TO INJECTING CHP INTO THE LOOP, DO NOT, UNDER ANY CIRCUMSTANCES, ARBITRARILY INCREASE THE SULFURIC ACID INJECTION WITHOUT FIRST TAKING A SAMPLE OF THE CIRCULATING LIQUID AND VERIFYING THAT THE CHP CONCENTRATION IS BELOW 0.2 WT%. IF ACID IS INJECTED INTO AN ACID-STARVED SYSTEM WITH A CHP CONCENTRATION GREATER THAN THIS, THERE IS A RISK OF RUNAWAY DECOMPOSITION AND POSSIBLE EXPLOSION OF THE DECOMPOSER CONTENTS.

THEREFORE, GET LAB VERIFICATION OF CHP AND ACID CONCENTRATION BEFORE PROCEEDING!

If proper attention has been paid to the required conditions for feed in, the decomposition reaction has begun and crude product is being sent to neutralization on level control. The remaining steps of the start-up are to route all of the flash column bottoms to the decomposer, set up the calorimeter and water injection functions, flush the long circulation line, then put the rest of the section into production (from start-up) mode.

- 5) Over a period of less than 5 minutes the long circulation flow should be reduced to zero.
- 6) The total flash column bottoms is now being fed to the decomposer. The conditions should be:
 - a. Temperature 70°C (160°F)
 - b. Sulfuric Acid Injection 40 wt ppm of fresh feed
 - c. Water Injection Zero at present
 - d. Dehydrator Outlet 135°C (275°F)
 - e. Crude Product 38°C (100°F)
 - f. Amine Neutralization Injection 2.3-2.7 pH
- 7) When the long circulation bypass FIC valve is completely closed, close the UV block valve upstream of the flow meter.

- 8) Items a), b), and c) below must take place concurrently with the normalization of operations of the decomposition section.
 - a) Depress the "Cumene Flush to Long Circulation Line Start" pushbutton. The HIC valve and flush valve will open for about 30 minutes. A light will indicate that flush is in progress.
 - b) Automatic rerouting of PFC and FC net overhead cumene to the oxidation feed wash column.
 - c) Check the operation of the neutralization, concentration and oxidation sections to ensure that the start of production has not adversely affected these sections.
- 9) With the sulfuric acid injection set at 40 wt ppm of fresh feed, and the temperature of the decomposer at 70°C (160°F), the operations of the decomposition section are now safe. There will not be a delta T across the first stage of the calorimeters until the sulfuric acid in the bulk liquid drops down to 40 ppm acid. The same reason (safety equates with the complete decomposition of CHP) that we start the unit with a high concentration of sulfuric acid in the bulk liquid keeps the calorimeters from performing their function. As the sulfuric acid concentration decreases and the water injection is started and then optimized, the delta T across the calorimeters will increase to a normal range. That the delta T's are very low or zero is normal for high acid/high temperature operation and is not a concern as long as the cooling water flow to the decomposer cooler indicates that the CHP feed is decomposing.
- 10) With the removal of concentrated CHP from the synthesis section, the oxidation section conditions might/will need to be adjusted to get CHP production in balance with the feed to the decomposition section.
- 11) The decomposition section is now started up. The next phase of the commissioning procedure is to adjust neutralization, and the fractionation train(s) to produce on spec products, while tuning the synthesis (mainly oxidation and decomposition sections) section to optimize performance. Finally, the entire unit is optimized to produce the product phenol and acetone in the most economical manner possible.
- 12) Since the sulfuric acid injection rate is currently at 40 wt ppm of the fresh feed rate, the inventory wt ppm of sulfuric acid will begin to decline to this level (exponentially, since the decomposer is essentially a continuous stirred tank reactor) as soon as feed is introduced. As the sulfuric acid concentration in the decomposer drops to 40 ppm, the delta T's in the first and second stage calorimeters will begin to show normal (i.e. non-zero) values. The first calorimeter delta T gives a measure of the residual CHP in the system. The second delta T gives a measure of the dicumylperoxide (hereinafter referred to as DCP) in the system. This DCP is formed by the reaction between CHP and DMPC in the decomposition section. Since DCP is more difficult to decompose by acidic means than CHP, it is expected that the second delta T will be the first to become non-zero. It is difficult to predict when this will occur because it is dependent upon too many things to be accurate.

- 13) When the sulfuric acid concentration of the decomposer inventory approaches 40 wt ppm (during start-up this sample should be taken and verified by the laboratory every 30 minutes, less if a **reliable** (verified by lab) analyzer is available. Even with an analyzer, the laboratory should periodically analyze the reaction mixture for sulfuric acid, CHP, H₂O and composition to verify the veracity of the on-line readings.
- 14) When the calorimeter delta T's have stabilized, the next items to attend to are the decomposer temperature (currently 70°C, 160°F) and the sulfuric acid concentration in the decomposer (currently 40 wt ppm), and the water injection rate (currently zero).
- The start-up temperature of 70°C (160°F) is used in order to ensure the complete reaction of CHP at the time of feed introduction. However, this temperature is not optimal for the best selectivities. The design temperature of 60°C (140°F) is the optimal temperature combining relative completeness of reaction with good selectivities. A secondary reason for this temperature is it is close to the shutdown temperature of 55°C. If the CHP decomposition reaction is suddenly lost (which is the usual way it occurs), the temperature will dip and safely shut the unit down before CHP has the chance to accumulate in the decomposer.
 - Begin decreasing the decomposer temperature 1°C (2°F) at a time, with about 15 minutes between each move to allow any changes in the system to become apparent. The delta T across the first stage calorimeter(s) may increase slightly as the decomposer temperature is decreased.
 - When the decomposer temperature has been decreased to 60°C (140°F), allow the system to equilibrate, and take a crude product sample to the lab to verify the effects of the changes made to the system. After the composition and operations are verified, the next optimization can begin.
 - The lab sample taken above should have verified that a 40 wt ppm concentration of sulfuric acid is present in the decomposer.
 - The sulfuric acid concentration normally present in the decomposer circulating inventory is 40 wt ppm. As the acid concentration decreases, the selectivities are enhanced at the sacrifice of the extent of CHP reaction. It is vitally important that the unit be monitored very carefully during such optimizations and the optimizations reversed if the plant seems to be responding in a dangerous manner such as poor temperature control due to variation in reaction rate. In this case, an example might be helpful.
 - The water injection tempers the decomposition reaction and promotes selectivity of CHP to phenol and acetone at decomposer condition. The design amount of water injected to the decomposer feed stream is about 0.6 wt% of fresh feed. However, the specification for water injection is the water concentration of the dehydrator outlet. It is possible, depending upon the DMPC concentration in the concentrated CHP (FC bottoms) that more or

less water will be necessary. The design water concentration is 1-1.5 wt% at the dehydrator outlet.

- g) Start the decomposer water injection pump, if applicable (it is assumed that the pump has already been calibrated and checked out). After checking the water concentration in the dehydrator cooler outlet, add water to the system **carefully!** (remember that water in too great an amount (>2 wt%) retards the decomposition reaction and that this could result in a dangerous situation). Allow the system to come to equilibrium. The calorimeter delta T's will respond (typically increasing) to the increased water concentration in the decomposer inventory.
- h) After allowing the system to come to equilibrium, check the water concentration of the dehydrator cooler outlet. If it is less than 1.0 wt%, and the decomposer AMS yield is low, repeat step b) above, allow the system to equilibrate, and check. Repeat as necessary.

During the course of optimization, if operations goes a little too far and while decreasing the acid injection rate, decreases it below the recommended 40 wt ppm and the decomposition of CHP is hindered CHP can build up in the system. How is this diagnosed and what should be done? The calorimeters are the means by which the problem is diagnosed. If the CHP builds up in the system, several things will happen:

- a) The first thing that will occur is that the increased CHP, combined with the reduced acidity, will result in some CHP getting into the second stage of the calorimeters. Remember that the calorimeter design is normally to have no CHP remaining at the sulfuric acid injection point. The result will be an **increasing second stage delta T** while the **first stage delta T may decrease**. A normal reaction to optimizing the CHP concentration in the decomposer is for the 1st stage calorimeter delta T to increase, while the second stage calorimeter also increases. Increasing CHP increase formation of DCP, which react in each stage. The amount of DCP that can be formed is finite, so the 2nd stage delta T is also finite and will reach a maximum. This can be calculated based on the DMPC and DCP in the decomposer feed. Assuming that 85 mol% of the DMPC will convert to DCP plus the native amount of DCP (0.2 wt%) in the feed, one can calculate the delta T based on 3.3°C/wt% DCP plus 3°C for acid heat of solvation.
- b) The next effect (much more subtle, but noticeable) is that since the dehydrator is using heating media to dehydrate the crude product stream, and any exothermic reaction has an effect on the amount of steam required for a certain temperature and flow rate, a decrease in the amount of steam is a reasonable thing considering that more CHP is being cleaved thermally.
- c) Therefore, a combination of stable or decreasing first stage calorimeter delta T, increasing second stage calorimeter delta T, and a decreasing steam usage in the dehydrator all point to the same problem-insufficient reaction of CHP-and steps must be taken to correct the problem immediately.

- i) If any optimization steps are being taken and it is suspected that these steps led to the problem, they should be reversed.
 - ii) If the acid injection is verified at a reasonable concentration (40 wt ppm) then the water concentration of the product should be checked to see if it is excessive. If so, the water injection rate should be reduced.
 - iii) If the acid concentration is low, **extreme care should be taken in increasing it. The sulfuric acid injection rate should be increased by no more than 5% of the current rate. The results of this change should be analyzed and the acid increased as required, but only in small steps.**
- d) Increasing the acid injection quickly when it is suspected or known that CHP has built up in the system is **extremely dangerous** one of the most certain ways to cause a relief incident, and a total shutdown of the complex.
- e) Fortunately, the UOP phenol plant shutdown system in the decomposition section is designed to keep the above scenario from occurring. High alarm/shutdown on the second stage calorimeter delta T would shut the decomposition section down in the event of a high delta T, thus ensuring the overall safety of personnel and equipment in the plant. In addition, the decomposer TIC has a shutdown of 55°C that will likely cause a shutdown if reaction is lost.
- 15) Optimization of the decomposition section is complete. Lab analyses and periodic readjustment of operating conditions will of course be required.

4. Neutralization

Compared to the start-up of the preceding sections, the start-up of the neutralization section is extremely simple since the flow of diamine should have already been lined up and there is minimal control and instrumentation in the section.

Establish the ratio control of crude product flow to diamine injection at a rate to keep the pH in a range of 2.3-2.7, as measured in the lab. Frequent adjustments of the ratio in a downward direction will be required as the sulfuric acid concentration of the dehydrator effluent falls to the 40 wt ppm normal operating range.

5. Fractionation

There are several configurations in which Fractionation could have been left following the Commissioning subsection. Regardless of which configuration was used, Fractionation should now be arranged as described in part "a" below.

The Fractionation section is circulating around the fractionation feed tank. The FAC product stream is circulating directly back to the fractionation feed tank; all other net streams are returning via the FCDH or going to their normal destination.

As the level in the fractionation feed tank reaches 30%, adjust the feed rate to the CAC to 50% of the normal design rate. Continue circulation through the return header cooler and line out the columns such that each stream meets specifications (see Section VII. Normal Operations).

With Fractionation operating to specifications:

- 1) Swing the FAC overhead product stream to a fresh acetone rundown tank.
- 2) The material from the FAC bottoms drum has been circulating back to the fractionation feed tank via the drum's overhead organic phase line. If the acetone concentration is low, then send the cumene to the C-AMSCWC in the effluent section.
- 3) Re-route the C-AMSC organic from the FCDH to the C-AMSCWC in the effluent section. Make sure the C-AMSCWC is currently ready for the flow.
- 4) Verify that the aqueous phase from the C-AMS column to the effluent section.
- 5) If the residue column bottoms have not been diverted to the residue tank, do so when the phenol concentration is below 10 wt%.
- 6) The phenol rectifier side-draw product should be recirculating to the fractionation feed tank via the FCDH.

6. Phenol Purification Reactors

The CTR and IXRT need to be brought online sequentially to minimize the formation of byproducts in the acidic IXRT.

- 1) Begin a flow of diamine at 1100 wt ppm of the C-AMSC bottoms flow rate or 25% over the stoichiometric consumption of carbonyl measured. The residence time of the reactor is about 1.5 hours.
- 2) At about one hour put one IXRT into service. Leave the second bed out of service until stable operation is established. This leaves one treater in reserve in case of an upset of high diamine injection rate.
 - a) The IXRT should have been filled and dried per the procedure detailed in Special Procedures Section XII, Chapter C.
 - b) Start-up Section VI, Chapter A.11.4 describes the procedure for filling and drying the IXRT. When the resin is dry it will become active to react 2-MBF.
- 3) When the second resin treater is brought online while the phenol unit is running send wet phenol to either the fractionation sump or the CAC bottoms, not to the phenol rectifier.

- 4) Check the carbonyl concentration of the CPC side-draw to the IXRT; it should be low. Adjust the diamine injection rate as necessary to get 5-15 wt ppm acetol in the CTR effluent.
- 5) Check the 2-MBF concentration of the Phenol Rectifier side-draw product several hours after the IXRT has been put into service. When the phenol is on specification, divert the flow to the phenol run down tank. Adjust the PR top and bottom flows accordingly to optimize the operation.
- 6) Put the lag IXRT into service and proceed with normal monitoring of the 2-MBF concentration between the treaters.

7. Hüls MSHP (AMS Hydrogenation)

When the C-AMSC organic phase material in the hydrogenation feed tank reaches a 30% level, proceed with the hydrogenation section start-up. The unit should be under hydrogen pressure at 2 kg/cm²g (30 psig) in standby, awaiting feed. Proceed as outlined below:

- 1) The unit must first be inventoried with cumene. Use the cumene start-up line at the hydrogenation recycle pumps provided to fill the unit. As the reactors fill, the top of the reactor must be vented to control the pressure and keep the cumene from draining into the product separator. Once inventory is complete, start the hydrogenation recycle pump and set at the design flow rate.
- 2) Check that the automatic shutdown of the hydrogen supply is active and not bypassed per the hydrogen pressure-up and catalyst reduction procedures. Set hydrogen flow to maintain the design H₂/HC mole ratio at the reactor inlet. The ratio as measured by the product separator vent will be 1/10 of the design inlet H₂/HC after the reaction has started.
- 3) Heat-up reactor #1 to 50°C (120°F). Maintain at least 2 kg/cm²(g) (30 psig) pressure at reactor #2.
- 4) Start the hydrogenation charge pump and utilize the automatic spillback to the MSHP feed tank. Confirm the pump discharge pressure is greater than the unit pressure before continuing.
- 5) Keep the feed flow control valve closed. To allow feed to go forward, the “MSHP Start-up” switch must be in the bypass position and the “Feed from Hydrogenation Charge Pump Reset” button must be pressed.
- 6) Line up the hydrogenation product back to the hydrogenation feed tank.
- 7) At this point, the reactor circuit should be at 2 kg/cm²(g) (30 psig) pressure, with a reactor inlet temperature of 50°C (120°F) and a design H₂/HC mole ratio. With these conditions established, proceed with the introduction of feed to the reactor.
- 8) Commission and put the feed flow control valve in automatic. This will cause the FIC to open the feed control valve. Monitor and adjust the reactor inlet temperature to maintain 50°C (120°F) and below a 35°C (63°F) both reactors delta T. If the delta T goes high increase the combined feed flow rate to dilute

the heat of reaction and decrease the inlet temperature to get the temperatures under control. Keep reactor inlet temperature below 80°C (175°F). An automatic shutdown at a bed temperature of 130°C (265°F) will stop the fresh feed and hydrogen. Without hydrogen, the saturation reaction will stop, but some polymerization reactions may continue and foul the catalyst.

- 9) When the product separator is at 50% level, commission the hydrogenation product separator pump and the separator level controller, if not done already, and slowly open the level control valve, sending material back to the MSHP feed tank.
- 10) Sample the product for IPCH and bromine index every hour. Aim for a bromine index from less than 100 mg/100g (AMS less than 600 wt ppm) and an IPCH concentration less than 300 wt ppm (preferably less than 100 wt ppm) by making the necessary process variable changes.

AMS concentrations below 600 wt ppm are generally acceptable from an overall phenol complex yield point of view. Concentrations of less than 100 wt ppm are possible with low IPCH formation and should be used as a goal. The oxidizers can accommodate a combined oxidizer feed AMS concentration of several hundred ppm, though the organic acid formation will increase and lead to sub-optimal oxidizer operation.

- 11) When the product meets specification, switch it from feed storage to the oxidizer feed section.
- 12) Increase feed rate as required.
- 13) Increase pressure if required to maintain the product separator control valve in range.

8. Phenol Recovery

The start of production of the unit will also start the production of byproduct streams that must be treated, and the treatment of the C-AMS column net overhead to remove phenol. In addition, the recovery of phenol (sodium phenate) from the caustic used to scrub the C-AMS column net overhead and other phenolic water streams is affected in this section.

8.1 Cumene-AMS Caustic Wash Column

The C-AMSCWC column has most likely been put either on internal circulation or simply shut down, awaiting feed forward of the C-AMS column overhead organic. Prior to this, the C-AMS column overhead was recirculating to the fractionation feed tank in order to maintain the fractionation feed composition during the line-out phase. The steps necessary to change the column to forward flow are:

- 1) Ensure that the cumene-AMS stream is lined up to the column correctly, and that the product is still lined up to the fractionation feed. Since the product may contain an unacceptably high concentration of phenol during the first few hours

of operation, it is best that the product be circulated to the C-AMSC or the FCDH until the phenol concentration is sufficiently low (<20 wt-ppm) to be routed to the MSHP section.

- 2) Start the C-AMSCWC recirculation pump if not already done, and set the total organic phase flow rate at **the design value**. The flow controller will alter the recycle flow control valve to maintain the correct jetting velocity in the C-AMSCWC. This is necessary to ensure optimal extraction of phenol.
- 3) Begin 20 wt% NaOH to the top of the C-AMSCWC and put the flow rate of caustic on cascade from the net feed flow meter (organic) to the C-AMSCWC. Set the globe valve to control the flow of caustic at the design value to the PDIC going to the caustic settler. Set the PDIC to 0.35 kg/cm² (5 psi). Set the caustic settler LIC on automatic.
- 4) When the cumene-AMS water wash drum fills with organic and begins to fill the sand filter, commission the LIC and begin water injection at the design rate.
- 5) Begin sampling the C-AMSCWC net effluent at the cumene-AMS sand filter outlet after about 1 hour of lined out operations and route the net overhead cumene-AMS to the hydrogenation feed tank when it is on spec (<20 wt ppm phenol); less than 10 wt ppm is easily attainable.
- 6) Line-up the spent caustic circulation pump back to the C-AMSCWC. Start the design flow and monitor the free caustic concentration and pH. Adjust the bias for the caustic injection to reach 100+ mol% caustic saturation with phenol. This optimization may take a number of shifts to attain.

8.2 Delay Pot, PRU Separator, OEC, SCWC

The start-up of this equipment is made necessary by the increasing level in the sodium phenate tank, which will begin as soon as the various sources of sodium phenate are produced. Sodium phenate comes from the SCWC, C-AMSCWC, and other sources within the fractionation section. In addition, the above equipment also recovers phenol from (and reduces the phenol concentration of) phenolic waters produced in the plant. Start-up will be necessary within a short time of the start-up of the rest of the plant, and if the sodium phenate and/or phenolic waste water tank(s) are practically full, it may already be started up, since this part of the effluent treating section does not really need any other part of the plant to be running in order to provide feed or some essential service. The following general steps are given to enable a particular start-up procedure to be formulated depending on what equipment needs to be put into service at what time.

When the sodium phenate (NaPh) make is sufficient to consider starting up the PRU and OEC, the following steps should be taken.

- 1) The OEC and SCWC should be circulating on total solvent (feed cumene). There will be no bottoms make from either column and no NaPH make from the SCWC. Verify that all line-ups are correct and that instrumentation (particularly

the pH meters) are functioning correctly and are calibrated. If the pH meters are not operating correctly, it will not be possible to achieve low ppm phenol concentrations in the waste water.

- 2) If there is no level in the bottom of the SCWC, begin a flow of 20 wt% caustic to this column on straight flow control until a 10% level is observed. Stop the caustic flow to the SCWC after this level is attained.
- 3) Inventory the delay pot with 50% cumene and 50% water. Establish circulation through the recycle pump and cooler. When the delay pot is filling, vent it off of the top PSV bypass and when full, start the PRU recycle pump. Begin recycle to the PRU delay pot at the design rate (FI and globe valve in field). The recycle stream has several purposes: to give good mixing in the static mixer, to reduce the response time in the delay pot, and to approach complete back mixing in the delay pot. The design recycle rate to fresh feed rate to the PRU is approximately 3:1. The recycle rate should be essentially fixed at the design rate unless operational experience dictates that it can be adjusted.
- 4) Ensure that H_2SO_4 is available for neutralizing the sodium phenate. When this is assured, begin a flow of sodium phenate to the delay pot. Add acid at a rate to completely neutralize the sodium phenate and to give a pH at the outlet of the delay pot of 4.5-5.0. It is best to take a representative sample of sodium phenate to the laboratory and have it neutralized in the above manner prior to start-up. This will allow the PRU bottoms to come on stream on spec. with respect to pH, rather than putting some basic water (which will not be scrubbed effectively) to the OEC. Set the FIC on the acid injection to give the same ratio of acid to phenate that the laboratory analysis provided. Set the acid/feed ratio controller in automatic after lining out.
- 5) Simultaneously, begin a flow of cumene to the recycle pump at 50% of the expected phenol from the PRU separator. This will minimize the quantity of the salt and water in the organic phase.
- 6) Put the PRU delay pot effluent temperature controller on auto controlling the delay pot outlet temperature at 40°C (105°F), and begin filling the PRU separator.
- 7) Vent the separator to the relief drain, and when full, allow the level control controlling sprung phenol to come into auto when an interface level is observed. Since there will be very little sprung phenol at this point, the level will be 100% and most of the liquid entering the PRU separator will leave the bottom. When a level **does** appear, route the material to the neutralization section and put the LIC in auto. The amine injection rate will need to be increased in the neutralization section.
- 8) If the pH of the material in the combined water to the OEC is still too basic (>5.0) there is one more chance to decrease it to the correct value, by the pH meter and acid injection point upstream of the oil extraction column. If the pH needs adjusting, put this acid injection point on line and put the controller in auto maintaining a pH of 4.5-5.0. Since phenolic waste water has to be injected into the stream at this point as well, pH adjustment may be necessary.

- 9) As water begins to flow to the oil extraction column, the inventory of cumene solvent in the OEC, SCWC and solvent drum will need to be decreased, to make up for the room in the columns that will now be taken up with phenolic waste water and caustic. An outside operator should be stationed at the solvent drum to decrease the cumene level by sending it to the fractionation feed tank as necessary when the level of solvent in the drum becomes too high.
- 10) Maintain solvent circulation as before.
- 11) When phenolic water is introduced to the OEC, re-introduce 20 wt% caustic to the SCWC at the rate mandated by the FFIC. This should be sufficient to wash essentially all of the phenol out of the solvent (less than 20 ppm phenol). The minimization of caustic necessary to achieve good washing of the solvent is one of the operations taken up in the optimization section.
- 12) Sample the caustic at the outlet of the SCWC; it should be at least 50-85% spent.
- 13) Line out and put these columns on cascade, to minimize phenol in the effluent, and to minimize the caustic necessary to effectively wash the two streams being washed (CAMSC overhead and solvent). Take the necessary laboratory samples and adjust flow rates of caustic and solvent as necessary to achieve the desired operation.

9. Caustic Blending Facilities

9.1 20 wt% Caustic

The maintenance of the 20 wt% caustic tank concentration and specific gravity should be totally automatic. Concentrated caustic from main storage is injected upstream of a static mixer, along with process water and also cumene recovery column net overhead aqueous material. The control of the tank level is on the process water flow. The control of the specific gravity of the 20 wt% caustic is responsible for adjusting the flow of concentrated caustic. The concentrated caustic flow is on ratio control to the water flow. The specific gravity analyzer on the combined stream puts out a signal that is added to (if the gravity is lower than setpoint) or subtracted from (if the gravity is above setpoint) the flow signal from the ratio controller. This allows periods of changing flow rate of water to be dealt with in a more consistent manner resulting in tighter control of the product gravity. As the demand for 20 wt% caustic changes, the water flow rate and concentrated caustic flow rates will change to keep the level consistent. A high level in the tank is appropriate in order to minimize any fluctuations in the tank specific.

VII. Normal Operations

This section includes information on how to evaluate, and optimize the process. Process monitoring and data evaluation are important parts of this section. Information from this section will describe the collection of data during normal operations. Data evaluation, calculations, and other programs are explained.

With the unit start-up completed, the operating conditions should be adjusted to give stable operation, within the range of the design operating conditions. The design operating conditions are those used in the design and specification of operating equipment and, therefore, will theoretically yield the optimum product quantity and quality. However, the actual conditions required for a stable operation may not coincide exactly with those stipulated by design. The ultimate criteria, of course, are the actual laboratory analyses and observed product rates. Nevertheless, until the unit is completely lined out and some solid basis for change is noted, the design operating conditions are good initial targets.

A. Calculations

1. Oxidation

1.1 Data Collection

When collecting daily data for performance evaluation and optimization, below is a recommended list of data to collect and/or calculate:

1.1.1 Inputs

- a. Date
- b. Hours of Operation
- c. Total Organic Rate to Feed Wash Column (FWC) Bottom Tray
- d. FWC Level
- e. FWC Pressure
- f. FWC Tray 2 pH
- g. FWC Aqueous Phase Outlet Temperature
- h. CHP Process Water Rate to FWC Top Tray
- i. FWC Circulating Caustic pH
- j. FWC Circulating Caustic Rate
- k. FWC Spent Caustic Flow Rate
- l. FWC Fresh Caustic Make-up Rate
- m. CHP Process Water Break Drum Level
- n. CHP Process Water Break Drum Pressure
- o. CHP Process Water Pump Discharge Rate

- p. Combined Feed Surge Drum Level
- q. Combined Feed Surge Drum Pressure
- r. Combined Feed Surge Drum Water Boot Level
- s. Combined Feed Pump Total Discharge Rate
- t. Air Compressor Discharge Temperature
- u. Compressed Air Temperature after Cooler
- v. Compressed Air Scrubber Level
- w. Compressed Air Scrubber Pressure
- x. Compressed Air Scrubber Circulating Dilute Caustic Flow Rate
- y. Compressed Air Scrubber Circulating Dilute Caustic pH
- z. Compressed Air Scrubber Spent Dilute Caustic Flow Rate
- aa. Compressed Air Scrubber Make-up Caustic Rate
- bb. Compressed Air Scrubber Make-up Water Rate
- cc. Compressed Air Scrubber Air Temperature
- dd. Compressed Air to Oxidizer #1 Flow Rate
- ee. Compressed Air to Oxidizer #1 Flow Control Valve Delta P
- ff. Nitrogen to Oxidizer #1 Air Line Flow Rate
- gg. Oxidizer #1 Feed Filter Delta P
- hh. Oxidizer #1 Temperature
- ii. Oxidizer #1 Spent Oxygen Concentration
- jj. Oxidizer #1 Spent Air Temperature
- kk. Oxidizer #1 Level
- ll. Oxidizer #1 Pressure
- mm. Oxidizer #1 Oxidate Temperature To Emergency Coolers
- nn. Oxidizer #1 Oxidate Temperature From Emergency Coolers
- oo. Oxidizer #1 Water Spray Line Pressure Between Isolation Valves
- pp. Oxidizer #1 Circulating CHP Concentration
- qq. Oxidizer #1 Circulating Flow Rate
- rr. Compressed Air to Oxidizer #2 Flow Rate
- ss. Compressed Air to Oxidizer #2 Flow Control Valve Delta P
- tt. Nitrogen to Oxidizer #2 Air Line Flow Rate
- uu. Oxidizer #2 Temperature
- vv. Oxidizer #2 Spent Oxygen Concentration
- ww. Oxidizer #2 Spent Air Temperature
- xx. Oxidizer #2 Level
- yy. Oxidizer #2 Pressure
- zz. Oxidizer #2 Oxidate Temperature To Emergency Coolers
- aaa. Oxidizer #2 Oxidate Temperature From Emergency Coolers
- bbb. Oxidizer #2 Water Spray Line Pressure Between Isolation Valves
- ccc. Oxidizer #2 Circulating CHP Concentration
- ddd. Oxidizer #2 Circulating Flow Rate
- eee. Oxidizer #2 Oxidate to Feed-Oxidate Exchanger Flow Rate
- fff. Oxidizer #2 Oxidate To Feed-Oxidate Exchanger Temperature
- ggg. Oxidizer #2 Oxidate From Feed-Oxidate Exchanger Temperature
- hhh. Oxidizer #2 Oxidate To Emergency Cooler Flow Rate

- iii. Fresh Feed after Oxidizer Feed Preheater Temperature
- jjj. Fresh Feed after Oxidizer Feed Heater Temperature
- kkk. Condensate from Oxidizer Feed Heater Flow Rate
- III. Decanter Level
- mmm. Decanter Water Pump Flow Rate
- nnn. Decanter Pump Flow Rate
- ooo. Oxidizer Spent Air Pressure
- ppp. Spent Air After Condenser Temperature
- qqq. Spent Air After Chilled Condenser Temperature
- rrr. Superheated Spent Air Temperature
- sss. Charcoal Adsorber Temperature
- ttt. Charcoal Adsorber Regeneration Steam Flow Rate
- uuu. Charcoal Adsorber Strainer Delta P
- vvv. CHP Sump Level
- www. CHP Sump Temperature
- xxx. Peroxide Destruction Reactor (PDR) Inlet pH
- yyy. PDR After Caustic Injection pH
- zzz. PDR Caustic Flow Rate
- aaaa. PDR Feed Temperature
- bbbb. PDR Nitrogen Flow Rate
- cccc. PDR Nitrogen Control Valve Delta P
- dddd. PDR Outlet Temperature
- eeee. PDR Level
- ffff. PDR Pressure
- gggg. PDR Vent Flow Rate
- hhhh. Peroxide Destruction Separator Level
- iiii. Peroxide Destruction Separator Vent Rate
- jjjj. Waste Water from Peroxide Destruction Separator Flow Rate

1.1.2 Stream Information (All Flows)

- a. Orifice Flow Rate (Raw and Corrected)
- b. Mass Flow Meter Flow Rate
- c. Mass Flow Meter Density
- d. Flow Meter Temperature
- e. Flow Meter Pressure

1.1.3 Calculated Values

- a. Phenol Unit Mass Balance
- b. Oxidizer Space Velocity
- c. Feed Wash Column Tray Jetting Velocity
- d. Oxidizer CHP Yield

1.2 Calculations

1.2.1 Phenol Unit Mass Balance

$$\text{Phenol Unit Mass Balance} = \frac{\text{Mass Flow Out} + \text{Accumulation}}{\text{Mass Flow In}}$$

Mass Flow Out

$$\begin{aligned} &= \text{Phenol Product} + \text{Acetone Product} + \text{Residue} \\ &+ \text{Vent Gas from Oxidizer 1} + \text{Vent Gas from Oxidizer 2} \\ &+ \text{Vent Gas from MSHP} \end{aligned}$$

Accumulation

$$\begin{aligned} &= \text{Change in Level of Oxidizers} + \text{Fractionation Feed Tanks} \\ &+ \text{MSHP Feed Tank} + \text{Phenolic Waste Water Tank} \\ &+ \text{Sodium Phenate Tank} \end{aligned}$$

$$\begin{aligned} \text{Mass Flow In} &= \text{Cumene Feed} + \text{Air to Oxidizer 1} + \text{Air to Oxidizer 2} + \text{Make} \\ &- \text{Up Hydrogen to MSHP} + \text{Diamine to Neutralization} \\ &+ \text{Sulfuric Acid to Calorimeters and Decomposer} \\ &+ \text{Diamine to Chemical Treatment Reactor} \end{aligned}$$

Please note that the various aqueous streams entering and leaving the unit are not included in the mass balance. The various streams and tanks indicated above should only include changes in organic material and be independent of water.

1.2.2 Oxidizer Space Velocity

$$\text{Oxidizer Space Velocity, hr}^{-1} = \frac{\text{Combined Cumene Feed Rate, m}^3/\text{hr}}{\text{Oxidizer Volume (Clear Liquid Height), m}^3}$$

1.2.1 Feed Wash Column Tray Jetting Velocity

$$\text{FWC Jetting Velocity, m/hr} = \frac{\text{FWC Combined Feed Rate, m}^3/\text{hr}}{\text{FWC Tray Open Area} = \# \text{ Holes} \times \text{Hole Area, m}^2}$$

1.2.2 Oxidizer CHP Yield

$$\text{CHP Yield, mol\%} = \frac{\text{wt\% CHP in Oxidizer Effluent} / \text{MW}_{\text{CHP}}}{\sum \text{wt\% X in Oxidizer Effluent} / \text{MW}_x(\text{CHP, DMPC, DCP, AP})}$$

The Honeywell Frankford Phenol Unit has empirically correlated their oxidizer CHP yield versus the operating parameters. On the next page is the formula relating air, CHP concentration, AMS concentration in the cumene feed and spent air oxygen concentration to CHP yield. The constants in the formula are unit dependent and would have to be derived based on operating data from each unit.

$$\begin{aligned}\text{CHP Yield, mol\%} = & 98.28 - 0.89 * (P/A) - 0.13 * (\% \text{ CHP at terminal oxidizer} - 30) \\ & - 0.23 * (\# \text{ AMS to oxidizers}/1000 \# \text{ CHP in oxidizers}) \\ & - ((7\text{-mole\% O}_2 \text{ in spent air})/(\text{mole\% O}_2 \text{ in spent air}))\end{aligned}$$

Where:

P/A = pounds O₂ consumed per area (square feet) of oxidizer. A is dependent on the oxidizer levels being constant.

= pounds

2. Concentration

2.1 Data Collection

When collecting daily data for performance evaluation and optimization, below is a recommended list of data to collect and/or calculate:

2.1.1 Inputs

- a. Preflash Column Feed Filter Delta P
- b. Preflash Column Feed Temperature
- c. Preflash Column Feed Flow Rate
- d. Preflash Column Overhead Pressure
- e. Preflash Column Overhead Temperature
- f. Preflash Column Flash Zone Pressure
- g. Preflash Column Reflux Flow Rate
- h. Preflash Column Net Bottoms Temperature
- i. Oxidate from Preflash Column Feed-Oxidate Exchangers Temperature
- j. Preflash Column Feed-Oxidate Exchanger Tub Temperature
- k. Preflash Column Steam Heater Tub Temperature
- l. Steam to Preflash Column Steam Heater Flow Rate
- m. Steam to Preflash Column Steam Heater Pressure
- n. Concentration Cumene Quench Drum Level
- o. Concentration Cumene Quench Drum Temperature
- p. Feed Flow Rate to Flash Column Vaporizer
- q. Steam to Flash Column Vaporizer Flow Rate
- r. Steam to Flash Column Vaporizer Pressure
- s. Flash Column Vaporizer Tube Temperature
- t. Flash Column Vaporizer Bulk Temperature
- u. Flash Column Overhead Pressure
- v. Flash Column Overhead Temperature
- w. Flash Column Flash Zone Pressure
- x. Flash Column Accumulator Liquid Temperature

- y. Flash Column Bottoms Temperature
- z. Flash Column Bottom Level
- aa. Flash Column Reflux Flow Rate
- bb. Flash Column Bottoms CHP Concentration
- cc. Flash Column Bottoms Pump Suction Temperature
- dd. Flash Column Bottoms Pump Spillback Flow Rate
- ee. Preflash and Flash Column Condenser/Receiver Level
- ff. Recycle Cumene to Oxidation Feed Wash Column Flow Rate
- gg. Recycle Cumene Temperature
- hh. Non-condensable Vent Line After Chiller Temperature
- ii. CHP Nitrogen to Vacuum System Flow Rte
- jj. Steam to Vacuum System Flow Rate
- kk. Steam to Vacuum System Pressure
- ll. Vacuum System Sealant Purge Flow Rate
- mm. Vacuum System Non-condensable Vent Flow Rate
- nn. Vacuum System Non-condensable Vent Oxygen Concentration
- oo. CHP Process Water to Vacuum System Flow Rte

2.1.2 Stream Information (All Flows)

- a. Orifice Flow Rate (Raw and Corrected)
- b. Mass Flow Meter Flow Rate
- c. Mass Flow Meter Density
- d. Flow Meter Temperature
- e. Flow Meter Pressure

3. Decomposition

3.1 Data Collection

When collecting daily data for performance evaluation and optimization, below is a recommended list of data to collect and/or calculate:

3.1.1 Inputs

- a. Cumene Flush Drum Pressure
- b. Cumene Flush Drum Level
- c. Decomposer Feed Flow Rate
- d. Long Circulation Flow Rate
- e. Decomposer Feed Flush Drum Pressure
- f. Decomposer Feed Flush Drum Level
- g. Decomposer Water Injection Rate
- h. Decomposer Water Injection Line Pressure
- i. Decomposer Pressure
- j. Decomposer Level
- k. Decomposer Temperature

- l. Decomposer Product Pump Suction Temperature
- m. Decomposer Product Pump Flow Rate
- n. Decomposer Cooler Delta P
- o. Decomposer Circulation Flow Rate
- p. Decomposer Circulation Cooler Process Inlet Temperature
- q. Decomposer Circulation Cooler Process Outlet Temperature
- r. Decomposer Circulation Cooler Cooling Water Inlet Temperature
- s. Decomposer Circulation Cooler Cooling Water Outlet Temperature
- t. Calorimeter #1 Flow Rate
- u. Calorimeter #1 First Stage Delta T
- v. Calorimeter #1 Second Stage Delta T
- w. Calorimeter #1 Total Delta T
- x. Calorimeter #1 Second Stage Inlet Temperature
- y. Calorimeter #1 Second Stage Outlet Temperature
- z. Calorimeter #2 Flow Rate
- aa. Calorimeter #2 First Stage Delta T
- bb. Calorimeter #2 Second Stage Delta T
- cc. Calorimeter #2 Total Delta T
- dd. Calorimeter #2 Second Stage Inlet Temperature
- ee. Calorimeter #2 Second Stage Outlet Temperature
- ff. Decomposer Acid Injection Drum Level
- gg. Decomposer Acid Injection to Decomposer Flow Rate
- hh. Decomposer Acid Injection to Calorimeter #1 Flow Rate
- ii. Decomposer Acid Injection to Calorimeter #2 Flow Rate
- jj. Dehydrator Inlet Temperature
- kk. Dehydrator Outlet Temperature
- ll. Dehydrator Delta P
- mm. Steam to Dehydrator Temperature
- nn. Steam to Dehydrator Pressure
- oo. Steam to Dehydrator Flow Rate
- pp. Dehydrator Reactor Loop In Service (True/False)
- qq. Crude Product Upstream of Cooler Temperature
- rr. Crude Product Upstream of Cooler Pressure
- ss. Crude Product Downstream of Cooler Temperature

3.1.2 Stream Information (All Flows)

- a. Orifice Flow Rate (Raw and Corrected)
- b. Mass Flow Meter Flow Rate
- c. Mass Flow Meter Density
- d. Flow Meter Temperature
- e. Flow Meter Pressure

4. Neutralization

4.1 Data Collection

When collecting daily data for performance evaluation and optimization, below is a recommended list of data to collect and/or calculate:

4.1.1 Inputs

- a. Diamine Injection Flow Rate
- b. Combined Crude Product and Sprung Phenol Temperature
- c. Static Mixer Delta P
- d. Neutralized Crude Product Sulfuric Acid Concentration

4.1.2 Stream Information (All Flows)

- a. Orifice Flow Rate (Raw and Corrected)
- b. Mass Flow Meter Flow Rate
- c. Mass Flow Meter Density
- d. Flow Meter Temperature
- e. Flow Meter Pressure

5. Fractionation

5.1 Data Collection

5.1.1 Inputs

1) Crude Acetone Column (CAC)

- a) Fractionation Feed Tank Level
- b) Fractionation Feed Tank Temperature
- c) CAC Charge Pumps Mixing Flow Rate
- d) CAC Charge Pumps Total Flow Rate
- e) CAC Feed Filter Delta P
- f) CAC Feed Flow Rate
- g) CAC Feed Temperature
- h) CAC Bottoms Flow Rate
- i) CAC Overhead Pressure
- j) CAC Overhead Temperature
- k) CAC Tray Temperatures
- l) CAC Reboiler Return Temperature
- m) CAC Bottom Pressure
- n) CAC Bottom Level
- o) CAC Reboiler Condensate Flow Rate
- p) CAC Steam Injection Flow Rate

- q) CAC Bottoms Temperature
- r) CAC Overhead Receiver Level
- s) CAC Reflux Flow Rate
- t) CAC Net Vapor Overhead Flow Rate

2) Finished Acetone Column (FAC)

- a) FAC Caustic Injection Flow Rate
- b) FAC Water Dilution Flow Rate
- c) FAC Overhead Temperature
- d) FAC Tray Temperatures
- e) FAC TDIC
- f) FAC Live Steam Injection Flow Rate
- g) FAC Bottoms Level
- h) FAC Reboiler Tub Temperature
- i) FAC Reboiler Condensate Flow Rate
- j) FAC Bottoms Temperature
- k) FAC Bottoms Pressure
- l) FAC Column Delta P
- m) FAC Bottoms Strainer Delta P
- n) FAC Bottoms to CPC Condenser Flow Rate
- o) FAC Bottoms Drum Level
- p) FAC Bottoms Drum Aqueous Phase Flow Rate
- q) FAC Bottoms Drum Organic Phase Flow Rate
- r) FAC Bottoms Drum Pressure
- s) FAC Bottoms Drum Temperature
- t) FAC Bottoms Drum Anthracite Coal Bed Delta P
- u) FAC Acetone Product Flow Rate
- v) FAC Acetone Product Temperature
- w) FAC Acetone Product Water Concentration
- x) FAC Overhead Pressure
- y) FAC Overhead Receiver Level
- z) FAC Vent Condenser Effluent Temperature
- aa) FAC Vent Condenser Liquid Flow Rate
- bb) FAC Reflux Flow Rate
- cc) FAC Reflux Temperature
- dd) FAC Aldehyde Reflux Flow Rate
- ee) FAC Vent Condenser Pressure
- ff) FAC Vacuum System Sealant In Flow Rate
- gg) FAC Vacuum System Sealant Purge Flow Rate
- hh) FAC Vacuum System Nitrogen Flow Rate
- ii) FAC Vacuum System Non-Condensable Vent Flow Rate
- jj) FAC Vacuum System Non-Condensable Vent Oxygen Concentration

3) Cumene-AMS Column (CAMSC)

- a) CAMSC Overhead Temperature
- b) CAMSC Tray Temperatures
- c) CAMSC Bottoms Pressure
- d) CAMSC Delta P
- e) CAMSC Bottoms Level
- f) CAMSC Reboiler Return Temperature
- g) CAMSC Reboiler Condensate Flow Rate
- h) CAMSC Bottoms Temperature
- i) Off-Spec Phenol to CAMSC Bottoms Flow Rate
- j) CAMSC Reflux Flow Rate
- k) CAMSC Reflux Phenol, Acetone and AMS Concentration
- l) CAMSC Net Overhead Organic Flow Rate
- m) CAMSC Reflux Flow Rate
- n) CAMSC Overhead Pressure
- o) CAMSC Overhead Receiver Level
- p) CAMSC Overhead Receiver Water Boot Level
- q) CAMSC Net Overhead Aqueous Flow Rate

4) Chemical Treatment Reactor (CTR)

- a) CTR Level
- b) CTR Outlet Temperature
- c) CTR Effluent Flow Rate
- d) Chemical Treatment Injection Drum Level
- e) Chemical Treatment Injection Flow Rate

5) Crude Phenol Column (CPC)

- a) CPC Overhead Temperature
- b) CPC Tray Temperatures
- c) CPC Accumulator Level
- d) CPC Reboiler Return Temperature
- e) CPC Reboiler Condensate Flow Rate
- f) CPC Bottoms Level
- g) CPC Bottoms Pressure
- h) CPC Delta P
- i) CPC Side-Cut Temperature
- j) CPC Side-Cut Flow Rate
- k) CPC Overhead Pressure
- l) FAC Bottoms From CPC Condenser Temperature
- m) CPC Overhead Receiver Level
- n) CPC Reflux Temperature
- o) CPC Reflux Flow Rate
- p) CPC Vent Condenser Vent Temperature

- q) CPC Vent Condenser Liquid Flow Rate
- r) CPC Net Overhead Flow Rate
- s) CPC Bottoms Flow Rate (RS Feed Flow Rate)

6) Residue Stripper (RS)

- a) RS Condenser Level
- b) RS Condenser Liquid Effluent Temperature
- c) RS Overhead Temperature
- d) RS Overhead Pressure
- e) RS Level
- f) RS Bottoms Pressure
- g) RS Bottoms Temperature
- h) RS Reboiler Return Temperature
- i) RS Reboiler Condensate Return Flow Rate
- j) RS Bottoms Pump Spillback Flow Rate
- k) RS Net Bottoms Flow Rate
- l) RS Net Bottoms After Cooler Temperature

7) Ion Exchange Resin Treaters (IXRT)

- a) IXRT Inlet Temperature
- b) IXRT Effluent Filter Delta P
- c) IXRT Filling and Start-up Bypass Flow Rate
- d) IXRT A Delta P
- e) IXRT B Delta P
- f) IXRT Emptying Flow Rate

8) Phenol Rectifier (PR)

- a) PR Feed Temperature
- b) PR Overhead Temperature
- c) PR Overhead Pressure
- d) PR Delta P
- e) PR Accumulator Tray Level
- f) PR Internal Reflux Flow Rate
- g) PR Bottoms Pressure
- h) PR Bottoms Level
- i) PR Reboiler Return Temperature
- j) PR Reboiler Steam Flow Rate
- k) PR Bottoms Temperature
- l) PR Bottoms Flow Rate
- m) PR Side-Cut Temperature
- n) PR Phenol Product Temperature

- o) PR Phenol Product Flow Rate
- p) CAC Feed Return Temperature
- q) PR Overhead Run Down Temperature
- r) PR Overhead Receiver Level
- s) PR Reflux Temperature
- t) PR Net Overhead Flow Rate
- u) PR Vent Condenser Vent Temperature
- v) PR Vent Condenser Liquid Flow Rate

9) Fractionation Miscellaneous Equipment

- a) Fractionation Vacuum System Suction Drum Pressure
- b) Fractionation Vacuum System Suction Drum Temperature
- c) Fractionation Vacuum System Suction Drum Level
- d) Fractionation Vacuum System Pressure
- e) Fractionation Vacuum System Sealant In Flow Rate
- f) Fractionation Vacuum System Sealant Return Flow Rate
- g) Fractionation Vacuum System Nitrogen Flow Rate
- h) Fractionation Vacuum System Vent Flow Rate
- i) Fractionation Vacuum System Vent Oxygen Concentration
- j) Phenolic Relief Drum Level
- k) Water to Phenolic Relief Drum Flow Rate
- l) Fractionation Sump Level
- m) Fractionation Sump Temperature
- n) Fractionation Sump Flow Rate
- o) Fractionation Return Header Cooler Effluent Temperature
- p) Phenolic Waste Water Sump Temperature
- q) Phenolic Waste Water Sump Level
- r) Phenolic Waste Water Sump Flow Rate
- s) Acetone Product Day Tank Levels
- t) Phenol Product Day Tank Levels
- u) Phenol Product Day Tank Temperatures
- v) Residue Tank Level
- w) Residue Tank Temperature

5.1.2 Stream Information (All Flows)

- a. Orifice Flow Rate (Raw and Corrected)
- b. Mass Flow Meter Flow Rate
- c. Mass Flow Meter Density
- d. Flow Meter Temperature
- e. Flow Meter Pressure

5.1.3 Calculated Values

- a. IXRT LHSV

5.2 Calculations

5.2.1 IXRT LHSV

$$\text{LHSV, hr}^{-1} = \frac{\text{Feed Rate in m}^3/\text{hr}}{\text{m}^3 \text{ of Resin Loaded in Treater}}$$

6. Hüls MSHP (AMS Hydrogenation)

6.1 Data Collection

6.1.1 Inputs

- a. MSHP Feed Tank Level
- b. MSHP Feed Tank Temperature
- c. Hydrogenation Charge Pump Discharge Flow Rate
- d. Hydrogenation Feed Filter Delta P
- e. MSHP Reactor Fresh Feed Flow Rate
- f. MSHP Reactor Fresh Feed Temperature
- g. MSHP Reactor 1 Total Flow Rate
- h. Make-up Hydrogen Flow Rate
- i. Make-up Hydrogen Pressure
- j. Make-up Hydrogen Temperature
- k. Catalyst Reduction Nitrogen Flow Rate
- l. Catalyst Reduction Steam Flow Rate
- m. MSHP Reactor 1 Inlet Temperature
- n. MSHP Reactor Temperatures
- o. MSHP Reactor 1 Outlet Temperature
- p. MSHP Reactor 1 Delta P
- q. Cooled Reactor 1 Liquid to Reactor 2 Flow Rate
- r. MSHP Reactor 2 Inlet Temperature
- s. MSHP Reactor 2 Temperatures
- t. MSHP Reactor 2 Outlet Temperature
- u. MSHP Reactor 2 Delta P
- v. MSHP Reactor Control Pressure
- w. Hydrogenation Separator Vent Flow Rate
- x. Hydrogenation Separator Pressure
- y. Hydrogenation Separator Level
- z. Hydrogenation Separator Bottoms Flow Rate
- aa. Hydrogenation Separator Bottoms Temperature
- bb. Hydrogenation Separator ButylBenzene Purge Flow Rate
- cc. Hydrogenation Separator Bottoms Start-up Flow Rate

6.1.2 Stream Information (All Flows)

- Orifice Flow Rate (Raw and Corrected)
- Mass Flow Meter Flow Rate
- Mass Flow Meter Density
- Flow Meter Temperature
- Flow Meter Pressure

6.1.3 Calculated Values

- Reactor 1 LHSV
- Reactor 1 Superficial Velocity
- Reactor 2 LHSV
- Reactor 2 Superficial Velocity
- Hydrogen/Hydrocarbon Ratio

6.2 Calculations

6.2.1 Reactor 1 LHSV

$$\text{Reactor 1 LHSV} = \frac{\text{Combined Feed Rate to Reactor 1, m}^3/\text{hr}}{\text{Volume of Catalyst in Reactor 1, m}^3}$$

6.2.2 Reactor 1 Superficial Velocity

$$\text{Reactor 1 Superficial Velocity} = \frac{\text{Combined Feed Rate to Reactor 1, m}^3/\text{hr}}{\text{Cross Sectional Area of Reactor 1, m}^2}$$

6.2.3 Reactor 2 LHSV

$$\text{Reactor 2 LHSV} = \frac{\text{Fresh Feed Rate to Reactor 1, m}^3/\text{hr}}{\text{Volume of Catalyst in Reactor 2, m}^3}$$

6.2.4 Reactor 2 Superficial Velocity

$$\text{Reactor 2 Superficial Velocity} = \frac{\text{Fresh Feed Rate to Reactor 1, m}^3/\text{hr}}{\text{Cross Sectional Area of Reactor 2, m}^2}$$

6.2.5 Hydrogen/Hydrocarbon Ratio

Hydrogen/Hydrocarbon Ratio, mol/mol

$$= \frac{\text{MakeUp Hydrogen Rate, } \frac{\text{Nm}^3}{\text{hr}} / 22.4 \frac{\text{Nm}^3}{\text{kmol}}}{\text{Fresh Feed Rate to Reactor 1, } \frac{\text{m}^3}{\text{hr}} \times 872 \frac{\text{kg}}{\text{m}^3} / 118.17 \frac{\text{kg}}{\text{kmol}}}$$

7. Phenol Recovery

7.1 Data Collection

When collecting daily data for performance evaluation and optimization, below is a recommended list of data to collect and/or calculate:

7.1.1 Inputs

- a. C-AMSCWC Recycle Pump Discharge Flow Rate
- b. C-AMSCWC Total Feed Flow Rate
- c. C-AMSCWC Total Feed Temperature
- d. C-AMSCWC Level
- e. C-AMSCWC Overhead Pressure
- f. C-AMSCWC Tray 5 pH
- g. Fresh Caustic to C-AMSCWC Top Tray Flow Rate
- h. C-AMSCWC Spent Caustic Pump Total Flow Rate
- i. C-AMSCWC Bottoms pH
- j. C-AMSCWC Bottoms to Feed Static Mixer Flow Rate
- k. C-AMSCWC Net Bottoms Flow Rate
- l. C-AMSCWC Fresh Caustic Recycle Flow Rate
- m. C-AMSCWC Mixing Control Valve Delta P
- n. Caustic Settler Coalescer Delta P
- o. Caustic Settler Level
- p. Make-up Caustic to Caustic Settler Flow Rate
- q. Make-up Caustic to Caustic Settler Temperature
- r. CAMS Water Wash Drum Level
- s. Water to CAMS Water Wash Drum Flow Rate
- t. Sand Filter Delta P
- u. Sand Filter Level
- v. Sand Filter Effluent Flow Rate
- w. Sand Filter Effluent Temperature
- x. Sodium Phenate Tank Level
- y. Sodium Phenate Tank Temperature
- z. Sodium Phenate Pump Discharge Flow Rate
- aa. Delay Pot Feed Flow Rate
- bb. Delay Pot Feed Density
- cc. Tank Skimming Settler Level
- dd. Tank Skimming Settler Water Boot Level
- ee. Sulfuric Acid to Delay Pot Flow Rate
- ff. Delay Pot Inlet Line Pressure
- gg. PRU Cooler Inlet Temperature
- hh. PRU Cooler Effluent Temperature

- ii. PRU Cumene Addition Flow Rate
- jj. PRU Recycle Flow Rate
- kk. PRU Recycle pH
- ll. PRU Separator Level
- mm. PRU Separator Organic Phase Conductivity
- nn. PRU Separator Aqueous Phase Flow Rate
- oo. PRU Separator Aqueous Phase Density
- pp. Sprung Phenol Coalescer Water Boot Level
- qq. Sprung Phenol Coalescer Effluent Flow Rate
- rr. Phenolic Waste Water Tank Level
- ss. Phenolic Waste Water Tank Temperature
- tt. Phenolic Waste Water Pump Discharge Flow Rate
- uu. Sulfuric Acid to OEC Flow Rate
- vv. OEC Feed Flow Rate
- ww. OEC Feed pH
- xx. OEC Feed Temperature
- yy. OEC Level
- zz. OEC Bottoms pH
- aaa. OEC Bottoms Phenol
- bbb. Solvent to OEC Flow Rate
- ccc. SCWC Bottoms Level
- ddd. Fresh Caustic to OEC Flow Rate
- eee. SCWC Overhead Pressure
- fff. Recycle Solvent Temperature
- ggg. Make-Up Cumene to Solvent Drum Flow Rate
- hhh. Solvent Drum Level
- iii. Solvent Drum Water Boot Level
- jjj. Combined Waste Water Drum Level
- kkk. Combined Waste Water Flow Rate
- lll. Combined Waste Water pH
- mmm. Process Water Drum Level
- nnn. Process Water Drum Pressure
- ooo. Process Water Pump Discharge Flow Rate

7.1.2 Stream Information (All Flows)

- a. Orifice Flow Rate (Raw and Corrected)
- b. Mass Flow Meter Flow Rate
- c. Mass Flow Meter Density
- d. Flow Meter Temperature
- e. Flow Meter Pressure

7.1.3 Calculated Values

- a. C-AMSCWC Jetting Velocity
- b. OEC Jetting Velocity

c. SCWC Jetting Velocity

7.2 Calculations**7.2.1 C-AMSCWC Jetting Velocity**

$$\begin{aligned} \text{CAMSCWC Jetting Velocity, m/hr} \\ = \frac{\text{CAMSCWC Combined Feed Rate, m}^3/\text{hr}}{\text{CAMSCWC Tray Open Area} = \# \text{ Holes} \times \text{Hole Area, m}^2} \end{aligned}$$

7.2.2 OEC Jetting Velocity

$$\text{OEC Jetting Velocity, m/hr} = \frac{\text{OEC Solvent Rate Rate, m}^3/\text{hr}}{\text{OEC Tray Open Area} = \# \text{ Holes} \times \text{Hole Area, m}^2}$$

7.2.3 SCWC Jetting Velocity

$$\text{SCWC Jetting Velocity, m/hr} = \frac{\text{OEC Solvent Rate, m}^3/\text{hr}}{\text{SCWC Tray Open Area} = \# \text{ Holes} \times \text{Hole Area, m}^2}$$

B. Evaluation and Optimization**1. Oxidation****1.1 Air Usage**

The object of the oxidation section is to produce the CHP required to make the required phenol and acetone, with minimum by-products (maximum cumene efficiency) **and** at the safest possible conditions. There is a simple algorithm for this that is described in the following paragraphs.

Given:

- a. A certain CHP production rate is required from an oxidizer (tons/hr of CHP).
- b. A safe O₂ concentration of the spent air is 5-7% (vol/vol). 5% will be used in the following examples. 7% gives a marginally higher yield compared to 5%, but cost more in utilities and generates more waste water via the charcoal adsorber regenerations.
- c. The outlet concentration of the oxidizer in question must remain constant **and** at its normal (design) level. This is typically required to minimize by-products and to ensure that the concentration section's ability to remove cumene from the oxidate is not exceeded.

- d. The temperature required to accomplish the above should be the minimum necessary.

Proceed as follows:

Perform an oxygen balance around the oxidizer in question when it is performing according to the outlines given above.

Since one mol of oxygen is required for each mol of CHP, and there are (1000 kg ton / 152 kg/kmol) = 6.6 kmols CHP per ton of CHP, it follows that the amount of air required (theoretical) for the formation of one ton of CHP will be:

$$6.6 \text{ kmols/ton} \times 22.4 \text{ Nm}^3/\text{kmol} = 147 \text{ Nm}^3 \text{ Oxygen/ton of CHP.}$$

To take things further, the following is noted:

$$\text{Air} = 21 \text{ vol\% oxygen (approx.)}$$

Air usage: Only 16 Nm³ of each 21 Nm³ of O₂ in each 100 Nm³ of air is used.

Therefore, for one ton of CHP production, the following **air** is required:

$$147 \text{ Nm}^3 \times 100/16 = 920 \text{ Nm}^3 \text{ of air per ton of CHP produced}$$

i.e. at a production rate of 100,000 MTA (220 mm lb/yr) of phenol, the (**theoretical!**) air requirement would be about 18,600 Nm³/hr. In an actual unit, the requirement would be more (about +3-4%), due to by-product formation, but the calculation would be essentially identical.

1.2 Oxidizer Temperature

The optimal oxidizer temperature is the lowest temperature at which CHP can be safely produced at the desired acetone and phenol production rate. The UOP-recommended minimum oxidizer temperature is 80 deg C. At an oxidizer temperature below 80 deg C UOP recommends to stop air to the oxidizer for safety. Low oxidizer temperature can result in a low reaction rate which could create a high spent air oxygen concentration which can potentially create an explosive atmosphere. For optimal oxidizer performance, at a given production rate, the oxidizer level should be at the highest value, within the design of the unit, that can achieve the lowest oxidizer temperature, without going below 80 deg C.

Finding the minimum temperature for the required CHP production rate usually done by a trial and error method which keeps the vent gas O₂ concentration out of the explosive range. The LEL for oxygen in cumene is 11 mol% oxygen. UOP specifies a shutdown at 9.5 mol% O₂ in the spent air and a high alarm at 8 mol% O₂. UOP recommends to remain below the alarm point to ensure safety. The following scenario is detailed to make the procedure clear.

The design rate of this hypothetical plant is 100,000 MTA (220 mm lb/yr). From the discussion above, the required air rate (theoretical) for such a production would be about 18,600 Nm³/hr, or 9,300 Nm³/hr to each of the two (hypothetical) oxidizers, operating in series. Assume, for the sake of this discussion, that the normal operating oxidizer temperatures, by design, are 86°C (187°F) and 83°C (181°F). When operating at design rates, the normal operating oxidizer temperatures, by design and detailed in the EDI, should initially be used.

However, start-of-production is typically done at a reduced rates, thereby requiring a reduced oxidizer temperature. To start production at 70% rates we must determine the new target oxidizer temperatures. We suspect that the required temperatures will be below the design temperatures. What is the safest and most efficient way to proceed?

ANSWER: Based on the constraints we have set for ourselves, the logical way to proceed is as follows:

Raise the oxidizer temperature to some value, which makes sense. A reasonable method to estimate the new oxidizer temperatures is to decrease the oxidizer temperature by 1°C (2°F) for every 10% of capacity under design that the plant is operating, and use this as a first guess to set the operating temperature. Since this hypothetical plant is operating at 70% of rated capacity, the temperatures to try for a first operation would be 83°C (181°F) and 80°C (176°F). These are only initial estimates. Final tuning of the required temperatures for the necessary CHP production would be carried out after the necessary laboratory analyses were performed.

1.3 Oxidizer Level

Note that in the example above the oxidizer 2 temperature is at the UOP-recommended minimum value of 80 deg C. If the oxidizer 2 temperature needs to be further decreased to achieve the 70% production rate then the oxidizer 2 level should be decreased. A decrease in level will allow the oxidizer 2 temperature to be increased (remain at 80 deg C) while achieving the desired production target.

When the unit is ready to increase production rates to 100%, the first step is to increase the oxidizer temperature, following the temperature rule of + 1 deg C / +10% in capacity. After the oxidizer temperature is increased then the air flow rate to the oxidizers can be increased linearly with the desired production rate increase. As operating conditions are changed watch the specific gravity analyzer on the oxidizer circulation pumps which are used to indicate CHP concentration to observe if it changes. If the CHP value changes from the desired value adjust operating conditions accordingly.

The production rate of crude product, set by the flow control valve feeding the decomposer, can be increased as well to the new production rate.

The items to watch while doing all the above are:

- a. Oxidizer CHP Concentration (Approximated by Online Specific Gravity Analyzer)
- b. Spent Air Oxygen Concentration (For Safety)

NOTE: The above calculations have not taken into account any by-product formation or other possible destinations for the air. Since vent gas O₂ concentrations will vary from unit to unit and even in different operations in the same unit, it must be understood that the above is meant only as a guideline to writing up the correct instructions for the optimization of the oxidizer operations. The air rates in the UOP EDI do take into account byproduct formation and can be referenced for initial actual air rate values.

1.4 Optimal Oxygen Concentration in the Spent Air

The optimal oxygen concentration in the spent air is the value that results in no possibility of an explosive mixture at the oxidizer outlet. Maintaining a low oxygen concentration for safety is balanced by the impact to yield at low spent oxygen concentrations. The lower the oxygen concentration in the spent air, the more DMPC that is formed in the oxidizers, increasing by-product AMS formation. The design value of 5-7% vol% O₂ in the oxidizer spent air is a good compromise between the explosive limits and optimal CHP yield. Oxygen concentrations below 5 vol% tend to form more DMPC, which results in more AMS and an overall lower phenol yield.

Please note that at lower oxidizer temperatures the explosive limit, LEL decreases. UOP recommends air not be charged to the oxidizers if the oxidizer temperature decreases below 80°C. An automatic shutdown is provided to prevent feeding air at low oxidizer temperature thereby creating an explosive mixture.

2. Concentration

2.1 Flash Columns

The objective of the concentration section is to produce concentrated CHP (80-84 wt%) at the safest possible conditions. In this case, with the concentration of the product (flash column bottoms) essentially fixed, safe operations are defined as running at the lowest temperature possible to produce the product, and minimizing the recycle of CHP to the oxidation section.

Given that the CHP concentration to the oxidation section is usually fixed at the design value (this is true except in cases of turndown over long periods of time, where it might be better to decrease the terminal oxidizer CHP concentration), the operation of the concentration section is optimized when the required CHP concentration at the bottom of the flash column is delivered at the lowest (safest) temperature possible at all points in the system (flash zone, preflash column tubs, flash column vaporizer, flash column bottoms).

The corollary to this is that the CHP in the overhead stream should be limited to less than 4 wt% typically. Excess CHP in the recycle cumene should be avoided as it will impact the yields when at high concentrations. Operate the flash columns with at least the design reflux flow rate, but not excessive, to maintain this specification. CHP concentrations in the recycle cumene of less than 1wt% have been consistently demonstrated in existing units at design rates.

2.2 Vacuum System

The vacuum system is critical to the operation of the concentration section. Because the steam-jet ejectors are critical to the multi-stage vacuum system in the concentration section they are discussed in more detail below. For liquid ring vacuum pump operation, please refer to vendor instructions.

The dimensions of an ejector are such that any given ejector will only operate over a relatively limited range. The effects of operating condition changes can be summarized as follows:

- a. Steam pressure must be maintained quite close to that for which the equipment was designed. If the steam pressure greatly exceeds that for which the nozzle was designed, the quantity of steam discharging into the diffuser will be greater than that which can pass through the diffuser, and steam will back into the suction header.
- b. Wet steam will cause random fluctuation in ejector performance and will erode the nozzle and diffuser.
- c. If the discharge pressure increases above design, the probability of reverse flow increases. This problem is only possible if the discharge is obstructed. On multi-stage ejector systems, high steam pressure will result in higher interstage pressure, immediately affecting the performance of the first stage.
- d. A decrease in load (mass rate of process vapor to the ejector) will result in a somewhat higher vacuum being obtained. If the load is increased above design, the vacuum will fall off quite dramatically.
- e. A moderate change in the temperature at which the steam is condensed in the inter- and after-condensers will generally have a minor effect on the vacuum. However, a higher temperature than design will substantially reduce

the load at which the ejector system breaks down since an increase in condensate temperature increases the interstage pressure.

To ensure flexibility and to conserve utilities, an ejector system will generally be constructed using two parallel sets of unequal size. The vapors drawn to the jets consist of air from leaks and light gases and vapors from the process. As some of the light gases and vapors will condense in the inter-condenser, the first stage ejectors can be heavily loaded under conditions which only lightly load the second stage ejectors. For this the reason there is flexibility in the capacity of each stage.

3. Decomposition

The objective of the decomposition section is to efficiently and safely decompose the product CHP from the concentration section into phenol and acetone at the highest selectivity possible, thus minimizing by-products. Due to the nonzero possibility of incidents in this section, deviations from design operational parameters are strongly discouraged. The operating conditions in the decomposer and dehydrator are such as to operate essentially on the limit of the reaction chemistry involved. The fact that there is a non-zero amount of CHP in the circulating decomposer liquid attests to this. Operating the decomposer with more than 1.5 wt% CHP in the circulating loop will create instability in the reaction that can cause the temperature control to oscillate, becoming unstable. This same situation can also occur if the water concentration as measured in the crude product increases above 2 wt%. Unstable operation can result in a loss of reaction and a sudden increase in CHP concentration, an undesirable and unsafe situation.

The result is that further optimization of the decomposer and dehydrator operations is not, in UOP's opinion, worth the risk of operating even closer to unstable conditions. The unit should be able to meet 80+ mole% AMS yield given the following operating conditions.

- a. Decomposer Temperature: 60 deg C
- b. Decomposer Pressure: 0.70 kg/cm²(g)
- c. Decomposer Sulfuric Acid Concentration: 40 wt ppm
- d. Decomposer CHP Concentration: 1 – 1.5 wt%
- e. Calorimeter First Stage Delta T: 7 – 10 deg C
- f. Crude Product DCP Concentration: 300 – 900 wt ppm; toward the upper end of the limit is recommended as limited by o-cresol in phenol product

The dehydrator temperature is recommended to be maintained at the minimum value to achieve the above crude product DCP concentration without going below 120 deg C. Below 120 deg C and the dehydration reaction rate becomes sufficiently slow that the target DCP in the crude product cannot be easily controlled. As DCP is allowed to increase in the crude product the concentration of o-cresol in the phenol product will increase. The limiting DCP concentration in the crude product is the maximum value, within 900 wt ppm, as allowed by the o-cresol concentration in the

phenol product. Licensees have demonstrated operation at 700 – 900 wt ppm DCP while still easily achieving their o-cresol specification in the phenol product.

4. Neutralization

The purpose of the Neutralization Section is to remove the sulfuric acid from the crude product, making it suitable for fractionation into high purity phenol and acetone. The removal of sulfuric acid from the crude product is necessary from both a corrosion standpoint and from a product quality standpoint. Optimization of this section is primarily economic.

By economic optimization, the actions and expense we are seeking to minimize is minimize the injection rate to that which produces a pH = 2.3-2.7. The ratio controller should be used maintain the pH within range. Adjustments to the ratio should be made based on regular laboratory analysis of the pH.

5. Fractionation

The emphasis on process control to enhance normal operations to achieve high product yield continues in the Fractionation Section. In addition, product purity is now an equally important concern, with expected acetone and phenol purities in excess of 99.6 wt% and 99.9 wt%, respectively. Still, the importance of safety should not be forgotten. The following points with regard to the fractionation section should be kept in mind at all times by the operating staff:

- a. Phenol is highly toxic, is readily absorbed by the skin, and can be fatal with only moderate exposure. When handling phenol, use extreme caution and follow recommended safety procedures.
- b. Much of the fractionation section is under vacuum, making possible the formation of dangerous hydrocarbon and air mixtures through difficult-to-detect equipment leaks.
- c. Phenol and residue are solids at room temperature. Consequently, all piping and equipment that handles crude phenol, phenol, or residue is at elevated temperature and/or steam-traced. Operators should be aware of the potential for line blockage that exists from solidification of these materials in the piping.

Four of the six columns in the fractionation section operate under a vacuum drawn by the fractionation section vacuum system. The Finished Acetone Column operates with its own vacuum system. Refer to the discussion of steam-jet ejector operation in the concentration section as required.

5.1 Crude Acetone Column

The primary goal in operating the crude acetone column (CAC) is to maximize the removal of acetone from the fractionation feed. Most of the acetone that is **not** recovered to the CAC overhead is **never** recovered. In addition, some of the organic acids in the feed should also be stripped out. Some recommended CAC stream specifications are the following:

- a. Column Bottoms: 0.05 wt% acetone or less, 0.02 wt%, if possible.
- b. Column Overhead: No detectable phenol or AMS corresponding to about 12-14% cumene in the overhead
- c. The maximum cumene concentration in the overhead represents the approximate concentration at which phenol begins to show up in the overhead.

The following process parameters have an effect on the quality and efficiency of the separation in the CAC:

5.1.1 Tray Temperature and Vapor Draw Flow Rate

The TI on the upper tray is the general indication of the overhead composition in this column. An increase in the temperature indicates a decrease in the vapor draw flow rate is required to maintain the overhead composition, all other conditions being equal. It is difficult to specify a normal temperature for this indicator as it is impacted by other parameters.

Once a vapor flow rate is determined, for a given feed flow rate and composition, the feed forward flow controller can be set to adjust the vapor draw flow rate for a given column feed flow rate.

5.1.2 Reboiler Vapor Return Temperature

This temperature should be approximately 130-160°C (300-320°F), depending on the live water/steam injection rate. This will ensure that organic acids are sent to the top of the column and removed with overhead material.

5.1.3 Stripping Steam Flow Rate

The design rate of stripping steam to the bottom of the CAC is dependent on the water concentration of the crude product. Normally, the fractionation feed will contain sufficient water to perform the stripping function. In some circumstances, additional steam may be necessary to adequately boost the relative volatility of acetone. When used, frequent adjustments of the steam rate should be avoided due to the large and immediate impact of the steam on the heat and water balances in the column. The steam flow rate is about 8-10% of the CAC bottoms flow rate.

5.1.4 Reflux Rate

The optimum separation of acetone is achieved at a reflux:feed ratio of approximately 0.6 wt:wt. Though the reflux rate cannot be independently varied, other column conditions, primarily the reboiler heat input should be set to maintain the reflux roughly equivalent to the feed rate.

5.1.5 Column Pressure

The overhead pressure controller regulates the rate of receiver vent gas to the FAC. The column pressure should be maintained as low and as constant as possible. The optimum pressure for acetone separation is the minimum pressure achievable while still being able to condense the overhead vapor with cooling water and is generally set at 0.7 – 0.8 kg/cm²(g) (10 psig) at the pressure controller in the overhead vapor line. Once this minimum pressure is set following start-up, there is no need to make further pressure adjustments. At a pressure of 1.1 kg/cm²(g) (15.5 psig) the overhead pressure controller will automatically stop heat input to the reboiler. This shutdown prevents the pressure from exceeding the design pressure limitation of the column. If pressure control becomes erratic, check the overhead condenser for a buildup of non-condensables.

A concern in operating the CAC is fouling of the reboiler tubes. Fouled reboiler tubes hinder the column's ability to achieve the desired acetone separation. Should fouling pose a problem, the fractionation section must be shutdown periodically for a boil out of the reboiler with water. This typically can be done in a 12-24 hour period. The fractionation feed tank will have to accumulate the crude product for this period. The decline in reboiler performance should not be so great that the boil out can not be planned ahead of time.

The use of a direct neutralization section instead of ion exchange or other competing neutralization technologies reduces the requirement for boil outs. The sprung phenol remains a potential source of sodium sulfate in the fractionation section. Sodium sulfate would only occur in the fractionation feed if the PRU separator is upset and entrained aqueous phase is recycled with the sprung phenol.

5.2 Finished Acetone Column

The primary objective of the finished acetone column (FAC) is to minimize the quantity of acetone in the column bottoms while maintaining less than the specified concentration of water in the acetone product. Another major concern is the organic acid, benzene, methanol and aldehydes in the FAC feedstock, as both must meet specification limits in the acetone product. The following operating parameters are of greatest importance in achieving necessary product specifications while maximizing product yield:

5.2.1 Differential Tray Temperature Control

This temperature differential automatically controls the flow rate of acetone product from the column. An increase in the setpoint temperature difference will increase the acetone product flow rate, which will draw more water/cumene up the column.

5.2.2 Reflux Rate

The reflux to the FAC will be strictly dependent on the concentration of water desired in the acetone product. A typical specification is 0.3 wt% water or less. A moisture analyzer on the side-draw product allows continuous monitoring of the water concentration in the column side-draw product. An increase in the reflux:feed ratio will generally reduce the water concentration of the acetone and vice versa.

However, too high a reflux rate can push acetone to the bottom of the column, increasing its loss rate. In addition to pushing acetone too far down the column, excessive reflux will dry-out the reaction zone in the FAC. Some water is required on the trays in the reaction zone to promote the aldol condensation reaction. Column stability is critical to maintaining the proper water concentration on the trays in the reaction zone to reliably achieve the aldehyde specification of the acetone product.

The reflux is adjusted by the heat input into the column. For a constant feed flow rate, increasing the heat input will increase the reflux. The stab-in reboiler heat input is cascaded to a heat input calculator from the vapor feed to the column via a feed forward flow controller and the acetone product rate. This advanced control system is designed to ensure column stability for optimum reaction zone and therefore acetone product quality.

5.2.3 Column Pressure

The overhead pressure controller regulates the FAC vacuum system. To keep the column base temperatures low enough to avoid decomposition of the aldol, the column is operated under a moderate vacuum. The FAC should be operated at the lowest pressure allowed by the vent condenser cooling water temperature, about 465 mm HgA.

5.2.4 Caustic Addition

The dilute sodium hydroxide solution that is injected to the column above the feed tray converts feedstock aldehydes and organic acids to heavy aldol and water-soluble salts, respectively. In this way, most of these potential contaminants are shifted to the bottom of the column, reducing their concentrations in the overhead product. The permanganate fade time of the acetone product indicates its aldehyde concentration. A fade time of 2-4 hours, or longer, is a standard specification for the acetone product. At caustic injection rates sufficient to meet the fade time specification the acidity of the acetone product is generally not a problem. 1 kg/hr of

dry caustic is required per 1000 kg/hr of acetone product. A 1 wt% solution of caustic is used to ensure there is an aqueous phase in the column to perform the reaction. Typically, this results in about 0.5 wt% caustic in the aqueous bottoms stream.

The aldol condensation reaction does not proceed to 100% across the reaction zone. For this reason there is a recycle of aldehydes from the FAC receiver to below the reaction zone. This recycle allows another opportunity for the aldehydes to react. The aldehyde recycle rate can be varied to impact the acetone product aldehyde specification and permanganate fade time.

Changes in the FAC feed quality can have major effects on the FAC operation. Both the FAC and CAC must be carefully monitored to achieve stable operation of the FAC. Mostly, if the cumene concentration of the feed is varying substantially, control of the FAC Bottoms Drum phase separation will be difficult. Increased heavies accumulation or fouling of the FAC bottoms strainers and/or drum can be countered by a lower FAC bottoms temperature, if possible and/or increased cumene to the FAC. Increased cumene to the FAC will decrease the residence time of the aldol in the hot parts of the column limiting its further reaction to a fouling heavy.

Should excessive aldehydes in the FAC feed be a problem, it may be the result of aldehyde accumulation in the CAC overhead system. Accumulated aldehydes can be eliminated by allowing the CAC receiver to warm up, venting the aldehydes to the relief header.

Methanol and benzene are typically below detectable limits in the FAC feed. Misoperation of the decomposer or operating for extended periods with the decomposer in start-up mode can generate high methanol or benzene. If methanol or benzene are found to be high in the acetone product decreasing the water concentration of the acetone product can limit their contamination of the acetone product. Methanol and benzene will tend to follow water in the FAC therefore operating with more-dry acetone can reduce the methanol or benzene concentrations sufficiently to meet product specifications. Careful attention should be given to the aldehydes and permanganate fade time during this time as the reaction zone will not be operating at optimum conditions while benzene or methanol are rejected from the FAC.

5.3 C-AMS Column

Because of the nearly 100% yield of cumene from AMS in the hydrogenation section, the value of AMS is essentially equal to that of cumene. The primary objective of the Cumene-AMS Column (C-AMSC) is to maximize AMS recovery in the column overhead. At the same time, however, it is very important that design concentrations of phenol in the column overhead not be exceeded. Too much

phenol in the overhead organic phase may overload the phenol extraction capability of the cumene-AMS caustic wash column. Eventually, unextracted phenol can get back to the oxidizers where it acts as a reaction inhibitor. The following operating factors are critical to the performance of the C-AMSC:

5.3.1 Tray Temperature and TDIC

The TDIC measures the difference between tray 8 and the overhead vapor temperature. The TDIC is cascaded to the reflux flow controller. When the setpoint temperature is decreased, the reflux is automatically increased and, generally, the phenol concentration of the overhead material will be reduced. Typically, the TDIC temperature should be in the range of 28-31°C (50-56°F). To avoid overloading the cumene-AMS caustic wash column with phenol, the phenol concentration of the overhead organic phase should be 10 wt% phenol or less. The phenol analyzer (if provided) on the reflux can be used to monitor the phenol concentration in the column overhead. Due to a minimum boiling azeotrope between AMS and phenol, a certain minimum fraction of phenol (4 wt%) in the overhead of the column is necessary to achieve maximum AMS recovery. Typically, a reflux:feed ratio of 1.3 is designed.

5.3.2 Heat Input

The steam flow rate to the C-AMSC reboiler is straight flow control, so it is essentially a heat input controller. This scheme is intended to keep a stable heat input to the column. At typical bottoms temperatures of 175-180°C (347-356°F), the AMS and cumene concentration of the bottoms material should be less than 100 wt ppm and 10 wt ppm, respectively, but the typical GC analysis goes to 0.1 wt% (100 wt ppm), so no measurable amount of AMS or cumene should be observed.

5.4 Chemical Treating Reactor

The primary concern in chemical treating reactor (CTR) operation is that sufficient conversion of C-AMSC bottoms carbonyl impurities, mostly hydroxyacetone (HDA or acetol), is achieved to meet specifications on carbonyl concentration in the final phenol product. If not converted to heavies, HDA will follow phenol through the crude phenol column and phenol rectifier column, eventually contaminating the phenol product. To optimize reactor performance, the following process parameters are important:

5.4.1 Chemical Injection Rate

The CTR chemical injection rate is set at the ratio needed to remove the carbonyls in the C-AMSC bottoms. This ratio is relatively constant, but can vary based on fractionation. 1100 wt ppm is the design value, but should be optimized at 25mol% over stoichiometric consumption after lab samples are taken.

5.4.2 Residence Time

The residence time is not an independent variable, but will affect the quantity of chemical required. A longer residence time at reduced capacity should require a lower ratio of diamine to feed.

5.4.3 Temperature

Temperature is also a dependent variable and is fixed by the fractionation section operating pressure and composition. A higher temperature is favorable and will aid in the reaction rate, but is not favorable for maximizing phenol yields. There is no significant change in CTR reaction rate in the operating range, 175-180°C (347-356°F).

5.5 Crude Phenol Column and Residue Stripper

The primary goal in these two columns is to maximize phenol recovery from the feed; i.e., minimize the phenol concentration of the residue stripper bottoms material at the minimum bottoms temperature.

In addition, maximum recovery of residual DMPC not dehydrated in the decomposer/dehydrator is desired as most of it should dehydrate to AMS in the C-AMSC bottoms. Unfortunately, AP has a boiling point similar to that of DMPC, and is therefore difficult to separate from DMPC by distillation. Some DMPC must be sacrificed to the RS bottoms in order to keep the overhead AP concentration below the suggested limit of 0.1 wt%. If the AP concentration is not strictly controlled in the CPC column, it will accumulate in the C-AMS column-CTR-CPC-IXRT-phenol rectifier column system because of the rectifier bottoms-to-CPC recycle. Eventually, AP will overload the phenol rectifier and result in unacceptable AP concentrations in the phenol product.

The following column operating variables affect CPC/RS performance:

5.5.1 Reboilers Vapor Return Temperatures

Reboiler vapor return temperature is maintained by cascade control to the condensate flow from the reboiler. Typically, this temperature is set between 165-188°C (330-370°F). At these conditions, the phenol concentration of the bottoms residue is reduced to its lowest practical concentration, determined by the maximum boiling azeotrope between acetophenone and phenol. As this azeotropic boiling point is exceeded, the AP concentration of the column overhead material will quickly increase.

5.5.2 Reflux Rate

The columns are designed for a reflux rate giving a reflux:feed ratio of about 1.3. Minor adjustments to the reflux rate can be made to keep the AP concentration of the distillate low.

5.5.3 Residue Stripper Heat Input

It is possible to have the residue stripper overloaded with phenol. This can occur if the CPC reboiler allows excessive phenol to the RS. The RS reboiler return temperature cannot be attained and maximizes the heat input. The RS reboiler is oversized for a fouling case and will then add more heat than the trays can handle and the column will flood. The RS condenser can then also flood and be overloaded resulting in phenol entrainment into the fractionation vacuum system. These problems can all occur simultaneously. The operator should be aware that the first action is put the RS reboiler heat input off of cascade to the TIC and reduce the condensate rate to design. The RS bottoms flow should then be routed back to the fractionation feed tank until the phenol is low again. The CPC reboiler then needs to be adjusted.

5.6 Ion Exchange Resin Treaters

The Ion Exchange Resin Treaters will convert remaining impurities in the phenol to heavy compounds by reaction with phenol. Carbonyls will be converted into 2-MBF and other compounds. The 2-MBF will be converted to a heavy hydrocarbon compound. Trace AMS will be converted to a heavy, cumylphenol or AMS dimer compounds.

Any unreacted amine or amine fragment will be ion exchanged on the resin and result in a loss in acid activity. Eventually amine contamination can cause the catalytic activity of the resin to fall to a level that 2-MBF will not completely react to a heavy compound. Water upsets can also cause the catalytic activity to be temporarily reduced. The 2-MBF concentration between the treaters is monitored because the 2-MBF will be the first compound to breakthrough due to loss of acid activity. When 2-MBF breakthrough occurs, the resin bed must be taken off-line and reloaded.

When 2-MBF starts to break-through, the inlet temperature to the resin treaters can be increased to recover some resin activity. However, increased temperatures will not gain significant resin life and a reload should be planned soon.

5.7 Phenol Rectifier

The phenol rectifier (PR) bottoms material is recycled back to the CPC column. This provides some liberty in the quantity of phenol that is allowed in the rectifier bottoms because this phenol will be fully recovered. The quantity of bottoms purge should always be sufficient to guarantee that overhead phenol product specifications are

met. However, it is not sensible to excessively surpass the minimum product quality requirements, as the higher purge rate of bottoms material to the CPC consumes excessive energy. The overhead drag rate is set to recycle any light-boiling compounds back to the C-AMSC for removal. The following process guidelines should put the column close to its optimum operating condition:

5.7.1 Bottoms and Overhead Purge Rate

The bottoms purge rate is set by local FIC control. Typically, a purge:feed ratio of 0.05-0.1 is sufficient to remove all of the heavy impurities. This will result in a bottoms purge material containing approximately 90% phenol and 10% heavies.

The overhead purge rate should initially be set to a purge:feed ratio of 0.05. This flow can be optimized as with the bottoms flow to minimize the recycled phenol. The overhead purge will need to be increased to keep the phenol product on-specification when a resin treater is reloaded and excessive water is fed to the PR.

There is no temperature control on the column as essentially pure phenol is present throughout the column. The heat input to the column is controlled by the bottoms level. Consequently, the column's composition profile and, hence, product quality are strictly dependent on the purge rates and pressure profile across the column. At design conditions, the proper heat input will give a bottoms temperature near 170°C (340°F).

5.7.2 Reflux Rate

The reflux rate is set by the operator with a direct flow controller. The column is designed for a reflux:feed ratio of 0.72. This reflux ratio should be established and maintained until operating experience suggests a change.

5.7.3 Column pressure

A pressure instrument on the overhead vapor line controls the rate of vent gas to the fractionation vacuum header. The column should be set and maintained at design vacuum, 280 mm HgA. The low operating pressure allows low operating temperatures, avoiding condensation reactions that consume phenol and create impurities.

6. Hüls MSHP (AMS Hydrogenation)

The AMS hydrogenation section is the only hydrogen atmosphere process in the UOP Phenol Process. Consequently, with respect to safety, this section should be held in a different regard by the operating staff. Hydrogen is an invisible, odorless, yet highly flammable gas. The operators should always be on the alert for signs of

hydrogen escaping to the atmosphere, such as unusual rates of hydrogen consumption or audible hissing of gas leaks at valve packings, flanges, gauges, etc. Hydrogen is also expensive, so less hazardous leaks should also be minimized, such as through safety valves or compressor seal vents.

Maintaining sufficient reaction severity to achieve full conversion of AMS to cumene in the hydrogenation reactor is important. If AMS is recycled to the oxidizers two primary reactions will take place:

- 1) AMS will form the byproduct hydroxyacetone (HDA) which combines with phenol in the phenol treating reactor to form 2-MBF, a non-recoverable byproduct. In short, passage of AMS unreacted through the hydrogenation unit reduces phenol yield.
- 2) AMS also will convert to form organic acids in the oxidizers. 300-400 wt ppm of AMS in the oxidizer combined feed will increase the oxidizer acidity to an unacceptable concentration. AMS concentrations above 600 wt ppm in the hydrogenation product will reduce the overall phenol yield a measurable amount.

As the hydrogenation catalyst slowly deactivates through its expected life, the severity of reaction conditions must be slowly increased to maintain full AMS conversion. The concentration of AMS in the reactor product, measured in terms of Bromine Index (BI), should normally be <100 mg/100g (<600 wt ppm AMS). The AMS concentration should never exceed a BI of 100 mg/100g. The following process variables determine the reaction severity and, therefore, product quality:

6.1 Reactor Pressure

The reaction pressure is set by controlling the back pressure control valve upstream of the product separator. The product separator pressure floats on the relief header. Initially, the reactor pressure should be set at approximately 2.0 kg/cm²(g) (30 psig) to ensure that the reactor effluent can flow to the product separator. As the catalyst deactivates the reactor pressure should be increased to the maximum of ~21 kg/cm²(g) (300 psig), the actual maximum reactor pressure is a function of hydrogen supply limitations and reactor design. The increasing pressure improve AMS hydrogenation by increasing the partial pressure of hydrogen. As well, increased reactor pressure, to a limited degree, will counter the increased coking tendency that results from operating at increased reactor temperature. Pressure is recommended to be increased to the end-of-run pressure before temperature to achieve maximum catalyst life.

6.2 Reactor Inlet Temperature

After the reactors are operating at near end-of-run pressure, then the reactor inlet temperature can be increased to maintain AMS conversion. A typical start-of-run reactor inlet temperature is 60°C (140°F). As the catalyst continues to deactivate

the reactor inlet temperature can be increased to the design end-of-run inlet temperature of ~87°C (190°F).

6.3 Reactor Delta Temperature

The feed to reactor 1 is diluted with reactor effluent recycle to maintain less than 6 wt% AMS in the combined feed to the reactor. This corresponds to a reactor delta temperature about 30-35°C (54-63°F) and should not be changed throughout the life of the catalyst.

While maintaining sufficient reactor severity is very important in achieving maximum AMS conversion, it is also important not to be too severe. It is possible, at high reactor temperature and/or high pressure, to saturate the aromatic ring in cumene producing isopropylcyclohexane (IPCH). IPCH should be maintained less than 100 wt ppm in the reactor product.

The phenol concentration in the hydrogenation reactor product must be 20 wt-ppm or less. While not significantly impacting the MSHP unit, increased phenol concentrations can severely inhibit the CHP-forming reactions in the downstream oxidizers. As phenol is neither created nor destroyed in the hydrogenation reactor, any phenol present in the product must have entered with the feed. Generally, high phenol in the feed is a problem relating to C-AMS caustic wash column operation. However, hydrogenation section operators should be aware that phenol can accumulate in a water layer at the bottom of the hydrogenation feed tank. This aqueous layer should be drained daily to ensure that it is not charged to the AMS hydrogenation reactors.

Butylbenzenes in the cumene feed are unreactive in the phenol unit. They will accumulate in the recycle cumene. Eventually, a cumene purge back to the Q-Max unit might be required to fractionate out the butylbenzenes. The purge line is found at the product pump discharge.

7. Phenol Recovery

The purpose of the phenol recovery section is to recover phenol from process streams where phenol is considered undesirable (C-AMS column overhead, circulating solvent, waste water). Phenol is also recovered for environmental concerns, for its value and to ensure that the contaminant does not cause undesirable side effects (phenol in particular can cause problems with depressed reaction rates in the oxidation section).

Minimization of phenolic waste water in this section is a function of the amount of caustic used to wash the C-AMS column net overheads and the circulating solvent. The specification for the caustic rate used in the washing of these columns is that

the bottoms (spent caustic) of the SCWC column should be more than 50 percent spent, that is, should not contain less than 10 wt% strong base as a fraction of total base. The C-AMSCWS bottoms should be controlled to 100-120% spent. Sodium phenate streams can become supersaturated with phenol.

The key to good performance is a low caustic/hydrocarbon ratio. To achieve this low ratio the column must be constructed well and carefully checked for leaks prior to commissioning. The fewer tray leaks the better. In this case, the following procedure can be implemented to optimize operations. Please note that this procedure will only be valid during a period of steady net flow to the column in question. start-up, shutdown or during an emergency are definitely not the times to optimize a column. An instrumentation engineer is required to help in the described process.

- 1) Put the caustic to the column being optimized on straight flow control (off of cascaded FFIC). Maintain the same caustic rate as it was when in cascade.
- 2) With the column lined out and all flows steady, have an instrument engineer reduce the ratio factor which determines the ratio of caustic to hydrocarbon (this is currently the ratio of the two flow rate meter maximums) by no more than 5%, i.e. if the ratio is 0.05, do not reduce it to less than 0.0475.
- 3) Decrease the flow to the new flow rate indicated by the ratio controller, line out, and put the controller in cascade at the new flow rate.
- 4) Check the hydrocarbon outlet (either C-AMS column overhead or circulating solvent) at the laboratory. As well, check the condition (% spent) of the caustic at the bottom of the column being optimized to ensure that it does not become so spent that column performance decreases.
- 5) It is not recommended to change the total flow rate through the column-this is set by design to a specified jetting velocity and is a function of the number of holes in each tray and the bulk liquid temperature. However, if the trays are suspected of possibly leaking through (that this does not occur is vitally important) then the total flow rate can be increased and the effects noted. If there is an increase in extraction efficiency, then the trays are probably leaking and should be inspected and repaired at the first available opportunity.
- 6) The above procedure can be repeated until the extraction is optimized.

The phenol recovery section is not necessarily optimized but instead is either operating to the design specifications or is not. The pH of the feed to the delay pot should be 4.5-5.0, with a target of 5.0. pH values below this do not improve unit performance and might increase corrosion. pH values above this will drastically increase the phenolic loading in the OEC bottoms water.

The OEC inlet pH should likewise be in the range of 4.5-5.0, but with a target that is more acidic, 4.5. The OEC must be run with very lean solvent (<20 wt ppm phenol) to provide the driving force to reduce the treated waste water to below 150 wt ppm phenol. Phenol concentrations in the waste water below 50 wt ppm are possible, but only with less than 5 wt ppm phenol in the lean solvent to the OEC and good OEC feed pH control.

VIII. Analytical

Information in this section lists laboratory analysis that may be included in the required analytical schedule for the process. Typical operating ranges for some of the analysis may be included.

A. General

Samples should be taken on a regularly scheduled basis by authorized personnel knowledgeable in the procedure. Operating company standards must be consulted in determining the proper personnel protective equipment required for the taking and handling of samples.

The general design and specification of the sample points is provided in the Project Specifications.

Sample cylinders should be received from the laboratory properly cleaned and purged consistent with the analysis being performed, e.g. if analysis for oxygenates are being performed then the sample cylinder must be purged of oxygen prior to use.

All samples leaving the process area for the laboratory should be tagged with the following information: sample description, sample source, analyses requested, date and time sampled.

Analytical test methods and standards by UOP and ASTM are available at www.astm.org.

B. Sample Procedures

Sampling procedures will need to be developed in conformance to the operating company's procedures.

1. Closed Flow-Through To Process Sampling (Liquid or Gas)

Sample points are provided that allow a portion of the process stream to be directed through a closed sampling system. The system is used to contain a portion of the stream and allow its transport to the laboratory. This type of system is used when the material sampled is considered to be toxic or when taking an open sample would compromise the required analysis.

Instructions for the use of the closed sample system are to be provided by the supplier of the system. The general arrangement of these sample points are include on the Piping and Instrument diagram provided for the project.

2. Closed Flow-Through to Vent/Drain Sampling (Liquid or Gas)

When it is impractical to take a sample through a closed sample system and return it to the process flow stream it may directed to the venting system of the unit or be sent into the drain system of the unit. The system is used to contain a portion of the stream and allow its transport to the laboratory. This type of system is used when the material sampled is considered to be toxic or when taking an open sample would compromise the required analysis.

Instructions for the use of the closed sample system are to be provided by the supplier of the system. The general arrangement of these sample points are include on the Piping and Instrument diagram provided for the project.

3. Sample on Vent to Safe Location - Type "DT"

Occasionally analysis is made with gas detection tubes. These devices require a low pressure at their connection to the process in order to provide an accurate reading. After connecting the sample tube to the "DT" connection point on the sample point a small purge of gas is vented through the vent line to atmosphere and the sample is taken following the procedure given by the vendor of the gas detection tube.

Care should be taken so the vent rate is not to high as this may cause a higher flow than intended through the gas detection tube and cause poor analysis accuracy.

4. Special Sampling Procedure(s)

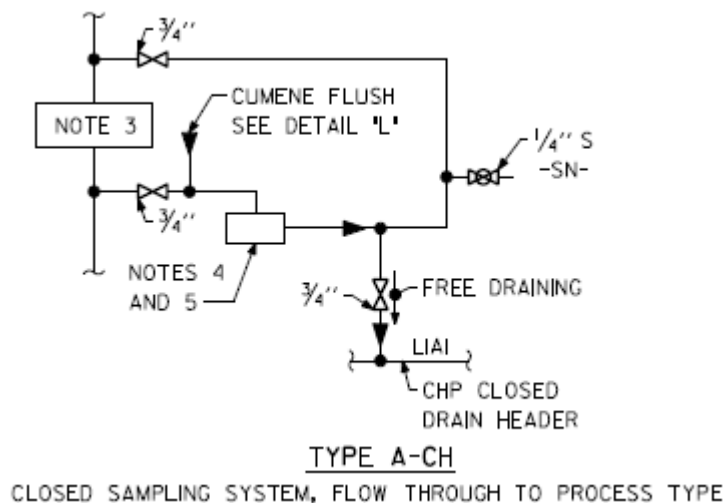
4.1 Sampling of CHP-Containing Streams

Samples containing CHP cannot be taken in a closed sample container because the CHP will continue to react in the container therefore impacting the accuracy of later analysis. To counter the continued reaction of CHP, these samples are 'frozen.' Frozen is a term for neutralizing the sample to prevent further reaction of CHP. There are number of approved agents, NaOH solutions, soda ash, sodium bicarbonate, Hexamethylenediamine (HMDA), and Dytek A (2-MPMD). Sufficient excess of the neutralizing agent is added to a sample container to combine all of the acid in the sample.

The sampling system below in Figure IX-1 shows how sampling for CHP is taken with an open-style sample system so that the CHP-containing liquid can be put directly into the sample container with one of the neutralizing agents described above.

Because CHP can rapidly decompose to phenol CHP should be treated just as if it were phenol and all of the same precautions taken.

Figure IX-1: Flow-Through To Process Sampling System for CHP



4.2 Sampling of Phenol-Containing Streams

The individuals who will be involved in collecting the various stream samples should be fully aware of the nature of the species involved with respect to health hazards, toxicity, etc. During the actual sampling two (2) operators will be required. One operator will perform the sampling while the other will provide any backup assistance.

The following procedural guidelines should be followed during the sampling process.

- 1) The specified personnel protection equipment should be worn. During sampling of phenol streams and any concentrated sulfuric acid streams, as a minimum, a protective facial shield and 14 inch long PVC heavy duty gauntlets should be worn. This is in addition to the prescribed safety hats and boots. PVC jackets should be worn over cotton clothing as an added protection.
- 2) The backup person should also be similarly attired in case any assistance is required.
- 3) Check direction of wind - stay upwind of the sample point if possible.

- 4) Check the safety shower before going to the sample point. **DO NOT** take sample unless the safety shower is working properly. Make sure a path to the safety shower is open and free of any debris or extraneous equipment.
- 5) Once the sampling is completed the sample bottle or cylinder **SHOULD NOT** be allowed to sit around but should be taken to the laboratory immediately.

As always, exercise extreme precaution to avoid spills and skin contact. If spills do occur the area of the spill should be thoroughly flushed down with water. Medical assistance should be obtained if even the slightest doubt exists with regard to having incurred skin contact.

C. Analysis

1. Oxidation

1.1 Cumene Feed

1.1.1 Bromine Index

Bromine index is measured by ASTM D1492 or ASTM D5776. Bromine index is a measure of olefins. As AMS is the only significant olefin in the cumene feed, the bromine index is used measure the concentration of AMS. AMS reacts to heavies and organic acids in the oxidizer which is a yield loss and oxidizer contaminant, respectively.

1.1.2 Composition

Composition is analyzed by ASTM D3760 or ASTM D7057 for Cumene, EB, nPB, ButylBenzene, Benzene, AMS, and non-aromatics. EB and nPB are important feed contaminants for FAC operation and acetone product quality as they will form aldehydes in the phenol unit which co-boil with acetone. Butylbenzene is non-reactive and in large concentrations could build-up in the phenol unit and could need to be dragged out in the MSHP unit. Benzene will be lost to the oxidation vent gas and flash column vacuum system thus representing a yield loss. Non-aromatics can form organic acids in the oxidizer negatively impacting oxidizer acidity and phenol concentration. As well non-aromatics can impact the acetone product permanganate fade time, though this relationship is not yet well understood.

1.1.3 Phenol

Phenol is measured in organic streams by Honeywell QCL-103-FR and QCL-031-FR or QCL-265-FR and QCL-197-FR or ASTM D3160. Phenol in the oxidizer feed is a poison to the auto-catalytic CHP formation reaction and must be minimized.

1.1.4 Total Peroxides

Total peroxides are measured by ASTM D2340. Peroxides in the cumene feed, such as CHP and MHP, are not a concern themselves to the oxidizers, however CHP can readily decompose to phenol which is an oxidizer contaminant.

1.1.5 Distillation Range

Distillation range is measured by ASTM D850 and is an indirect measure of cumene feed purity, specifically heavies.

1.1.6 Acid Wash Color

Acid wash color is measured by ASTM D848 and is another measure of olefins similar to bromine index. Acid wash color is generally no longer used.

1.1.7 Acidity

Acidity is measured by Honeywell QCL-118-FR or ASTM D1613. Acidity is important in both the cumene feed, the oxidizers and the flash columns as a measure of how readily CHP might exothermically decompose to phenol and acetone which is undesirable outside of the decomposer.

1.1.8 pH

The pH of organic streams is measured by Honeywell QCL-145-FR. Measuring the pH of an organic stream is a difficult analysis as the hydrocarbon will plug the pH probe quickly rendering the measurement inaccurate. For most streams water is added to the organic sample and the pH of the water is measured. For crude product, phenol buffers the pH of the water therefore the pH of crude product is measured directly without water addition.

1.1.9 Sulfur

Sulfur is measured by Honeywell QCL-224-FR or ASTM D5453 or D7183 or D4045. Sulfur in the cumene feed, typically propylthiophene is a contaminant further discussed in process principles.

1.1.10 Specific Gravity

The specific gravity of all samples is measured by ASTM D4052 and is used for mass balance flow corrections.

1.1.11 CHP Compatibility

The CHP compatibility test is analyzed by Honeywell V-180-FR and CDL-230-FR. This test adds concentrated CHP to the given sample to observe if a decomposition reaction takes place forming phenol and acetone. A negative result is the desired result indicating that there was no reaction and therefore the sample is not sufficiently acidic so as to decompose concentrated CHP.

1.1.12 Color

All samples are analyzed for Pt/Co color by ASTM D1209. Color is typically an indication of heavies in the sample.

1.1.13 Chlorides

Chlorides are measured by UOP 395 or ASTM D5194 or ASTM D7359 or D5808 or D7536 and their negative impact as acids in the oxidizers is further discussed in Process Principles Section II.

1.2 Feed Wash Column Tray 2 Water

1.2.1 NaOH

Caustic, or 'free caustic' is measured by UOP 209. There should be no caustic detected at this point in the FWC. Presence of caustic indicates carryover of caustic into the organic phase, likely due to insufficient jetting velocity.

1.2.2 Phenol

Phenol in ppm concentrations in the aqueous phase is measured by Honeywell QCL-103-FR and QCL-031-FR or QCL-266-FR or ASTM D1783. The different methods vary by amount of analytical work required. The 103 method is a wet-chemistry method while the 266 method utilizes a SmartChem analyzer. ASTM D1783 is based on the color reaction of phenol with 4-aminoantipyrine. Any other phenolic compounds will react and are reported as phenol. The FWC will scrub some phenol from the recycle cumene. High phenol in the aqueous phase is an indication of high phenol in the organic phase which is a problem for the oxidizers. As well high phenol here could indicate a contaminated water supply.

1.2.3 pH

The pH of all aqueous streams is measured by Standard Method 4500HB. The pH should be the same as the make-up water source which should be 6 – 8. Excessively basic pH is an indication of caustic entrainment up the FWC.

1.2.4 Alkalinity or Acidity to methyl orange at a pH of 4.9 and 20 deg C, meq/L

Alkalinity or acidity is measured by Standard Method 2310B or ASTM D1067.

1.3 FWC Bottoms

1.3.1 NaOH

See 1.2.1 The free caustic concentration at the bottom of the FWC should be 1 – 1.5wt% for proper removal of organic acids from the recycle cumene. The caustic concentration can be decreased as needed during ‘good’ operation with low oxidizer organic acids and phenol or increased with ‘poor’ operation with high oxidizer organic acids and phenol.

1.3.2 Phenol

See 1.2.2.

1.3.3 CHP

CHP in ppm concentrations in the aqueous phase is measured by Honeywell QCL-112-FR, QCL-004-FR and QCL-108-FR. In addition to phenol, CHP can be extracted from the recycle cumene in the FWC. Extracted CHP is a yield loss. Removal of organic acids in the FWC needs to be balanced with CHP losses to determine the economic optimum.

1.3.4 Alkalinity or Acidity to methyl orange at a pH of 4.9 and 20 deg C

See 1.2.4.

1.4 CHP Process Water Break Drum

All analyses are to look for contaminants. There should be no contaminants in the CHP process water.

1.4.1 CHP

See 1.3.3

1.4.2 Phenol

See 1.2.2.

1.4.3 Acetone

Acetone is measured by Honeywell QCL-193-FR and W-197-FR via headspace GC.

1.4.4 pH

See 1.2.3.

1.5 Combined Feed Surge Drum Water Boot

See 1.2. Note that any other feeds to the CFSD besides the FWC can impart some influence on any water collected in the CFSD water boot.

1.6 Combined Feed Pump Discharge

See 1.1. Analysis of the combined feed is the same as the fresh cumene feed. Typically the organic acids, CHP, AMS and phenol in the combined feed will be higher than the fresh cumene feed. Typically the combined feed is monitored more actively than the fresh feed since ultimately the contaminants present here will influence what goes to the oxidizer.

1.7 Compressed Air Scrubber Bottoms

See 1.3. The free caustic concentration at the compressed air scrubber bottoms should be 0.1 wt%.

1.8 Oxidizer #1 and #2 Pump Discharge

1.8.1 CHP

CHP is measured by Honeywell QCL-232-FR and QCL-197-FR or Honeywell QCL-112-FR, QCL-004-FR and QCL-108-FR. Method 232 is by gradient liquid chromatography. Method 112 is a wet-chemistry method using iodometric titration. Since DMPC and AP, below, need to be measured by Honeywell QCL-232-FR and QCL-197-FR, it is typical to measure CHP by the same method. Oxidizer #1 effluent is typically targeted at 12.5wt%. CHP from oxidizer #2 is typically 25wt%.

1.8.2 Dimethylphenylcarbinol (DMPC), Acetophenone (AP)

DMPC and AP are measured by Honeywell QCL-232-FR and QCL-197-FR. DMPC and AP are critical to operating the oxidizer at optimal yield. Low yields are an indication of operating at excessive CHP or temperature which can indicate misoperation or the presence of a contaminant.

1.8.3 Phenol

See 1.1.3. Phenol in oxidizer #1 is typically 5 – 15 wt ppm. Phenol in oxidizer #2 is typically 15 – 30 wt ppm.

1.8.4 pH

See 1.1.8.

1.8.5 Acidity

See 1.1.7. Acidity in oxidizer #1 is typically 20 – 40 wt ppm as benzoic acid. Acidity in oxidizer #2 is 50 – 80 wt ppm as benzoic acid.

1.9 Oxidizer Vent Lines

1.9.1 Oxygen Concentration

Oxygen in the spent air from each oxidizer is measured by UOP 539. This analysis is typically completed to confirm the online oxygen analyzer accuracy. The spent air oxygen concentration is typically between 4 – 7 mol%. High oxygen concentration in the spent air is a concern as approaching the LEL.

1.10 Decanter Aqueous Phase

See 1.4. High CHP in the decanter aqueous phase is an indication of excessive CHP carryover from the concentration section. High CHP in the aqueous phase is also an indication that the decanter aqueous phase is basic. This water should be acidic and all streams entering the decanter should be investigated accordingly. High phenol and acetone are indications that CHP is decomposing to phenol and acetone, likely in the concentration section overhead. This indicates the presence of an acid, such as sulfur from welding slag, in the overhead lines. This can also result from improper passivation during oxidizer and concentration section pre-commissioning.

1.11 Decanter Cumene

1.11.1 pH

See 1.1.8. The pH of the decanter cumene should be acidic, just as the decanter aqueous phase should be acidic per discussion in 1.11.

1.11.2 Composition

The composition of the decanter cumene is analyzed with ASTM D3760. The decanter cumene composition is important to understand what is being recycled back to the oxidizers via the FWC.

1.12 Charcoal Adsorber Outlet / Vent Stack

1.12.1 Total Organic Concentration

The concentration of total organics from the charcoal adsorbers is important for emissions (HS&E) and charcoal life monitoring. The concentration of total organics can be measured with a portable infrared type ambient air analyzer (e.g., Thermo Scientific Miran). The charcoal can typically remove more than 99% of organics fed to the charcoal. As charcoal performance decreases, regenerations should be completed more frequently, if possible and a reload planned.

1.13 Peroxide Destruction Reactor Feed, Upstream of Caustic Addition

1.13.1 CHP

See 1.3.3. All peroxides will be destroyed in this reactor and measurement of individual peroxides, such as CHP, into the reactor is important in understanding reactor performance.

1.13.2 MHP

MHP is measured by Honeywell QCL-195-FR and QCL-197-FR. All peroxides will be destroyed in this reactor and measurement of individual peroxides, such as MHP, into the reactor is important in understanding reactor performance.

1.13.3 Phenol

Phenol is measured by Honeywell QCL-194-FR or QCL-266-FR or QCL-266A-FR. The above methods are used, instead of QCL-103-FR or ASTM D1783 due to the potentially high phenol concentration in this stream. Phenol is important as it is a waste water contaminant and impacts the aerobic biological treatment performance.

1.13.4 Acetone

See 1.4.3. Acetone is important as it is a waste water contaminant and impacts the aerobic biological treatment performance.

1.13.5 pH

See 1.2.3. The feed to the PDR needs to be pH corrected up to a pH of 12-14 for proper performance of the PRD. The inlet pH will dictate how much caustic is added.

1.14 Peroxide Destruction Reactor Feed, Downstream of Caustic Addition

See 1.13.

1.15 Treated Waste Water from Oxidation Waste Water Pumps

See 1.13.

1.16 CHP Sump

See 1.12. If the pH of the sump is acidic it is an indication that peroxides are decomposing or will soon and the sump should be pumped out soon.

2. Concentration

2.1 Preflash and Flash Column Net Overhead

2.1.1 CHP, Cumene, DMPC and AP

See 1.8.1. CHP is measured to determine column performance. Excessive CHP in the overhead indicates insufficient 'reflux,' damaged internals or insufficient vapor for proper flow distribution.

2.1.2 Acidity

See 1.1.7. High acidity is an indication of CHP decomposition in the overhead, likely due to welding slag. The acidity will decrease with time, however if the degree of CHP decomposition is not acceptable a shutdown is required to inspect the overhead and correct the problem.

2.2 Preflash and Flash Column Bottoms

See 2.1.

2.2.1 CHP

The CHP concentration in the flash column bottoms is to be 80 – 85wt% for safe and optimal decomposer operation.

2.2.2 Acidity

Acidity in the oxidizers will concentrate in the flash column bottoms. Excess acidity is undesirable as it could catalyze CHP decomposition in the concentration section. High acidity should instigate oxidizer, MSHP and FWC optimization to remove additional organic acids. If necessary a reduction in production rates to allow lower oxidizer temperatures and therefore lower acid generation should be completed to allow organic acids to decrease.

2.2.3 DCP

DCP is measured by Honeywell QCL-261-FR and QCL-197-FR and is important to calculate CHP yield.

3. Decomposition

3.1 Dehydrator Inlet/Outlet

3.1.1 Sulfuric Acid

Sulfuric acid is measured by QCL-101-FR or QCL-260-FR. Sulfuric acid should be controlled at 40 wt ppm during normal operation and 300 wt ppm during start-up.

3.1.2 CHP

CHP in the decomposition section is measured with Honeywell QCL-217-FR, QCL-004-FR and QCL-108-FR. This method is different than analyzing for CHP in oxidation and concentration due to the presence of acid in the sample which will continue to decompose the CHP in the sample container if the acid is not previously neutralized. The sample is taken 'frozen' indicating that a neutralizing chemical is added to the sample jar prior to taking the sample to neutralize the acid and retain the CHP in the sample as it is taken to the lab for analysis.

CHP is to be controlled at 1 – 1.5wt% during normal operation for optimal AMS yield.

3.1.3 DCP

See 2.2.1. DCP in the dehydrator effluent is controlled at 300 – 900 wt ppm, with the upper limit preferred for yield, but limited by o-cresol in the phenol product. Achieving the target DCP in the crude product with low dehydrator temperature is critical to optimal cumene consumption.

3.2 Decomposer Calorimeter Inlet (Decomposer Circulation)

3.2.1 CHP

See 3.1.2

3.1 Decomposer Cooler and Crude Product Cooler Cooling Water

3.1.1 pH

See 1.2.3. If there is a leak from the process into the cooling water system the pH of the water should be low. Hydrocarbons in the cooling water is a safety concern.

4. Neutralization

4.1 Neutralized Crude Product

4.1.1 Composition

The crude product composition is analyzed by Honeywell QCL-261-FR and QCL-197-FR. The composition is critical in calculating yields and understanding the feed to the CAC.

4.1.2 o-Cresol

o-Cresol is measured by Honeywell QCL-263-FR and QCL-197-FR. o-Cresol is a phenol impurity that is difficult to remove by distillation. As well DCP in the crude product will decompose to additional o-cresol. Understanding the concentration of o-cresol in the fractionation feed will indicate the phenol product quality.

4.1.3 Acidity

See 1.1.7. Acidity is important to understand fractionation corrosion, most importantly in the CAC and C-AMSC. The neutralized crude product is still quite acidic with a pH of 2.3 – 2.7, primarily driven by organic acids.

4.1.4 Water

Water is measured by QCL-168-FR or ASTM D1364. Water in the crude product is an indicator of yield as water is injected to the decomposer and is produced in the dehydrator. The water in the crude product should be less than 2 wt%.

4.1.5 pH

See 1.1.8. When measuring the pH of the crude product the sample should not be diluted/extracted with water. The pH of the raw sample should be measured as phenol in the sample will buffer the pH of the solution. The pH of the crude product should be 2.3 – 2.7.

4.2 Acid Aromatics Sump

4.2.1 Composition

See 4.1.1. The contents of the acid aromatics sump is pumped to neutralization and will go to the fractionation feed tanks; understanding the composition is important for impact to fractionation.

4.2.2 Acidity

See 4.1.3.

4.2.3 Sulfuric Acid

See 3.1.1. The sulfuric acid in the sump will need to be neutralized. The rate of neutralization chemical might need to be increased when pumping out the sump and this analysis will indicate by how much.

5. Fractionation

5.1 Fractionation Feed Tanks

5.1.1 Composition

Composition is analyzed by Honeywell QCL-261-FR and QCL-197-FR. The composition is critical for stable column performance as it impacts advanced ratio controllers as well as can be used to determine which feed tray will be used for a given column. This is typically changed when re-processing off-spec product at increased rates.

5.1.2 Acidity

See 4.1.3.

5.1.3 Alkalinity

Alkalinity of the fractionation feed is measured by ASTM D1614.

5.1.4 pH

See 4.1.5.

5.2 CAC Bottoms

5.2.1 Acetone

Acetone is measured by Honeywell QCL-249-FR and QCL-197-FR. Acetone that is lost to the CAC bottoms is not recovered and is a yield loss. Acetone in the CAC bottoms should be less than 0.05 wt%.

5.3 CAC Reflux

5.3.1 Composition

The composition is analyzed by Honeywell QCL-248-FR, W-197-FR, and QCL-197-FR. Phenol and AMS in the CAC overhead shall be minimized. As much cumene shall be taken overhead as possible without causing phenol to go overhead. The amount of cumene taken overhead in the CAC is not critical hence the name 'crude' acetone column, the crude split is of cumene.

5.4 FAC Bottoms Organic Phase

5.4.1 Composition

The composition is analyzed by Honeywell QCL-249-FR and QCL-197-FR. High phenol or AMS in the FAC bottoms indicates a problem in the CAC split. High acetone in the FAC bottoms indicates a problem in the FAC split. Acetone, phenol and AMS in the FAC bottoms should be below detectable limits. High acetone will cause problems downstream in the C-AMSCWC by forming an emulsion.

5.5 FAC Bottoms Aqueous Phase

5.5.1 Composition

The composition is analyzed by Honeywell QCL-249-FR and QCL-197-FR. High phenol or acetone in the FAC bottoms aqueous phase is an indication of improper separation in the FAC. Increased acetone in the FAC bottoms will draw more cumene and phenol into the aqueous phase causing problems in waste water treatment. High phenol in the FAC bottoms aqueous will require the stream being processed through the PRU, however the high acetone in the stream will limit the rate at which the stream can be processed. In addition the solvent of the OEC and SCWC will need to be changed more frequently due to acetone.

5.6 Alkalinity or Acidity to methyl orange at a pH of 4.9 and 20 deg C

Alkalinity or acidity is measured with Standard Method 2310B or ASTM D1067. Alkalinity in the FAC bottoms aqueous phase is an indication of excessive caustic addition to the FAC reaction zone. Excessive caustic is both a utility waste and can cause fouling in the FAC bottoms.

5.7 FAC Aldehyde Recycle

5.7.1 Composition

Composition is analyzed by Honeywell QCL-236-FR, QCL-168-FR and QCL-197-FR. The goal is to maximize the concentration of unreacted aldehydes from the reaction zone to the aldehyde recycle so they do not accumulate in the FAC receiver. Proper balancing of condensing duty between the FAC condenser and the FAC vent condenser will ensure the highest concentration of aldehydes in the recycle.

5.8 FAC Acetone Product

5.8.1 Composition

Composition is analyzed by Honeywell QCL-236-FR and QCL-197-FR. Note there is no ASTM equivalent of this method. For most acetone, water is the primary contaminant to the product therefore only an ASTM method for water in acetone exists and is discussed further below.

5.8.2 Color

See 1.1.12.

5.8.3 Water

Water in acetone is measured by Honeywell QCL-168-FR or ASTM D1364. Water is primarily controlled by FAC reflux. Water in the acetone product is typically 0.1 – 0.3 wt%.

5.8.4 Permanganate Time

Permanganate time, often called permanganate fading time, of acetone product is generally an indication of aldehydes and is measured by ASTM D1363. Permanganate time is a measure of the concentration of oxidizing compounds in the acetone product. Other impurities in addition to aldehydes will oxidize resulting in a color change. DAA, non-aromatics, and others can also impact the fade time in sufficient concentrations. The method is older and was the primary judge of acetone quality before the analytical techniques of today allowed detailed speciation of the

acetone product impurities. A permanganate time of more than two hours is easily achievable and satisfies the needs of most acetone specifications.

5.8.5 Water Miscibility

Water miscibility is analyzed by ASTM D1722. Water miscibility is a qualitative determination of the acetone and any contaminants present to completely dissolve in water. Water-insoluble materials present in acetone might interfere with its uses.

5.8.6 Distillation Range

Distillation range of acetone is analyzed by ASTM D1078 and is an indirect measure of acetone purity, specifically for heavies or lights.

5.8.7 Non-Volatile Matter

Non-volatile matter is analyzed by ASTM D1353 and is a measure of heavies.

5.8.8 Acidity

Acidity is analyzed by ASTM D1613. Acidity in acetone would primarily come from organic acids. An upset in the CAC and FAC could introduce organic acids into the acetone product.

5.8.9 Alkalinity

Alkalinity is analyzed by ASTM D1614.

5.8.10 Peroxides and Oxygen

Peroxides and oxygen are analyzed by ASTM E299. This method is specifically for active oxygen. The FAC operates under vacuum and air leaks into the unit could result in oxygen contamination or peroxide formation.

5.8.11 Chlorides

Chlorides are analyzed by UOP 395 or ASTM D5194 or ASTM D7359 or D5808 or D7536. Chlorides in the acetone product can originate in the caustic or process water added to the FAC.

When having issues with chlorides, UOP recommends customers to purchase “Membrane Grade Caustic” (or also referred to as “Mercury Cell Grade”, or “Rayon Grade”). Membrane Grade Caustic typically has a NaCl limit of 100 wt ppm (typical is 40 – 60 wt ppm) and a limit on NaClO₃ of 5 wt ppm (typical 2 – 4 wt ppm).

Combined, this is equivalent to a total chloride limit of approximately 60 wt ppm in 30-50 wt% caustic.

5.8.12 Refractive Index

Refractive Index is measured by ASTM D1218. Refractive Index is used to characterize pure components when other analytical techniques are limited. Refractive Index was used more previously when the contaminants in acetone product could not be readily identified nor quantified.

5.8.13 Sulfur

Sulfur is analyzed by ASTM D3120. Sulfur in the acetone product can typically be traced back to sulfur from the cumene feed or air.

5.8.14 Iron

Iron is measured by ASTM E394. Iron is not a typical contaminant, but can come from carbon steel mistakenly being used in the unit instead of stainless steel. Common places to mistakenly find carbon steel are valve or vessel internals.

5.9 Chemical Treatment Injection Pump

5.9.1 Water

Water in the diamine chemical is measured by ASTM D6304. Water is used to dilute HMDA or other diamines to decrease their freeze temperature to below ambient conditions.

5.10 C-AMS Column Bottoms

5.10.1 Composition

Composition is analyzed by Honeywell QCL-258-FR and QCL-197-FR or ASTM D6142. The C-AMS column bottoms should have a cumene concentration of less than 15 wt ppm and an AMS concentration of less than 150 wt ppm. Cumene and AMS lost to the bottoms can be recovered overhead by the CPC, however an increased amount will be sent to the IX-resin treaters where AMS will be a yield loss as it will react across the resin. As well high AMS will more quickly deactivate the resin.

The C-AMS column bottoms sample is also analyzed for HDA (hydroxyacetone or acetol) to optimize the diamine inject rate to the CTR.

5.11 Chemical Treatment Reactor Effluent

5.11.1 Composition

See 3.10. HDA and other carbonyls are measured at the CTR effluent to calculate degree of reaction. Unreacted carbonyls will react across the IXRT and could potentially put the phenol product off-spec.

5.11.2 Water

Water is analyzed by Honeywell QCL-168-FR or ASTM D1631. Water at the C-AMS bottoms is an indication of how much water needs to be stripped in the CPC and PR and can alert operations to the phenol product going off-spec for water.

5.12 Cumene-AMS Column Overhead Organic Phase

5.12.1 Composition

Composition is analyzed with Honeywell QCL-249-FR and QCL-197-FR. The concentration of cumene and AMS is important in C-AMSC operations. The concentration of phenol is important in optimizing downstream C-AMSCWC operations and ensuring that excessive phenol is not being carried overhead.

5.13 Cumene-AMS Column Overhead Aqueous Phase

5.13.1 Phenol Concentration

Phenol is measured with Honeywell QCL-103-FR and QCL-031-FR or QCL-266B-FR or ASTM D1783. The 103 method is a wet-chemistry method whereas the 266 method utilizes the SmartChem analyzer. The phenol concentration in the aqueous phase is important in optimizing the sulfuric acid injection rate to PRU to spring out the phenol.

5.13.2 pH

pH is measured with Standard Method 4500HB. pH is important in understanding organic acids present which impacts C-AMSC corrosion.

5.14 Residue Stripper Bottoms

5.14.1 Composition

Composition is analyzed by Honeywell QCL-123-FR and W-197-FR. The phenol concentration in the RS bottoms is to be minimized for optimum yield.

5.14.2 Ash

Ash concentration is measured with ASTM D482. Ash is a measure of the non-volatile material present, primarily salts. The ash concentration is important in using the residue as fuel oil or other.

5.15 Crude Phenol Column Bottoms

5.15.1 Phenol

Phenol is measured with Honeywell QCL-123-FR and W-197-FR. Low phenol in the CPC bottoms is important to ensuring the RS can achieve low phenol in the RS bottoms.

5.16 Crude Phenol Column Overhead

5.16.1 Composition

Composition is analyzed by Honeywell QCL-258-FR and QCL-197-FR or ASTM D6142. Primarily the CPC overhead is analyzed to ensure low AP in the CPC side-cut.

5.16.2 Water

Water is measured by Honeywell QCL-168-FR or ASTM D1631 or ASTM D1364. The CPC overhead primarily rejects cumene, AMS and water back to the C-AMSC to ensure an on-spec phenol product.

5.17 Crude Phenol Column Side-Draw

See 5.16

5.18 Ion Exchange Resin Treater Outlet

5.18.1 2-MBF

2-MBF is measured by Honeywell QCL-258-FR and QCL-197-FR or ASTM D6142. The IXRT primarily convert 2-MBF to heavies for rejection in the PR. The resin in the treaters is reloaded with the IXRT effluent contains high 2-MBF risking the phenol product going off-spec.

5.19 Sulfuric Acid

Sulfuric acid is measured by Honeywell QCL-101-FR or QCL-260-FR. If the resin is not properly washed with water prior to use then residual free sulfuric acid can wash off the resin. Sulfuric acid can remain in the PR bottoms and can crack heavies

back to 2-MBF resulting in off-spec phenol product. Analyzing for sulfuric acid is important following reload of a resin treater.

5.20 S1.5 Steam Drum Water

5.20.1 Total Hardness

Total hardness is measured by ASTM D1126. Hardness is a measure of metals, calcium and magnesium accumulating in the drum. Increased hardness indicates increased blowdown rates or frequency are required.

5.20.2 Sodium

Sodium is measured by ASTM D4191. Increased sodium indicates increased blowdown rates or frequency are required.

5.20.3 Copper

Copper is measured by ASTM D1688. High copper can indicate tube corrosion.

5.20.4 Iron

Iron is measured by ASTM D1068. High iron can indicate tube corrosion.

5.20.5 Alkalinity or Acidity to methyl orange at a pH of 4.9 and 20 deg C

Alkalinity or acidity is measured by Standard Method 2310B or ASTM D1067. Alkalinity or acidity indicates if any additives are required to maintain the boiler feed water near neutral to avoid corrosion.

5.20.6 Chloride

Chloride is measured by ASTM D512. High chloride indicates increased blowdown rates or frequency are required.

5.20.7 Sulfate

Sulfate is measured by ASTM D516. High sulfate indicates increased blowdown rates or frequency are required.

5.20.8 Silica

Silica is measured by ASTM D859. High Silica indicates increased blowdown rates or frequency are required.

5.20.9 Total Suspended Solids

TSS is measured by Standard Method 2540C. High TSS indicates increased blowdown rates or frequency are required.

5.20.10 Total Dissolved Solids

TDS is measured by Standard Method 2540D. High TDS indicates increased blowdown rates or frequency are required.

5.20.11 pH

pH is measured by ASTM D1293. pH indicates if any additives are required to maintain the boiler feed water near neutral to avoid corrosion

5.20.12 Specific Conductance

Specific conductance is measured by ASTM D1125. Conductivity is a measure of salts. High conductivity indicates increased blowdown rates or frequency are required.

5.20.13 Dissolved Oxygen

Dissolved oxygen is measured by ASTM D888. Oxygen in the steam system can cause corrosion and fouling of downstream exchangers. High dissolved oxygen indicates insufficient preparation of the boiler feed water.

5.20.14 Phosphates

Phosphates are measured by ASTM D4327. High phosphates indicate increased blowdown rates or frequency are required.

5.20.15 Oil

Oil is measured by ASTM D3921. High oil can indicate a leak from a downstream exchanger or low quality boiler feed water.

5.20.16 Hydrazine

Hydrazine is measured by ASTM D1385.

5.20.17 Sulfite

Sulfite is measured by ASTM D5542. High Sulfite indicates increased blowdown rates or frequency are required.

5.20.18 Total Organic Compounds

TOC is measured by ASTM D4779. High TOC indicates increased blowdown rates or frequency are required or a leak from a downstream exchanger.

5.21 S1.5 Steam Drum Steam

See 5.20.

5.22 Phenol Rectifier Bottoms

5.22.1 Phenol

Phenol is measured by Honeywell QCL-123-FR and W-197-FR. The phenol rectifier bottoms is primarily to drag heavies generated from the IXRT. Low phenol concentration in the PR bottoms can indicate insufficient drag resulting in the build-up of heavies.

5.23 Phenol Rectifier Overhead Drag

See 5.16. The build-up of lights such as water, cumene or AMS indicates insufficient drag in the CPC or PR.

5.24 Phenol Rectifier Side-cut (Phenol Product)

5.24.1 Composition

Composition is analyzed by Honeywell QCL-258-FR and QCL-197-FR or ASTM D6142. The impurities in the phenol product dictate phenol fractionation, CTR and IXRT operations. Excessive o-cresol in the phenol product indicates high DCP in the crude product. High 2-MBF in the phenol product indicates the resin is approaching end of life or there is an acidic contaminant in the PR bottoms.

5.24.2 Water

Water is measured by Honeywell QCL-168-FR or ASTM D1631 or ASTM D1364. Water is the highest concentration phenol impurity. High water tends to only be a problem when a new IXRT is put online and can be corrected by increased PR overhead drag rates. Water in the phenol product is typically 100 – 300 wt ppm.

5.24.3 Color

Phenol color is measured by ASTM D1686 on the Pt/Co scale. High purity phenol typically has an APHA color less than 5. Though high color is typically a measure of

heavies, high purity phenol can change color for a variety of reasons as discussed further in the Section IX Troubleshooting.

5.24.4 Crystallization (Solidification) Point

Crystallization point is measured by ASTM D6875. Crystallization point is impacted by the contaminant of highest concentration, which for high purity phenol is water. Impurities in high purity phenol do not impact the crystallization point.

5.24.5 Water Solubility

Water solubility is measured by Honeywell QCL-036-FR or ASTM D1722. Water solubility is a measure of the phenol impurities that are not soluble in water.

5.24.6 Total Carbonyls

Total carbonyls are measured by Honeywell QCL-258-FR and QCL-197-FR or ASTM E411. Total carbonyls are reported as mesityl oxide. Insufficient diamine injection to the CTR or low resin activity would result in high carbonyls in the phenol product.

5.24.7 Residue On Evaporation

Residue on evaporation is measured by UOP 811. Residue on evaporation is a measure of non-volatile compounds, or heavies and is similar to the ash method of the acetone product. High residue indicates the need to drag more from the PR bottoms and adjust CPC and RS operations.

5.24.8 Total Sulfur

Total sulfur is measured by ASTM D5453 or ASTM D7183. Sulfur in the phenol product can come from loss of containment of the resin or, more likely, high sulfur in the cumene feed.

5.24.9 Alkalinity

Alkalinity is measured by ASTM D2896. Acids from the upstream IXRT could result in the phenol product being acidic.

5.24.10 Iron

Iron is measured by ASTM E394. Iron in the phenol product can be impacted by equipment, valves, pumps or other vessel internals, incorrectly being installed containing carbon steel.

5.24.11 Sulfuric Acid Discoloration

Sulfuric acid discoloration (SAD) is measured by Honeywell QCL-246-FR. SAD is a measure of impurities in the phenol product. Experience has shown that 'new' analytical grade sulfuric acid is required for the analysis to be accurate.

5.24.12 Peroxide and Oxygen

Peroxide and oxygen are measured by ASTM E299. As most of the phenol fractionation section operates under vacuum, leaks can introduce air (oxygen) to the system and those leaks need to be corrected.

5.24.13 Chloride

Chlorides are measured by UOP 991 or ASTM D5194 or ASTM D7359 or D5808 or D7536. Chlorides can be present in the phenol from the process water used in the resin treaters or elsewhere in the phenol unit.

5.25 Fractionation Ejector Condensate Drum Skimming Line

5.25.1 Composition

The composition is measured by Honeywell QCL-249-FR and QCL-197-FR. Any hydrocarbon accumulating on the water should be primarily phenol, but could be some cumene and AMS from the C-AMS column. Accumulation of excessive hydrocarbon indicates a problem in one of the phenol fractionation column's vent condensers. Ultimately high phenol can negatively impact PRU operations.

5.26 Fractionation Ejector Condensate

5.26.1 Phenol

Phenol is measured by Honeywell QCL-138-FR and QCL-108-FR and QCL-109-FR or QCL-266(A or B)-FR or ASTM D1783. See 5.25.1

5.26.2 Composition (if organics are present)

See 5.25.1.

5.26.3 pH

pH is measured by either Standard Method 4500HB or Honeywell QCL-145-FR depending if the phase collected is aqueous or organic. A pH different than the pH of the process water indicates the water in the drum needs to be replaced.

5.27 Fractionation Sump

See 5.1.1.

5.28 Phenolic Waste Water Sump

5.28.1 Phenol

Phenol is measured by Honeywell QCL-138-FR and QCL-108-FR and QCL-109-FR or ASTM D1783. The phenol concentration is important in understanding where the sump contents should be pumped to. If the phenol concentration is greater than 2wt% pump the sump contents to the sodium phenate tank. Otherwise pump the contents to the phenolic waste water tank.

5.28.2 pH

pH is measured by Standard Method 4500HB. The pH is important in understanding how the contents of the sump will impact the pH of the tank which ultimately impacts the sulfuric acid addition rates to the PRU or OEC.

6. Phenol Recovery

6.1 Cumene-AMS Caustic Wash Column Spent Caustic Circulation Pumps

6.1.1 Caustic (NaOH)

Caustic is measured by UOP 209. The concentration of caustic in the bottom of the C-AMSCWC should be approximately 8 - 10wt%. Low caustic will generally result in low phenol extraction in the C-AMSCWC and result in phenol carryover to the MSHP unit, which is undesirable as it will result in phenol to the oxidizers.

6.1.2 Phenol

Phenol is measured by Honeywell QCL-138-FR and QCL-108-FR and QCL-109-FR or QCL-266A or B or ASTM D1783. Similar to caustic as discussed in 6.1.1, low phenol in the C-AMSCWC bottoms generally means it is not being sufficiently extracted and is being lost to the overhead to the MSHP unit, which is undesirable as it will result in phenol to the oxidizers.

6.2 Cumene-AMS Caustic Wash Column Overhead Organic Phase

6.2.1 Phenol

Phenol is measured by Honeywell QCL-265-FR and QCL-197-FR or ASTM D3160. There should be very low amounts of phenol left in the C-AMSCWC overhead organic phase, typically 1 – 5 wt ppm. High phenol in the overhead is generally due to insufficient jetting velocity on the trays or the presence of acetone causing the formation of an emulsion and not allowing proper separation of the organic and aqueous phases. Over utilization of the caustic, beyond 150mol%, can also result in insufficient extraction.

6.2.2 Aqueous Phase

A sample of the C-AMSCWC overhead organic phase should be allowed to settle into a separatory funnel to observe if an aqueous phase forms and if so, quantify the amount by decanting the aqueous phase into a graduated cylinder. Insufficient separation of the organic and aqueous phase in the C-AMSCWC is an indication of insufficient or excessive jetting velocity. Insufficient separation from the C-AMSCWC can result in caustic carryover and deactivation of the MSHP catalyst.

6.3 Fresh 20 wt% Caustic

6.3.1 NaOH

See 6.1.1. This sample is simply to confirm the fresh caustic strength for proper metering to the C-AMSCWC.

6.4 Caustic Settler Aqueous Phase

6.4.1 NaOH

See 6.3.1.

6.4.2 Phenol

Phenol is measured with Honeywell QCL-138-FR and QCL-108-FR and QCL-109-FR or QCL-266 or QCL-266A or ASTM D1783. The different methods are for different analytical techniques as well as varying with phenol concentration. Most all of the phenol should be extracted from the organic phase in the C-AMSCWC. If significant phenol is detected in the overhead it means that the C-AMSCWC column is not operating properly and phenol might also be in the organic phase which will

impact oxidation upon recycle. Operations should confirm the jetting velocity and caustic utilization in the C-AMSCWC.

6.5 Caustic Settler Overhead Organic Phase

6.5.1 Phenol

See 6.2.1.

6.5.2 Aqueous Phase

A sample of the Caustic Settler overhead organic phase should be allowed to settle into a separatory funnel to observe if an aqueous phase forms and if so, quantify the amount by decanting the aqueous phase into a graduated cylinder. Insufficient separation of the organic and aqueous phase in the caustic settler will result in caustic carryover to the MSHP catalyst causing deactivation. This can be corrected by adjusting the level in the caustic settler and/or the fresh caustic recycle rate to the caustic settler.

6.6 C-AMS Water Wash Drum Aqueous Phase at Tri-lines

6.6.1 NaOH

Caustic is measured by UOP 209. Typically there should be no caustic detected in the water wash drum. The presence of increased caustic in the wash drum indicates caustic carryover from the caustic settler indicating the need for optimization of those conditions and might indicate the need for a higher water flow rate to the wash drum to ensure the water is clean of contaminants and can effectively remove trace caustic from the organic phase.

6.6.2 Phenol

See 6.4.2.

6.6.3 Alkalinity or Acidity to methyl orange at a pH of 4.9 and 20 deg C

Alkalinity or acidity is measured with Standard Method 2310B or ASTM D1067. Alkalinity is similar to measuring caustic or pH. The water wash water should be neutral. The presence of alkaline or acidic water in the wash drum indicates the need for optimization of the C-AMSCWC and caustic settler and might indicate the need for a higher water flow rate to the wash drum to ensure the water is clean of contaminants and can effectively remove trace caustic from the organic phase.

6.7 C-AMS Water Wash Drum Bottoms Aqueous Phase

See 6.6.

6.8 Treated Cumene-AMS To MSHP Feed Tank

See 6.5.

6.9 PRU Recycle Pump Mixed Phase

6.9.1 pH

pH is measured by Standard Method 4500HB or Honeywell QCL-145-FR. The pH is to be controlled between 4.5 and 5.0, with a target of 5.0. pH values below this do not improve unit performance and might increase corrosion. pH values above this will drastically increase the phenolic loading in the OEC bottoms water.

6.10 Sprung Phenol Coalescer Outlet

6.10.1 Composition

Composition is analyzed by Honeywell QCL-249-FR and QCL-197-FR. This stream is primarily phenol and cumene. The composition helps set the ratio controller setting the flow rate of cumene to the PRU to help improve the separation of phenol and water.

6.10.2 Sulfate by Ash Content

Sulfate by ash is measured by ASTM D482. Sulfates detected in the sprung phenol indicate the PRU separator and coalescer are not functioning properly. This could be due to insufficient cumene to the PRU, high separator temperature, high pH or decreased density of the aqueous phase resulting in phase inversion in the PRU separator.

6.10.3 pH

pH is measured by Honeywell QCL-145-FR. Proper pH is critical to the phase separation in the PRU separator. See 6.9.1.

6.11 PRU Separator Aqueous Phase

6.11.1 Phenol

Phenol is measured by Honeywell QCL-138-FR and QCL-108-FR and QCL-109-FR or QCL-266A or B or ASTM D1783. Complete removal of phenol from the aqueous phase in the PRU is not required. The remaining phenol in the aqueous phase will be removed in the OEC. High phenol in this stream indicates insufficient separation in the PRU separator. See 6.10.2.

6.11.2 pH

pH is measured by Standard Method 4500HB. Proper pH is critical to the phase separation in the PRU separator. See 6.9.1.

6.11.3 Sulfate by Ash Content

See 6.10.2.

6.12 Treated Oil Extraction Column Water

6.12.1 Phenol

Phenol is measured by Honeywell QCL-138-FR and QCL-108-FR and QCL-109-FR or QCL-266 or QCL-266A or ASTM D1783. 50 wt ppm or less phenol in the OEC water should be achievable with low phenol in the solvent. High phenol in the OEC water is typically due to improper jetting velocities in the OEC and/or SCWC or high phenol in the solvent.

6.12.2 Composition

Composition is analyzed by Honeywell QCL-193-FR. The composition is analyzed to understand the load on the downstream aerobic biological waste water treatment plant.

6.12.3 Alkalinity or Acidity to methyl orange at a pH of 4.9 and 20 deg C

Alkalinity or acidity is measured by Standard Method 2310B or ASTM D1067. The OEC water should be acidic with a pH of 4.5. Too low of a pH can cause corrosion problems and a too high of a pH will result in insufficient phenol removal and increased load on the downstream aerobic biological waste water treatment plant.

6.13 Solvent Caustic Wash Column Aqueous Bottoms

6.13.1 NaOH

Caustic is measured by UOP 209. This analysis is to ensure sufficient caustic is injected to the SCWC to remove phenol from the solvent.

6.13.2 Phenol

Phenol is measured by Honeywell QCL-138-FR and QCL-108-FR and QCL-109-FR or QCL-266A or ASTM D1783. This analysis is to ensure phenol is being removed

from the solvent. Insufficient phenol removal is typically due to insufficient jetting velocity in the SCWC or insufficient caustic injection to the SCWC.

6.14 Solvent Pump

6.14.1 Composition

Composition is analyzed by ASTM D3760. Impurities in the solvent such as phenol and acetone will negatively impact the removal of phenol from the OEC waste water and need to be actively monitored. The solvent can be purged and fresh cumene added as needed to remove contaminants.

6.14.2 Phenol

Phenol is measured by Honeywell QCL-265-FR and QCL-197-FR or ASTM D3160. See. 6.14.1.

6.15 Phenolic Waste Water Static Mixer Outlet

6.15.1 pH

pH is measured by Standard Method 4500HB. See 6.12.3.

6.15.2 Phenol

See 6.12.1. The phenol concentration is to be less than 2 wt%. The phenol concentration is important as excessive phenol can make the separation in the OEC difficult resulting in increased load on the downstream aerobic biological waste water treatment plant.

6.16 Phenolic Waste Water Tank

See 6.15.

6.17 Sodium Phenate Tank

6.17.1 Composition

Composition is analyzed by Honeywell QCL-193-FR. The composition is important for PRU operations, setting the cumene injection rate and ensuring there is sufficient phenol. At phenol concentrations below 2wt% should be transferred to the phenolic waste water tank.

6.17.2 NaOH

Caustic is measured by UOP 209. The caustic concentration is important in calculating the sulfuric acid injection requirements of the PRU and confirming sufficient caustic is being added to the C-AMSCWC.

6.17.3 Phenol

Phenol is measured by Honeywell QCL-138-FR and QCL-108-FR and QCL-109-FR or QCL-266A or B or ASTM D1783. See 6.17.1.

6.18 Sodium Phenate and Phenolic Waste Water Tank Skimming

6.18.1 Composition (Organic phase)

Composition of the organic phase is analyzed by Honeywell QCL-249-FR and QCL-197-FR. This sample is primarily informational, but as it will be injected to the PRU will help understand its impact on PRU performance.

6.18.2 Phenol (Organic Phase)

Phenol is measured by Honeywell QCL-265-FR and QCL-197-FR or ASTM D3160. See. 6.18.1.

6.19 OEC Overhead Organic

6.19.1 Phenol

Phenol is measured by QCL-265-FR and QCL-197-FR or ASTM D3160. The phenol concentration in the OEC overhead will give an indication of how much caustic is required in the SCWC to achieve clean recycle solvent.

6.20 98 wt% Sulfuric Acid Tank

6.20.1 Sulfuric Acid

The sulfuric acid concentration is determined by concentration with a known base, such as caustic. The analysis is to confirm the sulfuric acid strength for use in the process. As well it needs to be confirmed the caustic is not becoming dilute due to contamination from another stream. Dilute caustic is more corrosive than concentrated caustic and can result in corrosion and leaks.

6.21 Combined Waste Water

See 6.12.

6.22 Process Water Break Drum

See 1.4.

7. Hüls MSHP (AMS Hydrogenation)

7.1 Fresh Feed

7.1.1 Composition

Composition is analyzed by Honeywell QCL-248-FR and W-197-FR and QCL-197-FR. The composition is analyzed to determine the concentration of AMS to calculate conversion, dilution requirements and expected reactor exotherm.

7.1.2 Phenol

Phenol is measured by Honeywell QCL-265-FR and QCL-197-FR or ASTM D3160. Phenol is not a concern for the MSHP unit, but will pass through unreacted and can impact oxidation at sufficiently high concentrations. High concentrations of phenol measured in the feed should start troubleshooting efforts at the C-AMSCWC.

7.2 Hydrogen Make-Up

7.2.1 Composition

Composition is analyzed by UOP 539 or ASTM D1946. The composition is used to determine the hydrogen concentration to calculate hydrogen partial pressure in the MSHP reactor and H₂:HC ratio.

7.2.2 Oxygen

Oxygen is measured by UOP 759 or an on-line analyzer. The presence of oxygen in the hydrogen is a safety concern.

7.2.3 Nitrogen

Nitrogen is measured by UOP 539 or ASTM D1946. Nitrogen is a diluent in the hydrogen. See 7.2.1.

7.2.4 Hydrogen Sulfide

Hydrogen Sulfide is measured by a detector tube or UOP 212. The MSHP catalyst utilizes palladium metal which is severely impacted by sulfur in the feed.

7.2.5 Carbon Monoxide And Carbon Dioxide

Carbon monoxide and carbon dioxide are measured by UOP 603 or a detector tube or ASTM D2504 and D2505. Carbon monoxide can impact the palladium metal function of the catalyst as all carbonyls can.

7.2.6 Water

Water is measured by a detector tube or a dew point method. Water itself is not a concern, but as water should not be present, water in the make-up hydrogen can indicate the presence of other contaminants.

7.2.7 Hydrogen Chloride

Hydrogen chloride is measured by a detector tube. Hydrogen chloride is a contaminant to the MSHP catalyst, impacting its activity and selectivity. As well it can impact corrosion rates in the unit and potentially impact catalyst strength.

7.2.8 Mercury as Hg

Mercury is measured by ASTM D6350. Mercury is a contaminant to the MSHP catalyst.

7.3 Reactor #1 Effluent Liquid

7.3.1 Composition

Composition is analyzed by Honeywell QCL-248-FR and W-197-FR and QCL-197-FR. The composition is used to calculate AMS conversion in the MSHP reactor and determine if sufficient effluent recycle is present to dilute the AMS in the fresh feed.

7.3.2 Bromine Index

Bromine index is measured by ASTM D1492 or D5776. As AMS is an olefin and bromine index is a measure of olefins, the bromine index of the reactor 1 effluent can be used to assess AMS conversion when the feed composition does not change significantly.

7.4 Hydrogenation Product Separator Pump Outlet

7.4.1 Composition

See 7.3.1.

7.4.2 Phenol

See 7.1.2.

7.4.3 Bromine Index

See 7.3.2.

7.5 Separator Off-Gas

7.5.1 Composition

Composition is analyzed by UOP 539. The hydrogen concentration of the off-gas is used in the calculation of excess hydrogen which sets the make-up hydrogen rate to the unit.

D. Sample Analysis at UOP or Third Parties

In order to comply with U.S. regulatory laws outlined by the Environmental Protection Agency (EPA) and other regulations, UOP must assure that all chemical substances received at our facilities adhere to certain procedures.

Please contact you UOP Regional Service or UOP Sales representative for the latest shipping instructions. Failure to ship samples in accordance with our instructions may result in delay of analysis, refusal of the sample, or confiscation by government agency.

IX. Troubleshooting

Troubleshooting of any process, should be an ongoing part of normal operations. Daily records of operations are collected and should be reviewed to observe any changes before they become problems. Any unexplained changes should be followed-up on to identify their source and the necessary corrections should be made. This section is focused on process related problems such as poor product quality or contamination and operating problems for unique equipment. Troubleshooting of general equipment such as pumps, compressors, and fired heaters is not included since this information is generally available from the equipment vendors.

A. Table of Problems and Solutions

Process Symptoms	Possible Causes	Corrective Actions
Oxidation:		
Oxidizer High Acidity	Fresh or recycle cumene feed contamination with one or more of the following: AMS, phenol, organic acids	Check feed wash column bottoms (aqueous phase) for pH and free caustic. pH should be ~14 and free caustic should be 1 – 1.5wt%. Caustic will neutralize organic acids and extract phenol and peroxides from the cumene feed
		Check fresh cumene for contaminants and remove, if present
		Check recycle cumene for contaminants and remove, if present
		It is possible that cooling water is leaking into the oxidizers. Check dedicated cooling tower water for organics, the leak might go

		both ways.
		Sodium phenate carryover from feed wash column. Check cumene feed surge drum water boot for sodium phenate. Drain water boot to remove any phenate and add additional water to the CFSD to wash phenate from the coalescer medium.
	Oxidizer air is contaminated with acids.	Check air scrubber bottoms for free caustic. Free caustic concentration should be ~0.1 wt%. Ensure water pH is basic. Check for particle entrainment.
Oxidizer High Phenol	High phenol in combined feed to the oxidizer	Remove phenol from oxidizer by scrubbing out in feed wash column as described above.
	Acidity in the oxidizer causing decomposition of CHP	Reduce acidity by scrubbing out in feed wash column as described above.
CHP concentration increasing in the oxidizers	Air consumption is not in balance with decomposer CHP consumption (e.g. decomposer feed rate)	Decrease air injection rate and oxidizer temperatures to decrease CHP production rate while maintaining spent air oxygen concentration
CHP concentration decreasing in the oxidizers	Air consumption is not in balance with decomposer CHP consumption (e.g. decomposer feed rate)	Increase air injection rate and oxidizer temperatures to increase CHP production rate while maintaining spent air oxygen concentration

Oxidizer recirculation flow rate unstable	Possible air entrainment to the pump suction	Check that the oxidizer pump-out line block valve to the circulation pump suction is closed.
		Adjust circulation pump suction and discharge block valves to minimize fluctuation
		Consider to change pump impeller or new pump to handle air entrainment. Review oxidizer disengaging well design.
	Possible pump problem	Put spare pump online and immediately fix the problem pump
High oxidizer spent air oxygen concentration	Oxidizer temperature not sufficient to react oxygen at given air flow rate	Increase oxidizer temperature and/or increase oxidizer level
Concentration, Decomposition and Neutralization:		
High DMPC concentration to the decomposer	Oxidizer spent air concentration is low (<5 vol% O ₂) causing excessive byproduct formation.	Either increase air flow rate or decrease oxidizer temperature to reduce spent air oxygen concentration while maintaining CHP production balance.
High acidity in the flash column bottoms	High acidity in the terminal oxidizer will concentrate in the flash column bottoms	Reduce oxidizer acidity with feed wash column as discussed above
Poor AMS yield in decomposer	Low CHP concentration in decomposer circulation loop	Check laboratory analysis versus calorimeter delta temperatures. See Section XI, Emergency Procedure for detailed description of

		causes of calorimeter delta temperatures.
	Excessive temperature or residence time in dehydrator and effluent	Target 300 - 700 wt ppm DCP in the crude product. If DCP is below this range then AMS yield will be low. The dehydrator temperature or residence time (bypass expansion loop) or both need to be reduced to reduce residue formation. Do not decrease dehydrator temperature below 120 deg C.
Frequent plugging of dehydrators causing increased maintenance (typical cycle is 1 – 3 months)	Excessive temperature or residence time in the dehydrator and effluent (might be accompanied by low yields)	See Above
	Frequent start-ups causing excessive residue formation. During start-up acid concentration is 300 wt ppm causing residue formation	Nothing can be done within the decomposition/dehydration section. The source of the frequent shutdowns resulting in start-up should be minimized.
	Improper dehydrator cleaning procedure	Hydroblasting with high pressure water or mechanically cleaning with a brush aided by water flushing are the most common ways of cleaning the dehydrator tubes.
Breakthrough of sulfuric acid to the fractionation feed tank	High acid from decomposer	Could be normal if start-up period is extended. (300 wt ppm sulfuric acid during start-up and 40 wt ppm during normal operations)

	Insufficient diamine injection	Check rate and adjust to achieve crude product target pH. When analyzing crude product do not dilute crude product in water.
High static mixer delta P limiting crude product flow to storage	Sodium sulfate carryover from PRU separator fouling static mixer	Bypass static mixer. Remove, clean and replace. Check PRU separator performance (interface LIC, temperature, pH, cumene addition).
Fractionation:		
Frequent CAC boil-out due to fouling of the reboiler (more than every 6 months)	Sodium sulfate carryover from PRU separator fouling reboiler	Check PRU separator performance (interface LIC, temperature, pH, cumene addition).
High acetone in CAC bottoms	Insufficient stripping steam injection to the bottom of the CAC due to low water concentration.	Increase water/steam injection.
Solids in FAC bottoms aqueous phase causing plugging in equipment	Excessive caustic injection will result in precipitation of aldol condensation products which can combine with rust to form a solid that varies from soft to hard	Reduce caustic injection to the minimum required to achieve the permanganate fade test result.
	Excessive residence time on trays resulting in further reaction to heavies of aldol condensation product	Increase cumene concentration of FAC feed to reduce organic phase residence time in FAC.
Permanganate fade time failing for acetone product	Insufficient reaction of aldehydes and ketones with caustic	Verify that the laboratory is cleaning the test equipment properly. Poorly cleaned equipment will

		result in poor fading times.
		Increase caustic contacting through one or more of the following: Increase caustic injection rate Inject the caustic at a higher point in the FAC Increase the TDIC to allow the reactive aqueous phase higher in the FAC Increase aldehyde recycle rates. Adjust duty on FAC condenser to allow more liquid to the FAC vent condenser which will better concentrate aldehydes in the recycle.
Excessive corrosion in the C-AMSC overhead system (trays, reflux lines)	High organic acid concentration such as formic and oxalic acids in the aqueous phase. The aqueous phase separating in the reflux line due to cold ambient temperatures. Colder than normal reflux temperature resulting in "shock condensation" of extremely acidic aqueous phase on the reflux tray.	Reduce the organic acid concentration by diverting the acidic overhead water going to the CAC to the effluent section.
		Regularly check the stagnant areas in the reflux piping such as low points and spare reflux pump for water and drain periodically

		If reflux tray corrosion occurs, then increase the reflux temperature to at least 11 deg C above the water dew point for the reflux tray. Alternatively, running the column vapor top temperature 20 deg C above the water dew point is acceptable.
		Increase the tray/piping metallurgy to improve more corrosion resistance.
Frequent plugging of the residue stripper reboiler (more than once per 6 – 12 months)	Excessive fouling due to residue and insoluble salts.	If excessive fouling of the CAC reboiler is also occurring then see above for troubleshooting salts.
		High reboiler temperatures/low bottoms phenol concentration (~2wt%) will produce a heavier residue. Reduce the reboiler outlet temperature, which will increase the phenol concentration
		Take the fouled reboiler out of service and flush with acetone followed by water to dissolve the residue from the reboiler tubes. This can be done in between mechanical cleaning of reboiler tubes before the tubes plug off completely.
High phenol product carbonyl concentration	Insufficient chemical injection to CTR	Increase chemical injection flow rate

High phenol product MBF	Poor reaction in IX resin treaters	Reload IX resin treater
	Some residual HDA is reacting in the IX resin treater forming 2-MBF	Increase chemical injection to CTR to react HDA before they get to the IX resin treater.
High phenol product GC impurities	Poor fractionation	Adjust C-AMSC and CPC operation and PR purge rates to improve the separation
High phenol product water concentration	Poor fractionation	If the C-AMSC is refluxing water, then water will go to the column bottoms. Check C-AMSC receiver water boot operation and confirm LT is reading properly. Check C-AMSC bottoms water concentration.
		Increasing the overhead drag rate of the CPC and PR will drag more water from the side-cuts and decrease the water concentration.
	Resin treater was just reloaded	The PR overhead drag rate must be increased to the maximum following a new IX resin treater being put online to drag off the water from resin treater flushing.
Off-spec Phenol Color	Phenol color can be off-spec for a variety of reasons	See separate phenol color troubleshooting section below the table.
Phenol product fails water lights test (water solubility at 20 deg C/68 deg C),	Incomplete dehydration of DCP in the dehydrator and fractionation section	Check phenol product for DCP. If positive, increase dehydrator temperature

cloudy result		and/or residence time.
	Poor fractionation in fractionation section resulting in insoluble organic contamination	Check fractionation column operation and stability. Increase CPC and PR drag rates.
Phenol Recovery:		
High phenol in treated waste water (OEC bottoms)	Erratic pH control	Check the pH control in the delay pot/PRU and OEC feed
	High phenol in lean solvent (cumene) from SCWC	Improve extraction in the SCWC by checking jetting velocities and sufficient caustic.
	Caustic or sodium phenate carryover reducing the pH in the OEC bottoms	Check operation of the SCWC for caustic carryover and check the solvent drum water boot for excessive aqueous phase accumulation.
High phenol in C-AMSCWC or SCWC organic phase	Excessive phenol to caustic ratio	Increase caustic rate
	Inadequate jetting velocity	Check combined feed flow rate and calculate the jetting velocity. It should be in the range of 1.2 – 1.5 m/s.
Poor separation of organic and aqueous phases in C-AMSCWC, possibly foaming	Excessive jetting velocity	Check combined feed flow rate and calculate the jetting velocity. It should be in the range of 1.2 – 1.5 m/s.
	Contamination of organic with substance that promotes foaming such	Adjust CAC and FAC operation to minimize

	as acetone	acetone in column bottoms
	Unstable pressure and/or combined feed flow rate resulting in sudden surging of organic phase flow up the column. This leads to severe overhead caustic entrainment.	Be sure controllers are in auto and tuned well. The PIC is set by design and should not be adjusted during normal operation. Make adjustments to flow rates slowly.
Poor caustic utilization in the C-AMSCWC or SCWC	Caustic injection is excessive for phenol flow rate.	Check C-AMSCWC or SCWC jetting velocity. Reduce caustic injection slowly and optimize utilization versus phenol concentration in the overhead. For the C-AMSCWC maximize the bottoms rate flow back to the feed line upstream of the static mixer.
MSHP:		
High IPCH in MSHP product	High reactor severity	Check reactor 1 effluent composition. Decrease reactor temperature until IPCH reduces without resulting in high AMS in reactor 2 effluent.
		High severity can also be caused by operating at too high system or hydrogen partial pressure. Decrease reactor pressure to decrease IPCH without causing high AMS in reactor 2 effluent.
		MSHP catalyst might be very active at start-of-run. Within 1 week of operation any IPCH initially formed should go away.

High AMS in MSHP product	Low reactor severity	Check reactor 1 effluent AMS and adjust temperature and pressure to achieve 1% AMS in the reactor 1 effluent.
	Catalyst deactivation by salts	Water wash and re-reduce the catalyst per the procedures in Section XII, Special Procedures, to restore activity
High phenol in MSHP product	High phenol from C-AMSCWC	Phenol is generally non-reactive across the MSHP catalyst therefore any phenol in the product was also in the feed. Optimize C-AMSCWC performance to eliminate phenol in the organic phase.

B. Phenol Product Color Troubleshooting

High purity phenol typically has an APHA color less than 5. Yellow phenol tends to have a color between 15 and 20, and brown phenol tends to have a color greater than 40. Though high color is typically a measure of heavies, high purity phenol can change color for a variety of reasons:

- 1) High phenol product rundown and storage temperatures, greater than 55°C
- 2) Phenol contact with non-stainless steel metals (carbon steel, copper or chromium)
- 3) Long storage times (> 1 month) with 304 SS
- 4) Phenol contact with oxygen (air)
- 5) Freezing and melting of phenol in storage
- 6) Contact of phenol with color producing contamination:
 - a) Air makes phenol yellow
 - b) Amines makes phenol pink
 - c) Sulfuric acid makes phenol pink
 - d) Copper makes phenol green
 - e) Chrome plated metal makes phenol red

1. Phenol Storage Temperature

Phenol product storage temperature is critical to maintaining color stability. Phenol will go off color if the steam coils in the phenol storage tanks are too hot. Overheating phenol can cause discoloration. Minimize the issue of storage temperature by using tempered water or very low-pressure steam with circulation of the phenol to reduce localized hot spots. This problem worsens as the organic impurity concentration increases due to the inter-reactions that take place. Optimum storage temperature for phenol should be 50 – 55°C. Higher temperatures will accelerate the color developing to yellow and pink. Even low temperature storage will not prevent phenol discoloration if the impurity concentration is high.

2. Phenol Contamination

2.1 Storage Tank Metallurgy

Storage vessel metallurgy is also important. The order of increasing phenol color stability for storage vessel metallurgy is 304SS, 316SS, phenolic resin lined carbon steel. Based on Honeywell Frankford experience, and all other contaminants being low, phenol in 304SS will tend to develop color after 1 month, 316 SS will be stable for 3 months, and phenolic resin lined carbon steel will be stable much longer.

‘Unseasoned’ stainless steel can cause phenol to turn red. Reprocessing the phenol can remove the color bodies and after a few batches of phenol the stainless steel is then ‘seasoned’ and will no longer change the color of the phenol.

2.2 Carbon Steel

Iron contamination from carbon steel can result in phenol changing from yellow to brown and even to black. Carbon steel can often mistakenly be installed in the unit from valve, pump or vessel internals. The metallurgy of all components should be checked for correct metallurgy prior to installation.

2.3 Copper

Copper makes phenol green.

2.4 Chromium

Chromium makes phenol red.

2.5 Acids and Amines

The color of the phenol product can be influenced by many trace impurities. Sulfuric acid and diamine will cause the phenol color to be pink or red.

Conditions of the IXRT may affect the amount of color bodies getting into the phenol product. IX-resin at the end of its life can be more vulnerable or a possible breakdown of the resin can allow color bodies to form in the phenol product. Understanding when the phenol product starts to change color can be used to correlate color to a given set of fractionation operating conditions or IXRT condition. If the resin is changed out on a more frequent basis (and properly loaded and washed), then the color problem may be avoided.

2.6 Oxygen

Exposure of the phenol product to oxygen, from air during storage, might also influence phenol color. Ensure storage tanks have been properly air-freed before use and nitrogen blanketing is functioning as designed. Oxygen will cause the phenol color to be yellow-brown.

3. Correcting High Phenol Color

Although high GC impurities in the phenol product can be reduced by blending high phenol color cannot typically be reduced by blending. Some color bodies will react across the ion exchange resin treaters to form heavies that exit in the residue stripper bottoms. To support this, a reprocessing line from the phenol storage tanks to the inlet of the IX resin treaters is available for reprocessing phenol that is off specification on color.

A high temperature in the CPC or PR reboiler may lead to the formation of heavy components and color bodies. Color bodies in the part per billion (ppb) range can cause the phenol color to be off-spec. Additionally, running the CPC or PR bottoms temperatures too high may cause carryover of heavy components and color bodies into the phenol product.

In troubleshooting, take samples of the PR overhead and bottoms. If one or both has a high color, increase the drag rate from that part of the column.

As discussed above in the phenol rundown or product temperature is in excess of 60 deg C, decrease the phenol product temperature to 50 – 55 deg C or change the heating medium.

X. Normal Shutdown

The procedures given in this section are intended for normal shutdown situations, when no emergency or special situations exist. The purpose of the shutdown procedures is to provide a safe and reliable way to take the process off stream. This section will address special process considerations, if any, specific to shutdown conditions which may be required to assure optimum performance of the process during subsequent operations.

The procedures outlined below encompass a total plant shutdown, i.e. shutdown and de-inventory. Because normal shutdowns do not require the same procedures as do total plant shutdowns, many of the items can be omitted from the procedures in shutdowns of shorter duration, e.g. utility failures, equipment failures, or any production stoppage without de-inventory. The following procedures are easily adaptable to most of the types of shutdowns that will be met during the operation of the unit and the basics in them should be made use of in preparing more specific shutdown instructions. Following an emergency shutdown (as described in section XI, Emergency Procedures), some variation of the following procedures may be necessary to further shut down the unit.

As a standard practice, other operating units and personnel in the refinery/chemical plant should be informed beforehand of the upcoming shutdown activities. Changes in feed demand, steam demand, make-up hydrogen, etc., will affect other units.

For the complete plant shutdown, inventories must be worked down to prevent excessive by-product formation in the tankage. Also, tankage must be available to recover the plant inventory for holding until the subsequent re-start.

All equipment that is to be entered or worked on should be drained, thoroughly flushed, and purged. A procedure for flushing should be prepared for each section of the unit. Standard refinery safety practices should be adhered to when making any vessel safe for entry.

A. Oxidation

There are two points to keep in mind during the shutdown of the oxidation section. The first is of course, safety. The temperature of the oxidizer inventory and the O₂ concentration of the vent gas must be kept in safe ranges at all times.

Secondly, the eventual restart of the oxidation section will be made as easy as possible if there is sufficient CHP to get the oxidation reaction.

The general method for shutting down the oxidation, concentration and decomposition sections will be as follows:

- a. Notify other units and utility stations of the shutdown
- b. Prepare tankage for plant inventories
- c. Stop production of CHP
- d. Reduce unit CHP concentration
- e. Shutdown decomposition and de-inventory
- f. Begin Oxidation shutdown and de-inventory
- g. Shutdown concentration and de-inventory
- h. Complete Oxidation shutdown and de-inventory

There will be some operations which overlap. Each of the process sections will be described separately, however, and where operations overlap a reference will be made.

1. Prepare Tankage

Some CHP in the plant must be retained for the subsequent start-up. The unit is designed to have this material held in an oxidizer. In so doing, it is important to end with a CHP concentration that is sufficient to easily re-initiate cumene oxidation, typically 5-9%. Therefore, only one oxidizer can be worked on in a given turnaround while the other holds the CHP required for start-up. Typically, this is not a problem because the oxidizers themselves rarely require maintenance once they have been commissioned. If maintenance of the oxidizer upstream equipment such as the oxidizer day tank is required then the contents will have to be pumped into an oxidizer after reducing the levels to the minimum.

2. Reduce Production Rates

- 1) With tankage ready, reduce the plant production rate to 60% of the normal operating level. Make the reduction slowly, typically 10%/hr, to allow the overall operation to maintain a stable condition.
- 2) Reduce oxidation air rates accordingly, making sure to hold the spent air oxygen concentration at 7 mol %. With the production rate at 60%, the air rates should similarly be about 60% of normal.
- 3) Adjust the oxidizer temperatures to maintain the design CHP concentration.
- 4) Monitor the sulfuric acid concentration in the decomposition section and maintain it at its normal operating concentration by reducing the acid injection rate accordingly. Check the acid and CHP concentrations, as time permits, with laboratory analyses.

3. Shut Down Air to the Oxidizers

Before air is stopped, take the oxidizer level controllers off "cascade" and set each oxidizer feed on flow control. At this time, slowly block in fresh cumene to and from the oxidizer day tank. Proceed with the "air-out" as follows:

- 1) Slowly close the oxidizer #1 level control valve.
- 2) Over a span of 10 minutes reduce both oxidizer air rates to 15% of design, then stop the air entirely. Shut down the air compressor. Bring in blanketing nitrogen as required to keep the oxidizer overhead inert.
- 3) With no new production of CHP in the oxidizers and with CHP being purged to the decomposition section, the CHP concentration in the oxidation section will begin to decrease. In addition, the oxidizer No. 1 level will begin to decrease very slowly. Given that the current objective is to dilute the CHP concentration in the oxidizers, fresh cumene can be added intermittently to oxidizer No. 1 to maintain its overall level of about 60%. This will also ensure that enough material is available for the next start-up.
- 4) Allow both oxidizers' CHP concentrations to decrease to about 6-9 wt%.
- 5) As the terminal oxidizer CHP concentration falls, the temperature will also fall due to the lack of heat generated. The preflash column oxidate-feed exchanger will provide less heat. This will require more heat input from the steam exchanger. This transition may upset the preflash and flash column and should be closely monitored so the concentration section does not automatically shut down.
- 6) Monitor the flash column bottoms CHP concentration while that of the oxidizers decreases. When the flash column bottoms CHP concentration decreases to 70 wt%, place the oxidation and concentration sections on long recycle to terminal oxidizer #2. It is not practical and potentially unsafe to operate the decomposer at less than 70 wt% CHP. The increased cumene content will inhibit the decomposition reaction. Then shutdown the decomposition section per the instructions in this section of the manual.
- 7) Continue to circulate cooling water through the oxidizer coolers and allow the oxidizer temperatures to decrease to less than 50°C (120°F). After the decomposition section is shut down, and the unit is on long recycle, if it is required, the pump out of the oxidizers can begin.

4. De-inventory of the Oxidizers

This procedure is provided only in the case that the need is there. It is not normally necessary to completely de-inventory an oxidizer and if it is done, the inventory should be pumped to other oxidizer so that sufficient CHP or DCP for restart of the complex can be achieved. If it ever becomes necessary to completely remove all CHP from the plant, provisions should be made to purchase CHP in drums or road tanker from an appropriate source to ensure that the unit can be restarted. Since this procedure varies depending upon the different schemes used for each oxidation section, only the roughest outline will be given here, in order to communicate the ideas. Detailed instructions based upon actual equipment at site will have to be generated by operations personnel in concert with maintenance. The following should not be construed as detailed instructions under any circumstances.

5. Pump-out Remaining Oxidation Contents

- 1) With cooling circulation still on, begin pumping material from the oxidizer to be emptied to another oxidizer that is capable of receiving the material. In order for this to be possible, the receiving oxidizer must have a low level, and the unit cannot be on long recycle, since the excess CHP must be disposed of in some manner. Clean temporary lines may need to be added to transfer the material from a lag oxidizer to the lead oxidizer. Take the oxidizer that is being pumped out off of any level cascade, bypass any oxidate coming into it from upstream or any cumene feed to the oxidizer. Switch the oxidizer pump suction to the pump-out line.
- 2) When the circulating pumps lose suction, shut them down and complete the pump-out of the oxidizer to the receiving vessel.
- 3) Bring in a couple feet of cumene into the oxidizer and pump it out to the receiving vessel batch-wise to remove as much CHP as possible. When the CHP is diluted below 3%. Float the remaining hydrocarbon by adding water to the oxidizer and then pumping out the mixture. This two phase material should be sent through the oxidizer pretreatment section (FWC, CFSD, and coalescer) to remove the water.
- 4) Steam out the vessel to the closed sewer and make safe for entry according to standard plant practices.

At this point, one of the oxidizers has been flushed and emptied, and blanketed with nitrogen. Proceed with the removal of the remaining inventory in the process piping per normal plant procedures i.e. Nitrogen purge and steam-out. Do not leave any areas of stagnant CHP.

The oxidizer is now available for maintenance. All vessels should be washed with treated water and steamed to remove traces of hydrocarbon. Give the vessels 24-48 hours open to the atmosphere before any internal inspections are made. The circulation pumps for each oxidizer can be used to circulate treated water, in order to remove any sodium salts from the exchangers and oxidizer inventory.

B. Concentration

It is assumed in this procedure that the feed rate to the concentration section is at 60% of normal operating and that the section is on "long recycle". Long recycle was initiated by the shutdown of the decomposition section via the decomposer shutdown switch. The following steps involve the shutdown of the preflash/flash column, followed by a pump-out of the concentration section inventory to the oxidation section. If the shutdown is to be of a short duration (<24 hours) the circulation between the oxidation section and the concentration section can be left running. Extract the necessary information from the following instructions to effect this short shutdown. Remember that the most important point here is to not allow a

high concentration of CHP (anything over 20 wt% is high) to sit stagnant in lines for any length of time. Flush out dead end piping with cumene from the quench drum.

- 1) Slowly stop heat to the preflash column steam exchanger and flash column vaporizer. Maintain the preflash column feed rate to prevent the automatic low flow shutdown to trip while not overloading the bottoms pumps. If the pumps are overloaded the concentration section will shut down due to high flash column bottoms level. Block in the steam control valve and let the feed material cool off the exchanger to the terminal oxidizer temperature. Do not stop the feed to the concentration section yet.
- 2) There will probably be enough heat latent in the terminal oxidizer effluent at 80°C (175°F) to give a net overhead rate sufficient to run the net overhead pump. If this is not the case however, there is no problem. Simply stop and isolate the pump, and allow the receiver to fill and start backing up into the flash column. This will dilute the flash column bottoms to the point that its concentration is the same as the feed.
- 3) Pump the contents of the flash column receiver to the CFSD, and then shut down the flash column net overhead pump.
- 4) Reduce the feed to the flash column to the minimum flow, and then stop feed to the preflash column. This action will introduce cooling water to the flash column vaporizer.
- 5) Use the emergency shutdown button to start cumene quench to the PFC and FC. Stop cooling water flow to the feed vaporizer after sufficient flushing has occurred, generally at least two times the exchanger volume. Open the external drains valves on the preflash column tubs to increase the draining rate.
- 6) Draw down the hydrocarbon level in the decanter and stop the decanter pump.
- 7) Shut down the concentration vacuum system. When there is no more net water from the decanter shut down the decanter water pump. Attach a nitrogen hose to a convenient location in the concentration section and increase the pressure of the PFC, FC and related equipment to a slight (0.2 kg/cm²(g), 3 psig is sufficient) positive pressure, to prevent ingress of air during the shutdown.
- 8) Pump the contents of the flash column bottoms to the oxidation section via the long recycle line after flushing with cumene from the quench drum several times. Shut down the FC bottoms pump when its operation is no longer necessary. Drain the remainder of the PFC and FC bottoms to the CHP sump and then pump the CHP sump contents to the oxidation section.

At this point, the concentration section is now available to be prepared for maintenance. If necessary, flush equipment to be opened with treated water prior to steam-out to remove any hydrocarbon left in the vessels prior to entry. Leave any vessel to be entered open to the atmosphere for 24-48 hours.

C. Decomposition

The shutdown procedure outlined for decomposition is relatively simple, with safety being the primary consideration. Before starting, confirm that cumene flush is available. It is assumed that the decomposition section is currently at 60% of the normal operating feed rate.

- 1) Before starting the shutdown procedure, ensure that all personnel are familiar with the decomposition safety systems. Also, personnel in the fractionation area should be aware that they would no longer have crude product after this section has been shut down. In other words, if the fractionation section is not in the process of shutting down at this time, there is not much time left to have an orderly shutdown of that section while still leaving sufficient inventory in the fractionation feed tank to allow proper start-up. In other words, PLAN.
- 2) Reduce feed past the low flow trip. The automatic isolation valves should all close, isolating feed to the decomposer. In this manner, the low feed trip can be checked during each shutdown.
- 3) Check immediately to ensure that the isolation valves have closed, and that the solenoid valve putting dehydrator effluent to the decomposer has opened, allowing internal circulation. Flush the decomposer feed line with cumene to remove any residual CHP. The flush should last for at least 2-4 minutes, and the flow rate of cumene should be such as to flush the feed line at least 10 times its volume.
- 4) **DO NOT** decrease the decomposer temperature at this time. After taking a sample of the decomposer and confirming its contents, restart sulfuric acid injection at the rate that was in effect when the feed was stopped.
- 5) Begin increasing the temperature in the decomposer to react residual CHP. Increase the decomposer temperature to 70°C (160°F) in 30 minutes with sulfuric acid injection continuing.
- 6) The result of steps 4) and 5) above will be the decomposition of all CHP and DCP in the decomposer inventory. This will, in turn, result in all calorimeter delta T's going to zero. Monitor this process.
- 7) When the sulfuric acid concentration in the circulating stream reaches 300 wt ppm (minimum) the sulfuric acid injection can be stopped. This is the only valid criterion for stopping the sulfuric acid injection. Even if the calorimeter delta T's have gone to zero, do not stop the sulfuric acid injection until the circulating liquid has 300 wt ppm sulfuric acid, verified by the laboratory.
- 8) When the calorimeter delta T's have gone to zero, take a sample of circulating liquid to the laboratory to analyze for CHP and sulfuric acid. If the CHP is nil and the sulfuric acid is at least 300 wt ppm, respectively, then proceed to step 9) If not, continue to circulate the decomposer until there is no CHP in the system.
- 9) Circulate for another 4 hours minimum (at 70°C, 160°F) after the laboratory has verified that the residual CHP is zero.
- 10) At this point, the temperature of the decomposer liquid can be decreased to cooling water temperature by opening the cooling water flow control valve to the

decomposer cooler. Stop steam to the dehydrator and isolate the steam line. Allow the entire system to cool.

- 11) If the shutdown is to be of a short duration, and maintenance (i.e. opening) of equipment is not necessary, the circulating pump(s) can be kept on, but typically they are shut off after the system has cooled down.
- 12) If it is required to enter the decomposer or to perform maintenance on equipment, perform the standard procedures mandated by refinery practice for the safe entry of vessels.

D. Neutralization

Since no dangerous chemical reaction occurs in the neutralization system, its shutdown is a relatively simple affair.

- 1) Isolate the chemical injection pump when the decomposer is shutdown.
- 2) Take the sulfuric acid analyzer out of service and block it in.
- 3) If there is any inventory in the acid aromatics sump, this should be pumped out to the fractionation feed tank via the ion exchange chambers at this time. This would be the case if the decomposer or any associated equipment had to be drained for maintenance.

E. Fractionation

The main ideas to keep in mind during a shutdown of the fractionation section are the following:

- a. Safety
- b. Effective removal of pure phenol streams from equipment
- c. Maintain fractionation feed tank(s) nominal composition

The ideas presented above may seem obvious, but by keeping them in mind during a shutdown the restart of the equipment will be made much easier.

The following discussions, while presented in a serial manner, can in fact be implemented at the same time and actually one action (blocking in CAC bottoms) will in fact be referred to again (block in C-AMSC feed) in the next section. Therefore, it should be realized that shutting down any one column will of necessity shut down (or at least put on total reflux) any and all columns downstream of the affected column that cannot get feed from an alternate source. In general, it is possible to produce phenol from the crude phenol tank, if available, for a short time while the rest of fractionation is being shut down. It is similarly possible to hydrogenate AMS without regard to the rest of fractionation (as long as there is

crude AMS in the charge tank) or produce crude phenol while a problem with the PR is being straightened out.

Also, please remember that columns that do not need to have maintenance done do not have to be pumped out if the material in the column is not prone to freezing up. This is true for the materials in the front end of the fractionation system.

1. Crude Acetone Column

- 1) Decrease feed to the CAC at a rate of 15% of design per hour.
- 2) Stop diamine injection to the chemical treating reactor. Allow all the diamine to react out of the system.
- 3) Arrange to have a high level in the receiver and a low level in the bottom of the column. When a feed of 50% of design is reached, stop the feed to the column and shut down the CAC feed pump.
- 4) If in use, stop stripping steam (or condensate) to the column and isolate.
- 5) Stop feed to the FAC and C-AMSC. Place them, along with the CPC, on total reflux.
- 6) Slowly reduce the flow of steam to the CAC reboiler. Isolate the steam control valve and leave the reboiler steam supply isolation valve open to allow the steam to contract as it cools.
- 7) Allow the column to cool and stop reflux. Bring in nitrogen to the column as needed to prevent pulling a vacuum. Pump all the material in the receiver into the column, then shut down the overhead pumps.
- 8) Pump the contents in the bottom of the column to one of the fractionation feed tanks through the return header cooler. All material that is evacuated from equipment in the fractionation section should be sent to the same tank. The resulting blend can be used for re-inventorying upon start-up.
- 9) Remove the nitrogen connection from the column after the column is cooled off and there is no possibility of pulling a vacuum in the column.
- 10) Wash the feed line to the column with treated water, sending it to the fractionation feed tank.
- 11) If the column is to be entered, connect low pressure steam to the column and steam it out. Drain condensate to the closed drain header. When the acetone concentration in the condensate is below acceptable limits, the condensate can be sent to the sewer. Open the top vent. Shutdown the overhead condenser and vent gas condenser.

2. Finished Acetone Column

When the CAC is shut down, the FAC is placed on total reflux. To do this, discontinue acetone product to the acetone rundown tank, organic phase bottoms to the fractionation feed tank, and aqueous phase bottoms to the ejector condensate drum. Reduce the steam rate to the reboiler, while maintaining reflux. Complete the column shutdown as follows:

- 1) Line up the FAC bottoms drum to the fractionation feed tank only (keep aqueous outlet blocked in).
- 2) Stop steam flow to the reboiler and allow the column to dump and cool. When the receiver is empty, stop the overhead pumps. Stop caustic injection.
- 3) Raise the column pressure and isolate the column from the vacuum ejectors. Pressure with nitrogen to 0.2 kg/cm²g (3 psig).
- 4) Connect treated water to the receiver and pump the treated water into the column to flush. The water will displace the residual hydrocarbon present in the bottoms drum.
- 5) Connect low pressure steam and steam out the column. Drain condensate to the closed drain header or the ejector condensate drum until the acetone is below acceptable plant effluent levels, then drain to the plant sewer system.
- 6) Shut down the condenser and vent gas condenser.

3. Cumene-AMS Column and Chemical Treating Reactor

With the CAC shut down, the C-AMS column should be on total reflux. The shutdown of the C-AMS Column will always be accompanied by the shutdown of the chemical treating reactor (CTR). The procedure that follows assumes preparation for entry to both vessels:

- 1) Line up the CTR effluent to the fractionation feed tank through the return header cooler. Bypass reactor and begin to drain reactor. Stop feed to the C-AMSC, if not already done.
- 2) Maintain a high receiver level. Shut down the cumene-AMS caustic wash column (see instruction next section).
- 3) Stop steam flow to the column reboiler and open its bypass. Allow the column to cool and dump.
- 4) Isolate from the ejector system and break the column vacuum with nitrogen raising the pressure to 0.2 kg/cm²g (3 psig).
- 5) Open the tie line between the receiver boot and the overhead pump suction line. Continue to send reflux to the column until the receiver is empty.
- 6) When the reboiler steam temperature is less than 100°C (212°C), empty the reboiler to the appropriate system.
- 7) Shut down the condenser.
- 8) Connect treated water to the receiver and flush into the tower. Send the flush water through the CTR.
- 9) Connect steam, open the top vent, and steam out along with the treated water flush. Include in the steaming all contiguous lines which could contain phenol.
- 10) Shut down the column; steam out and flush. Pump the water through the CTR. Leave the column to cool.

- 11) Connect treated water to the feed line and flush through the PTR to the fractionation feed tank until the phenol level is below 200 wt-ppm. Afterwards, send it to the closed drain header.

4. Crude Phenol Column and Residue Stripper

These columns will be on total reflux with the shutdown of the C-AMSC. Net bottoms export will be stopped as will overhead export to the IXRT. Bottoms recycled from the Phenol Rectifier should be directed to the fractionation feed tank through the return header cooler. Steam the CPC feed line for 5-10 minutes, taking care not to break vacuum on the column otherwise some trays may be dislodged by the sudden change in pressure. Proceed with the shutdown of the column:

- 1) Leave a high level in the receiver and low level in the column base. Slowly close steam condensate flow to the reboiler, removing heat input. Do not isolate the reboiler supply, to allow for contraction.
- 2) Isolate the tower from the ejector system and break the vacuum with nitrogen. Raise the pressure to 0.2 kg/cm²g (3 psig).
- 3) Empty the contents of the receiver to the column and then shut down the overhead pump.
- 4) Pump the column bottoms to the fractionation feed tank through the residue column bottoms cooler. Do not reduce the bottoms level below 10% while steam is in the reboiler. A drain line is available at the bottom of the CPC column to allow draining to the FCDH.
- 5) Shut down the condenser and bottoms cooler.
- 6) When the steam temperature is less than 100°C (212°F), empty the reboiler to the appropriate system. Afterwards pump the remainder of the bottoms material to the fractionation feed tank. Drain any residual reboiler material to the closed drain header.
- 7) Connect treated water to the receiver and flush into the column. Send the wash water to the fractionation feed tank or the closed drain header.
- 8) When the column is cold (less than 80°C, 175°F), open the top vent and steam. Continue this operation and the treated water flush for 2-3 hours, removing wash water.
- 9) After stopping the treated water flush, continue to steam, draining condensate to the closed drain header. When the wash water meets the allowable plant effluent concentrations, switch to the plant effluent system.

5. Ion Exchange Resin Treaters

The resin treaters must be taken off line and emptied prior to unloading. The procedure is simply a subset of the regeneration procedures. The treater is isolated and then drained and blown down with nitrogen. Process water is used to flush the phenol from the resin. A shoot (e.g. trough) is attached to the side-

manway. The side manway is then opened and the resin flows out. Remaining resin must be shoveled.

6. Phenol Rectifier

At the time the CPC is shut down the PR will be placed on total reflux. Proceed with shutdown of the column as follows:

- 1) Leave a high receiver level and a low bottoms level. The bottoms should be lined up to the fractionation feed tank.
- 2) Isolate steam to the reboiler. Allow the column to cool and dump.
- 3) Stop the net phenol product and isolate from the phenol rundown tank.
- 4) Isolate the column from the ejector system and break the vacuum with nitrogen, raising the pressure to 0.2 kg/cm²g (3 psig).
- 5) Empty the bottoms to 10% and wait until the reboiler condensate has cooled to below 100°C (212°F). Pump the condensate to the appropriate system and the process material to the fractionation feed tank.
- 6) Shut down the condenser (steam generation system).
- 7) Connect treated water to the receiver and flush into the column. Steam the column out at the same time, sending the wash material to the fractionation feed tank. Continue for 5-6 hours to assure the column is cleaned.

F. Hüls MSHP (AMS Hydrogenation)

This procedure describes the steps required to prepare the hydrogenation unit for work in a hydrocarbon-free atmosphere. For shutdown of shorter duration such as feed shortage, off-spec feed from the cumene caustic wash column, off-spec product, etc., simply cool the reactor below 60°C (140°C), stop feed, maintain reactor effluent recycle and M/U hydrogen. Proceed with the full shutdown as follows:

- 1) Reduce the reactor feed slowly. Reduce the feed temperature while reducing the feed rate. Isolate the feed line and close the feed flow control valve.
- 2) Switch hydrogenated product from the oxidation section pre-wash to the hydrogenation feed tank.
- 3) Slowly reduce the reactor pressure to 0.5 kg/cm²g (7 psig) above the relief header pressure to avoid back flow of the vent gas.
- 4) Close and isolate the hydrogen supply. Shutdown the make-up gas compressor, if applicable. Purge hydrogen from the reactor section to the separator with nitrogen until the hydrogen concentration is less than 0.3 mol%.
- 5) Shutdown the reactor recycle pumps and continue nitrogen purging.
- 6) Drain residual hydrocarbon to the fractionation closed drain header. When no hydrocarbon liquid remains in the reactor loop, decrease the separator level to a

minimum. Maintain a small amount of liquid in the separator to avoid damage to the product pumps. When a minimum level is reached, isolate the separator and manually drain it to the fractionation closed drain header.

- 7) Depressure the reactor through the separator to the flare header, down to a pressure of 0.2 kg/cm²g (3 psig). Isolate the reactor.
- 8) Before entry to the separator steam out the vessel and then leave open to the atmosphere for 24-48 hours.
- 9) If the catalyst is to be changed, the reactors should be steamed to the separator to remove residual aromatic hydrocarbons before dumping the catalyst.

G. Phenol Recovery

Since the PRU, phenol springing and oil extraction systems are somewhat independent of the other phenol extraction columns, its shutdown can be delayed if necessary (i.e. if there is a high level in the sodium phenate tank or if fractionation is still operational). The following procedure should be used when it is desired to shut down this equipment.

1. PRU and Associated Equipment

- 1) Stop sulfuric acid injection to the delay pot. 1 minute later, stop sodium phenate to the Delay Pot.
- 2) Maintain Delay Pot recycle for at least 10 minutes to clear out any strongly acidic material from the delay pot inlet line.
- 3) When the sodium phenate to the delay pot is stopped, the PRU separator effluent to the OEC and sprung phenol will also stop. Block in the respective control valves on the aqueous and organic outlets of the PRU separator.
- 4) When aqueous effluent is stopped from the PRU to the Oil Extraction Column, unless it is desired to maintain the treatment of phenolic waste water, shut down the acid to the OEC and then (after a delay of at least 1 minute) shut down and block in phenolic waste water to the OEC.
- 5) Shut down the oil extraction system. (see next section)
- 6) If it is desired or necessary to enter equipment, it should be drained, purged, steamed out to the appropriate sewer, and made safe for entry per established plant practices. Otherwise, the system can be left as is for a quick restart of the PRU.

2. Oil Extraction Column and Solvent Caustic Wash Column

These columns are to be shut down after the PRU has shut down, thus ensuring that aqueous streams are treated or stored for subsequent removal of any phenol before discharge to secondary effluent treatment. Cumene solvent should still be circulating through the OEC and SCWC. Proceed as follows:

- 1) With the OEC feed rate at zero, close the feed isolation valve.
- 2) Maintain solvent circulation for 1 hour. Put the caustic to the SCWC on manual at the previous rate and wash the solvent of phenol for this hour.
- 3) Stop solvent circulation.
- 4) Stop caustic to the extraction oil caustic scrubbing column. Block in sodium phenate export to the sodium phenate tank. The bottoms level will increase when the caustic in the column falls down. This is not a problem.
- 5) Reduce the pressure on the system to 0.2 kg/cm²g (3 psig). Line up the solvent pump to the fractionation feed tank and pump the solvent drum empty. For the restart, use a fresh batch of solvent.
- 6) If it is desired or necessary to enter equipment, it should be drained, purged, steamed out to the appropriate sewer, and made safe for entry per established plant practices. Otherwise, the system can be left as is for a quick restart of the section.

3. Cumene-AMS Caustic Wash Column

Flow of hydrocarbon to this column will have stopped with the shutdown of the C-AMS column. Continue to inject caustic for about 10 minutes after feed is stopped to ensure total phenol recovery, and then proceed with the shutdown.

- 1) After feed to the column has been stopped, isolate caustic to the top of the column, and sodium phenate from the column bottom. The caustic in the column will drop down to the bottom and the level will increase. This is not a problem and will take care of itself during the next start-up. Stop the circulating pump and block it in. Prepare it for maintenance if necessary.
- 2) If the shutdown does not require entry into this column, isolation is sufficient to shut it down. They can be restarted when necessary. If entry of the column is necessary, make the column safe for entry per established plant procedures. The following points should be kept in mind.
 - a) The cumene caustic wash column and caustic settler hydrocarbon should be routed to the hydrogenation feed tank.
 - b) This column will have to be steamed out and/or flushed very carefully. The inlet weirs of this column have been seal welded and it may be necessary to flush these individually from the top with a water hose to remove the caustic from these wells. If this is not done, there is a possibility of personnel entering the vessel encountering sodium phenate or strong caustic, which should be avoided at all costs.

XI. Emergency Procedures

The purpose of the procedures in this section is to describe a safe manner to shut the process down during emergency situations while minimizing damage to the process. The procedures given in this section are general instructions which cannot be specific because of the variations in the design and construction of the equipment and the variation in operations philosophy of each operating company. The methods should be used by each operating company to develop a detailed set of emergency procedures for the process.

A. General Procedure to put the Unit on Standby

In an emergency, there usually isn't enough time to shut down the unit in an orderly fashion. It is most important to stop production and put the unit in a safe standby condition, so that full attention can be paid to the emergency situation. Variations on this procedure should be made depending on the nature and location of the emergency. Intermediate storage tanks allow some sections to be operated normally while other sections are put on standby. The following procedures consist of five parts (a through e), any or all of which may be executed to put sections of or the entire unit on standby.

1. Oxidation, Concentration, Decomposition, Neutralization

- 1) Stop air injection to the oxidizers and shut down the air compressor if necessary. Maintain all circulation until there is no significant heat of reaction being removed from the oxidizers. This assumes that the emergency is of such a nature as to stop production for several hours. It is not a good idea to let the synthesis of CHP continue in the oxidizers for any length of time without any removal of CHP. The result would be a rising CHP concentration in the oxidizers, which could be extremely hazardous. It is better to have to go through a restart including heating up of the oxidizers rather than risk an increased CHP concentration with such a large volume.
- 2) Stop decomposition charge via the emergency shutdown button. Verify that this is so. Flush concentrated CHP out of the feed line.
- 3) When feed to the decomposition section is stopped, the concentrated flash column bottoms automatically go to the oxidation section. All forward flow to the dehydrator cooler stops, and the dehydrator effluent is brought back to the decomposer. As soon as this happens, the following should be done to maintain the system ready for a quick restart.
- 4) Bring the decomposer temperature up to 70°C (160°F), normal start-up temperature.
- 5) Bring the circulating decomposer sulfuric acid concentration up to 300 ppm minimum. This will decompose all of the remaining CHP in the system and will

result in the calorimeter delta T's going to essentially zero. This is the normal feed in condition.

- 6) Flush the feed line between the isolation valves to the long circulation line with cumene for at least 45-60 seconds; this should automatically shut-off. The decomposition and concentration sections are now in a safe standby mode and can be left in this mode for at least 24 hours while the problem is being sorted out. Any more time than this in a non-producing mode should dictate that the oxidizers should be cooled down 50°C (120°F) or less, since CHP will decompose into DMPC in the oxidizer inventory in an oxygen deficient atmosphere.

If it is **not** possible to establish Long Circulation:

- 1) Stop feed to the preflash column.
- 2) Steam to the preflash column steam exchanger and flash column vaporizer should stop automatically when feed to these vessels is stopped. Verify that steam has been stopped.
- 3) Maintain recirculation through the flash column bottoms pump.
- 4) Stop sodium phenate and sulfuric acid injection to the PRU, unless it is desired (if the sodium phenate tank level is high, for instance) to maintain operation. Unless there is a problem in the effluent treatment section, the operation of this section is at the discretion of operations. This holds as well for the caustic washing columns. Unless there is a problem in fractionation, there is no specific reason to stop the operation of these columns, if the oxidation section can accept the cumene caustic wash column effluent. If this is not the case, this column will have to be bypassed and the material routed to the fractionation feed.

2. Fractionation

Stop charge to the CAC. Do not shut down the CAC charge pump, but let it spillback to the fractionation feed tank. Shut down all bottoms pumps and close all column feed flow control valves. Put columns on total reflux, with minimal heat input to the column reboilers. Stop stripping steam to the CAC and caustic injection to the FAC. Close the diamine injection. Bypass IX Resin Treaters. Maintain levels in all columns and receivers. If the environment is cold, then the IXRT must be emptied to prevent freezing of the phenol in the vessel. The phenol should be blown down with nitrogen. Wet phenol resin can be allowed to freeze without any negative impact to the resin or phenol.

3. Hüls MSHP (AMS Hydrogenation)

Stop liquid charge to the hydrogenation reactor and shut down the hydrogenation charge pump. Maintain reactor effluent recycle and cool reactors and hydrogen purge lag reactor of hydrocarbon.

4. Phenol Recovery

Stop the liquid feeds and products to and from the scrubbing and extraction columns and hold the appropriate vessels liquid full. Try to maintain the temperature in the delay pot to prevent precipitation of the sodium phenate or sodium sulfate. If this is not possible, shut down the circulation pump and entry the delay pot and PRU and flush with water.

B. Oxidation

1. Loss of Fresh Cumene Feed

Loss of fresh cumene feed could result from several actions, either

- a. Loss of cumene in the day tank (someone forgot to keep it full)
- b. Shutdown of the fresh feed pump (most likely)

Since there is a low flow auto-start feature built into the cumene fresh feed pumps, coupled with emergency power, there should not be an instance of the fresh feed pump failing without the spare coming on. If maintenance has to be performed on the spare pump, this should be done on an emergency basis, and the spare put back on line as soon as possible.

Since the fresh feed pump delivers cumene to many other places besides the oxidizers, it is recommended to consider suspending operations if only one pump is available for an extended period. The lack of cumene flush to the decomposer feed line, to the decomposer, to any pumps that might require flush and, most importantly, to the cumene quench drum, should the quench be required, are strong incentives to maintaining both pumps available during normal operations.

2. Loss of Recycle Cumene (from Concentration and/or Fractionation Sections)

If cumene recycle is lost the spare cumene recycle pump (decanter pump or flash column overhead pump) should start automatically. If it does not, have operations start the pump and re-establish flow. It should be mentioned that without any reflux, there will be an increased concentration of CHP in the overhead liquid from the preflash and flash columns. In a short period of time, if the spare overhead pump does not come on or there is no spare available, for units with internal condensers the accumulator will overflow back into the preflash and flash column and the result will be inability to concentrate CHP and a shutdown due to high level in the flash column bottoms. Alternatively for units with external condensers the receiver will fill

until sufficient condensing duty is lost by liquid-covered tubes to cause the concentration section to shut down on high pressure.

The fresh feed rate from the oxidation day tank will increase substantially, and there will not be sufficient cumene feed to maintain production. The loss of recycle cumene will cause a large increase in inventory of cumene in the preflash and flash column internals within a few minutes and soon after it will not be possible to operate the synthesis section. The units will have to go on long circulation until the problem can be solved.

The loss of recycle cumene from the fractionation section (C-AMSC overhead) is addressed in that section. If the reflux pump of the C-AMSC shuts down and there is no spare and/or the pump cannot be restarted, the primary fractionation section will have to be shut down.

3. Loss of Combined Feed Pumps

Loss of the combined feed pumps will immediately cause the combined feed surge drum level (CFSD) to increase and the fresh feed to stop on level control. The decomposition section will have to be shut down in a short period of time, 5-10 minutes due the concentration section needing to be put on long recycle as the high level in the CFSD will not allow re-processing of recycle cumene. The concentration section can be maintained on long recycle back to the terminal oxidizer. If the problem is not fixed in a reasonable time, the concentration section should be shut down and oxidizers cooled. The concentration section should not be operate on long recycle for more than 24 hours.

4. Loss of Air Compressor

Loss of oxidizer air can result from compressor failure or by action of the emergency trip system. If the compressor can be promptly restarted and/or the solenoid shutdown valves quickly reset, air can be reintroduced immediately. However, if compressed air cannot be regained immediately, the oxidizer temperatures may fall below suggested air-in temperatures, 80°C (175°F). In this case, put the unit on standby and refer to the air-in portion of Start-up Section VI.

The oxidation section should not be kept in a heated condition (greater than 50 deg C) for more than 24 hours with no air input. Doing so will result in CHP decomposing to DMPC and causing problems downstream in the decomposer. If any one of the problems mentioned in this section occurs to the degree of keeping the oxidizers out of production mode for more than 24 hours, UOP recommends to cool the oxidizers to 50°C or less to minimize side reactions.

5. Loss of Regular (Reliable) Cooling Water

This is an extremely serious situation.

The design of the UOP phenol plant has been carefully thought out to eliminate the possibility of total loss of cooling to the oxidizers for any length of time. Reliable cooling water, reliable and emergency power supplies and reliable cooling towers are all provided to ensure that the oxidation section never has a chance to operate without sufficient cooling. Even the oxidizer philosophy, which might at first seem less than optimal in terms of total CHP inventory, is actually **safer** than systems that operate at higher temperatures with reduced inventory. The typical operating temperatures employed in the UOP Phenol process give operations enough time (on the order of 20-24 hours at least) to get emergency cooling available. It is assumed, given the urgency of the situation, that the refiner would do anything possible to get cooling to the unit as quickly as possible.

The reliable cooling water supply should come on immediately if there is a loss of regular cooling water from the reliable cooling tower. The operators' first action if there is a loss of cooling water is to ensure that the air has been shut off to the oxidizers, and that heat input to the oxidizers (feed preheat) is minimized. The oxidizers should cool down sufficiently with the emergency cooling water if no heat is input to the system. If it is desired for the oxidizers to cool off more quickly, the oxidizers can be deluged with cumene from the oxidation day tank via manual block valves. There is usually plenty of time for operations to decide if this would be necessary or desirable. The excess cumene can be sent back to the combined feed surge drum when the unit is restarted.

If the problem is local, that is, a problem with the oxidation section cooling water that is not causing any problems with the rest of the plant, the oxidizers can be made more safe by continuing to remove oxidate to the concentration section, and sending flash column bottoms to the decomposition section, while not making any more CHP in the oxidizers. The resulting lower CHP concentration in the oxidizers will cause the temperature to increase more slowly, giving operations and/or maintenance more time to get the emergency and regular cooling water back in service. In addition firewater can be connected to the emergency coolers to cool the oxidizers.

6. Loss of Regular (Reliable) and Emergency Cooling Water

This is an extremely serious situation.

If the loss of regular cooling water from the reliable cooling tower is serious, the loss of both regular and emergency cooling water supplies could be disastrous.

Fortunately, between the cumene deluge provided for (described in the section above) and the extra time available for action by virtue of running the oxidation section at a lower temperature, it is anticipated that the loss of both regular and emergency cooling capability for a period of time long enough to result in an uncontrolled decomposition of the oxidizer contents is statistically unlikely. Fire water can also be used in the oxidizer emergency coolers if other sources are not available. It is UOP's recommendation that between the checking that must be a part of routine maintenance of the safety systems, and the favorable operating conditions employed in the UOP phenol process, the above scenario is statistically so unlikely that further safeguards other than the ones employed are unnecessary.

Nevertheless, if both cooling water supplies fail for an extended period and all of the operational actions taken to prevent CHP decomposition are in vain, then the area surrounding the oxidizers for a radius of at least 200 meters should be evacuated. If fired heaters in close proximity have not been shut down by this time, this should be done prior to evacuation. If firefighting water is available and if it has not been trained on the oxidizers prior to evacuation in an attempt to cool them down, it should be trained on the oxidizer area and left in that position prior to evacuation.

If the oxidizer temperature increases to 100°C (212°F), then the water spray system should be prepared to spray water into the top of the oxidizer. Water spray should only be used if other measures to cool the oxidizer are not available. The water will cool the oxidizers by generating steam from the hot oxidate. The steam will exit the top of the oxidizer and go to the condensing system. The water spray should be added when the oxidizer temperature reaches 110°C (230°F) and removed before reaching 100°C (212°F), preferably at 105 deg C. If water is added at 100°C (212°F) or below, then full vaporization of the water will not occur and water will accumulate in the oxidizer bottom. Liquid water in the oxidizer bottom is a very dangerous situation as there is the potential for rapid and violent vaporization when the oxidizer temperature does increase above 100°C (212°F) potentially resulting in loss of containment.

7. Loss of Oxidizer Circulation Pumps

This is an extremely serious situation.

The loss of flow through an oxidizer emergency cooler will require a field operator to commission and start the spare pump. If the spare pump cannot be started for some reason there are several options available.

The air should be shut off to the oxidation section to minimize heat generation, unless it is certain that the problem will be remedied shortly. If this is the case, then the air can be shut off to just the oxidizer in question. Otherwise, go on long circulation, shutting down decomposition. Do not try to do too many things at once.

Cumene deluge to the affected oxidizer can be initiated. This will cool the oxidizer to a large degree-especially if the material can be circulated to the other oxidizer and quench repeatedly introduced to the affected oxidizer until the circulation pumps can be restarted. This has the doubled effect of cooling the affected oxidizer and at the same time, reducing its CHP concentration. Directionally this is the way to proceed.

Nitrogen can be used to cool the oxidizer as well as mix the contents. Nitrogen will evaporate cumene which will absorb heat and cool the oxidizer. This operation should be combined with adding fresh cumene to prevent concentration of CHP through cumene evaporation.

If the loss of a circulation pump is limited to one oxidizer, then the cooling capacity of the others can be used to cool the affected oxidizer. Net feed cumene and recycle cumene (with absolute minimum feed preheat) can be bypassed around the unaffected oxidizer(s) to the oxidizer which has lost the circulation pump. The oxidizer pumps for identical size oxidizers are intentionally designed to be the same size. This design concept makes them interchangeable on an emergency basis.

8. Explosion or Fire

Do the following, if possible:

- 1) Notify the plant firefighting squad and other available emergency crews. The operating staff should concentrate on putting the unit into a safe standby condition, leaving the firefighting to the emergency crews.
- 2) The operating superintendent must quickly assess the seriousness of the situation. Except in the case of a small, remote fire that does not threaten the main process, the operations staff should, at the very least, attempt to put the endangered sections of the unit into standby mode. Following this, the operators should shut down all of the equipment in that section and wait for the firefighting crews to do their job. Every **SAFE** attempt to evacuate flammable materials from this threatened section of the plant should be taken to minimize the damage to equipment and help the firefighting crews. As time allows, the rest of the unit can be put on standby or shut down, depending on the seriousness of the incident.
- 3) The danger posed by a fire or explosion can be greatly alleviated by evacuating flammable organic materials from equipment in the accident vicinity. Some rules for evacuation of material from equipment are:
 - a) Only board-mounted controllers and switches should be used. OPERATORS SHOULD NEVER ENTER THE ACCIDENT VICINITY, UNLESS ASSISTING THE EMERGENCY CREWS.

- b) Often, however, block valves at battery limits can be safely reached. If this is possible, any flow of flammable material into the accident site should be blocked in after conferring with the firefighting crew.
- c) All other affected equipment in the phenol plant should be emptied by blocking feed to the equipment, while maintaining forward flow from the equipment. This method is safe, requiring only control room maneuvers. Evacuation of equipment to sump systems usually requires opening of manual valves and swinging of blinds, field operations that should not be attempted during a fire unless operations personnel are absolutely certain that the attempt can be made safely. Equipment can be replaced.

9. Line Rupture or Serious Leak

In the event of a line rupture or other serious leak of flammable or toxic material, do the following as quickly as possible:

- 1) Notify the refinery fire squad and other available emergency crews. For flammable materials, the likelihood of an ensuing fire or explosion should be the primary concern.
- 2) Identify the exact location of the rupture or leak. What material is involved? Is it flammable? Is it toxic?
- 3) For leaks of flammable materials, shutdown all fired heaters in the vicinity.
- 4) As quickly as possible, isolate the vessel or section of line that has failed, using auto control valves or manual block valves. Put nitrogen or steam purges upstream and downstream of the isolated equipment to prevent air from entering the system.
- 5) In the case of a vessel rupture, the amount of material spilled to the ground may be reduced by evacuating the vessel to drain, sump, or blowdown condenser.
- 6) If the leak or line rupture is inside a fired heater, add snuffing steam to the firebox.
- 7) Only that area of the unit containing the rupture or leak should be put in a standby condition, according to the standby procedure presented earlier. Often, even serious leaks can be repaired in a matter of hours. Ruptures generally require more time and a shutdown of the entire phenol plant may be carried out, as time allows, according to the total plant shutdown procedure.

A spill of material, flammable or otherwise, will not usually be an immediate threat to the entire phenol unit. Normal operation in the unaffected areas can be maintained to the extent of the capacity or supply in intermediate tankage. However, a spill of flammable material can easily ignite, so the operating staff must be prepared for the contingencies outlined in the previous discussion concerning explosions and fire.

C. Concentration

It should be noted that the oxidation section should not be kept in a heated condition for any length of time with no air input. This will result in CHP decomposing to DMPC and causing problems downstream in the decomposer. When any of the problems mentioned in this chapter occur to the point of keeping the oxidizers out of production mode for more than 24 hours, serious consideration should be given to cooling the oxidizers to at most 50°C (120°F) to minimize side reactions.

This applies to the concentration section as well. Long operation without making product will cause DMPC formation across the concentration section and should be avoided.

1. Loss of Feed

This will initiate the cumene quench to flush out the preflash column feed line, and will initiate the short recycle sequence, taking feed out of the concentration and decomposition sections. Since there will be no preflash column feed, the net material recycled to oxidation will decrease to zero based on flash column bottoms and receiver level control. When the feed can be restarted (terminal oxidizer pump) put material back to the flash column and return to normal start-up. Bring the decomposition section to "ready for feed in" status.

2. Loss of Cooling Water

This will cause a shutdown on high pressure due to lack of vacuum, condensing and cooling in the flash columns. The shutdown sequence will be initiated, sending the section on long recycle. Depending on the amount of vacuum available, there may or may not be flashing of cumene overhead. If not, shut down the net overhead pump, if applicable, and continue to circulate oxidate through the system and back to the oxidizers. Bring the decomposition section to "ready for feed in" status.

3. Loss of Vacuum System

See loss of cooling water, above.

4. Loss of Heating Media

This will force the concentration of CHP in the flash column bottoms to decrease to unacceptable levels for proper decomposition operation. Put the unit on long recycle, and bring the decomposition section to "ready for feed in" status. If the

vacuum in the unit is sufficient, there will probably be enough vaporization from flashing to operate the net overhead pump, if applicable; if not, shut it down.

5. Loss of Decanter Pump(s)

For units with the concentration section recycle cumene fed to the decanter pumps this will quickly (within a few minutes) result in filling the decanter with cumene and a high decanter level shutdown. Put the unit on long recycle, stop heat to the feed vaporizer, and bring the decomposition section to "ready for feed in" status. When the pump(s) are ready for service, proceed with normal start-up of the concentration section.

6. Loss of Flash Column Bottoms Pumps

This is an extremely serious situation.

Loss of the flash column bottoms pumps is also a loss of cooling of the concentrated CHP in the flash column bottoms pumps suction and discharge piping. The cooler is stabbed-in to the bottom of the flash column so the concentrated CHP in the column will retain cooling, but the CHP in the piping and any liquid above the level of the tubes will no longer be cooled.

Loss of the flash column bottoms pumps will initiate a shutdown of the concentration and decomposition sections on low decomposer feed flow. Without the flash column bottoms pumps all of the cumene quench to the flash columns will accumulate in the flash column bottoms and the level will go off-scale high. If the spare flash column bottoms pump cannot be restarted quickly the flash column bottoms and flash column bottoms suction and discharge piping need to be flushed with cumene and drained to the CHP sump. The contents of the CHP sump can then be pumped back to the oxidizers or feed wash column as needed. Additional cumene should be quenched into the preflash and flash columns to dilute and purge CHP from the flash column bottoms to the CHP sump.

7. Explosion or Fire

Please refer to previous section (oxidation).

8. Line Rupture or Serious Leak

Please refer to previous section (oxidation).

D. Decomposition

In many ways, the decomposition section is the most dangerous section of the plant. However, if normal precautions and attention to detail are maintained, this section's operation should be trouble free. Most of the emergency situations seen in the decomposition section will be the result of loss of sulfuric acid leading to high calorimeter delta T's, and problems related to instrumentation.

In the following subsections, the phrase "ready for feed in" is used. This refers to the following conditions:

- | | |
|---|---------------------|
| a. Decomposer Temperature | 70°C (160°F) |
| b. Sulfuric Acid Concentration | 300 wt ppm |
| c. Decomposer Circ. Water Concentration | <2 wt% |
| d. Water Injection Rate | 0 kg/hr |
| e. Decomposer Feed Line | Flushed with cumene |

1. Loss of Feed

This is not really a problem for the decomposition section. The main problem is that if the flash column bottoms pump is lost there will be no flash column bottoms circulation and cooling. This incident is covered in the concentration section emergency procedures.

Return the decomposition section to "ready for feed in" conditions.

If feed is lost, the decomposition section will go on internal recycle and the dehydrator effluent will return to the decomposer. Stop water injection, maintain sulfuric acid injection at the normal rate in effect at the time of the loss of feed until the sulfuric acid concentration of the circulating liquid is 300 wt ppm. Then increase the decomposer temperature (via the TIC) to 70°C (160°F). These are the conditions necessary for restart of the section when the feed capability is restored.

2. Loss of Decomposer Circulation

When the circulation flow rate is low the spare decomposer circulation pump will start automatically. If this fails and decomposer circulation decreases to the shutdown value the unit will shut down immediately due to low pressure drop across the decomposer cooler, and then high temperature in the decomposer. This will activate the automatic trip system, putting flash column bottoms back to the oxidation section on long circulation.

Return the decomposition section to "ready for feed in" conditions.

3. Loss of or Excessive Calorimeter Delta T's

This can be a serious problem, depending upon the specific circumstances. Since there are two delta T's to consider here, the various combinations and actions to be taken will be discussed separately. Since, in some circumstances, the loss of or increase in delta T in any calorimeter can be the symptom of a very dangerous situation developing, it should be checked out **immediately** and the corrective action taken.

It should also be stated that the use of the word excessive means high, but less than shutdown values. If the plant is shut down due to excessive total delta T caused by the second stage, the section should be put into the "ready for feed in" mode, and try again.

In each of the following scenarios, it is assumed that a current lab analysis of the circulating liquid has been taken, and the results either are available or will be available soon. If action is necessary immediately, it is indicated.

4. Loss of First Stage Delta T: Second Stage Lost

This is an indication of a lack of reaction in the calorimeters. In particular, since the sulfuric acid concentration (local) at the second stage should be greater than 4000-6000 wt ppm, it indicates that the decomposition reaction is either hindered very severely (dangerous) or that the decomposition reaction has taken place to such an extent that no CHP or DPC is in the system. This latter is not so dangerous and is in fact the normal state of affairs for decomposition section start-up. Therefore, check that the sulfuric acid injection system is functioning correctly and that the sulfuric acid concentration of the circulating liquid is at least 20 wt ppm. Check the decomposer liquid for residual CHP and DCP, and water concentration.

If the sulfuric acid concentration is OK, and all organic components are in reasonable ranges, the problem is either too much water in the system (reduce water injection) or the CHP concentration of the feed is too low (excessive cumene in the decomposer feed hinders the reaction-increase the CHP concentration of the flash column bottoms).

If the sulfuric acid concentration is below 20 ppm, this is an extremely dangerous situation, since the re-introduction of sulfuric acid to such a system in an uncontrolled fashion could result in the uncontrolled decomposition of CHP in the system, typically resulting in a relief event. This also implies that the sulfuric acid flow meters, if they appear to be working, are in fact erroneous, and the indication is not to be trusted. In such a case, the best recourse is to go on long recycle, flush out the feed line, recalibrate the instruments, and try again.

SLOWLY return the decomposition section to "ready for feed in" conditions while the instruments are being worked on.

5. Loss of First Stage Delta T: Second Stage OK

The most common reason for this happening in normal operation is an unnoticed increase in the acid concentration in the circulating system. Verify by the laboratory, check instrumentation and ensure that the water concentration of the circulating inventory is not excessive (reduce the injection if it is) and carefully lower the acid concentration to raise the first stage delta T to normal values.

6. Loss of First Stage Delta T: Second Stage Excessive

This could imply several things. The most dangerous is that the sulfuric acid concentration in the circulating liquid is too low. The relatively high local sulfuric acid concentration in the second stage is now seeing the residual CHP in the circulating liquid in addition to the DCP that is normally converted there. In this case, the proper action is to gradually increase the circulating sulfuric acid concentration by increasing the sulfuric acid injection carefully. Laboratory analyses should confirm any operational actions.

This also could happen during start-up. The DCP might be formed in slightly excessive amounts (DMPC might be in excess during a normal oxidation start-up) and it would form before the CHP concentration in the circulating liquid reaches a normal concentration. As long as the reaction is only due to slightly excessive DCP concentration in the circulating liquid, this is not really an emergency but a normal start-up. It should be stressed that this situation should not occur for very long before the normal CHP concentration is established in the circulating liquid.

7. First Stage Delta T OK: Second Stage Lost

The most common reason for this is that the sulfuric acid injection to the calorimeter has been cut off or lost somehow. The CHP decomposition taking place in the first stage is relatively unhindered, but there is insufficient sulfuric acid in the second stage to decompose the DCP. In this case, the sulfuric acid injection to the circulating liquid should be increased to make up for the loss of the calorimeter (if this has not already been done), and the problem (usually a plug in the line or flow meter or pump failure) should be rectified immediately. Do not operate without the calorimeters in operation.

This could also imply that the DCP concentration in the circulating liquid is low, which would result from a low DMPC concentration (relative to normal) in the flash column bottoms. This could happen in turndown situations where the operator has

opted to lower the oxidizer CHP concentrations. This would directionally result in lower DMPC formation. This is not really an emergency but the system should be monitored to ensure that sulfuric acid supply has not been cut off.

8. First Stage Delta T OK: Second Stage Excessive

One of the reasons for this might be a high DMPC concentration in the concentrated oxidizer product. Since the DMPC concentration in the feed to decomposition is directly related to the DCP concentration in the circulating liquid, high DMPC implies high DCP, which would result in directionally higher second stage delta T's without affecting the first stage.

9. Excessive First Stage Delta T: Second Stage Lost

This could also imply that the residual CHP concentration in the circulating liquid is high (a dangerous situation) and that either the DCP concentration is low or negligible (possible) or (more likely) that the sulfuric acid injection has been lost to that calorimeter, resulting in low or zero second stage delta T. This is analogous to item below, with the added complication of having one sulfuric acid injection point out of service for some reason. This should be handled by increasing the sulfuric acid concentration in the circulating liquid carefully to decrease the delta T in the first stage and bring up the delta T in the second stage to be able to monitor the DCP concentration.

With all of the sulfuric acid injection parameters set correctly and the water injection at the correct rate, there would be no other explanation for this behavior.

10. Excessive First Stage Delta T: Second Stage OK

This implies that the CHP concentration in the circulating liquid is increasing. This is a very serious situation if not rectified. The CHP concentration in the circulating liquid could be rising because the reaction is being hindered by water (analyze and reduce injection if necessary) or low feed CHP concentration (analyze and increase if required) or low circulating liquid sulfuric acid concentration. If the circulating acid concentration is low (less than 20 ppm) the flow rate of sulfuric acid to the bulk liquid should be increased very gradually to bring up the bulk acid concentration to greater than 40 ppm, at which time the delta T should come back in line. The key thing is to increase the sulfuric acid injection slowly.

11. Excessive First Stage Delta T: Second Stage Excessive

This implies that the decomposition reaction is not proceeding to a sufficient degree of completion, and that unreacted CHP (past the 1-1.5% that is normally in the circulating liquid) is beginning to build up in the system. This is a dangerous

situation, and the operational response is to slightly increase (i.e. by 5 to 10% of the current flow rate) the sulfuric acid injection to increase the circulating acid concentration up to 40 wt ppm. Remember that if the sulfuric acid concentration is very low (i.e. less than 20 ppm) the addition of sulfuric acid to increase the acid concentration should be done with great care, in order to increase the sulfuric acid concentration gradually and not so quickly that the decomposition of the excess CHP in the system is also gradual and not so quick that the relief system is involved.

12. Loss of Sulfuric Acid Injection

Partial loss of sulfuric acid injection (i.e. to one of the calorimeters) can be compensated for by increasing the sulfuric acid injection to the bulk liquid. If sulfuric acid injection is lost to both calorimeters then an automatic unit shutdown will be initiated. It is UOP's recommendation that the decomposition section not be operated without at least one calorimeter online. While it is literally possible to run the unit with only the bulk sulfuric acid injection, indications of circulating CHP concentration are impossible, and upsets in the chemistry of the decomposition reaction may not be visible until it is too late. It is better to shut the unit down and restart when sulfuric acid is available to at least one calorimeter.

Complete loss of sulfuric acid will be accompanied by immediate automatic shutdown. This is to ensure that the acid concentration of the circulating liquid does not decrease to the point that the CHP concentration of the decomposer inventory increases to dangerously high (or inconveniently high) value.

13. Loss of Dehydrator Heating Media

Two options are available: If the heating media is not going to be available for an extended period of time, the decomposition section can be run in a "complete conversion" mode, similar to start-up. By increasing the severity of the decomposition reaction (done by returning to feed-in conditions) the CHP in the circulating stream can be reduced to zero. However, residue make will be high and the economic performance of the unit will decrease considerably. This should only be the course of action if dehydrator heating media will not be available shortly. Otherwise, the section should be put on long recycle and the decomposer circulating stream returned to "ready for feed in" conditions, to await the return of the dehydrator heating media.

**UNDER NO CIRCUMSTANCES SHOULD APPRECIABLE
QUANTITIES OF CHP BE SENT TO THE FRACTIONATION
SECTION.**

14. Loss of Cooling Water to Dehydrator Cooler

High temperatures to the direct neutralization section will result in increased byproduct make and potential yield loss as the diamine reacts with acetone at increased temperatures. Additionally there can be increased losses of organics to the fractionation feed tank vent at high temperatures. To avoid this, stop feed to the decomposition section and put the synthesis section on long recycle. Put the decomposition section back at a "ready for feed in" status to await the return of cooling water to the dehydrator cooler.

15. Explosion or Fire

Please refer to oxidation section emergency procedures.

16. Line Rupture, or Serious Leak

Please refer to oxidation section emergency procedures.

E. Neutralization

The neutralization section is not subject to emergencies with the exception of loss of mechanical integrity of equipment, leading to fire or serious leak. Normal operational problems in the neutralization section are not classed as emergencies. However, lack of neutralization should not be taken lightly. Excess yield loss and corrosion will result if the crude product is not neutralized. To avoid this, stop feed to the decomposition section and put the synthesis section on long recycle. Put the decomposition section back at a "ready for feed in" status to await the return of diamine to the neutralization section.

1. Inability to Neutralize Crude Product

If, despite all effort, the pH of the crude product cannot be brought into the desired range, there are several possibilities, each of which should be investigated.

- a. The amine injection pump is not pumping correctly.
- b. The amine tank is empty.

Check the calibration of the field pH transmitter to ensure that it is operating properly. Check the sulfuric acid concentration of the crude product to see if it is higher than normal.

2. Explosion or Fire

Please refer to oxidation section emergency procedures.

3. Line Rupture, or Serious Leak

Please refer to oxidation section emergency procedures.

F. Fractionation

Emergencies in the fractionation section are to be handled in a routine manner, with the exception of the fact that most of the liquids being handled here are toxic (phenol) or extremely flammable (acetone). The more typical problems and usual responses are discussed below.

1. Explosion or Fire

Please refer to oxidation section emergency procedures.

2. Line Rupture, or Serious Leak

Please refer to oxidation section emergency procedures.

3. Loss of Cooling Media

If one of the columns loses overhead cooling, the entire section will have to be shut down. Since overhead cooling is typically cooling water, if cooling water is lost, the most important aspect of emergency procedures is to take care of the synthesis section.

If the cooling media failure is local however, put the columns before and after the affected column on total reflux, and take the heat out of the affected column until cooling media is again available. If cooling media will not be available for an extended period, proceed with shutdown.

4. Loss of FAC Bottoms Pumps

Loss of the FAC bottoms pumps is a concern because these pumps feed the FAC partial reboiler / CPC condenser. Loss of FAC bottoms will result in increased vaporization and therefore increased fouling of the FAC Reboiler / CPC Condenser. Efforts should be made to start the spare FAC bottoms pump as soon as possible. If the FAC bottoms pump cannot be restarted the entirety of the fractionation section will need to go on circulation or total reflux. Specifically the CPC should be shutdown to avoid additional vaporization of FAC bottoms in the CPC condenser.

5. Loss of Vacuum

This will force a shutdown of the fractionation section. Stop heats to the columns, reflux the contents of the receivers into the columns and pump out the column bottoms per normal shutdown procedures.

G. Phenol Recovery

The real emergencies that this section is likely to encounter are few; typically they would be those brought on by emergencies in other parts of the plant.

1. Explosion or Fire

Please refer to oxidation section emergency procedures.

2. Line Rupture, or Serious Leak

Please refer to oxidation section emergency procedures.

3. Loss of Caustic for Scrubbing

The PRU and associated equipment for recovery of phenol from sodium phenate should be shut down immediately to avoid putting excessive phenol to secondary treatment. This would happen since no caustic would be available to wash the phenol from the SCWC rich solvent. Solvent circulation can be maintained, but the OEC outlet water should be isolated to await the return of caustic service.

This will result in the loss of scrubbing ability for C-AMS column overhead organic stream. If the loss of caustic is temporary, these streams can be routed to the fractionation feed tank for a short time. However, the resulting fractionation feed will quickly become so enriched with cumene and AMS that it will start to upset and/or overwhelm the fractionation section's ability to remove it. Recycle should not be instituted except for short periods. How short depends upon the current capacity and the size of the fractionation feed tank(s).

If caustic is not going to be available for an extended period, the fractionation section should be put on total reflux or shut down.

H. Hüls MSHP (AMS Hydrogenation)

Since this section is comprised of a reactor, a compressor (if required), and hydrogen, the emergencies anticipated can be different.

1. Line Rupture, or Serious Leak

Please refer to oxidation section emergency procedures.

2. Explosion or Fire

Please refer to oxidation section emergency procedures.

3. Loss of Cooling

Stop heat and feed to the reactor immediately to avoid venting cumene and AMS to the relief header from the separator.

4. Loss of Make-up Hydrogen

Start spare hydrogen compressor, if applicable. Fresh feed and make-up hydrogen will automatically stop. If the reactor temperature begins to run away maximize cooling water and recycle flow to the cooler and depressure the unit to the separator to remove hydrogen and stop the reaction.

I. Service System Failures

This discussion refers to plant wide failures of service systems, consisting of internal systems (e.g., refrigerated water) and external or battery limits services (e.g., utility nitrogen). Certain emergency considerations that are peculiar to each service system are discussed. However, there are certain actions that are suggested for any service system failure:

- a. Immediately assign part of the operating staff to determining the extent and cause of the service failure. The remaining staff should concentrate on handling the process consequences of the failure.
- b. For loss of services at battery limits, cooperate with neighboring units to determine the extent and cause of the failure. Ensure that an adequate attempt to correct the problem is being made.
- c. For loss of internal services, contact the appropriate maintenance people, such as electricians, instrument specialists, etc. Cooperate with them in returning the service system to operation.

For a particular service system failure, refer to the appropriate discussion below:

1. Power Failure

Power failures come in any manner and degree. A total, long term power failure is the most serious situation, shutting down the entire unit for the duration of the power outage. A more common type of power failure is the "power blip." A power blip can also shut down the entire phenol plant, or it can severely upset operations by tripping out just one or more critical pieces of electrical machinery. The effect of a power blip is unpredictable, as any number or variety of different pieces of electrical equipment will be affected by each occurrence. The operators should be mindful of the following points concerning all power failures:

- 1) In the event of an extended power failure, an emergency power generator is available for restarting critical electrical equipment. This emergency power system may not be capable of supporting normal unit operation, but it will be sufficient to allow the operators to get the unit into a safe, standby condition. As a minimum, the following equipment will be provided with an emergency power supply:
 - a) Oxidizer Circulation Pumps
 - b) Concentration Flash Column Bottoms Pumps
 - c) Oxidizer Fresh Feed Pumps
 - d) Emergency Cooling Water Supply
- 2) For any power loss longer than a blip, the operators should immediately begin procedures for starting the emergency power generator, unless the start of this generator is automatic (it usually is). Refer to the manufacturer's instructions for these procedures. As soon as primary or emergency power is provided, restart the above equipment, in the given sequence.
- 3) Batteries provide emergency power to all electrical instrumentation. This auxiliary power is supplied automatically and instantaneously upon failure of the primary instrumentation power supply.
- 4) A power blip will always cause the overhead lighting in the control room to flicker. For this reason, this lighting should be left on continuously, day and night. Also, any time the lights are seen to flicker, the entire unit should be quickly surveyed to locate any isolated pieces of equipment that may have tripped out. The loss of electrical equipment, especially fin-fan motors, will not always be immediately signaled by an alarm. Isolated pieces of equipment that have been tripped by a power blip should be restarted as quickly as possible.
- 5) A power failure in the phenol plant may or may not affect the other service systems. In the event that the power failure does shut down other services, the operators should refer to the appropriate service system failure in the following discussion.
- 6) It is impossible to give separate procedures for every possible combination of equipment losses following a power blip. The following procedure should be used as a guideline for a total and extended power failure:

- a) Start the emergency power generator immediately. Refer to the generator manufacturer's recommended start-up procedure.
- b) As soon as auxiliary power is available, restart the equipment listed in item (1) above, in that order, unless auto-start has been provided.
- c) With step (b) above completed and any imminent danger eliminated, a few additional steps will bring the unit to the previously defined standby condition. From here, either start-up or shutdown activities can be initiated.

2. Instrument Air Failure

Loss of instrument air, if not regained almost instantaneously, will bring production to a complete halt. However, unlike a power failure that can create some critical situations requiring immediate attention, an instrument air failure will automatically put the unit into a safe condition. This safe condition is predetermined by specification of control valves and solenoid-operated shutdown valves as fail-open or fail-close type valves. The operators should memorize the fail action of all control valves and solenoid-operated shutdown valves in the unit. The following general rules regarding the fail actions of valves in the phenol plant may prove helpful:

- 1) Control valves on heating mediums (such as hot oil or steam) or fired heater fuels will fail closed.
- 2) Control valves on cooling mediums (such as all cooling waters) and refluxes and spillbacks will fail open.
- 3) Feed to the decomposition section will be blocked in, and long recycle will be automatically initiated.
- 4) Air to the oxidation section will be cut off. This is essentially a heating medium, since the oxidation reaction is exothermic.

In the event of an instrument air failure, these guidelines should be followed:

- 1) If the instrument air failure is due to a malfunction in an air dryer system, simply bypass the drying system until it is repaired. This is a stop-gap measure: Putting wet instrument air to instrumentation for any length of time (i.e. more than 6-10 hours) will probably result in water in the instruments and maintenance headaches for instrumentation engineers.
- 2) Ensure that all valves have failed to their proper fail-safe position.
- 3) If necessary, the unit can be fully shut down according to the Total Plant Shutdown procedure without instrument air by using valve hand wheels and control valve bypasses.

3. Cooling Water Failure

Any cooling water system is susceptible to failure of two sorts: failure of the water cooling system, and failure of the cooling water circulation pumps. Generally, the latter is more critical as it means sudden loss of cooling capability. There are three cooling water systems in the phenol plant. These are: cooling water, tempered water, and refrigerated water. Since each has different causes and consequences of failure, each will be discussed separately.

4. Normal Cooling Water System

This water is generally supplied to the unit's battery limits from an external cooling water system, which usually contains a cooling tower. Failure of the cooling tower circulation pumps or failure of the draft fans will cause total or partial loss of the cooling capability of the cooling water. The more severe of the possible consequences of cooling water failure are:

- a. Shutdown of air compressor in oxidation section
- b. Loss of vacuum in concentration section
- c. Shutdown of charge to decomposition section
- d. Loss of vacuum in fractionation section

5. Tempered Water System

Partial or total loss of the tempered water cooling capability will result from failure of the circulating pump or the cooler. In the event of tempered water failure, these possible consequences are most serious:

- a. Loss of vacuum in phenol columns (where phenol is an overhead product)
- b. Loss of cooling for phenol products and residue stripper bottoms.

Item a. will probably necessitate that the phenolic fractionation section be shut down immediately. The lack of pressure control in the phenol rectifier and crude phenol column will put a great deal of phenol in the vacuum system purge, which might cause effluent problems later. In addition, the higher temperatures necessary to reboil the phenol will put the product off spec with respect to color anyway, so those columns should be shut down with only enough heat on the bottom to maintain a temperature above the freezing point of phenol. The columns can be restarted when tempered water is again available.

6. Chilled Water System

Failure of the circulation pump or failure of the refrigeration unit (e.g., refrigerant compressor) will mean partial or total loss of the refrigerated water cooling capability. This will affect the following:

- a. Oxidizer Spent Air System
- b. Preflash-Flash Column Vacuum System
- c. FAC Venting Losses and possibly Vacuum Control
- d. Acetone Tankage Venting Losses

6.1 Oxidizer Spent Air System

Loss of chilled water to the oxidizer spent air system will result in excessive cumene going to the charcoal absorbers, resulting in excessive cumene discharge to the atmosphere. This is undesirable from economic, environmental and safety considerations, though it would not necessarily shut down the plant, the presence of large amounts of cumene vapor anywhere near a fired heater (hot oil heater) is of course something to be avoided, lest an explosion occur. For this reason, a lack of availability of the chilled water for any period of time should probably result in cutting the air out of the oxidation section, to await the restart of chilled water.

6.2 Preflash-Flash Column Vacuum System

Loss of chilled water to the preflash and flash column vacuum system may or may not result in necessitating either turndown of the concentration section or shutdown, depending on how much the load on the vacuum system in the concentration section increases. If it increases too much, and high pressure in the system cuts heat to the flash column, then the synthesis section should be put on long recycle to await the return of the chilled water. Otherwise, the load can be decreased by turning the unit capacity down. It is not known (all systems are different and this can depend upon ambient conditions) at this time if turning down system capacity would allow the unit to continue production.

6.3 FAC Venting Losses and possibly Vacuum Control

Loss of chilled water to the FAC vent condenser will result in increased losses of acetone to the vent system, and possibly problems with the vacuum system pressure control that may affect the fractionation section as well. This may or may not (depending upon the current capacity and particular configuration) result in having to shut down the fractionation section as well.

6.4 Acetone Tankage Venting Losses

Loss of chilled water will result in increased acetone in the vent gas from the acetone storage tanks. This is an environmental problem, and depending on the duration of the chilled water shutdown, may result in a need to shut down the fractionation section.

Chilled water is generally used for reduction of contaminant or product concentrations in vent gases. Consequently, failure of the chilled water system may increase pollutant discharge and slightly decrease product yields until chilled water can be regained. Though the failure should not severely upset unit operation or interrupt production, it should be restored as quickly as possible.

7. Emergency Cooling Water System

If, in time of necessity, the emergency cooling water fails, the return of this utility should be viewed as a matter of the utmost urgency. If this service fails in a time of need (regular cooling water to the oxidation section has failed) there is little to do except everything possible to:

- a. Get the service restored as soon as possible.
- b. Cool off the oxidizers by alternate means.

Item a. needs no further discussion. The availability and reliability of the emergency cooling water system should be checked on a periodic basis by operations.

The cooling off of the oxidizers by alternate means can be helped by many things. It is important to realize that the concentration of CHP in any vessel is directly related to how much time there is before spontaneous decomposition occurs. The lower the CHP concentration, the longer the time available for action to restore cooling. The steps shown in the oxidizer loss of cooling section above detail the procedures required to stop the decomposition of CHP in the oxidation section. They should be followed and if any other procedures are found that have merit even in very specific circumstances, UOP would like to hear about how the procedure is implemented and the results of the implementation (ideally the theoretical results. We hope that any refiner never has an occurrence of loss of all cooling water).

8. Steam Failure

Steam is generally supplied to the phenol plant's battery limits from an external boiler system. Usually, steam at several different pressures is provided to cover a wide range of heating or process requirements. It is possible to lose any or all of the available steam systems. However, steam failure is usually a result of control valve or relief valve failure or the rupturing of a steam header or other serious leak. In these instances, only one of the steam systems is likely to be lost.

The most critical steam service in the unit is in the concentration ejectors. Failure of ejector steam will require immediate shutdown of the concentration section. Depending on the duration of the shutdown, loss of steam may require a partial or total shutdown of the phenol plant:

9. Nitrogen Failure

The nature and cause of a nitrogen failure depend on the type of nitrogen system that the refiner has. Systems that evaporate liquid nitrogen do not generally require a compressor. Refiners that produce their own gaseous nitrogen will normally require a booster compressor to deliver nitrogen at standard nitrogen header pressures. This latter system is susceptible to compressor failures. In any case, the operators should be familiar with the nitrogen system manufacturer's recommendations concerning system emergencies.

Unlike the other service systems, the nitrogen system is not extremely critical to normal operations. It is used primarily as a purge for various vent lines, a precaution against back flow of air or header gases into process equipment. Failure of the nitrogen system will not, by itself, interfere with normal unit operation. However, nitrogen serves a very important emergency function, especially in the oxidation system where it may have to be used to fluff the oxidizer contents if the conditions warrant. In this and other emergencies, it is vital that nitrogen be available. So, for primarily safety purposes, it is very important that the nitrogen system be returned to operation as quickly as possible following a system failure.

J. Automatic Shutdowns

Below is a brief description of the automatic shutdowns in the UOP phenol unit. While some details may change for a specification unit design, these descriptions should give the reader a sense of the purpose of the shutdowns. In Table XI-1 is the combined concentration/decomposition shutdown logic.

1. Oxidizers

A total of 18 temperature sensing points for each oxidizer are tied into the automatic shutdown system. High oxidizer temperature, high spent air oxygen concentration, high oxidizer pressure, high oxidizer level, low air control valve delta P, low circulation rate to the emergency coolers and a high level in the decanter will all cause an automatic shutdown of air to one or both oxidizers.

All oxidation shutdowns resulting in air stopping to the oxidizers are accompanied by a trip of the charcoal adsorber regeneration system since some flow from the

oxidizers to the charcoal adsorbers is required for the cool down steam following steaming.

If a high temperature condition persists in either oxidizer then, in addition to stopping air to the oxidizers, reliable cooling water is started to the emergency coolers to cool the oxidizers down to cooling water temperature. As well the concentration and decomposition sections are tripped. A high temperature in the oxidizers is very serious as there is limited to cool the oxidizers to a safe condition. The concentration and decomposition sections are shutdown so that operations can focus on the problem which is resulting in a high oxidizer temperature so they are not also trying to manage other critical operations.

For units wherein the circulating oxidate from oxidizer #2 provides heat to the preflash column feed-oxidate exchanger, then a high oxidizer temperature will also stop the flow of oxidate to the preflash column feed-oxidate exchanger. Although the preflash column removes heat from oxidizer #2 and therefore it seems reasonable to leave the preflash column operational, as discussed above, for safety and to ensure the focus of the operations staff only on oxidation, the concentration section is shutdown. Oxidizer #2 can be fully cooled by the emergency cooler and does not require the heat duty of the preflash column to aid in its cooling.

An auto-start circuit starts the spare fresh feed pump upon a low pressure switch signal on the cumene feed header pressure transmitter.

An auto-start circuit starts the spare decanter pump upon a low flow switch signal on the pump discharge flow meter.

2. CHP Sump

The CHP sump's main purpose is to collect the liquids containing CHP from the CHP closed drain header. CHP must be handled separately from the other streams to maintain safety. The CHP sump pump in the sump then sends the CHP/cumene liquid back to the lead oxidizer automatically by a high level pump start switch. An automatic quench of cumene fresh feed is added on high sump temperature in case of thermal decomposition of the CHP in the sump. CHP is added to the sump for initial seeding of the oxidizers.

3. Preflash and Flash Columns

Concentrated CHP at elevated temperatures is considered unstable over time. Even though the UOP design minimizes the time that the CHP is at elevated temperatures, there is some possibility that decomposition could take place in the flash column. A cumene quench drum is provided in this section for emergency quenching of the concentration section equipment to cool the columns' contents and dilute the CHP.

The cumene quench drum has automatic filling via the cumene fresh feed pumps and an LIC on the cumene quench drum level. The drum is elevated above the equipment it is protecting, so it does not have to rely on a pump during an emergency. The cumene gravity flows from the cumene quench drum. The control valves regulating the flow of quench are fail open. The cumene quench drum is divided in four sections. One section flushes the preflash column feed line upstream of the control valve and the preflash column bottoms. Another section quenches downstream of the feed flow control valve including the preflash column heater tubs. Another section quenches the flash column vaporizer. The last section quenches the flash column accumulator tray and the concentrated CHP in the bottom section.

Low feed flow, high temperature, high pressure, and high bottoms level will all shutdown the decomposer and flash column vaporizer and put the flash column overhead and bottoms on long recycle back to the terminal oxidizer. Simultaneously, the preflash column feed flow control valve closes, and a small circulation continues back to the oxidizer on short recycle.

In addition, a high temperature in a specific area of the flash column will result in cumene quench being sent to that area. Automatic line size valves will open upon high temperature to quench the high temperature area. Typically the flushing is stopped after the temperature is reduced below the shutdown point so excessive quantities of cumene are not sent to the oxidizer, diluting their CHP concentration.

There is also an emergency purging of the condensate in the flash column vaporizer with reliable cooling water as a result of a high temperature in and around the vaporizer. There are two switches that will cause the cooling water to start, a high temperature with a time delay and a higher temperature without a time delay. The two switches are added to minimize the chance of reliable cooling water being mixed with condensate unless a high temperature persisted or was excessively high.

Further shutdown cause and effects are shown below in Table XI-1.

4. Decomposer

Two automatic shutdown valves isolate the concentrated CHP feed from the decomposer; one valve is located upstream and the other downstream of the decomposer feed flow control valve. Upon decomposer shutdown concentrated CHP from the flash column bottoms will mix with the flash column overhead recycle cumene and circulate back to the terminal oxidizer. All other stagnant decomposer feed piping is flushed with fresh cumene during a shutdown by the decomposer feed flush drum. Cumene from the cumene flush drum flushes the isolated piping around the decomposer feed control valve back to the terminal oxidizer via the long

circulation line. As well the cumene flush drum flushes the long circulation line clear of CHP following decomposer start-up. The flush drums act as a cumene surge volume for gravity flow and positive isolation from the fresh cumene supply.

The decomposer can be shut down by a shutdown in the concentration section, by pushing the decomposer shutdown button, low decomposer feed flow rate, high and low decomposer temperature, high decomposer level, low process flow to the calorimeters, high calorimeter first stage or total delta temperature, low acid injection flow rate, low decomposer circulation flow rate and high dehydrator outlet temperature. When the decomposer is shut down, the decomposer water injection flow is stopped to prevent a build-up of water in the decomposer. Too much water will inhibit the cleavage reaction.

An auto-start circuit starts the spare decomposer circulation pump upon a low flow switch signal on the decomposer circulation elbow tap flow meter.

Further shutdown cause and effects are shown below in Table XI-1.

Table XI-1: Concentration/Decomposition Interlock & Shutdown Logic System

TAG	SERVICE	ACTION
FSLLA/B	Feed to Preflash Column	1, 2, 3, 11
TSHH	Preflash Column Feed-Oxidate Exchanger Tub Liquid	1, 2, 3, 11
TSHH	Preflash Column Steam Heater Tub Liquid	1, 2, 3, 11
TSHH	Preflash Column Bottoms	1, 2, 3, 11
LSHH	Flash Column Bottoms	1, 2, 3, 11
LSHH	Preflash Column Bottoms	1, 2, 3, 11
PSHH	Preflash/Flash Column Overhead Vapor	4, 11
TSHH	Flash Column Vaporizer Vapor Space/Outlet	1,2,3,5,6,9,11
TSHH	Flash Column Vaporizer Vapor Space/Outlet	1, 2, 3, 5,6,8,11
TSHH	Flash Column Bottoms	1,2,3,5,6,7,9,11
TSHH	Flash Column Bottoms Pump Suction	1,2,3,5,6,7,11
TSHH	Oxidizers (for cascaded reliable CW only)	3, 10

LSHH	Oxidation Decanter	3
HS	Concentration Section Emergency Shutdown	1,2,3,5,6,11
FSLLA/B	Decomposer Feed	12, 13
LSHH	Decomposer	12, 13
TSHH	Decomposer	12, 13
TSL	Decomposer	12, 13
TSHH	Dehydrator Effluent	12, 13
FSLLA/B	Acid Injection	12, 13
PDSLLA/B	Decomposer Cooler (Circulation)	12, 13
TDSHH	Calorimeter 1 Total Delta T	12, 13
TDSHH	Calorimeter 1 Inlet Delta T	12, 13
TDSHH	Calorimeter 2 Total Delta T	12, 13
TDSHH	Calorimeter 2 Inlet Delta T	12, 13
FSLLA/B	Circulating Liquid to Calorimeter Systems	12, 13
UC	Concentration Section Shutdown	12, 13
HS	Decomposition Section Shutdown	12, 13
PSLL	Process Water to Decomposer	14
HS	Steam to Dehydrator Enable/Shutoff	15

Action 1: Stop feed to PFC, drain & flush upstream of PFC FCV, cumene quench to PFC heater tubs and FC vaporizer

Action 2: After time delay close cumene flush upstream of PFC FCV

Action 3: Immediately stop steam to PFC heater and FC vaporizer, vent steam/condensate from PFC heater and FC vaporizer and trip decomposition section. Vent condensate pot and stop condensate pump if applicable.

Action 4: After time delay stop steam to PFC heater and FC vaporizer, vent steam/condensate from PFC heater and FC vaporizer and trip

decomposition section. Vent condensate pot and stop condensate pump if applicable.

Action 5: Start cumene quench to flash column collector tray

Action 6: After time delay stop cumene quench to flash column collector tray

Action 7: Start Cumene quench to flash column bottoms

Action 8: Immediately start reliable cooling water quench to/from flash column vaporizer and close condensate from flash column vaporizer

Action 9: After time delay start reliable cooling water quench to/from flash column vaporizer and close condensate from flash column vaporizer

Action 10: Stop reliable cooling water to concentration section

Action 11: Trip oxidation section

Action 12: Stop feed to decomposer, stop steam to dehydrator, divert dehydrator effluent back to decomposer, stop water to decomposer, divert recycle cumene and flash column back to terminal oxidizer via long circulation line, stop cumene flush to long circulation line.

Action 13: After time delay cumene flush to decomposer feed line and cumene flush of decomposer feed control valve between the isolation valves.

Action 14: Closes decomposer water injection isolation valve

Action 15: Manually shuts off steam to dehydrator even when the rest of logic allows it

5. Fractionation Crude Acetone Column

The CAC column pressure is run at a minimum positive pressure to minimize the temperature severity in the reboiler while still allowing the net overhead vapor to be pressured to the FAC. If the column pressure goes high, then a switch will automatically close the condensate flow control valve from the reboiler, thus stopping the primary column heat input and avoid the column exceeding its design pressure.

There are also shutdowns on the acetone and phenol fractionation vacuum systems. A high level in the FAC condenser and a high temperature from the FAC vent condenser will isolate the vacuum system from the process to protect the vacuum pump.

In phenol fractionation, similarly a high temperature from the vacuum system suction drum or a high level in the suction drum will isolate the vacuum system from the process.

6. Hüls MSHP (AMS Hydrogenation)

Loss of make-up hydrogen, low fresh feed flow, low reactor effluent recycle flow and high reactor temperature will cause a shutdown of MSHP, stopping both make-up hydrogen and fresh feed to the unit.

7. Tempered Cooling Water Section

An auto-start circuit starts the spare tempered cooling water circulation pump upon a low flow switch signal on the pump discharge flow meter.

8. Chilled Water Section

An auto-start circuit starts the spare chilled water circulation pump upon a low flow switch signal on the pump discharge flow meter.

XII. Special Procedures

This section includes procedures not covered in other sections of the General Operating Manual or other documentation provided by UOP. Typically, these special procedures are performed as a normal course of action in the operation of the unit. The following procedures will be described in this section:

- a. Charcoal Adsorber Loading
- b. Charcoal Adsorber Unloading
- c. Ion Exchange Resin Treater Loading
- d. Ion Exchange Resin Treater Unloading
- e. MSHP Catalyst Loading
- f. MSHP Catalyst Unloading
- g. Sand Filter Backwash Procedure
- h. Crude Acetone Column Boil-out Procedure
- i. Hüls MSHP (AMS Hydrogenation) Shutdown, Water Wash, Dry-out and Hydrogen Reduction Procedure
- j. Routine Maintenance

A. Charcoal Adsorber Loading

As in the reactors, it is important for a successful and efficient loading to have the equipment and material ready prior to beginning the loading. Due to the fragile nature of the charcoal, the adsorbers should be loaded into a level of treated water. After final inspection of the bottom distribution grids, the adsorber should be half filled with treated water. If required, add or remove water during the loading to maintain a significant liquid depth for the charcoal to fall through before reaching the top of the bed. Utilize the multiple top manways and remove multiple top grids to ensure an even loading across the entire vessel. Drain and purge each adsorber with nitrogen after the loading.

B. Charcoal Adsorber Unloading

Ensure the charcoal has been through one complete regeneration cycle of steaming and cool down prior to unloading the charcoal. Note that following the charcoal steam regeneration and cool down hydrocarbon remains on the charcoal and proper precautions for handling must be taken. Prior to unloading the charcoal adsorber can be filled with water using the water spray lines available in each adsorber. Be sure to provide a vent for the adsorber to avoid over-pressuring it as the adsorber fills. After the adsorber is filled, drain the water from the adsorber to an appropriate location. The charcoal will not be saturated with water and be less reactive as it contacts air during unloading. Please contact your charcoal vendor for their recommendations on safe handling and disposal of spent charcoal.

C. Ion Exchange Resin Treater Loading

Due to the fragile nature of the resin the resin treaters are recommended, but not required, to be loaded into a level of treated water. Amberlyst 36 Wet resin is not sensitive to free water and can be loaded in free water like charcoal. After final inspection of the bottom collector grids, the treater is recommended to be half filled with treated water. As required, water can be added or removed during the loading to maintain a significant liquid depth for the resin to fall through before reaching the top of the bed.

After resin loading is complete and the vessel is closed, if the treater will not be used for a long period of time, the water can be drained and the vessel purged with nitrogen. Alternatively, the resin treater can continued to be filled liquid full targeting 50% fluidization with water in preparation for downflow water washing.

Fresh resin is water washed to remove excess acidity present from the manufacturing process. Water will extract the free acid species easily. The resin bed should be filled upflow with water at 50% fluidization to ensure the bed is expanded during filling and settles properly. Once the resin treater is liquid full, the flow should be reversed to downflow targeting a LHSV = 2.0 hr^{-1} and the effluent water sent to an appropriate destination. The pH of the effluent water should increase to the same as the fresh water pH within 3 - 4 bed volumes. Check the water pH every 15-30 minutes with pH paper during the downflow wash.

If the treater will remain offline after water flushing UOP recommends to wash the resin until the effluent water pH is within 1 pH unit of the inlet water pH. It is not desired to hold low pH water in the treaters and piping for an extended period of time. However, if the resin treater will be put online immediately after water flushing then it is not necessary to rinse to a pH specification, but instead wash 3 – 4 bed volumes. Further washing of the resin will be completed with phenol during the subsequent dry-down as detailed in Start-up Section VI, Chapter A.11.4.

On a recent design, the bottom profile wire screen distributor has been replaced by layers of ceramic balls. It is permissible to have a loading operator load the balls with the sock and stand on the smaller ceramic balls as long as his weight is distributed with a board of at least 0.2 m^2 (2 ft^2). The purpose is to prevent the ball layers from mixing during the loading. The ceramic balls should be either replaced or removed and screened each time the resin is changed otherwise they will eventually plug with resin fines.

D. Ion Exchange Resin Treater Unloading

Following hydrocarbon freeing of the ion exchange resin, the side manway of the resin treater can be opened and the resin removed. The procedure is very similar to how a clay treater is unloaded. Alternatively the resin can be vacuum unloaded from

the top manway. All spent resin from the previous load shall be confirmed as removed prior to loading the new resin.

E. MSHP Catalyst Loading

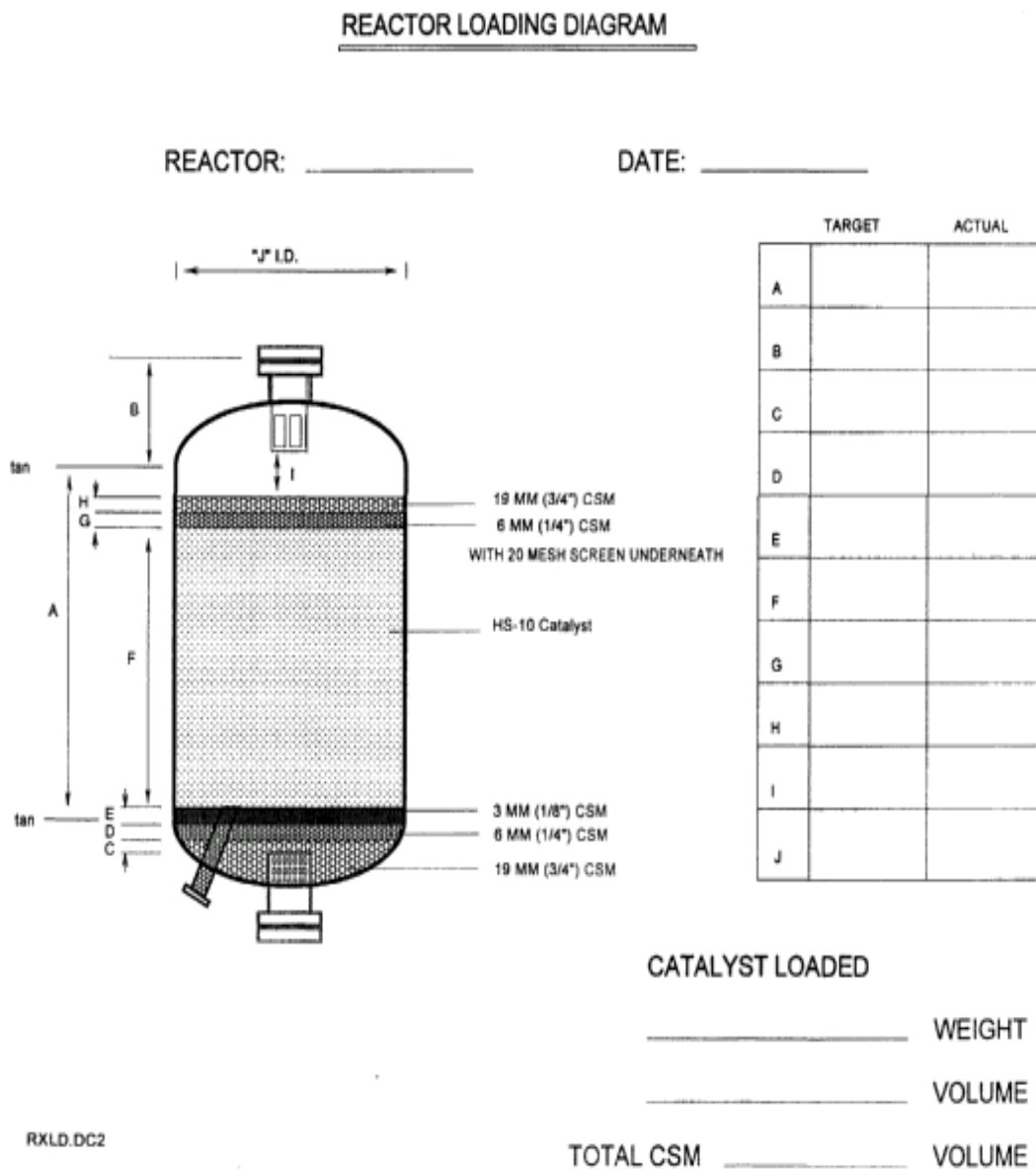
There are two catalytic reactors in the UOP Phenol Process: the hydrogenation reactors. Properly loading the MSHP catalyst is a major factor in obtaining the expected performance of the reactor. Experience has shown that reactor performance can be severely diminished by loading errors, those typically resulting in low ABD and therefore a lower mass of catalyst loaded compared to design. Loading errors can be greatly diminished by careful attention to proper records and measurements. The following set of general guidelines will also help ensure correct catalyst loadings.

- 1) Take an inventory of catalyst, support material, and any of the vessel internals not installed at the time of vessel inspection (inlet baskets and outlet screens, e.g.). The inventory should be made well in advance of the catalyst loading, as suggested in previous pre-commissioning steps.
- 2) The major reactor inspection should have been performed long before the catalyst loading, but the following items need to be rechecked:
 - a) The bottom distributors of each vessel should be correctly positioned and securely attached to the reactor base.
 - b) The screens at the top should be checked to verify conformance to design specifications, if not previously done.
 - c) The catalyst unloading nozzles must be checked to ensure that the closure plate is well positioned. Some coils of ceramic rope should be placed on top of the closure plate to prevent the spinning and possible release of the closure plate during the loading of catalyst support material into the nozzle.
- 3) All loading equipment should be checked for availability and operability:
 - a) Catalyst lifting device: winch or crane.
 - b) Transfer hopper or catalyst drum holster.
 - c) Loading hopper and platform.
 - d) Explosion-proof lamp inside the reactor, also flashlights.
 - e) Materials for protection of flange face and ring groove where loading is taking place: high temperature ceramic rope and wooden doughnut.
 - f) Canvas or plastic sheet for covering catalyst drums and open reactor inlets during inclement weather or loading intervals.
 - g) Rope ladder or other ladder for entering and exiting the reactor.

- h) Loading sock of correct diameter and sufficient length to run from bottom of loading hopper to bottom of reactor. Also need means of attaching sock to hopper.
 - i) Measuring tape and outage gauge.
 - j) Chalk, crayon, or grease pencil.
 - k) Air hose to keep ambient moisture out of reactor during loading.
 - l) Wood boards to stand on while leveling the catalyst bed or catalyst support material.
 - m) Flat leveling board and carpenter's level.
 - n) Dust masks for all personnel in loading area. Fresh air masks may be preferred by those inside the reactor.
 - o) Safety harness.
 - p) Whatever tools are necessary.
- 4) Transportation and manpower for supply of full catalyst drums and removal of empties should be planned to prevent delay and confusion during loading.
- 5) Staff and personnel involved in the loading activities should be properly informed as to the reasons and requirements for correct loading through use of pre-planned loading diagrams. Procedures for follow-up to loading details should be established.
- 6) Keep an inventory of all materials in the loading area. Do not leave any extraneous material in the vessels. Aside from the embarrassment of unloading to retrieve an object left behind during the loading period, serious problems during start-up or normal operation can result from an oversight in this matter.
- 7) Of special importance in all loadings is the finishing or "topping off" stage. At this point careful measurement should be taken to ensure that the proper amount of catalyst has been loaded and that the allowance for hold down material has been verified. Good practice is to have the loading supervisor personally verify the measurements and initial the loading diagrams.
- 8) Perform the following tasks in preparation for the loading:
- a) Ensure that the feed line and gas circuit in the hydrogenation section are fully dry before loading. If necessary, blow through lines with dry air or nitrogen and check low points for water.
 - b) Blind the reactor vessel from all process lines.
 - c) Obtain a final check from the safety department to ensure that relevant safety standards and procedures have been met and followed.
 - d) Protect the outlet flange by putting asbestos rope in raised face and covering flange surface with wood doughnut.
 - e) Position all loading equipment and attach the loading sock to the loading hopper.
 - f) Introduce a dry instrument air purge of the reactor to keep out ambient moisture.

Refer to the Project Specifications for the exact depth requirements of catalyst beds, catalyst support material layers, and hold-down material layers.

- g) Mark the reactor walls to indicate the fill height of each layer of support material, catalyst, and hold-down material. For each height, marks should be made at three or more points around the reactor circumference.
- 9) Load the bottom support materials and level each successive layer with a flat board and a carpenter's level. The persons loading the support material should stand on boards to distribute their weight and avoid disturbing the layers. Catalyst support material should also be loaded into unloading nozzles. Record the actual depth of the support layers.
- 10) While the first 300 mm to 400 mm of catalyst is loaded, the flow of catalyst through the loading tube should be restricted so that a crater is not produced in the top layer of support material from the impact force of the falling catalyst. An effort should be made to limit the free fall of the catalyst to no more than 1 meter.
- 11) After the catalyst and additional alumina layer has been loaded and leveled, the layers of hold-down material should be carefully loaded and leveled on top of the catalyst. Record the actual depth of catalyst and hold-down layers.
- 12) Summarize the loading data on loading diagram such as shown in Figure XII-1 on the next page.

Figure XII-1: MSHP Reactor Loading Diagram

When the loading operation is completed, verify that no extraneous material has been left inside. Check that the vessel outlet flange was not damaged during the loading and that a new and proper class gasket is used. Install the reactor outlet elbow. All the connections leading to the reactors should remain blinded as a precaution against accidental introduction of water or premature introduction of hydrocarbons. Purge the reactors with nitrogen to an acceptable concentration, generally less than 1.0 mol-%, then leave a slight positive pressure.

F. MSHP Catalyst Unloading

After depressuring, hydrocarbon freeing and cooling the MSHP catalyst and reactor the catalyst withdrawal nozzle can be opened. Take care when opening the nozzle as some liquid hydrocarbon can remain in the unloading nozzle and be sitting against the blind flange. After removing the support plate and ceramic rope some ICBs and all of the catalyst will flow from the reactor. After all free-flowing catalyst is removed from the reactor the side manway or hand hole can be opened to assist in pulling the remaining catalyst from the reactor. After confirming the reactor is safe for entry personnel can enter the reactor and assist in final catalyst removal. It is necessary to reload some ICB's that were lost during catalyst unloading.

G. Sand Filter Backwash Procedure

The normal operating criteria for the C-AMS sand filter is approximately 0.1 kg/cm² (1.0 psi) delta P. The screen which supports the sand is designed for a higher pressure of course. However, the sand filter should be backwashed when the delta P reaches 0.2 kg/cm² (3.0 psi). The procedure follows:

- 1) If the C-AMS column is operating at this time, route net organic overhead to the fractionation feed tank. Isolate and block in the C-AMS sand filter.
- 2) Open the C-AMS sand filter top vent connection and drain the aqueous phase to the phenolic waste water closed drain header. The organic phase can be routed to the FCDH using a temporary hose connected to the blank-off connection at the bottom of the vessel.
- 3) Attach a temporary water connection to the backwash connection and fill the sand filter upflow at a rate estimated to be approximately 0.3 m³/hr (1.3 gpm). At first, let the water push the remaining cumene and AMS out to the oily water sewer. When only water is issuing from the outlet of the sand filter, open the overhead vent, and collect the water in a measuring cylinder to adjust the rate (the line to the sewer will have to be blocked in temporarily). When the rate is lined out to the spec, open the water back to the oily water sewer, and flush upflow for 1 hour.
- 4) After one hour, examine the effluent visually for any suspended material (sand dust, etc.) and if it is clear, proceed to step #5. Otherwise, repeat the one hour flush/examine cycle until the effluent is clear.
- 5) Flush upflow for one more hour to the oily water sewer, then stop. Block in the water line and disconnect. Take a lab sample of the water and if it is sufficiently clean, it can be dumped to the open sewer. Otherwise, pressure the water in the sand filter to the sodium phenate tank prior to restart.
- 6) If it is necessary to enter the sand filter, it should be steamed out and made safe according to standard practices.

H. Crude Acetone Column Boil-Out Procedure

The crude acetone column (CAC) reboiler will not normally need to be washed except to prepare for maintenance turnaround. This is due to the recent inventions of adding cumene to the sprung phenol and the use of diamine for neutralization of the sulfuric acid in the crude product. However, upsets can occur which will result in fouling. The symptoms of the salt build-up will be loss of LMTD resulting in inadequate heat input to run the column. Salt deposits can also occur in the bottoms piping to the bottoms pumps. The fractionation section must be shut down for part of a day to wash the column. This wash out is termed "boil out" because the reboiler heat is used to reflux water up the column to aid washing. The following procedure describes the boil out.

- 1) Stop feed to the fractionation section. Put all columns on total reflux. Bypass the IX resin treaters and stop diamine injection.
- 2) Stop heat input to the CAC reboiler. Pump CAC receiver into column to cool and pump CAC bottoms to fractionation return header cooler, FRHC.
- 3) Once column is empty, positively isolate column from relieve header by blocking in PCV "E" assembly block valve and opening drain.
- 4) Open boil out vent valves from the CAC receiver to atmosphere.
- 5) Inventory the CAC with process water of similar quality as used for injection to the CAC during normal operation. Note: there are several ways of doing this,
 - a) Use existing water injection to the blind tray at the bottom of the column, but this flow may be too low to inventory quickly.
 - b) Connect water to the steam out nozzles at the bottom of the column and on the reboiler return.
 - c) Connect water to the steam out nozzle to the CAC receiver and pump water back into the column. While this may take longer than two to get the reboiler inventoried with water, this location has the advantage of washing the trays and downcomers with water.
- 6) Once a water level is established in the bottom of the column, put heat back to the reboiler via steam condensate FIC.
- 7) Heat up column bottoms around 100°C (212°F), temperature will vary depending on bottoms pressure and amount of super heating.
- 8) Reflux water back into the tower if possible and continue flushing with fresh water out the bottoms to sewer. Reflux column for several hours then remove heat from reboiler and drain the column completely.
- 9) Close atmospheric vent while simultaneously adding nitrogen to the column via the PIC. Close PCV "E" assembly drain and open block valves to flare.
- 10) Start column up using normal start-up procedures.

I. Hüls MSHP (AMS Hydrogenation) Shutdown, Water Wash, Dry-out, and Hydrogen Reduction Procedure

This procedure describes the steps required to prepare the hydrogenation unit to remove sodium phenate and sodium sulfate salts that result from carryover from the caustic scrubbing system. Proceed with the full shutdown as follows:

- 1) Reduce the reactor feed slowly. Reduce the feed temperature while reducing the feed rate. Isolate the feed line and close the feed flow control valve.
- 2) Switch the flow of separator bottoms from the oxidation FWC to recycle back to the hydrogenation feed tank.
- 3) Slowly reduce the system pressure to 0.5 kg/cm²(g) (7 psig) above the relief header pressure to avoid back flow.
- 4) Close and isolate the hydrogen supply. Shutdown the make-up gas compressor if one is provided. Purge hydrogen from the reactor section to the product separator with nitrogen until the hydrogen concentration is less than 0.3 mol%.
- 5) Shutdown the recycle pumps and continue nitrogen purging.
- 6) Drain residual hydrocarbon to the FCDH. When no hydrocarbon liquid remains in the reactor loop, pump the separator level down to a minimum. Maintain a liquid seal in the separator to prevent sending vapor to the product pumps and causing cavitation. When a minimum level is reached, isolate the separator from the product pumps.
- 7) Drain the remaining product separator contents to the FCDH and isolate the product separator level control valve.
- 8) Depressure the unit through the separator to the relief header down to a pressure of 0.2 kg/cm²g (3 psig).
- 9) Steam out the separator and then blanket with nitrogen. Optionally, the separator can be purged with nitrogen.
- 10) Purge reactor section with nitrogen and drain hydrocarbon from all low points.
- 11) Set reactor pressure at 2 kg/cm²(g) (30 psig).
- 12) When hydrocarbon and hydrogen are removed to acceptable concentrations, add de-mineralized water or cold clean condensate to the bottom of reactor #1 and fill at a superficial velocity of 6 m/hr (20 ft/hr). The fill rate is based on the empty cross sectional area of the reactor and the velocity. Vent nitrogen from the top of the reactor.
- 13) When a level appears in the separator, start the hydrogenation recycle pump. Remember to throttle the flow to avoid overloading the pump motor.
- 14) Circulate water in reactor #1 and vent the top of the reactor until the system is full of water.
- 15) Drain water from the separator to sewer.
- 16) Continue make-up water until drain water has a similar sodium concentration as the make-up water.
- 17) Stop water and drain the system of water.

- 18) Use nitrogen to purge the system of free water. Start flow at 1.6 Standard Cubic Feet (SCF) per pound catalyst ($80 \text{ Nm}^3/\text{hr}$ per m^3 catalyst) by the start-up line to the reactor recycle cooler. Vent nitrogen to relief header.
- 19) Add steam to the cooler and increase the reactor bed temperatures to a minimum 120°C .
- 20) Purge reactors until the Draeger tube (detector tube) reaches 100 mol ppm water.
- 21) Switch nitrogen gas to hydrogen gas and set the pressure controller to maintain $3.9 \text{ kg/cm}^2\text{g}$ (55 psig).
- 22) Begin reduction of the catalyst at 120°C at 1.6 SCF per pound catalyst ($80 \text{ Nm}^3/\text{hr}$ per m^3 catalyst). Add nitrogen, as required, if catalyst temperatures start to fall due to heat losses in the piping and reactor.
- 23) When hydrogenation reactor #2 outlet reaches 120°C (250°F), begin a 2 hour hold. The water produced should be drained at the separator.
- 24) After the 2-hour period, cool reactor to 60°C (140°F) to ready the reactor for feed. If feed is not available, then cut out hydrogen and maintain system pressure after cooling to ambient temperature.

J. Routine Maintenance

1. Oxidation

- 1) The oxidizer coolers have cooling water control, which will limit the flow compared to the design case. These exchangers require periodic cleaning on the cooling water side or change water source to tempered water.
- 2) The oxidizer filters will require periodic changing as they become plugged. The feed filters plug less often compared to the PFC feed filters.
- 3) If caustic is injected into the compressed air scrubber, there is the possibility that some of the caustic will entrain into the air lines to the oxidizers and then dry out. While this is not a corrosion problem, the sticky residue can come off and go into the oxidizer. Turnaround checking of the inside of the air line to the oxidizer is recommended.
- 4) All check valves in CHP service should be periodically checked to ensure they are working.
- 5) The charcoal in the charcoal adsorbers will maintain 99% efficiency for about 1 year. The charcoal replacement is dependent on local regulation. Charcoal dust can find its way into the decanter and should be checked during turnarounds.
- 6) The large butterfly valves around the charcoal adsorbers may require periodic maintenance to the bearings (e.g. every 5 years). Check with the manufacturer.
- 7) If charcoal adsorbers have screens they should be checked to prevent fines from clogging downstream pumps, etc.
- 8) Charcoal adsorber condensate piping from regeneration should be checked for corrosion.

2. Concentration/Decomposition/Neutralization

- 1) The decomposer feed check valves and the feed valve system should be inspected periodically to ensure it is in good working order.
- 2) The decomposer cooler has regulated cooling water supply and may foul at an increased rate compared to the other cooling water exchangers.
- 3) The ejectors in the vacuum system may require periodic replacement due to erosion of the nozzle by steam.
- 4) The preflash and flash column structured packing should be periodically inspected to ensure they are not damaged or fouled. The low operating pressure can create high vapor velocities damaging the packing. Salts from leaking cooling water from the condenser can foul the packing.
- 5) The dehydrators and calorimeters foul on a regular basis and must be cleaned to maintain production and safety. The foulant from the calorimeters has very little effect on the fouling in the decomposer circulation loop. However, the fouling from the dehydrators can go downstream if it is severe. Periodic inspection of the dehydrator and downstream piping is recommended. The easiest way of inspecting the piping is by removing the expansion piping and inspecting it.
 - a) UOP recommends procuring a Goodway (www.goodway.com) heat exchanger tube cleaner to sufficiently remove the residue buildup in the dehydrator tubes for routine maintenance. The Goodway cleaning machine utilizes a rotating steel head in combination with high pressure water to remove foulants. For more details about suitable cleaning equipment, contact a Goodway heat exchanger tube cleaning representative or your UOP technical service specialist.
- 6) The static mixer in the neutralization section has the potential of fouling due to either e) above or salt from the PRU.
- 7) Preflash and flash column overhead condensers should be pressure checked at each turnaround for tube leaks.

3. Fractionation

- 1) The crude acetone column can foul at the reboiler if there is carryover of salts from the PRU. One customer reported tray vibration damage at high rates (more than design) in the middle of the column. Corrosion is not normally observed.
- 2) The finished acetone column can produce a foulant that lays down in the low flow areas of the trays. Periodic “bumping” or start-up of the column can cause this material to drop to the bottoms suddenly. High caustic rates and concentration can also produce extra foulant. The foulant can plug up the anthracite coal in the bottoms drum if it is severe.

- 3) There are no reports of problems with the FAC stab-in reboiler, but this should be periodically checked for fouling and corrosion.
- 4) The Cumene-AMS column removes water using the boot in the overhead receiver. Organic acids such as formic and oxalic can concentrate in the water as well. Corrosion is possible in the overhead lines where water can stagnate.
- 5) The crude phenol column overhead condenser can foul on the FAC side if the FAC bottoms filters are overwhelmed and bypassed or if vaporization is excessive due to low FAC bottoms flow. It is possible that some foulant could form in the exchanger as well. Periodic checking of the condenser duty will indicate fouling is occurring. The exchanger should be cleaned when this is observed. It is possible that caustic or chlorides from the caustic can concentrate in a fouled tube and corrode the stainless steel tube.
- 6) The residue stripper reboiler can foul due to the heavies in the system and any inorganic material that ends up in the heavies. While inorganic materials such as salts are not normally found, upsets can occur that cause them to end up here. The reboiler can foul. The reboiler should be checked periodically and cleaned.
- 7) The ion exchange resin treaters' inlet and outlet distributors should be inspected when the resin is changed. The critical isolation valves that would result in potential bypassing should periodically inspected to ensure good working order. The associated filters should also be checked periodically to ensure they are working.
- 8) The phenol rectifier does not have any fouling or corrosion issues, as it is the area where the most pure phenol is produced. Periodic inspection of the phenol rectifier bottoms to check for resin particles should be performed.

4. Phenol Recovery Unit

- 1) The delay pot inlet has the highest potential for corrosion in the unit. The lines are Teflon lined to prevent corrosion. The delay pot and cooler should be inspected periodically for signs of corrosion.
- 2) The PRU separator has the potential for the precipitation of salts if any nozzles or attached instruments get cold (<35°C). The tracing and insulation should be checked periodically.
- 3) The OEC, SCWC and C-AMSCWC all use jetting of organic streams through small holes on trays to be effective. These holes can wear over time and must be inspected periodically to ensure they are at the proper size.
- 4) Acid is mixed into the feed to the OEC and has the potential of corroding the OEC mix pot and associated lines. Periodic inspection is warranted.
- 5) The sand in the sand filter will be attacked by the entrained caustic over time. The caustic causes the surface of the sand to smooth and reduce its effectiveness in separating entrained water droplets. This sand should be replaced routinely (every year).

5. Hüls MSHP (AMS Hydrogenation)

- 1) There is the potential for some fouling in the feed lines to MSHP reactor #1. These lines should be inspected as well as the inlet of the reactor. A black foulant has been reported. It is suspected that air has contaminated the MSHP feed. Oxygen and AMS will react with heat to form a gum.
- 2) The MSHP feed tank should be checked periodically for water that may entrain from the PRU.
- 3) There have been no reports of corrosion in this area.

6. General

- 1) Inspect floating roof seals, tank vapor scrubbers, pressure/vacuum safety valve
- 2) Check steam traps for blow through
- 3) Check heat tracing for effectiveness
- 4) Sumps and drains - make sure they are protected from debris, rust and flow freely

XIII. Safety

The process unit and its operation may present hazards to the operators and the environment. This section outlines some of the proper controls expected to be implemented. Since the use of UOP products by others is beyond UOP control, no guarantee, expressed or implied, is made and no responsibility assumed for the use of this material or the results obtained therefrom. Moreover, the recommendations contained in this manual are not to be construed as a license to operate under, or a requirement to infringe, any existing patents, nor should they be confused with state, municipal or insurance requirements, or with national safety codes.

The owner is responsible for establishing and following all pertinent safety procedures for the start-up, shut down, operation and maintenance of this process unit. As UOP does not perform the final design work for the unit, it is expected that the owner will perform a process hazard analysis or similar investigation of the final process design.

A. UOP Provided Information

This process unit may contain hazards due to chemicals and materials including flammable hydrocarbons, hydrogen and other materials. A listing of the chemicals and materials present and recommended safety equipment can be found in the UOP Engineering Project Specification 903, "Safety Equipment" issued for this process unit.

Safety Data Sheets (SDS) for UOP supplied catalysts and adsorbents may be obtained by UOP at www.uop.com. For the SDS of all other catalysts and chemicals used in this process, contact the vendor.

The General Operating Manual provided by UOP is not considered sufficient operational guidance and the owner must develop operating instructions for the plant.

For safety procedures specific to the technologies presented in this manual, if applicable, please refer to Section XII Special Procedures.

B. General Elements of Process Safety Management

The owner is expected to develop and establish policy and procedures for the following:

- a. Operating procedures
- b. Employee training related to the process and its operation

- c. Contractors
- d. Mechanical integrity of equipment
- e. Preparing equipment or piping for introduction of hazardous materials
- f. Removal of hazardous materials from equipment or piping in preparation for inspection or maintenance of said equipment or piping.
- g. Line Breaking Procedures
- h. Confined Space Entry
- i. Energy control (Lock-out and Tag-out of Equipment)
- j. Fall Protection for Personnel.
- k. Hot Work Procedures

Operating procedures presented in this manual are general in nature. The steps required to start-up, operate, and shut down the unit will broadly apply to units for which this manual applies. Stepwise procedures that identify specific equipment are to be developed by the owner. Operating procedures may be reviewed with the UOP representative present for start-up and vendor representatives present for commissioning of specific equipment. The owner of the unit may also implement procedures and policies that govern the modification of or deviations from the normal operating procedures.

The owner will manage the training of the personnel and contractors who enter and operate the plant.

A regular plan of inspection and testing will be required to maintain the physical integrity of the equipment of the unit. Process equipment has an expected design life which can be altered by process and other environmental conditions. Procedures and practices should monitor the process equipment to determine if it is fit for service.

Many materials can be considered toxic or hazardous. Procedures and policy must be implemented to insure the safe handling of the materials used in the plant.

Confined spaces are a recognized hazard in most process units. Procedures for access control, entry, and work in confined spaces are implemented by the owner to protect personnel from hazardous encountered in confined spaces.

Energized equipment is also a recognized hazard in most operating units as unexpected operation of equipment may cause injury. The control of equipment and sources of other energy will need to be managed for the purposes of maintenance and inspection.

Falls constitute a hazard in most units when work is performed outside of the normal platforms and walk ways designed to be occupied by workers. Policy to reduce injuries as a result of falls is normally a part of the safety management system of the unit.

As most units contain flammable materials, a system to reduce accidental ignition of flammable materials is implemented. Such systems are commonly referred to as “hot work” systems to protect against ignition of flammable materials by open flames and other ignition sources.

Additional information regarding certain industrial hazards is available at the below website:

www.osha.gov – The website for the United States Department of Labor Occupational Safety and Health Administration.

www.cdc.gov/niosh – The website for the Centers for Disease Control and Prevention National Institute for Occupational Safety and Health.

C. Technology Specific Potential Hazards

In addition to the 903 project specification in the Schedule A from UOP which applies to this process unit, please refer to the information below.

1. Application and Scope

This section concerning safety procedures has been prepared to acquaint all personnel engaged in the operation and maintenance of a UOP Phenol Process with the safety procedures and regulations, which have been developed to insure a safe, operable unit.

Phenol is a hazardous chemical with properties peculiar to itself which call for specific handling and treatment. CHP must be treated as phenol because it may cleave to phenol in the presence of acids, which are present on the body.

With improper handling, phenol can be lethal.

Accordingly, it is essential that these Safety Standards be thoroughly understood and strictly followed. Under these conditions, the use of phenol will be no more hazardous than that of common hydrocarbons.

Before working in the UOP Phenol Process, all operators and mechanics must be familiar with the contents of this section and must know the location and use of all safety equipment on the unit. This safety equipment includes protective clothing, breathing equipment, first aid kits, eye washes, and emergency showers, and firefighting apparatus, containers of body wash solvent and water hoses.

2. Safety Procedures

All personnel working in the UOP Phenol Process must be instructed in the nature of CHP and phenolic materials so they will know how to recognize it, how it will affect them, how to use the safety equipment, and what to do if they encounter it.

All personnel working or coming within the battery limits are required to wear protective clothing as prescribed by the supervisor for the unit. Typical protective clothing is sealed vinyl type of overalls, overcoats and gloves with face shield and helmet. Other resistant materials can also be used. When identifying suitable materials they should also be resistant to the other major chemicals in the plant, cumene, acetone, benzene and phenolic residue.

No persons are allowed within the battery limits without the knowledge of the UOP Phenol Process supervisor.

A Phenolic Safety Work Permit is required before any maintenance work can be done. This permit lists personnel assigned and the specific work to be done on designated equipment. The phenol process supervisor must certify in the permit that all sources of pressure are blocked and blinded, that equipment to be worked on has been completely vented and flushed (by phenol unit personnel), the type of protective clothing which must be worn, and what special precautions are to be taken. The supervisor of the personnel assigned to do the work must also certify in the permit that a conference with the phenol unit supervisor and refinery safety supervisor on the hazards peculiar to the specific job has been completed, indicating what is required of the workers, and that the two are in agreement on how to go about the work; also that the crew was thoroughly briefed on the hazards. The supervisor must report if anyone contacted CHP or phenolic material, whether or not the job site was cleaned up afterwards; whether tools were washed before returning them to storage; and that any equipment removed from the site was washed. Finally, the maintenance supervisor is asked if they had a brief discussion with the phenol unit supervisor on how the job could be done in a safer manner. The last line in the permit states in large type "Always make initial opening of any equipment as though it was under pressure."

The use of impact wrenches is not permitted. This type of wrench can throw CHP or phenolic material a considerable distance.

When working on equipment an upwind position should be assumed if possible.

Initial equipment openings should be made carefully as if the equipment was under pressure and contained CHP or phenolic material and hydrocarbon.

Safety showers, eye wash stations, and breathing air supply will be tested daily.

A running water hose will always be available for flushing.

Neutralization or "body wash" solutions will be tested on a scheduled basis by the operator.

All assigned tools in the UOP Phenol Process must be painted with a distinctive paint. The operating personnel shall see that no tool leaves the unit.

All tools and equipment must be thoroughly washed off and returned to the tool rack at the Change House after each use.

Porous materials such as rags, wood, pipe coverings and packings, which could have been exposed to CHP, and phenolic material, should be placed in specially painted disposal cans and marked "PHENOLIC TRASH." Wood staging and other use of wood on the unit should be kept to a minimum. Metal staging must be washed off before being removed from the unit.

Open lines are not to be permitted on the UOP Phenol Process. All open lines must be blinded or plugged.

Following is the procedure for preparing pumps for maintenance:

- 1) For pumps with seal flushing, close the discharge valve and flush out pump through the suction line, using the seal flushing connection while closing the suction valve.
- 2) Crack open discharge valve slightly and flush out the discharge line for five minutes. Close the pump discharge valve.
- 3) Block in flush. For all CHP or phenolic pumps connect a flexible hose to the sump header from the pump. Depressure to the header. Drain pump to the closed drain header.
- 4) Attach steam, water, or nitrogen to a second hose connection and flush pump through the drain valve downstream of the pump discharge check valve to the drain header for 5 minutes.
- 5) Disconnect the hose and open the vent to allow air through the pump.
- 6) Blind off the pump.
- 7) If possible, draw air through the pump during maintenance work.

All safety guards, line braces and other safety devices must be kept in place and in operable condition when the equipment is in service or contains CHP or phenolic material. If it is necessary to remove a safety device to make repairs, it must not be removed before the equipment is safe to open and must be replaced again when the repairs are complete and before the equipment is returned to service.

Firefighting on the UOP Phenol Process should be handled in the same manner as on any other unit. Firefighting personnel will approach the fire from upwind wearing the necessary protective clothing. Several water monitors will be situated around

the perimeter of the unit, permitting every spot on the unit to be reached by one of the monitors.

The unloading area will be appropriately marked. Safety devices in the form of wheel chocks will be used to prevent the truck from moving, and the necessary safety equipment will be close by. Once unloading is complete, all tools and equipment should be washed-off.

CHP and phenol material spills will be hosed down immediately.

In the event of a CHP or phenolic material leak, supervisory personnel will be immediately notified. No one will approach the leak alone. Warning signs are to be posted immediately. Steps to stop the leak will immediately be initiated. In case of an excessive CHP or phenolic material leak which cannot be blocked in and vented from outside the tank area, two persons should immediately be dressed in appropriate phenol protection PPE. The leak should be approached from the upwind side and the equipment isolated and vented.

All shutdowns will be safely planned and a pre-shutdown safety meeting held with all participating personnel present.

Even though the UOP Phenol Process has been completely flushed, extreme care will still be followed to guard against burns from CHP or phenolic material trapped in scale, joints, etc. The protective equipment prescribed for the job must be worn.

Laboratory samples will be taken by an operator accompanied by a second operator. If special techniques are required a laboratory specialist can assist. Class "C" clothing will be worn for taking CHP and phenolic containing samples. Class "B" clothing will be worn for taking all other samples. Samples will not be taken into the control room. Samples will be taken only in the proper sample container. Samples will be transported to the laboratory in a manner prescribed by the chief chemists.

Maintenance personnel other than supervisory are not to enter the control room unless their work requires that they enter.

Operating personnel are responsible for making equipment safe for maintenance.

Before maintenance work is performed on any piece of equipment, that piece of equipment must be completely flushed. With the exception of pumps and instruments, this flushing must be done completely within the UOP Phenol Unit battery limits before the equipment is removed from the unit.

When the UOP Phenol Process is running, the amount of maintenance work done within battery limits should be minimized. If possible, pieces of equipment should be flushed and taken from the unit for required work.

Instrument specialists and machinists may wear protective clothing outside the unit if and only if they are taking a contaminated instrument or pump from the unit. The exterior surfaces of any such pumps or instruments should be flushed before removal from the unit. Pumps and instruments removed from the unit must be tagged with a tag marked "Contaminated with Phenol." The instrument and machinist shops will have facilities for flushing pumps and instruments as they are disassembled.

Arrangements must be made with a nearby clinic to have phenol-burn treatment medicines on hand at all times. The doctors there should be aware of the latest treatment methods and arrangements are to be made for immediate treatment of patients. The arrangement also should ensure that the doctors have been notified of the type of treatment that the patient will receive at the plant before being sent to the clinic.

Each person in operations should have a supply of body solvent in his own home. This reduces natural concern over a burn, which might not be noticed while at the plant, and increases the confidence of the person. A 4-ounce polyethylene container with spout application containing 70% polyethylene glycol 300 and 30% Methylated Spirit (V/V) should be maintained in his home with proper precautions for keeping it out of the reach of children.

3. Emergency Booths and Related Items

3.1 Emergency Shower

This shower sprays water from all directions and is activated by stepping onto a treadle. Its purpose is to flush any CHP or phenolic materials off an individual or their protective clothing.

3.2 Emergency Face and Eye Wash

The face and eye wash fountain directs a small stream of water upwards at an angle to the face and eyes. It is located about one meter above the floor and the wash unit is activated either by a lever or treadle. Both shower, face and eye wash are to be checked daily.

3.3 Alarms

There should be an alarm provided to indicate to other unit personnel when the emergency shower or eye wash is used to assure early assistance.

3.4 First Aid Cabinet

This cabinet contains some of the initial first aid supplies which will be required in the treatment of any injuries, i.e., blankets, eye wash solution, etc.

3.4.1 Eye Wash Solution

A 2.5 wt% Boric Acid and 2.5 wt% Sodium Borate solution is recommended with lint free medical swabs for application.

3.4.2 Body Wash Solvent

A 70% polyethylene glycol 300 (or 400) and 30% Methylated Spirit (V/V) is to be made available at strategic locations in 3 gallon bowls with sponges for application.

4. Toxicity and Related Safety Information

In general, operating and laboratory personnel involved in obtaining samples should wear chemical-type safety goggles and shield, solvent-resistant gloves, and approved respiratory protective equipment where ambient concentration exceeds allowable limits. This protective equipment is not, however, a substitute for safe working conditions, proper ventilation, good personal practices, and safety equipment. In all cases, skin contact (especially eyes) and inhalation must be avoided. Sampling guidelines are included in Analytical Section VIII.

While it is unequivocally the responsibility of the refiner and the local authorities to maintain adherence to the appropriate national and local safety standards, UOP suggests the following outline as a starting point for the refinery to build their own optimal safety program for dealing with aromatics:

- 1) All employees newly assigned to work dealing with aromatics, either in the plant or the laboratory, should be subject to a thorough and complete examination, such examination to include a past medical history, a complete physical examination, and particularly, a complete blood examination consisting of a red cell count, white cell count, hemoglobin determination, differential count obtained from a smear, and an accurate platelet determination. This complete blood examination should be repeated periodically at least every thirty days, and if the employee shows any changes in any of these elements, he or she should be reassigned to other types of work where there will be no exposure to aromatics. If any employee works in this environment over a year without any blood element changes, the examination could be carried out every ninety days.
- 2) All employees, new and current, should be alerted as to the early signs and symptoms of excessive exposure to the hazardous materials in use, and all workers should report such symptoms to the medical department.
- 3) Several additional points must be stressed regarding safety in the UOP Phenol Process:

- a) Appropriate clothing, depending on the type of work undertaken, is to be worn at all times. No contact lenses are to be worn in the plant area.
- b) The plant's emergency rescue procedures and the locations of the rescue equipment must be known before the need arises.
- c) In case of an emergency where a victim has been overcome, move the affected person away from the hazardous exposure. Notify someone else and put established emergency procedures into effect. Do not become a casualty.

The following is a brief information on some of the chemicals frequently encountered in the UOP Phenol Process. This information contains the key physical properties and emergency actions associated with each chemical. It should not, however, be used as a substitute for thoroughly studying the more detailed materials included at the end of this subsection.

4.1 Phenol

At ambient conditions, phenol ranges from a colorless to pink solid, to a thick liquid with a characteristic sweet, tarry odor. It has a freezing point of 40.9°C. In the vapor state, phenol forms a highly inflammable mixture with air at concentrations ranging from 1.7 to 8.6 vol%. Phenol is soluble in water to 8.4 g/100 g water at 20°C (60°F).

Phenol is highly reactive with strong oxidizers, and may cause fires and explosions. Toxic gases such as carbon monoxide may be released in a fire involving phenol. Fires are best put out with alcohol foam, carbon dioxide, or dry chemical. Liquid phenol will attack some forms of plastics, rubber, and coatings. Hot liquid phenol will attack aluminum, magnesium, lead, and zinc metals. Sealed vinyl clothing is resistant to phenol adsorption and attack.

Phenol odor is detectable at very low limits. Do not let this fact put anyone off guard, however, since phenol is an extremely hazardous substance. The TLV-TWA(1) is 8 ppm, while the TLV-STEL(2) is 10 ppm.

In general, phenol in the vapor form or in solution is an irritant to the eyes, mucous membranes, and skin. Systemic absorption causes central nervous system effects as well as liver and kidney damage. Sudden collapse is characteristic of gross overexposure. Phenol solutions have a marked corrosive action on any tissue on contact. On skin, there is local anesthesia, a white discoloration and the area may subsequently become gangrenous. Severe dermatitis will result from contact with dilute solutions.

4.2 Cumene

At ambient conditions, cumene is a colorless liquid with a sharp, penetrating, aromatic odor. In the vapor state, cumene forms an inflammable mixture with air in concentrations between 0.9 and 6.5 vol% cumene.

Cumene is highly reactive with strong oxidizers, and may cause fires and explosions. Toxic gases such as carbon monoxide may be released in a fire involving cumene. Fires are best treated with water spray, dry chemical, foam, or carbon dioxide.

Cumene odor is detectable at very low limits, generally less than 3 ppm. Since the odor threshold of cumene is below the permissible exposure limit, cumene is treated as a material with good warning properties. As with phenol, however, cumene is a very hazardous chemical and should be treated as such.

Short term overexposure can cause dizziness, drowsiness, loss of coordination, and unconsciousness. Long-term overexposure can also cause skin rash.

4.3 Acetone

At ambient conditions, acetone is a colorless liquid with a sweet odor. In the vapor phase it forms inflammable mixtures with air in the 2.6 to 12.8 vol% range. It is highly soluble in water.

Acetone is not highly reactive with common materials. Should an acetone-related fire occur, however, alcohol foam, dry chemical or carbon dioxide should be used to put it out. A stream of water should not be used, as it will scatter and spread the fire.

The odor threshold of acetone is about 100 ppm. Acetone vapor can be irritating to the eyes, nose, and throat. If inhaled in large doses it can cause difficult breathing, anesthesia, and loss of consciousness. In the liquid state, acetone is irritating to the eyes and mucous membranes but not necessarily to the skin, as it evaporates quickly off the skin's surface. Repeated exposure to the skin, however, can cause defatting and possible dermatitis.

4.4 Sodium Hydroxide (Caustic Soda)

In the UOP Phenol Process, sodium hydroxide (NaOH) is utilized in a 45-48 wt% solution as its "raw" form, and diluted to 20 wt% and lower solutions for different purposes. All solutions are clear, odorless, colorless liquids of different viscosities. In its solid form, NaOH is a white, odorless material. NaOH does not form flammable mixtures with air.

Sodium hydroxide solutions are reactive with acids and organic halogen compounds. More concentrated solutions produce a greater probability of fires and explosions. As sodium hydroxide itself is not combustible, fires should be treated

according to the combustible material(s) involved. Contact with metals such as aluminum, tin, and zinc causes formation of flammable hydrogen gas. Contact with nitro compounds causes formation of shock-sensitive salts.

Since caustic solutions are odorless, yet can be very hazardous, they are considered as having poor warning properties. Standing solutions of any concentration are therefore very hazardous unless properly labeled, as they are not discernible as caustic by sight either.

Sodium hydroxide is a strong alkali. The mist, dust, and solutions of this compound can cause severe injury to the eyes, mucous membranes and skin. Although inhalation is usually of secondary importance in industrial exposures, the effect from dust or mist will vary from mild irritation of the nose to severe pneumonia, depending on the severity of the exposure. The greatest industrial hazard is rapid tissue destruction of eyes or skin upon contact with either the solid or concentrated caustic solutions. Extreme eye damage and severe skin burns and ulceration can easily result. Skin irritation is felt within minutes for concentrated solutions and within hours for mild solution of approximately 4 wt%. These are not considered as sufficient warning signs and any suspected caustic contact should be treated immediately. Ingestion produces severe pain in the esophagus and stomach, corrosion of the lips, mouth, tongue and pharynx, and severe vomiting.

4.5 Sulfuric Acid

Sulfuric acid (H_2SO_4) is used in the UOP Phenol Process in highly concentrated form, a 98 wt% aqueous solution. At ambient conditions this material is an oily, viscous, light brown, fuming liquid. The odor is quite acrid. Sulfuric acid does not form combustible mixtures with air. In its solid form, highly concentrated sulfuric acid is also a brown, odorless material.

Concentrated sulfuric acid is highly reactive. Contact with organic materials may cause fires and explosions. Although it is not combustible by itself, sulfuric acid is also capable of igniting finely divided combustible materials on contact. Fires involving small amounts of combustibles may be smothered with dry chemical. Alcohol foam or carbon dioxide can be used with larger fires. Water applied directly to sulfuric acid causes evolution of heat and spattering.

Fuming sulfuric acid has an acrid odor and is irritating to the eyes and throat. Highly concentrated sulfuric acid is not considered to have good warning properties, as its fumes are generally hazardous, even in the low range of human delectability.

Sulfuric acid mist or vapor severely irritates the eyes, respiratory tract, and skin. Concentrated sulfuric acid destroys human tissue due to its severe dehydrating

action, whereas the dilute form acts as a milder irritant due to its acid properties. Splashed in the eye, concentrated acid can cause extremely severe damage.

Ingestion of the liquid can produce serious mucous membrane burns of the mouth and esophagus. As with exposure to the vapor as mist, erosion and discoloration of the teeth can also occur.